

A day without new knowledge is a lost day.

- nologin.in/1
- https://codeshare.io/saleel

Database Technologies - MySQL

In this module we are going to learn SQL, PL/SQL and NoSQL(MongoDB)

- `sudo apt install build-essential`

MySQL is case-insensitive

SQL keywords → Case-insensitive

Table names → Depends on operating system

Database/schema names → can also be OS-dependent

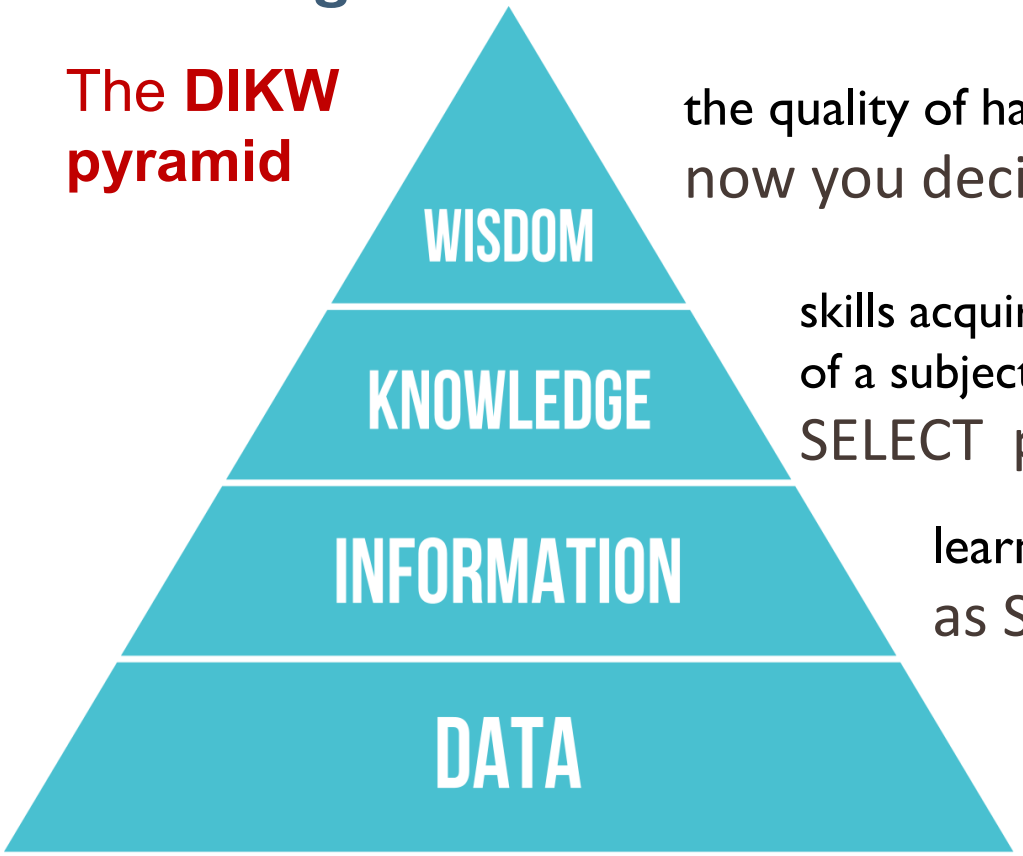
Column names → Case-insensitive

String values → Case-insensitive / Depends on collation (Collation defines how text (characters) are compared and sorted in a database.)

If A and a, B and b, ... are treated in the same way then it is case-insensitive.
MySQL is case-insensitive

DIKW Model describes how the data can be processed and transformed into **information**, **knowledge**, and **wisdom**.

The **DIKW pyramid**



the quality of having experience, knowledge, and good judgement.
now you decide when to use and how to use(**wisdom**)

skills acquired through education, it can be theoretical or practical understanding of a subject.
SELECT potential (**knowledge**)

learned about something or someone.
as SELECT (symbol: σ) (**information**)

Sigma(σ) (**data**)

Let's take another example.

wisdom

- Our employees Ben, Jo, and Tom are young.
- Our employees Ben, Jo, and Tom are in 20's
- Toy is going to retire next year.
- ….

knowledge

Age of employees:

R = { Ben is 25 yrs. Old, Jo is 29 yrs. Old, Kim is 45 yrs. Old, Tom is 23 yrs. Old, Toy is 60 yrs. Old , …. }

Information

Now we arrange the data and make some sense and create information.

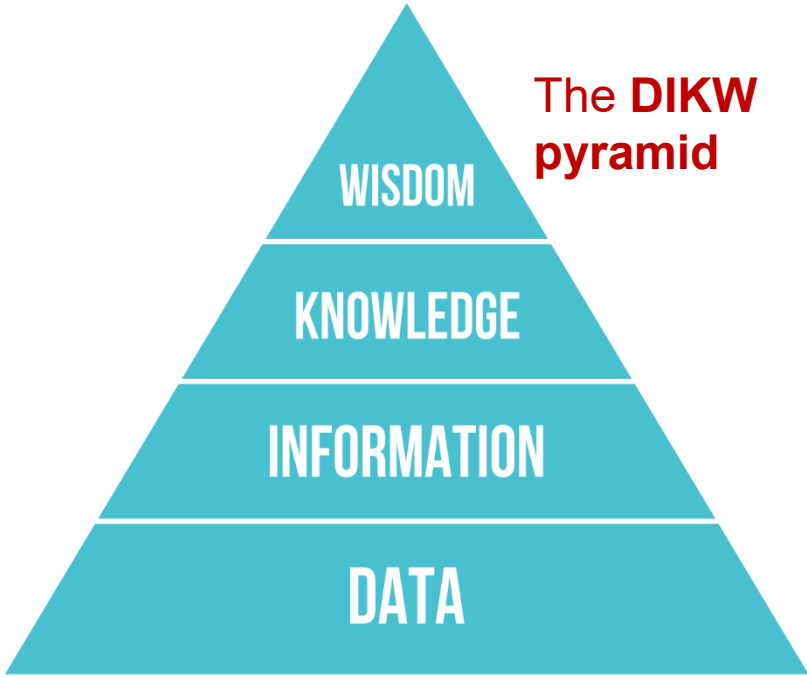
set A is **Age**, set B is **Name**

Age – { 25, 29, 45, 23, 60, 51, 35, …. }
Name – { Ben, Jo, Kim, Tom, Toy, Sam, Don, …. }

data

set A – { 25, 29, 45, 23, 60, 51, 35, …. }

set B – { Ben, Jo, Kim, Tom, Toy, Sam, Don, …. }



The **DIKW pyramid**

Class Room

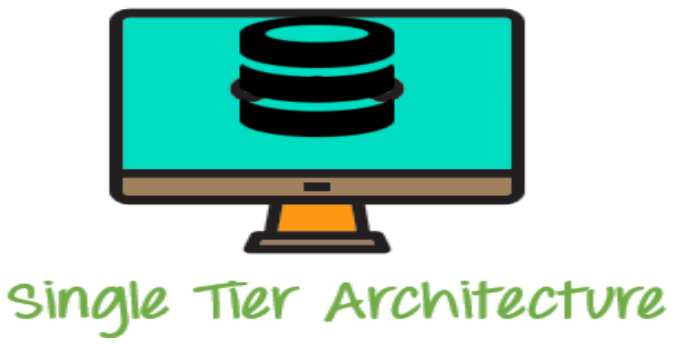
Session 1

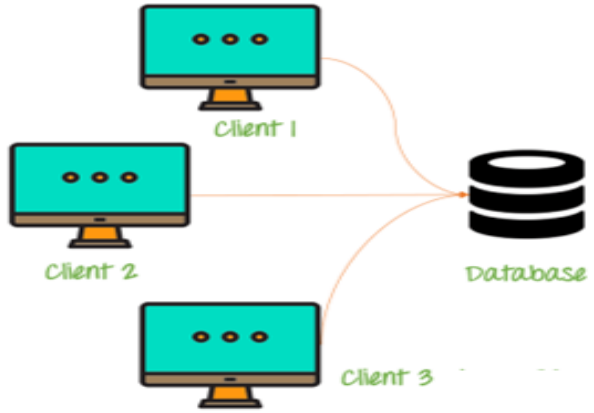
Introduction

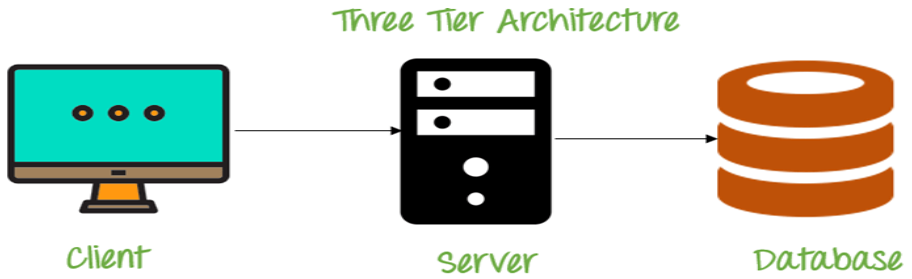
- If anyone who wants to develop a good application
  then he should have the knowledge three major components.

They are . . . . .

- Presentation Layer [**UI**]
- Application Layer [**Server Application** and **Client Application**]
  1. A Server Application is a software application that runs on a server and provides services to other applications.
  2. A Client Application is a software application that runs on the user's device and allows the user to interact with a server or service.
- Data Layer [Data Access Object (DAO) / Data Access Layer (DAL)] { Flat Files | RDBMS | NoSQL }







Types of Database Architecture

Single-Tier Architecture

1. The database and application (UI, Business Logic, and Data are all combined) reside on the same system.
2. No network communication is required since everything runs locally.
3. Used for small-scale applications.

Two-Tier Architecture (Client-Server)

1. The (client) application (Frontend/UI + Business Logic) communicates with the database server.
2. The client sends queries, and the server processes them and returns results.
3. Used in medium-scale applications.

Three-Tier Architecture

1. Introduces a middle layer (Application Server) between the client and database.
2. The middle layer handles business logic, security, and processing before accessing the database.
3. Used in large-scale web applications.

Introduction

- A layer refers to pieces of software that are logically separated, but typically live within the same process and machine.
- A tier, instead, refers to pieces of software that live in distinct processes or AppDomains or machines.
- A tier refers to physical separation; a layer is about logical separation.

Introduction

Why do we need databases (Use Case)?

We **need databases** because they organize data in a manner which allows us to store, query, sort, and manipulate data in various ways. Databases allow us to do all this things.

Many companies collects data from different resource (like Weather data, Geographical data, Finance data, Scientific data, Transport data, Cultural data (the ideas, customs, and social behaviour of a particular people or society), etc.)

role of database engineer

- Database Engineer is also known as software application developer or computer software engineer.
- Database Engineer is responsible for developing the database.
- Database Engineer handles the physical and logical models of a Database.
- Database Engineer identifies and handles the errors in a database system.
- Database Engineer creates the logical Design of the database.
- Database Engineers are not concerned much with the end users.
- The Database Engineer designs the database system to cater to the needs of the users.
- The Database Engineer guides the Database Administrator for the effective management of the databases.
- Responsible for designing, developing, and testing the database
- Implements new database features and upgrades.
- Collaborates with other teams to integrate database with other applications and systems.
- Works with developers to ensure database design meets application requirements.

Introduction

4 Important Roles of Database in Industry.

- It is needed for data access within the company.
- It is needed to maintain strong relationships between data.
- This system allows newer(latest) and better updates.
- It helps to search data in a better manner.

- **A foreign key constraint is also known as a referential constraint or referential integrity constraint.**
- A foreign key is a column or group of columns in a relational database table that establishes a link between data in two tables.
- It references a ***primary key*** in another table and can **cascade changes** or **delete** related data if the ***primary key is updated or deleted.***

What is Relation and Relationship?

Reference / Referential key

Remember:

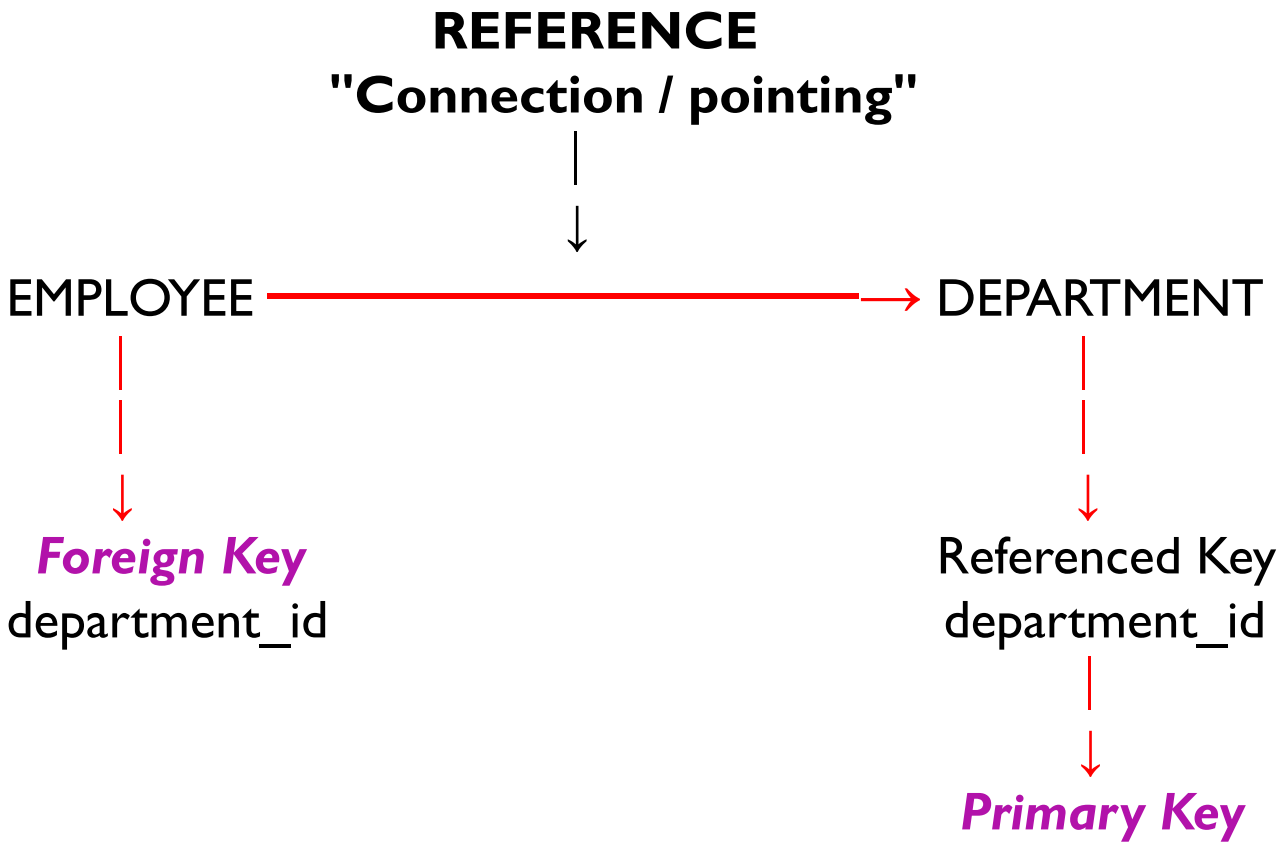
- A ***reference*** is a relationship between two tables where the values in one table refer to the values in another table. This is usually enforced using a ***foreign key*** constraint to maintain referential integrity.
  - ***a) The referencing column is called the Foreign Key. (***A referencing column is a column in a child table that refers to a primary key or unique key in another table.***)***
  - ***b) The referenced column is usually the Primary Key of the parent table.*** *(*A referenced column is the column in the parent table that is referred to by a foreign key in another table.*)*

- A ***referential key*** is a column or set of columns in a table that refers to the ***primary key*** of another table. It establishes a relationship between two tables, where one table is called the parent table, and the other is called the child table.

reference / referential key

To understand the difference is to think as

1. ***"reference"*** as the relationship and
2. ***"referential key"*** as the key used to create that relationship.



employee.department_id

REFERENCES

department.department_id

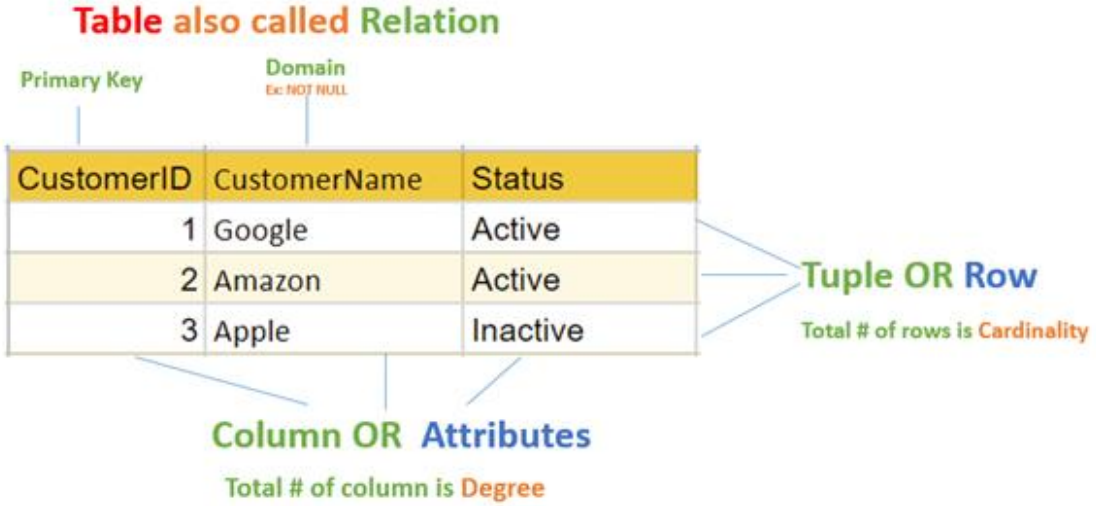
relation and relationship?

Relation ***(in Relational Algebra "R" stands for relation)*****:** In Database,

- a relation represents a **table** or an **entity** than contain attributes.
- In Relational Algebra, a relation is a table with rows and columns, just like in a RDBMS.
- It represents a set of tuples (records) that share the same structure. Relation is a Logical Instantiation/Model of a TABLE.

Relationship: In database, relationship is that how the two entities are **connected** to each other, i.e. what kind of ***relationship type*** they hold between them.

Primary/Foreign key is used to specify this relationship.



Remember:

Foreign Key is also known as

- referential constraint
- referential integrity constraint. (Referential Integrity is a constraint (rule) in a relational database that ensures the relationship between two tables remains consistent.)

Note:

- **Table -** The physical instantiation of a relation in the database schema.
- **Relation -** A logical construct that organizes data into rows and columns.

relation, table, and entity in RDBMS

Term	Definition	Example
Relation	A **concept** representing a **set of tuples** (rows) with attributes (columns). Used in **Relational Algebra**.	Employee(EmpID, Name, Dept, Salary) defines a relation.
Table	A **physical structure** in an RDBMS that stores data in **rows and columns**. It is an **implementation** of a relation.	Employee table
Entity	A **real-world object** that has attributes and is represented as a table in a database. Entities are later transformed into **tables in an RDBMS**. Used in **ERD**	An "Employee" is an entity with attributes like { EmpID (Primary Key), Name, Dept, Salary, . . . }

RELATIONAL MODEL

↓

RELATION

↓

implemented as

↓

TABLE

↓

rows + columns

Employee **Relation**

Attributes

↓

(employee_id, employee_name, salary)

Tuples

↓

(101, Rahul, 50000)
(102, Neel, 60000)
(103, Riya, 55000)

File Systems is the traditional way to keep your data organized.

File System
vs
DBMS

A File-Oriented System is the traditional way of storing and managing data before databases (like MySQL, Oracle, . . .) were introduced.

- Data is stored in separate files (text files, spreadsheets, binary files).
- Each application manages its own files independently.
- There is no central control over data.

file-oriented system

File Anomalies

```
struct Employee {
   int emp_no;
   char emp_name[50];
   int salary;
} emp[1000];
```

```
struct Employee {
   int emp_no;
   char emp_name[50];
   int salary;
};
struct Employee emp[1000];
```

c:\employee.txt

```
1   suraj 4000
2   ramesh 6000
3   rajan 4500
.       |
.       |
.       |
500 sam 3500
.       |
.       |
.       |
1000 amit 2300
```

c:\employee.txt

```
1   suraj 4000
2   ramesh 6000
3   rajan 4500
.       |
.       |
.       |
500 sam 3500
.       |
.       |
1000 amit 2300
.       |
.       |
2000 jerry 4500
.       |
.       |
```

c:\employee.txt

```
1   suraj 4000
2   ramesh 6000
3   rajan 4500
.       |
.       |
500  sam 3500
.       |
3   rajan 4500
.       |
500  sam 3500
.       |
.       |
1000 amit 2300
```

c:\employee.txt

```
1   suraj 4000
2   ramesh 6000
3   rajan 4500
.       |
.       |
sam 500 3500
.       |
ram  550 5000
.       |
1000 amit 2300
```

c:\employee.txt

```
1   suraj 4000
2   ramesh 6000
3   rajan 4500
.       |
500 sam 3500
.       |
600 neel 4500
```

- Create/Open an existing file
- Reading from file
- Writing to a file
- Closing a file

file-oriented system

File Anomalies

c:\employee.txt

1 suraj 4000
2 ramesh 6000
3 rajan 4500
.
.
.
500 sam 3500
.
.
.
1000 amit 2300

file attributes

- File Name
- Type
- Location

file permissions

- File permissions
- Share permissions

search empl ID=1

1 suraj 4000
2 ramesh 6000
3 rajan 4500
.
.
.
500 sam 3500
.
.
.
1000 amit 2300

search emp_name

1 suraj 4000
2 ramesh 6000
3 rajan 4500
.
.
.
500 sam 3500
.
.
.
1000 amit 2300

file-oriented system

A **flat file** database is a database that stores data in a plain text **file (**e.g. ***.txt, *.csv** format**)**. Each line of the text **file** holds one record, with fields separated by delimiters, such as **commas** or **tabs**.

1 rajan MG Road Pune MH 34500
2 rahul patil SSG Lane Pune MH 54000
3 suraj raj k Deccan Gymkhana Pune MH 22000

4, S M Kumar, Mg Road Pune MH, 32000
5, S M Kumar, Mg Road, Pune, MH, 32000

1,raj,k,1984-06-12,raj.kumar@gmail.com
2,om,,1969-10-25,om123@gmail.com
3,rajes,kumar,1970-10-25,
4,rahul,patil,1982-10-31,rahul.patil@gmail.com
5,ketan,,,ruhan.bagde@gmail.com

The Zen of Python,

Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
Readability counts.
Special cases aren't special enough to break the rules.
Although practicality beats purity.
Errors should never pass silently.
Unless explicitly silenced.
In the face of ambiguity, refuse the temptation to guess.
There should be one-- and preferably only one --obvious way to do it.
Although that way may not be obvious at first unless you're Dutch.
Now is better than never.
Although never is often better than *right* now.
If the implementation is hard to explain, it's a bad idea.
If the implementation is easy to explain, it may be a good idea.
Namespaces are one honking great idea -- let's do more of those!

advantages & disadvantage of file-oriented system

advantages of file-oriented system

The biggest advantage of file-based storage is as follows.

- **Backup**: It is possible to take faster and automatic back-up of database stored in files of computer-based systems.
- **Data retrieval:** It is possible to retrieve data stored in files in easy and efficient way.
- **Flexibility:** File systems provide flexibility in storing various types of data, including text documents, images, audio, video, and more
- **Cost-Effectiveness**: File systems often do not incur licensing costs, making them cost-effective for basic data storage needs.
- **Editing***:* It is easy to edit any information stored in computers in form of files.
- **Remote access***:* It is possible to access data from remote location.
- **Sharing***:* The files stored in systems can be shared among multiple users at a same time.

disadvantage of file-oriented system

The biggest disadvantage of file-based storage is as follows.

- **Data redundancy**: It is possible that the same information may be duplicated in different files. This leads to data redundancy results in memory wastage.
  (Suppose a customer having both kind of accounts - **saving** and **current** account. In such a situation a customer detail are stored in both the file, saving.txt- file and current.txt- file , which leads to Data Redundancy.)
- **Data inconsistency**: Because of data redundancy, it is possible that data may not be in consistent state.
  (Suppose customer changed his/her address. There might be a possibility that address is changed in only one file (saving.txt) and other (current.txt) remain unchanged.)
- **Limited data sharing**: Data are scattered in various files and also different files may have different formats (for example: .txt, .csv, .tsv and .xml) and these files may be stored in different folders so, due to this it is difficult to share data among different applications also if the saving account department wants to share data with loan department, they need to manually copy files, leading to delays because File Systems do not support multi-user environments.
- **Data Isolation:** Because data are scattered in various files, and files may be in different formats (for example: .txt, .csv, .tsv and .xml), writing new application programs to retrieve the appropriate data is difficult.
- (Suppose a loan data is in one file and account holder data in another, there is no easy way to analyze account holder data with his loan status.)
- **Data security:** Data should be secured from unauthorized access, for example a account holder in a bank should not be able to see the account details of another account holder, such kind of security constraints are difficult to apply in file processing systems.

disadvantage of file-oriented system

The biggest disadvantage of file-based storage is as follows.

Disadvantage of File-oriented system

- **Data Integrity**: Data integrity refers to the accuracy and consistency of data. In a file-oriented system, enforcing data integrity is difficult because there are no built-in mechanisms to ensure that data is valid or consistent across multiple files.
  (the balance field value must be grater than 5000.)
- **Concurrency Issues:** When multiple users or applications try to access and modify a file at the same time, concurrency problems can arise.
  (if two users attempt to update the same file simultaneously, it can lead to data corruption or loss of data.)
- **Lack of Flexibility:** Modifying the structure of files, such as adding new fields or changing data formats, can be difficult and time-consuming. Changes might require manual updates to each file or even rewriting entire applications that interact with the files.
- **Poor Scalability:** As the amount of data grows, file-based systems become less efficient and more difficult to manage. Searching through large files can be slow, and as more files are added, the complexity of managing the system increases.

A **database application** is a computer program whose primary purpose is entering and retrieving information.

What is database?

A major purpose of a database system is to provide users with an ***abstract view*** of the data.

abstract means existing in thought or as an idea but not having a physical or concrete existence.



what is database?

A database is a system to **organize, store** and **retrieve** large amounts of data easily, which is stored in **one** or **more data files** by **one** or **more users**.

Each database is a collection of tables, which are called **relations**, hence the name "**relational database**".



***Relation Schema*:** A relation schema represents name of the relation with its attributes, every attribute would have an associated domain.

e.g.

- **Student**(rollNo:INT, name:VARCHAR(20), address:VARCHAR(50), phone:VARCHAR(12), age:INT, PRIMARY KEY(rollNo)) is relation schema for STUDENT
- **Customers**(CustomerID:INT, Name:VARCHAR(50), Email:VARCHAR(100), City:VARCHAR(50), PRIMARY KEY(CustomerID)) is relation schema for CUSTOMERS

DBMS

- **database:** Is the collection of **related data** which is **organized,** database can store and retrieve large amount of data easily, which is stored in one or more data files by one or more users, it is called as **structured data.**
- **management system**: it is a software, designed to **define**, **manipulate**, **retrieve** and **manage** data in a database.



Difference between File System and DBMS

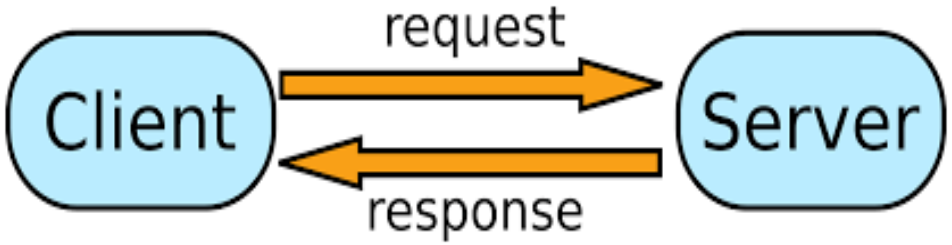
File Management System	Database Management System
• File System is easy-to-use system to store data which require less security and constraints.	• Database Management System is used when security constraints are high.
• Data Redundancy is more in File System.	• Data Redundancy is less in Database Management System.
• Data Inconsistency is more in File System.	• Data Inconsistency is less in Database Management System.
• Centralization is hard to get when it comes to File System.	• Centralization is achieved in Database Management System.
• User locates the physical address of the files to access data in File System.	• In Database Management System, user is unaware of physical address where data is stored.
• Security is low in File System.	• Security is high in Database Management System.
• File System stores unstructured data. "unstructured data" may include documents, audio, video, images, etc.	• Database Management System stores structured data.

relational database management system?

A RDBMS is a database management system (DBMS) that is based on the **relational model** introduced by Edgar Frank Codd at IBM in 1970.

RDBMS supports

- *client/server Technology*
- *Highly Secured*
- *Relationship (PK/FK)*







- A server is a computer program or a device that provides service to another computer program, also known as the client.
- In the client/server programming model, a server program awaits and fulfills requests from client programs, which might be running in the same, or other computers.

object relational database management system?

An object database is a database management system in which information is represented in the form of objects.

PostgreSQL is the most popular pure ORDBMS. Some popular databases including Microsoft SQL Server, Oracle, and IBM DB2 also support objects and can be considered as ORDBMS.

Advantage of ORDBMS

- Function/Procedure overloading.
- Extending server functionality with external functions written in C or Java.
- User defined data types.
- Inheritance of tables under other tables.

object relational database management system?

- CREATE or REPLACE TYPE ***address*** AS OBJECT(city VARCHAR2(10), state VARCHAR2(2));
- CREATE TABLE person(id INT, name VARCHAR2(10), addr ***address***);
- INSERT INTO person VALUES(1, 'saleel', address('baroda', 'GJ'));
- SELECT id, name, ***n***.addr.city FROM person ***n***;

- CREATE or REPLACE TYPE ***city*** AS VARRAY(3) OF VARCHAR2(10);
- CREATE TABLE x (id INT, ename VARCHAR2(10), c ***city***);
- INSERT INTO x values(1, 'saleel', ***city***('baroda', 'surat', 'bharuch'));
- SELECT n.id, n.ename, nn.column_value FROM x n, TABLE(n.c) nn;

difference between dbms and rdbms

DBMS	RDBMS
• Data is stored as file.	• Data is stored as tables.
• There is no relationship between data in DBMS.	• Data is present in multiple tables which can be related to each other.
• DBMS has no support for distributed databases.	• RDBMS supports distributed databases.
• Normalization cannot be achieved.	• Normalization can be achieved.
• DBMS supports single user at a time.	• RDBMS supports multiple users at a time.
• Data Redundancy is common in DBMS.	• Data Redundancy can be reduced in RDBMS.
• DBMS provides low level of security during data manipulation.	• RDBMS has high level of security during data manipulation.

Codd's Rules

Rule 1: Information Rule

The data stored in a database, may it be user data or metadata, must be a value of some table cell. Everything in a database must be stored in a table format.

Rule 2: Guaranteed Access Rule

Every single data is guaranteed to be accessible with a combination of table-name, primary-key (row value), and attribute-name (column value).

Rule 3: Systematic Treatment of NULL Values (absence of a value)

The NULL values in a database must be given a systematic and uniform treatment. This is a very important rule because a NULL can be interpreted as one the following − data is missing, data is not known, or data is not applicable.

Rule 4: Active Online Catalog

The structure description of the entire database must be stored in an online catalog, known as **data dictionary**, which can be accessed by authorized users.

Rule 5: Comprehensive Data Sub-Language Rule

A database can only be accessed using a language having linear syntax that supports data definition, data manipulation, and transaction management operations. This language can be used directly or by means of some application. If the database allows access to data without any help of this language, then it is considered as a violation.

Rule 6: View Updating Rule

All the views of a database, which can theoretically be updated, must also be updatable by the system.

Rule 7: High-Level Insert, Update, and Delete Rule

A database must support high-level insertion, updation, and deletion. This must not be limited to a single row, that is, it must also support union, intersection and minus operations to yield sets of data records.

Rule 8: Physical Data Independence

The data stored in a database must be independent of the applications that access the database. Any change in the physical structure of a database must not have any impact on how the data is being accessed by external applications.

Rule 9: Logical Data Independence

The logical data in a database must be independent of its user's view (application). Any change in logical data must not affect the applications using it. For example, if two tables are merged or one is split into two different tables, there should be no impact or change on the user application. This is one of the most difficult rule to apply.

Rule 10: Integrity Independence

A database must be independent of the application that uses it. All its integrity constraints can be independently modified without the need of any change in the application. This rule makes a database independent of the front-end application and its interface.

Rule 11: Distribution Independence

The end-user must not be able to see that the data is distributed over various locations. Users should always get the impression that the data is located at one site only. This rule has been regarded as the foundation of distributed database systems.

Rule 12: Non-Subversion Rule

If a system has an interface that provides access to low-level records, then the interface must not be able to subvert the system and bypass security and integrity constraints.

relational model concepts
and
properties of relational table

relational model concepts

Relational model organizes data into one or more tables (or "relations") of columns and rows. Rows are also called records or tuples. Columns are also called attributes.

- **Relation (Table)** – In relational model, relations are saved in the form of Tables. A table has rows and columns.
- **Attribute (Column)** – Attributes are the properties that define a relation. **e.g.** (**roll_no, name, address, age, . . .)**
- **Tuple (Row/Record)** – A single row of a table, which contains a single record for that relation is called a tuple.
- **Relation schema** – A relation schema describes the Relation Name (Table Name), Attributes (Column Names), Domain of Attributes (Data Types & Allowed values), Constraints (Primary Key, Foreign Key, etc.).

  e.g. **Customers**(CustomerID:INT, Name:VARCHAR(50), Email:VARCHAR(100), City:VARCHAR(50), PRIMARY KEY(CustomerID)) is relation schema for CUSTOMERS
- **Attribute domain** – An attribute domain in a relational database refers to the set of allowed values for an attribute (column). It defines the data type and constraints that restrict the values an attribute can take.

Remember:

- In database management systems, **null** (***absence of a value***) is used to represent **missing** or **unknown** data in a table column.

properties of relational table

ID	job	firstName	DoB	salary
1	manager	Saleel Bagde	yyyy-mm-dd	••••••
3	salesman	Sharmin	yyyy-mm-dd	••••••
4	accountant	Vrushali	yyyy-mm-dd	ABC
2	salesman	Ruhan	yyyy-mm-dd	••••••
5	9500	manager	yyyy-mm-dd	••••••
5	Salesman	Rahul Patil	yyyy-mm-dd	••••••

Properties of Relational Tables:

- Values are atomic (no multivalued cells).
- Column values are of the same kind. (***Attribute Domain*: Every attribute has some pre-defined datatypes, format, constraints of a column, and defines the range of values that are valid for that column known as attribute domain.)**
- Each row is unique.
- The sequence/order of columns is **irrelevant** – (unimportant).
- The sequence/order of rows is **irrelevant** – (unimportant).
- Each **table name**, attribute/column must have a unique name.
- Attributes may contain **NULL** (unknown/missing values)

What is data?



what is data?

Data is any facts that can be stored and that can be processed by a computer.

Data can be in the form of Text or Multimedia

e.g.

- number, characters, or symbol
- images, audio, video, or signal



Remember:

- A **Binary Large Object (BLOB)** is a MySQL data type that can store binary data such as multimedia, and PDF files.
- A **Character Large Object(CLOB)** is aa MySQL data type which is used to store large amount of textual data. Using this datatype, you can store data up to 2,147,483,647 characters.
- A number is a mathematical value used to count, measure, and label.

ACID

ACID properties of transactions

Atomicity. In a transaction involving two or more separate pieces of information, either all of the pieces are committed or none are.
For example, in an application that transfers funds from one account to another, the atomicity property ensures that, if a debit is made successfully from one account, the corresponding credit is made to the other account.

Consistency. A transaction either creates a new and valid state of data, or, if any failure occurs, returns all data to its state before the transaction was started.
For example, in an application that transfers funds from one account to another, the consistency property ensures that the total value of funds in both the accounts is the same at the start and end of each transaction.

Isolation. A transaction in process and not yet committed must remain isolated from any other transaction.
For example, If multiple users attempt to transfer money at the same time, the transactions should be isolated from one another to prevent conflicts or inconsistencies. Each transaction should be treated as if it is the only transaction being executed at that time.

Durability. Committed data is saved by the system such that, even in the event of a failure and system restart, the data is available in its correct state.
For example, in an application that transfers funds from one account to another, the durability property ensures that the changes made to each account will not be reversed.

ACID properties of transactions

Atomicity – **"All or nothing"** :- If any part of the order fails (e.g., payment gateway timeout), the entire transaction must be rolled back. **i.e.** Inventory should not reduce, Order should not be created, Customer should not be charged.

Consistency – **"Maintains valid data"** **:-** The system should enforces constraints. **i.e.** Product quantity ≥ 0, Only valid payment methods are used, Only in-stock items can be purchased

Note:- The system enforces constraints: After a successful order, 1. the stock decreases by 1. 2. It should never show negative stock.

Isolation – **"Concurrent transactions don't interfere"** Imagine 5 customers try to order the last phone at the same time. Isolation ensures: **i.e.** Only one transaction succeeds in reducing stock to zero, Others are blocked or rolled back with "Out of Stock" message

Durability – **"Once saved, it stays saved"** After a successful order: The order confirmation and payment details are stored permanently, **i.e.** Even if the site crashes or database restarts, The customer still sees their order in "My Orders"

ACID properties of transactions

Atomicity – **"All or nothing"** :- If the system crashes after booking the appointment but before processing payment, everything is rolled back. **i.e.** No appointment is confirmed, Doctor's time slot is not blocked, No bill is generated

Consistency – **"Maintains valid data"** **:-** Ensures appointments are booked only with existing patients and available time slots , Booking without checking doctor availability violates consistency. **i.e.** No double-booking of the doctor's slot, Payment amount matches consultation fee, A patient must be registered before booking

Isolation – **"Concurrent transactions don't interfere "** **:-** Two patients booking the same 10:00 AM slot with the same doctor at the same time **i.e.** One transaction succeeds, the other fails or chooses another time slot, Isolation prevents race conditions like double-booking

Durability – **"Once saved, it stays saved"** **:-** If appointment and payment are successfully done: **i.e.** Even if the server crashes or restarts, The appointment, payment, and receipt remain in the system, No data loss after confirmation

What is Entity Relationship Diagram?

Entity Relationship Diagram (ER Diagram)

Use E-R model to get a high-level graphical view to describe the **"ENTITIES"** and their **"RELATIONSHIP"**

The basic constructs/components of ER Model are **Entity**, **Attributes** and **Relationships**.

An entity can be a **real-world object.**

What is Entity?

An entity in DBMS is a real-world object that has certain properties called attributes that define the nature of the entity.

entity

In relation to a database , an entity is a

- Person(student, teacher, employee, client, department, …)
- Place(classroom, building, …) --a particular position or area
- Thing(computer, lab equipment, …) --an object that is not named (represents a tangible object)
- Concept(course, batch, student's attendance, …) -- an idea,

about which data can be stored. All these entities have some **attributes** or **properties** that give them their **identity**.

Every entity has its own characteristics.

What is Entity Type?

entity type

The entities that have the **common attributes** is called an **entity type.**

Each entity type in the database is described by a **name** and **a list of attributes.**

e.g. an entity Person is an entity type that has *Age, Name* and *Address* attributes.

Eg.

Entity TYPE

Person (Age, Name, Address ,. . .)

Entity

17 , Sharmin, Paud Road, …

When you are designing attributes for your entities,

- **you will sometimes find that an attribute does not have a value**.
- For example, you might want an attribute for a person's middle name, but you can't require a value because some people have no middle name. **For these, you can define the attribute so that it can contain null (*absence of a value*) values.**

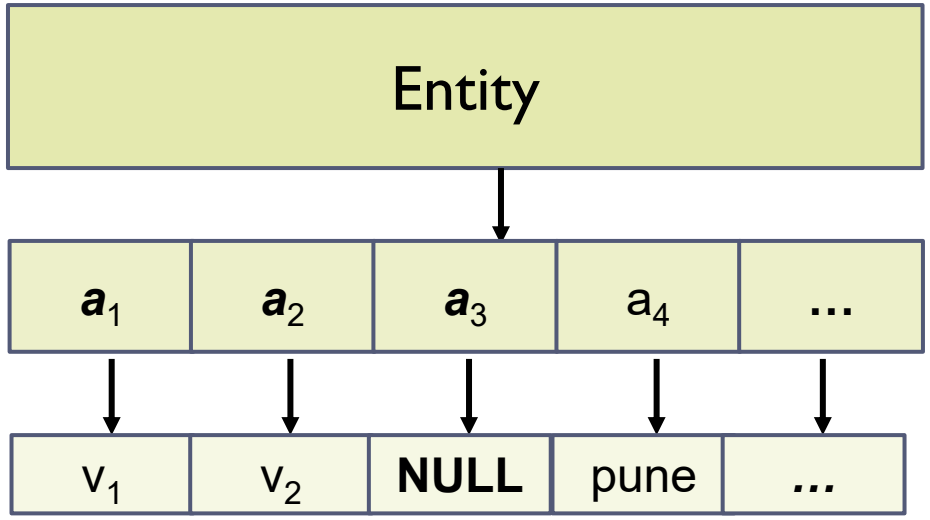
In database management systems, **null** (***absence of a value***) is used to represent **missing** or **unknown** data in a table column.

What is an Attribute?

Attributes are the properties that define a relation.

e.g. **Student**(***rollNo***:INT, ***name***:VARCHAR(20), ***address***:VARCHAR(50), ***age***:INT)

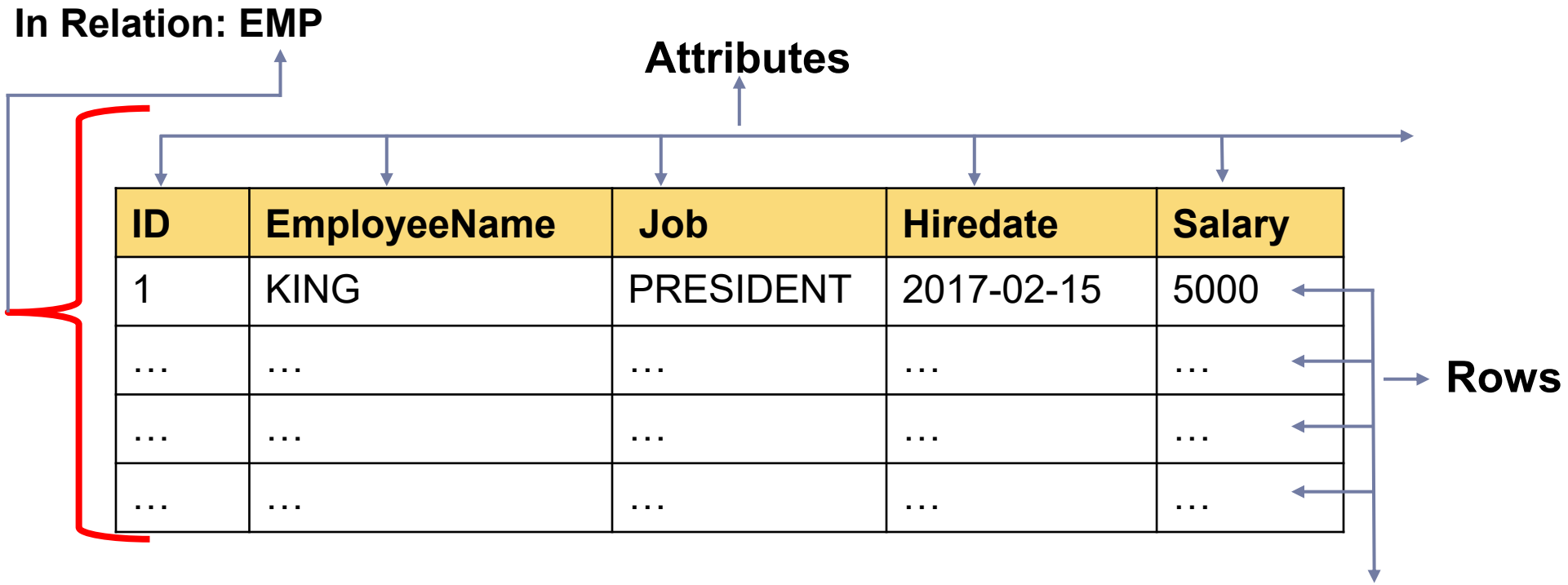
In some cases, you might not want a specific attribute to contain a NULL value, but you don't want to require that the user or program always provide a value. In this case, a default value might be appropriate. **A default value is a value that applies to an attribute if no other valid value is available.**



attributes

A table has rows and columns

In RDBMS, a table organizes data in rows and columns. The **COLUMNS** are known as **ATTRIBUTES / FIELDS** whereas the **ROWS** are known as **RECORDS / TUPLE**.



In Entity Relationship(ER) Model attributes can be classified into the following types.

- Simple/Atomic and Composite Attribute
- Single Valued and Multi Valued attribute
- Stored and Derived Attributes
- Complex Attribute

Remember:

In SQL, the same name can be used for two (or more) attributes as long as the attributes are in different relations.

attributes

- **Simple / Atomic Attribute** --VS-- **Composite Attribute**
  (Can't be divided further) (Can be divided further)
- **Single Value Attribute** --VS-- **Multi Valued Attribute**
  (Only One value) (Multiple values)
- **Stored Attribute** --VS-- **Derived Attribute**
  (Only One value) (Virtual)
- **Complex Attribute**
  (Composite & Multivalued)

Employee ID: An employee ID can be a composite attribute, which is composed of sub-attributes such as department code, job code, and employee number.

- **Atomic Attribute:** An attribute that cannot be divided into smaller independent attribute is known as atomic attribute.
  e.g. ID's, PRN, age, gender, zip, marital status cannot further divide.
- **Single Value Attribute:** An attribute that holds exactly one value for a given record at any point in time is known as single valued attribute. Single-valued attributes are typically used to provide a unique identifier for a record.
  e.g. manufactured part can have only one serial number, voter card ID, blood group, branchID can have only one value.
- **Stored Attribute:** The stored attribute are such attributes which are already stored in the database and from which the value of another attribute is derived.
  e.g. (HRA, DA…) can be derive from salary, age can be derived from DoB, total marks or average marks of a student can be derived from marks, TotalPrice can be derived from Quantity × UnitPrice.

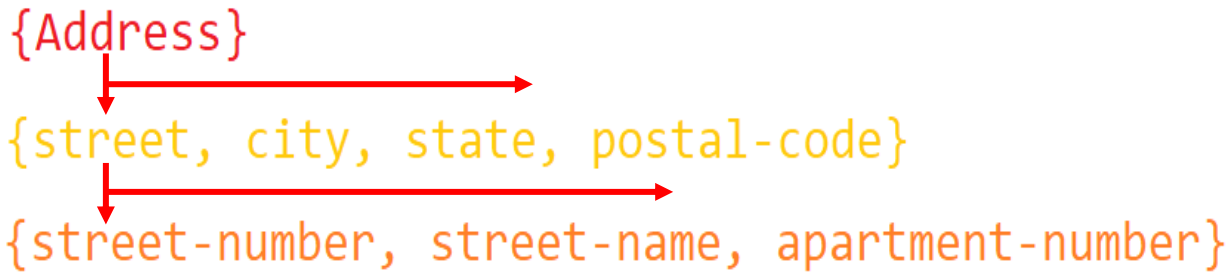
Composite **VS** Multi Valued Attribute

Composite Attribute

composite / multi valued attributes

Person Entity

- *Name* attribute: (`firstName` + `middleName` + `lastName`)
- *PhoneNumber* attribute: (`countryCode` + `cityCode` + `phoneNumber`)
- *Date* attribute: (`Day` + `Month` + `Year`)
- *productDimensions* attribute: (`Length` + `Width` + `Height`)
- *carRegistration* attribute: (`State_Code` + `Series` + `Number`)



Multi Valued Attribute

Person Entity

- *Hobbies* attribute: [reading, hiking, hockey, skiing, photography, . . .]
- *SpokenLanguages* attribute: [Hindi, Marathi, Gujarati, English, . . .]
- *Degrees* attribute**:** [10th, 12th, BE, ME, PhD, . . .]
- *emailID* attribute**:** [saleel@gmail.com, salil@yahoomail.com, . . .]
- *Skills* attribute: [MySQL, Oracle, Redis, MongoDB, Java, . . .]

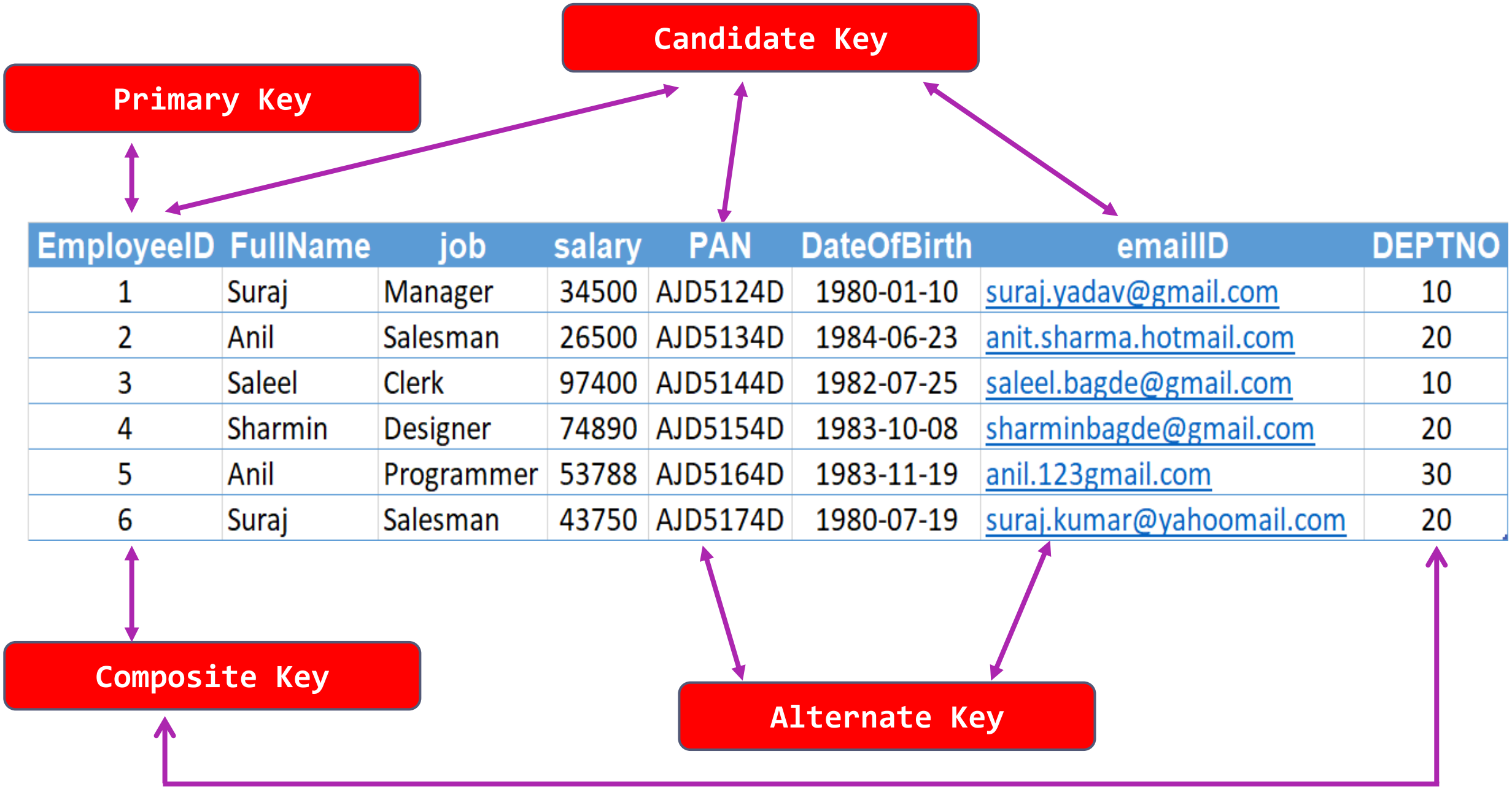
types of Keys?

Keys are used to establish relationships between tables and also to uniquely identify any record in the table.

types of Keys?

r = Employee(EmployeeID, FullName, job, salary, PAN, DateOfBirth, emailID, deptno)

- **Candidate Key:** are individual columns in a table that qualifies for uniqueness of all the rows. Here in Employee table EmployeeID, PAN or emailID are Candidate keys.
- **Primary Key**: is the columns you choose to maintain uniqueness in a table. Here in Employee table you can choose either EmployeeID, PAN or emailID columns, EmployeeID is preferable choice.
- **Alternate Key**: Candidate column other the primary key column, like if EmployeeID is primary key then , PAN or emailID columns would be the Alternate key.
- **Super Key**: If you add any other column to a primary key then it become a super key, like EmployeeID + FullName or EmployeeID + deptno is a Super Key.
- **Composite Key**: If a table do not have any single column that qualifies for a Candidate key, then you have to select 2 or more columns to make a row unique. Like if there is no EmployeeID, PAN or emailID columns, then you can make FullName + DateOfBirth as Composite key. But still there can be a narrow chance of duplicate row. Ensures data uniqueness in many-to-many relationships. e.g. in order_details table we can have multiple products OrderID + ProductID


Candidate Key
Primary Key
EmployeeID FullName job salary PAN DateOfBirth emailID DEPTNO
1 Suraj Manager 34500 AJD5124D 1980-01-10 suraj.yadav@gmail.com 10
2 Anil Salesman 26500 AJD5134D 1984-06-23 anit.sharma.hotmail.com 20
3 Saleel Clerk 97400 AJD5144D 1982-07-25 saleel.bagde@gmail.com 10
4 Sharmin Designer 74890 AJD5154D 1983-10-08 sharminbagde@gmail.com 20
5 Anil Programmer 53788 AJD5164D 1983-11-19 anil.123gmail.com 30
6 Suraj Salesman 43750 AJD5174D 1980-07-19 suraj.kumar@yahoomail.com 20
Composite Key
Alternate Key

What is a Prime, Non-Prime Attribute?

Prime attribute (*Entity integrity*):- An attribute, which is a **part of the prime-key** (candidate key), is known as a prime attribute.

Consider a relation Student(StudentID, Name, Email, Phone).

- ***Candidate Keys:*** {StudentID}, {Email}, {Phone}
- ***Prime Attributes:*** StudentID, Email, Phone (since they are part of a Candidate Key).

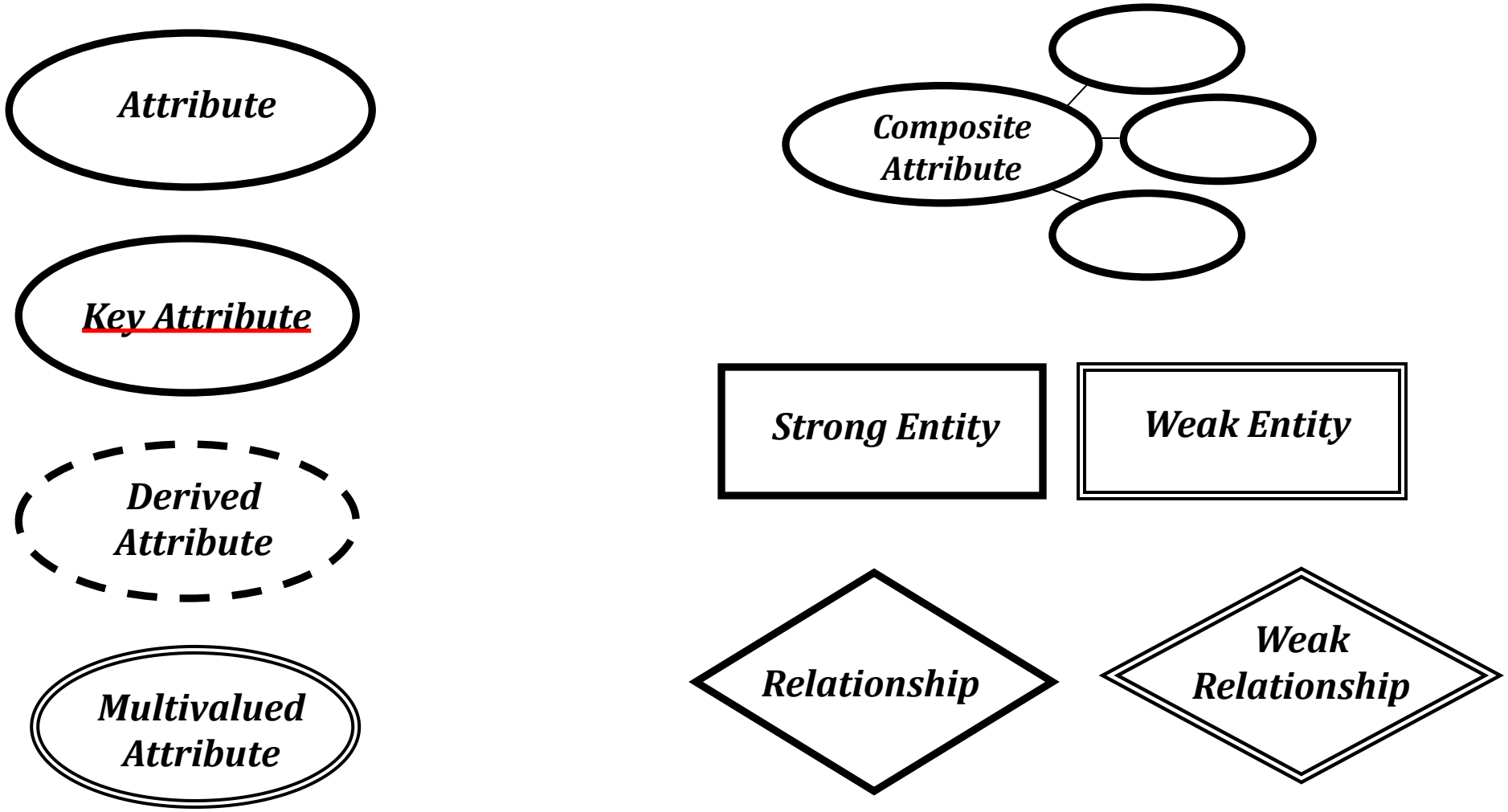
Non-prime attribute:- An attribute, which is **not a part of the prime-key** (candidate key), is said to be a non-prime attribute.

In the Student(StudentID, Name, Email, Phone) relation:

- ***Candidate Keys:*** {StudentID}, {Email}, {Phone}
- ***Prime Attributes:*** StudentID, Email, Phone
- ***Non-Prime Attribute:*** Name (because it is not part of any Candidate Key).

Entity Relationship Diagram Symbols

entity relationship diagram symbols



strong and weak entity

Strong Entity: A strong entity is not dependent on any other entity in the schema. A strong entity will always have a primary key. Strong entities are represented by a single rectangle.

Weak Entity: A weak entity is dependent on a strong entity to ensure its existence. Unlike a strong entity, a weak entity does not have any primary key. A weak entity is represented by a double rectangle. The relation between one strong and one weak entity is represented by a double diamond. This relationship is also known as ***identifying relationship***.

Example 1 – A loan entity can not be created for a customer if the customer doesn't exist

Example 2 – A payment entity can not be created for a loan if the loan doesn't exist

Example 3 – A customer address entity can not be created for the customer if the customer doesn't exist

Example 4 – A prescription entity can not be created for a patient if the patient doesn't exist

strong and weak entity

Strong Entity	Weak Entity
— Order (OrderID)	— OrderItem (ItemID, OrderID)
— University (UniID)	— Scholarship (ScholarshipID, UniID)
— Patient (PatientID)	— MedicalRecord (RecordID, PatientID)
— Account (AccountID)	— Transaction (TransactionID, AccountID)
— Student (StudentID)	— Grade (GradeID, StudentID)
— Vehicle (VehicleID)	— InsurancePolicy (PolicyID, VehicleID)
— Hotel (HotelID)	— RoomBooking (BookingID, HotelID)
— Product (ProductID)	— WarrantyClaim (ClaimID, ProductID)
— Student (StudentID)	— AttendanceRecord (RecordID, StudentID)
TODO	TODO
TODO	TODO
TODO	TODO
TODO	TODO

participation constraints

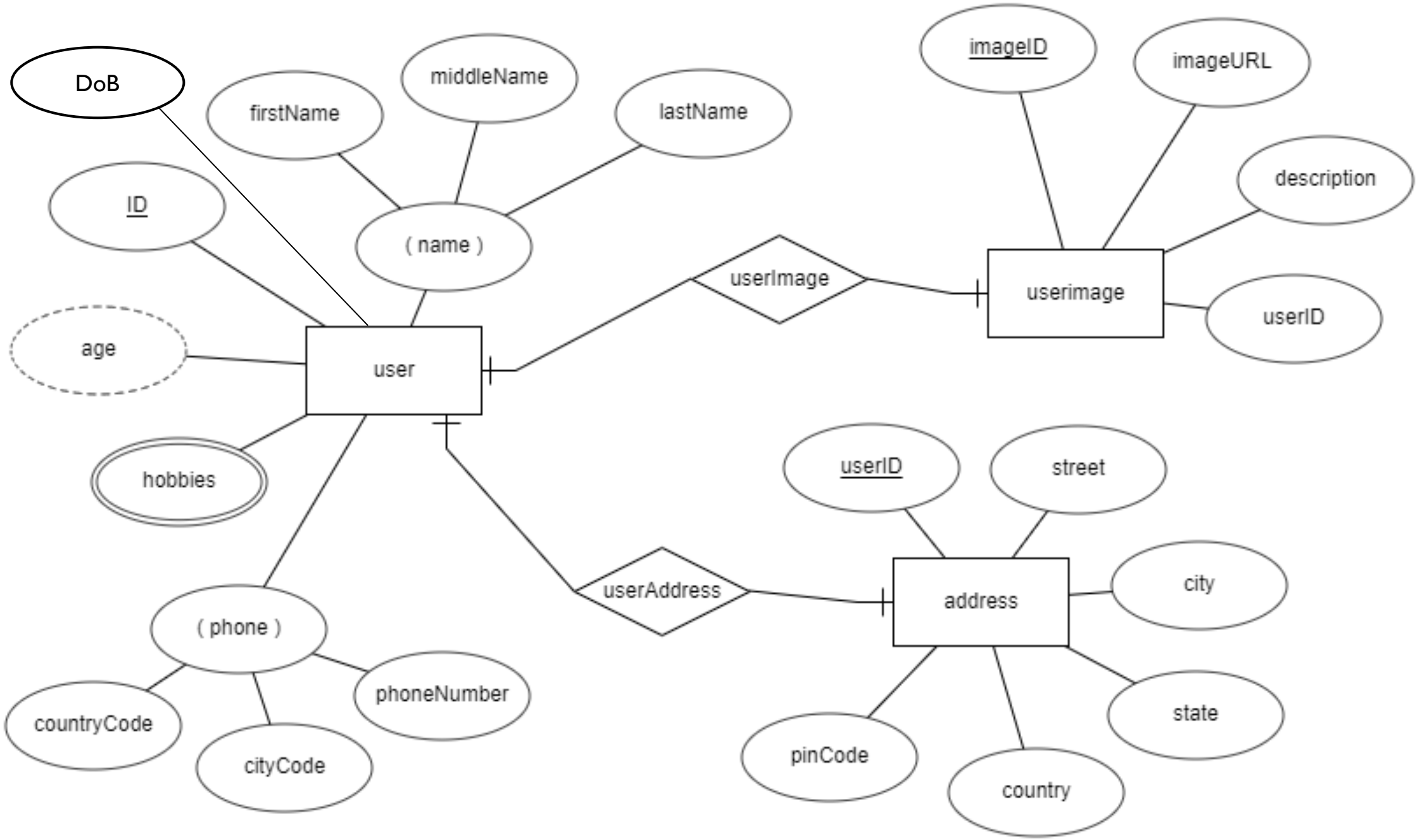
An entity may participate in a relation either totally or partially.

Total participation means that every entity is involved in the relationship, e.g., each student must be guided by a professor (there are no students who are not guided by any professor. This kind of relation is represented as a double line.

Partial participation means that not all entities are involved in the relationship, e.g., not every professor guides a student (there are professors who don't). A partial participation is represented by a single line.



entity relationship diagram



What is a degree, cardinality and union in database?

What is a degree, cardinality and union in database?

r = Employee(EmployeeID, FullName, job, salary, PAN, DateOfBirth, emailID, deptno)

- **Degree d(R) / Arity**: Total number of **attributes/columns** present in a relation/table is called **degree of the relation** and is denoted by **d(R)**.
  1. **Degree / Arity =** ***Number of columns***
  2. **d(Employee) =** *8*

- **Cardinality |R|**: Total number of **tuples/rows** present in a relation/table, **is called cardinality of a relation** and is denoted by $|R|$, which changes dynamically as rows are inserted or deleted.
  1. **Cardinality |R| =** ***Number of rows/records in relation |R|.***

  Cardinality is the numerical relationship between rows of one table and rows in another. Common cardinalities include ***one-to-one, one-to-many,*** and ***many-to-many.***

- **Union Compatibility**: Two relations ***R*** and ***S*** are set to be Union Compatible to each other if and only if:
  1. ***They have the same degree d(R).***
  2. ***Domains of the respective attributes should also be same.***
  3. ***Column names don't need to match, but positions must match.***

What is domain constraint and types of data integrity constraints?

Data integrity refers to the correctness and completeness of data.

A domain constraint and types of data integrity constraints

- **Domain Constraint** = data type + constraints **(not null/unique/primary key/foreign key/check/default)**

  e.g. custID INT, constraint pk_custid PRIMARY KEY(custID)

Three types of integrity constraints: **entity integrity, referential integrity** and **domain integrity**:

- **Entity integrity:** Entity Integrity Constraint is used to ensure the uniqueness of each record the table. There are primarily two types of integrity constraints that help us in ensuring the uniqueness of each row, namely, UNIQUE KEY constraint and PRIMARY KEY constraint.
-
- **Referential integrity:** Referential Integrity Constraint ensures that there always exists a valid relationship between two tables. This makes sure that if a foreign key exists in a table t_2 relationship then it should always reference a corresponding value in the second table t_1 :- t_1**[PK]** = t_2**[FK]** or it should be null.

- **Domain integrity:** A domain is a set of values of the same type (data type, range, and format).

Data integrity refers to the correctness and completeness of data.

A domain constraint and types of data integrity constraints

- **Domain Constraint** = data type + Constraints (**not null/unique/primary key/foreign key/check/default**)

  e.g. custID INT, constraint pk_custid PRIMARY KEY(custID)

Domain integrity is enforced using the following constraints:

Constraint	Description	Example
Data Type	Ensures that values match a specific type (e.g., INT, VARCHAR, DATE, . . .).	age INT NOT NULL (Only integers allowed)
NOT NULL	Prevents NULL (empty) values in a column.	name VARCHAR(50) NOT NULL
PRIMARY	Unique and Not Null; identifies rows	id INT PRIMARY KEY
UNIQUE	No duplicate values	id INT UNIQUE
CHECK	Restricts values based on a condition.	salary DECIMAL(10, 2) CHECK (salary > 0)
DEFAULT	Sets a default value if none is provided.	status VARCHAR(10) DEFAULT 'Active'
ENUM	Limits a column to predefined values.	gender ENUM('Male', 'Female', 'Other')
SET	Allows multiple predefined values.	roles SET('Admin', 'Editor', 'User')

Degrees of relationship

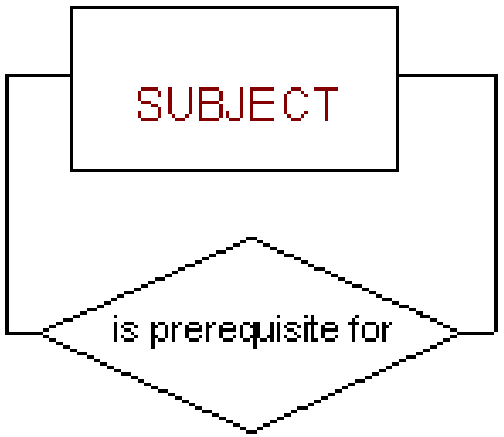
In DBMS, **a degree of relationship** represents the number of entity types that associate in a relationship.

1. Unary
2. Binary
3. Ternary
4. N-ary

relationships - unary

The three most common relationships in ER models are **Binary, Unary**, and **Ternary**

A ***unary relationship*** is when both participants in the relationship are the same entity.

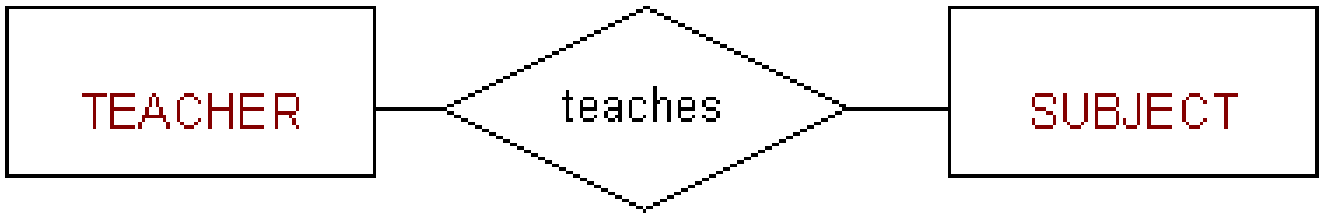


Employee —— manages ——> Employee

relationships - binary

The three most common relationships in ER models are **Binary, Unary**, and **Ternary**

A ***binary relationship*** is when two entities participate and is the most common relationship.

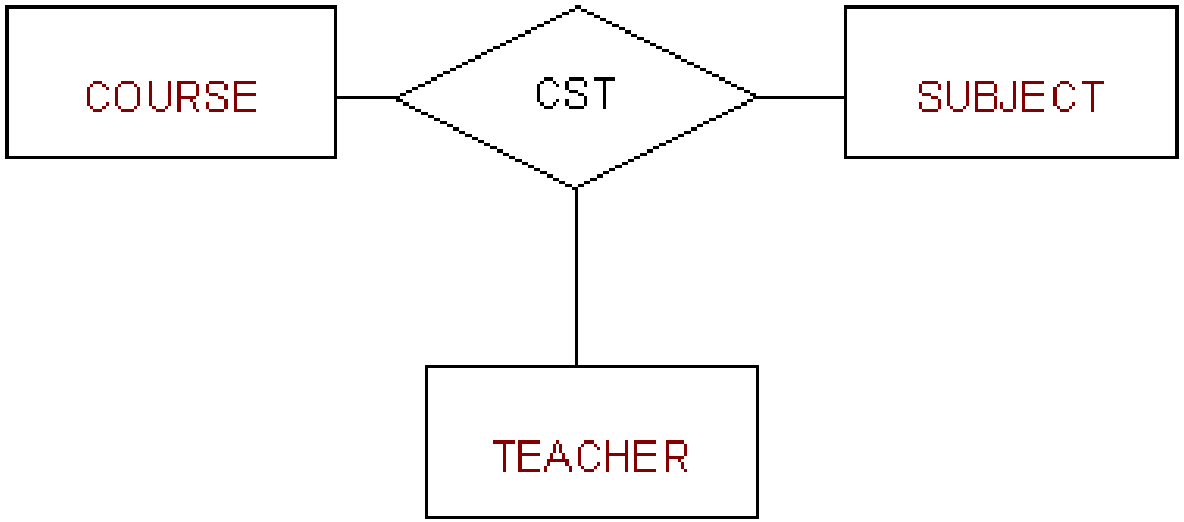


STUDENT —— enrolls ——> COURSE

relationships - ternary

The three most common relationships in ER models are **Binary, Unary**, and **Ternary**

A ***ternary relationship*** is when three entities participate in the relationship.



DOCTOR —— prescribes ——> PATIENT —— uses ——> MEDICINE

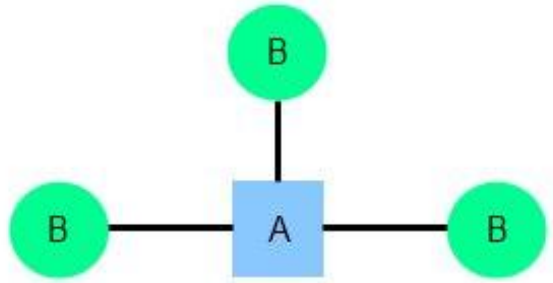
Common relationships

Common relationship

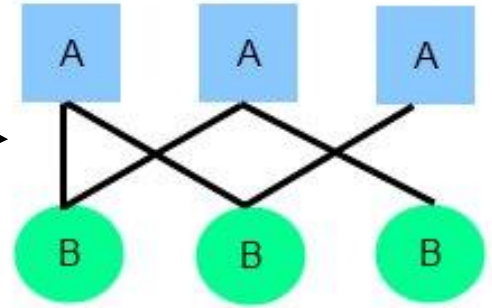
1. one-to-one (1:1)



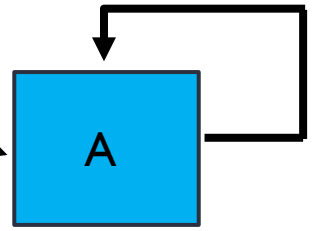
2. one-to-many (1:M)



3. many-to-many (M:N)

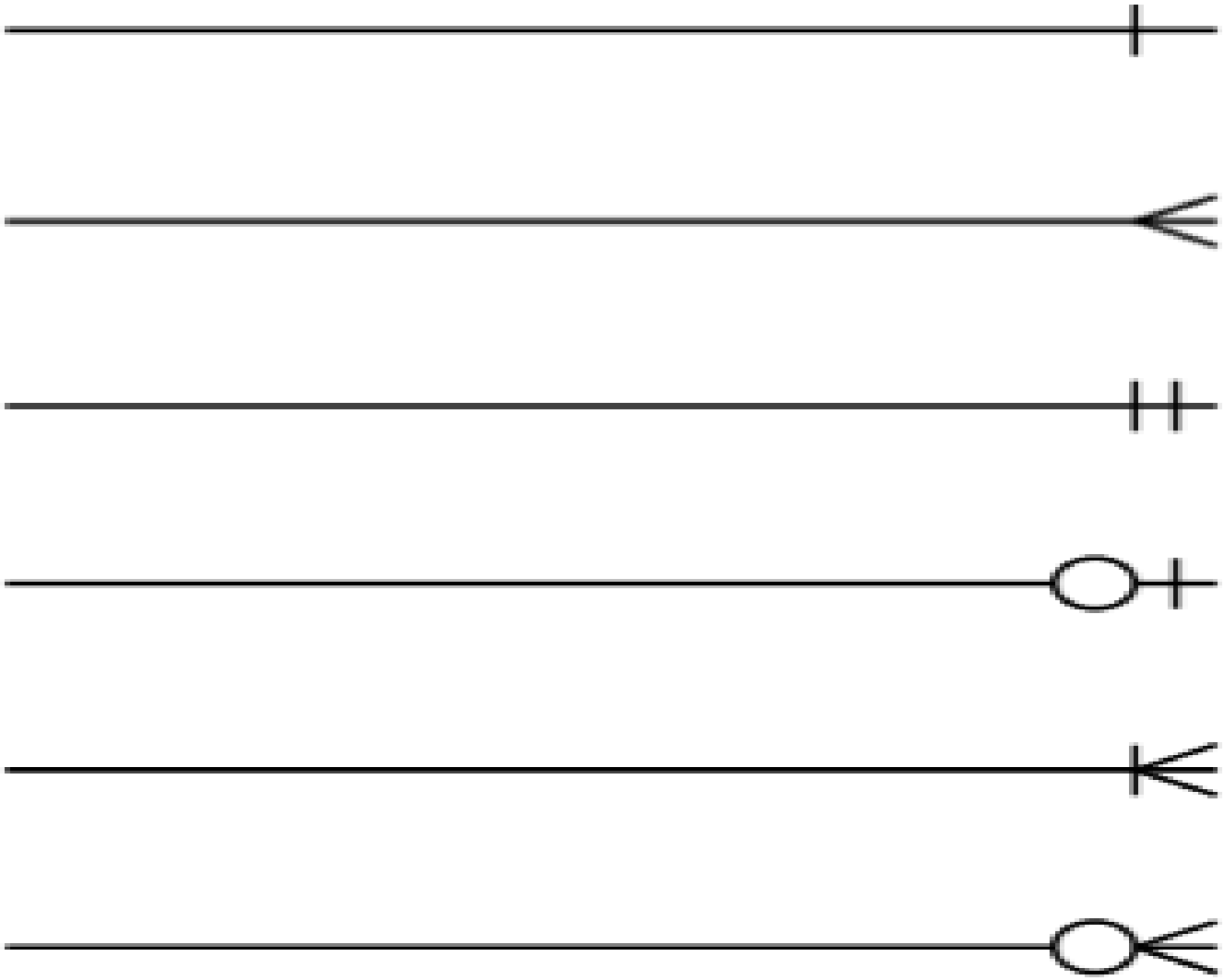


4. Self-Referencing (Recursive)



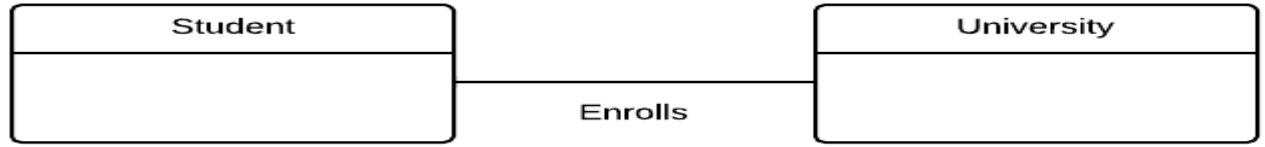
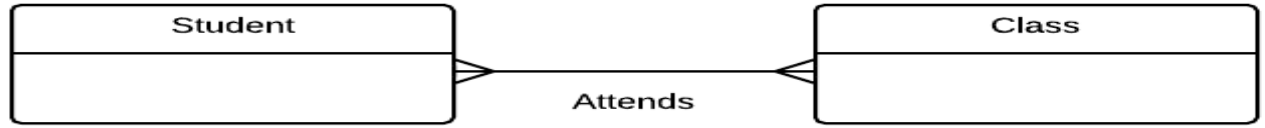
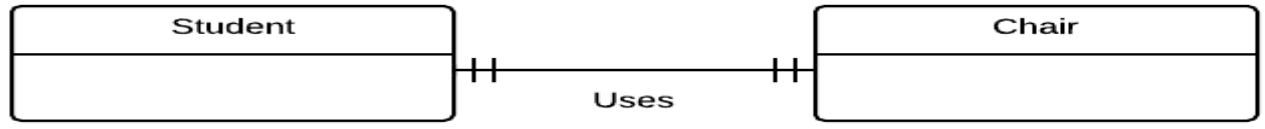
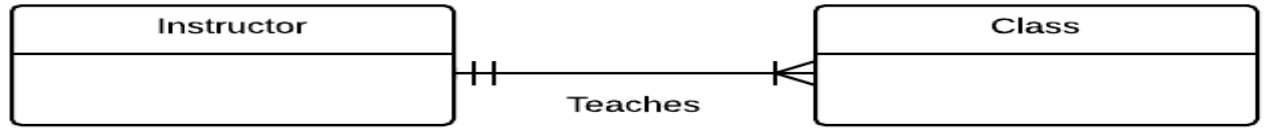
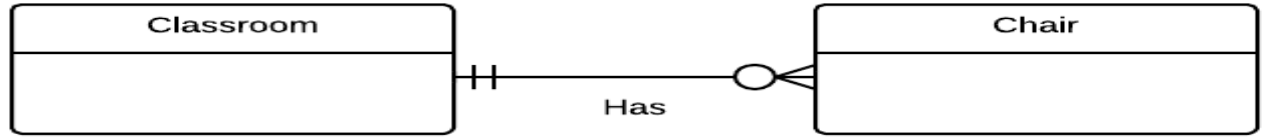
- ONE patient is allocated ONE bed
- ONE bed is occupied by ONLY ONE patient

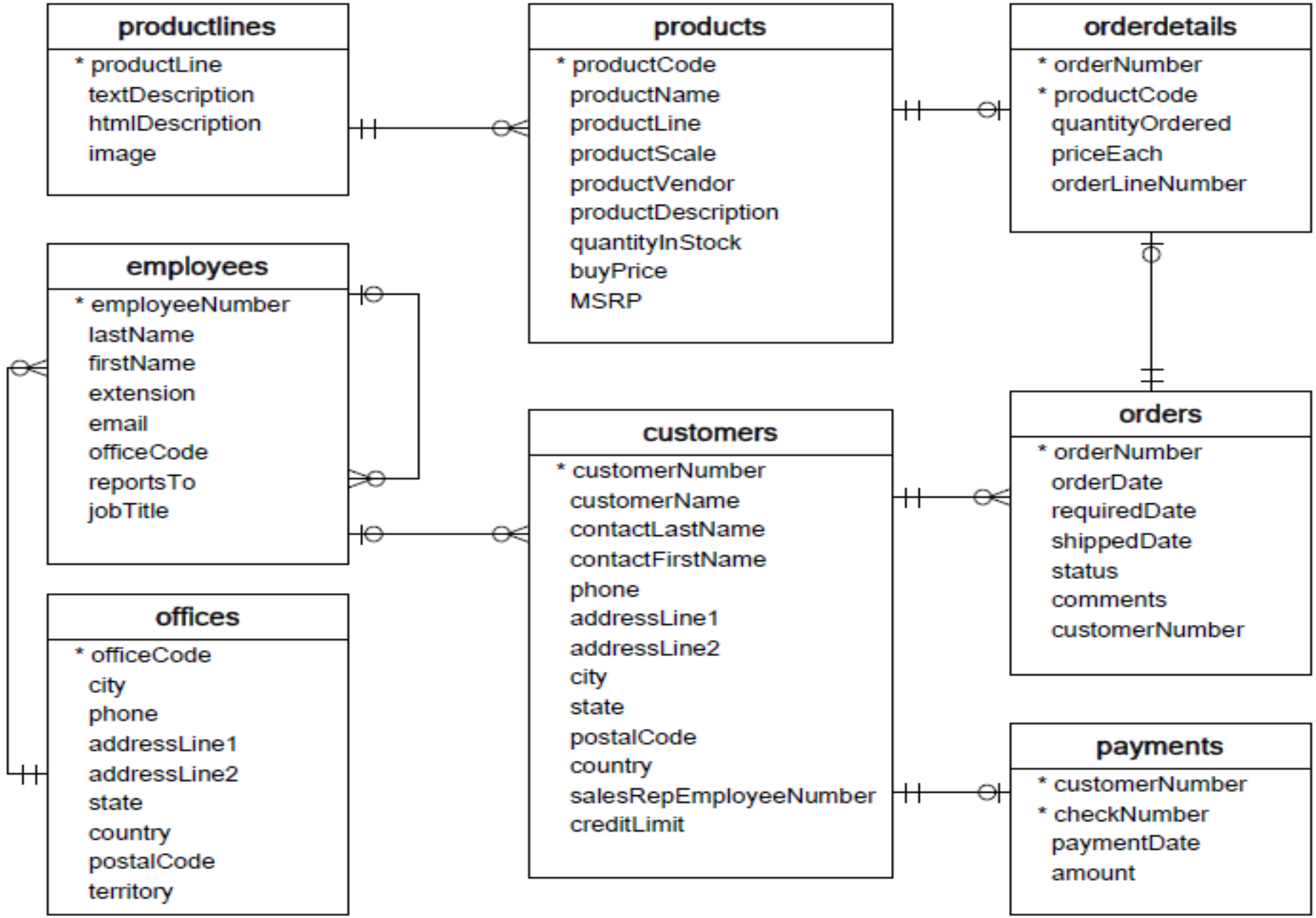
relationships



One and ONLY One (||) Min & Max only One

relationships

Notation	Meaning	Example
	Relationship	
	One	
	Many	
	One and ONLY One	
	Zero or One	
	One or Many	
	Zero or Many	


productlines
* productLine
textDescription
htmlDescription
image
products
* productCode
productName
productLine
productScale
productVendor
productDescription
quantityInStock
buyPrice
MSRP
orderdetails
* orderNumber
* productCode
quantityOrdered
priceEach
orderLineNumber
employees
* employeeNumber
lastName
firstName
extension
email
officeCode
reportsTo
jobTitle
customers
* customerNumber
customerName
contactLastName
contactFirstName
phone
addressLine1
addressLine2
city
state
postalCode
country
salesRepEmployeeNumber
creditLimit
orders
* orderNumber
orderDate
requiredDate
shippedDate
status
comments
customerNumber
offices
* officeCode
city
phone
addressLine1
addressLine2
state
country
postalCode
territory
payments
* customerNumber
* checkNumber
paymentDate
amount

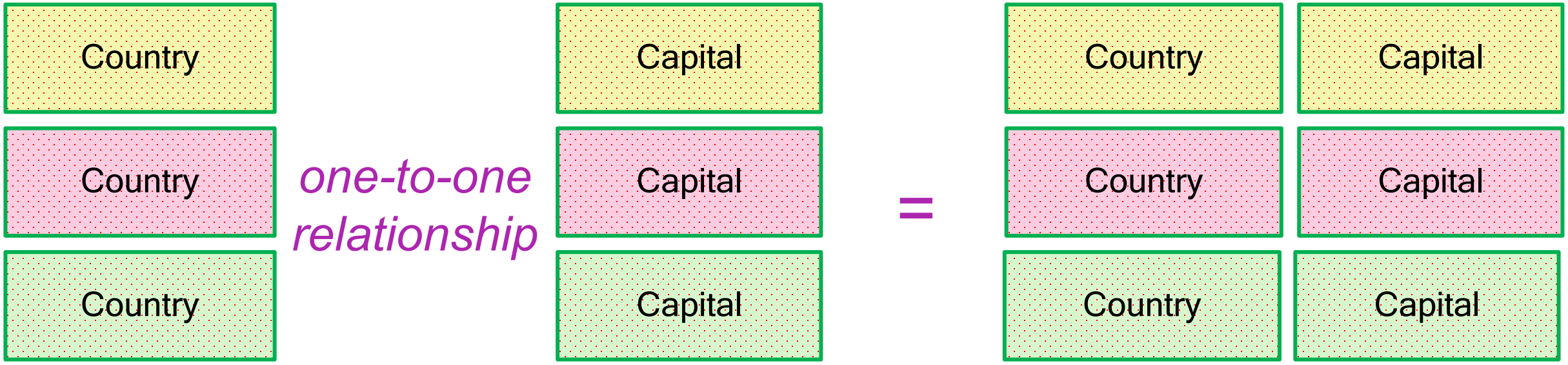
one-to-one relationship

One row in **Table A** corresponds to **one and only one** row in **Table B**.

one-to-one relationship

A *one-to-one* relationship between two tables means that a row in one table can only relate to zero/one row in the table on the other side of their relationship. This is the least common database relationship.

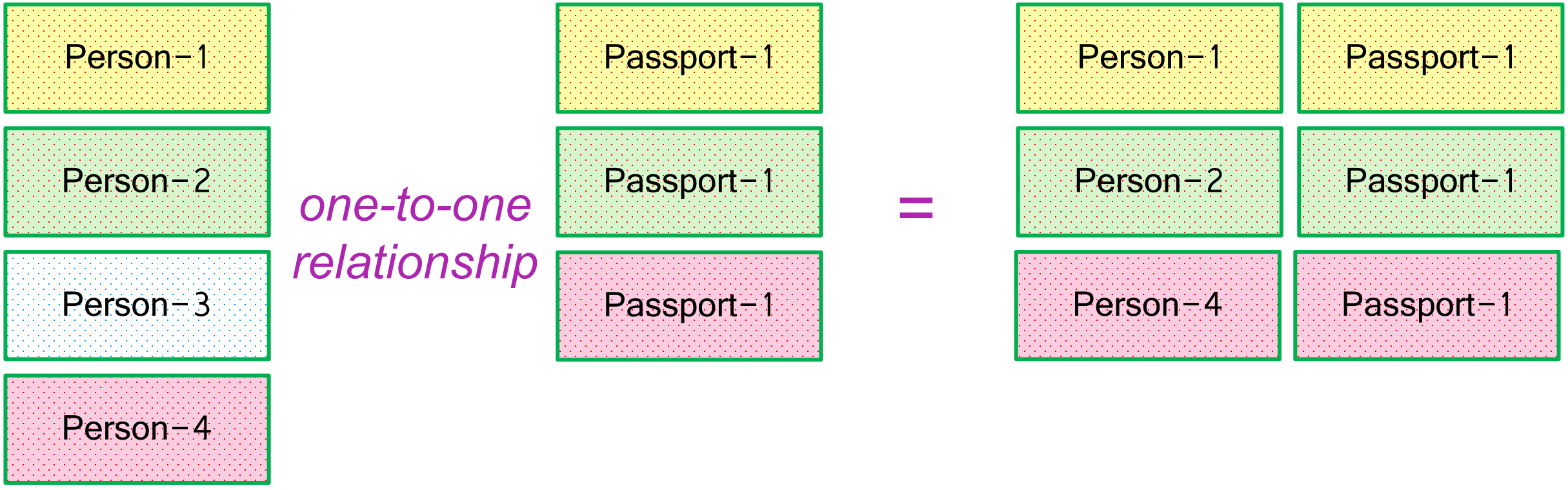
A *one-to-one* relationship is a type of cardinality that refers to the relationship between two entities ***R*** and ***S*** in which one element of entity ***R*** may only be linked to zero/one element of entity ***S***, and vice versa.



one-to-one relationship

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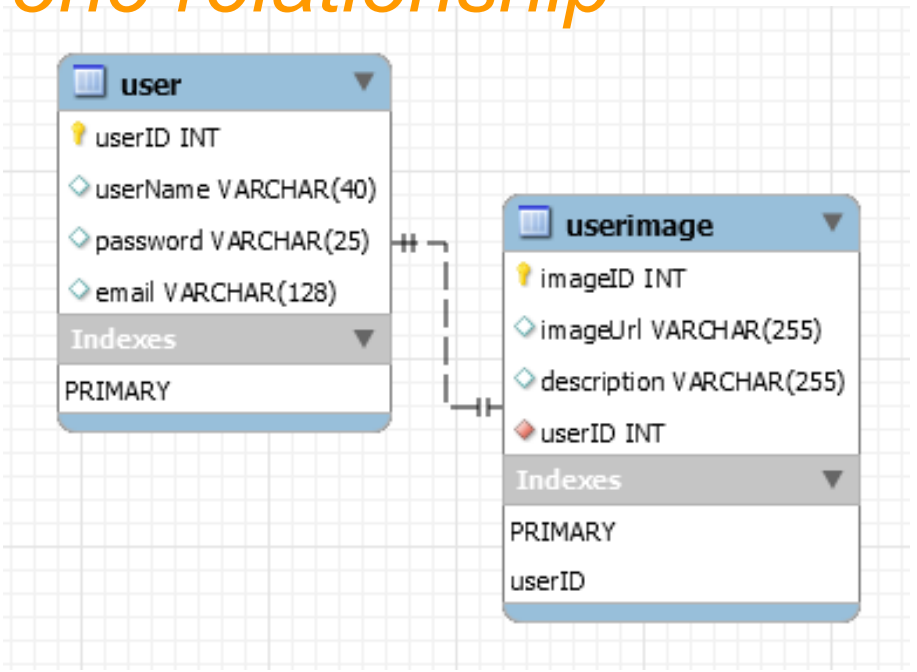
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how to create one-to-one relationship

```
CREATE TABLE user (
   userID INT PRIMARY KEY,
   userName VARCHAR(40),
   password VARCHAR(25),
   email VARCHAR(128)
) ;

CREATE TABLE userImage (
   imageID INT PRIMARY KEY,
   imageUrl VARCHAR(255),
   description VARCHAR(255),
   userID  INT NOT NULL UNIQUE,
   FOREIGN KEY(userID) REFERENCES user(userID)
) ;
```



userID	userName	password	email
1	Ramesh	******	ramesh@gmail.com
2	Rajan	******	rajan.hotmail.com
3	Kumar	******	kumer112@yahoomail.com
4	Suraj	******	suran@gmail.com
NULL	NULL	NULL	NULL

imageID	imageUrl	description	userID
1001	c:/images/img1.jpeg	For passport	1
1002	c:/images/img2.jpeg	For Voter Card	2
1003	c:/images/img3.jpeg	For AADHAR Card	3
1004	c:/images/img4.jpeg	For Licence	4
NULL	NULL	NULL	NULL

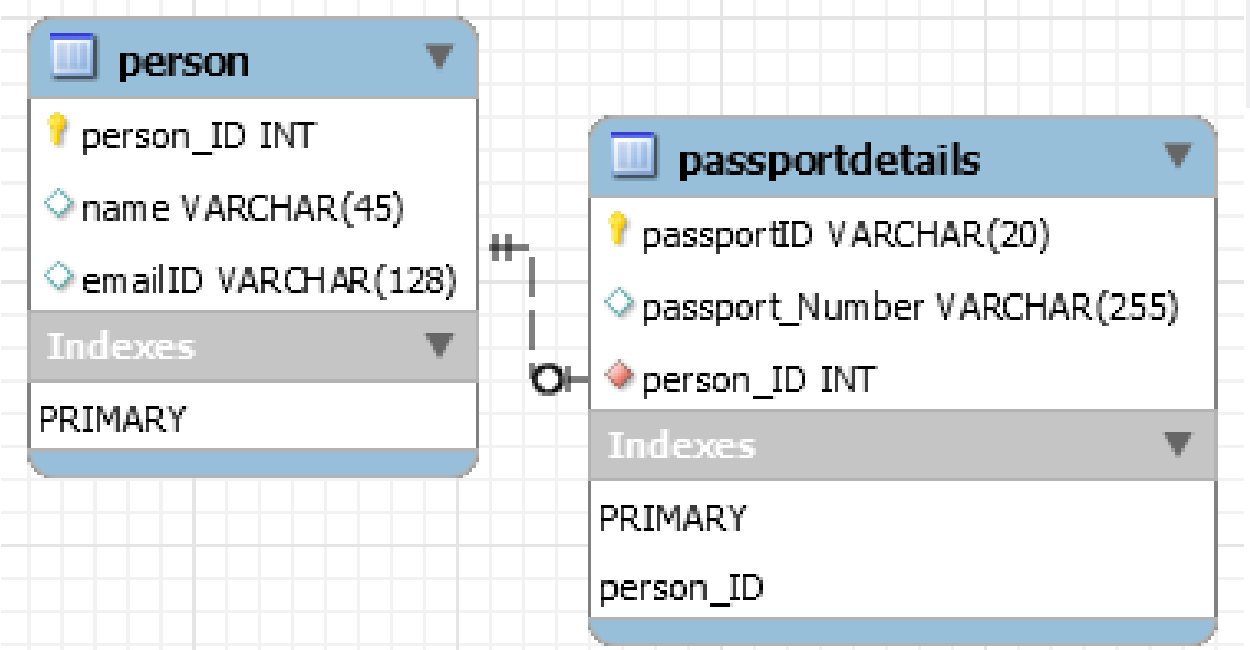
how to create one-to-one relationship

```
CREATE TABLE person (
    person_ID INT PRIMARY KEY ,
    name VARCHAR(45),
    emailID VARCHAR(128)
);
```

person_ID	name	emailID
1	Ramesh	ramesh@gmail.com
2	Rajan	rajan.hotmail.com
3	Kumar	kumer112@yahoomail.com
4	Suraj	suran@gmail.com
NULL	NULL	NULL

```
CREATE TABLE passportDetails (
    passportID VARCHAR(20) PRIMARY KEY,
    passport_Number VARCHAR(255),
    person_ID INT UNIQUE,
    FOREIGN KEY(person_ID) REFERENCES person(person_ID)
);
```

passportID	passport_Number	person_ID
IN-zx001	IN-XAJS1028S	1
IN-zx002	IN-XBDH1738S	2
IN-zx003	IN-XCKE1933S	3
NULL	NULL	NULL



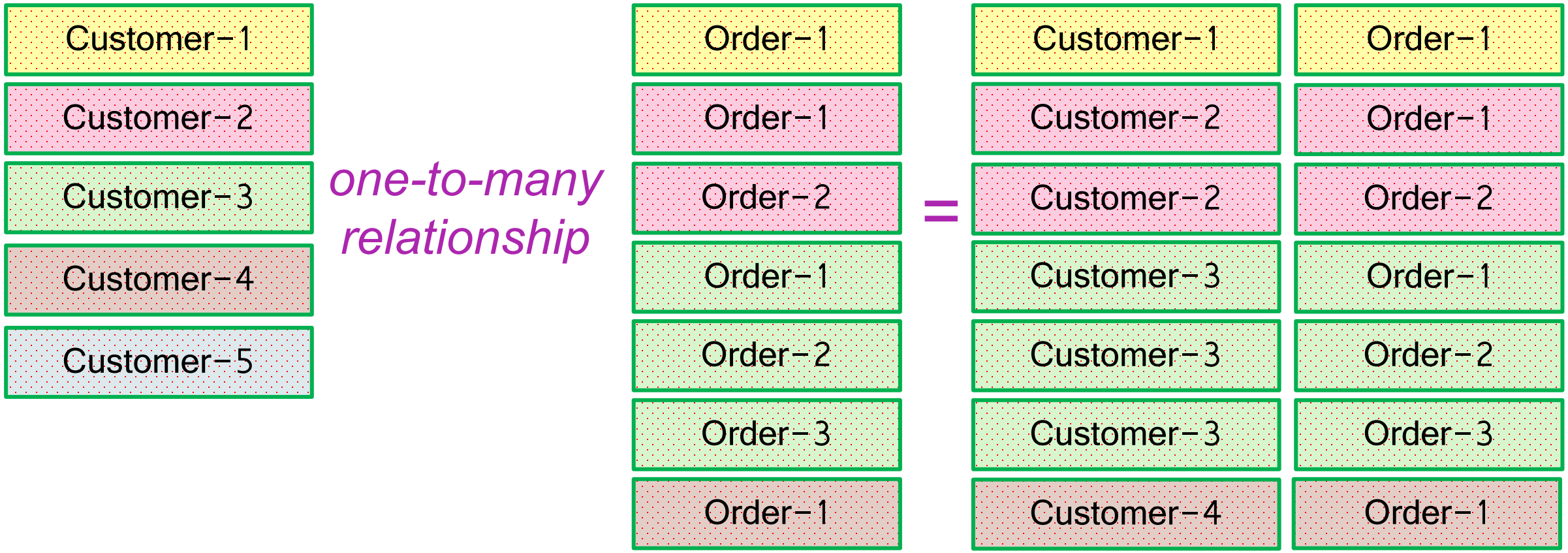
one-to-many relationship

One row in **Table A** can relate to **many rows in Table B**, but **many rows in Table** B **belongs to only one Table A**.

one-to-many relationship

A *one-to-many* relationship between two tables means that a row in one table can have zero or more row in the table on the other side of their relationship.

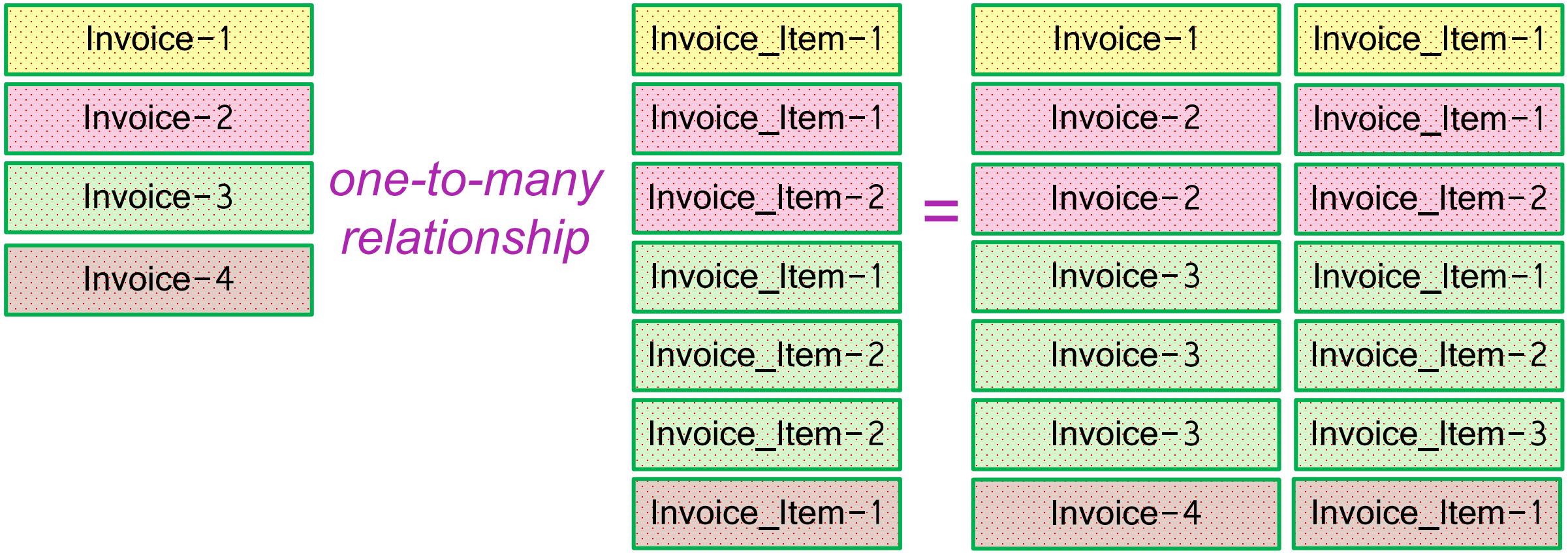
a *one-to-many* relationship is a type of cardinality that refers to the relationship between two entities ***R*** and ***S*** in which an element of ***R*** may be linked to many elements of ***S***, but a member of ***S*** is linked to only one element of ***R***.



one-to-many relationship

A *one-to-many* relationship between two tables means that a row in one table can have one or more row in the table on the other side of their relationship.

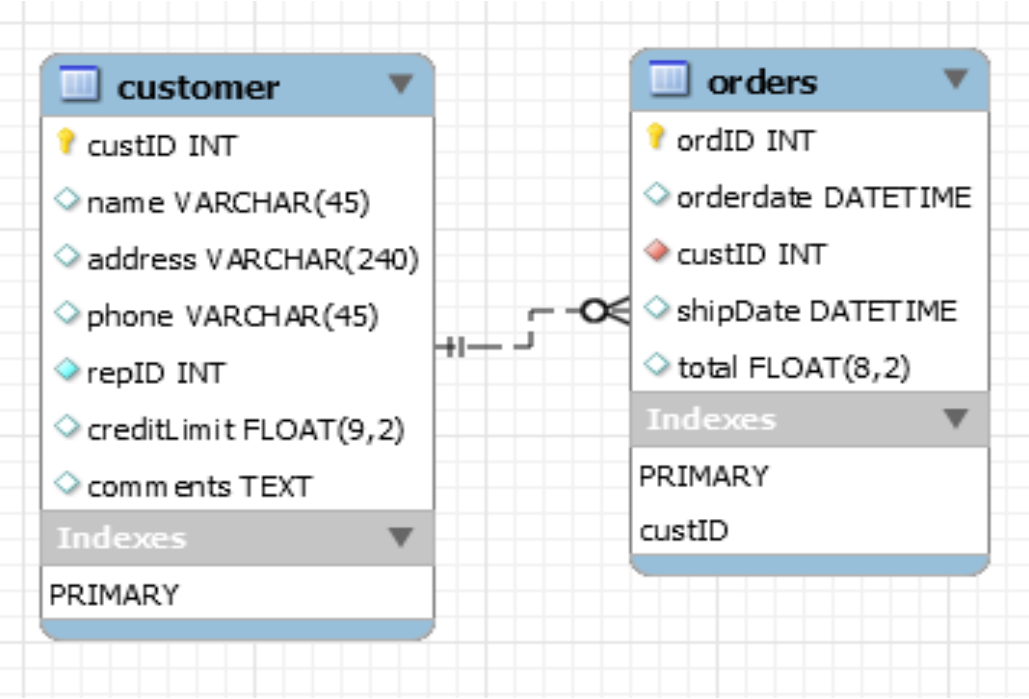
a *one-to-many* relationship is a type of cardinality that refers to the relationship between two entities ***R*** and ***S*** in which an element of ***R*** may be linked to many elements of ***S***, but a member of ***S*** is linked to only one element of ***R***.



how to create one-to-many relationship

```
CREATE TABLE customer (
    custID INT PRIMARY KEY,
    name VARCHAR(45),
    address VARCHAR(240),
    phone VARCHAR(45),
    repID INT NOT NULL,
    creditLimit FLOAT(9,2),
    comments TEXT,
    constraint custid_zero CHECK(custID > 0)
);
```

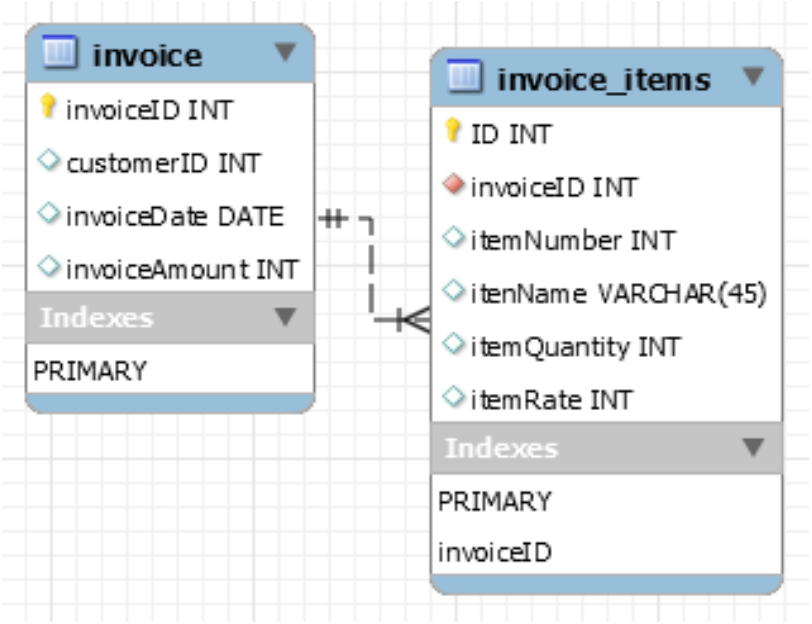
```
CREATE TABLE orders (
    ordID INT PRIMARY KEY,
    orderdate DATETIME,
    custID INT,
    shipDate DATETIME,
    total FLOAT(8,2),
    FOREIGN KEY(custID) REFERENCES customer(custID),
    constraint total_greater_zero CHECK(total >= 0)
);
```



how to create one-to-many relationship

```
CREATE TABLE invoice (
  invoiceID INT PRIMARY KEY,
  customerID INT,
  invoiceDate DATE,
  invoiceAmount INT
);
```

invoiceID	customerID	invoiceDate	invoiceAmount
1	235	2020-01-13	1750
2	235	2020-02-28	5000
3	778	2020-03-10	2000
4	778	2020-03-16	2300
NULL	NULL	NULL	NULL




```
CREATE TABLE invoice_items (
  invoiceID INT,
  itemID INT,
  itenName VARCHAR(45),
  itemQuantity INT,
  itemRate INT,
  PRIMARY KEY(invoiceID, itemID),
  FOREIGN KEY(invoiceID) REFERENCES invoice(invoiceID)
);
```

```
CREATE TABLE invoice_items (
  invoiceID INT NOT NULL,
  itemID INT NOT NULL,
  itenName VARCHAR(45),
  itemQuantity INT,
  itemRate INT,
  UNIQUE(invoiceID, itemID),
  FOREIGN KEY(invoiceID) REFERENCES invoice(invoiceID)
);
```

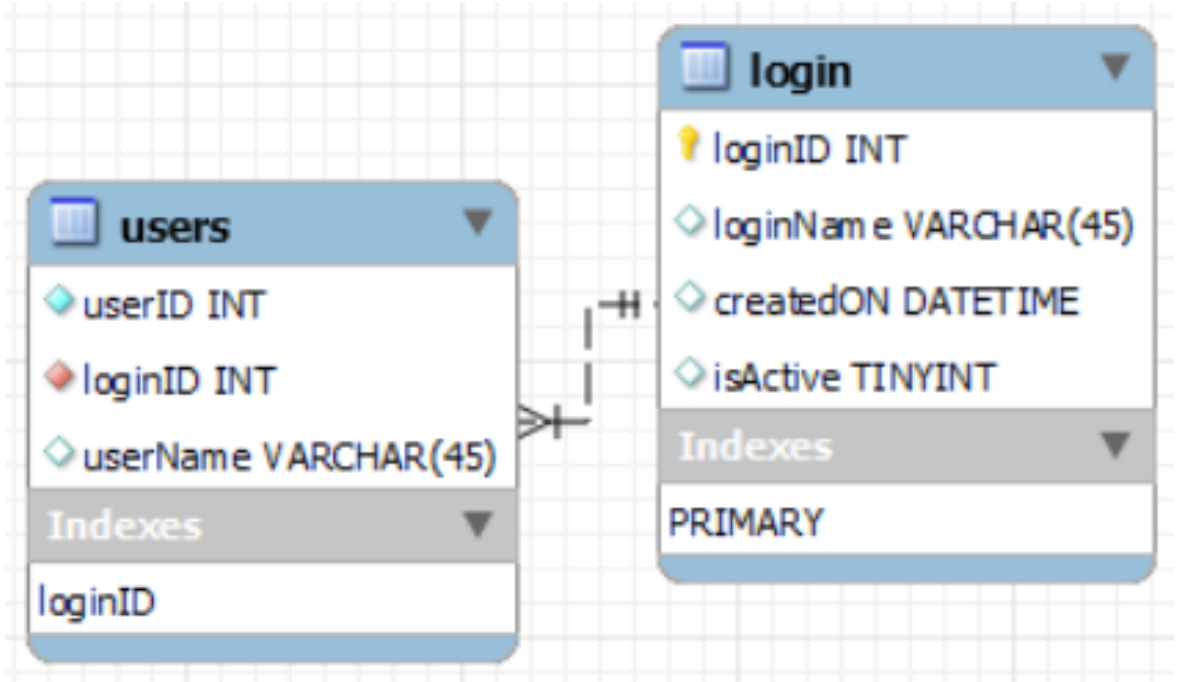
many-to-one relationship

many-to-one relationship

```
CREATE TABLE users (
  userID INT,
  loginID INT,
  userName VARCHAR(45),
  PRIMARY KEY(loginID, userID),
  constraint fk_users_login_loginID1 FOREIGN KEY(loginID)
  REFERENCES login(loginID)
);

CREATE TABLE users (
  userID INT NOT NULL,
  loginID INT NOT NULL,
  userName VARCHAR(45),
  UNIQUE(loginID, userID),
  constraint fk_users_login_loginID2 FOREIGN KEY(loginID)
  REFERENCES login(loginID)
);

CREATE TABLE login (
  loginID INT,
  loginName VARCHAR(45),
  createdON DATETIME,
  isActive TINYINT,
  PRIMARY KEY(loginID)
);
```

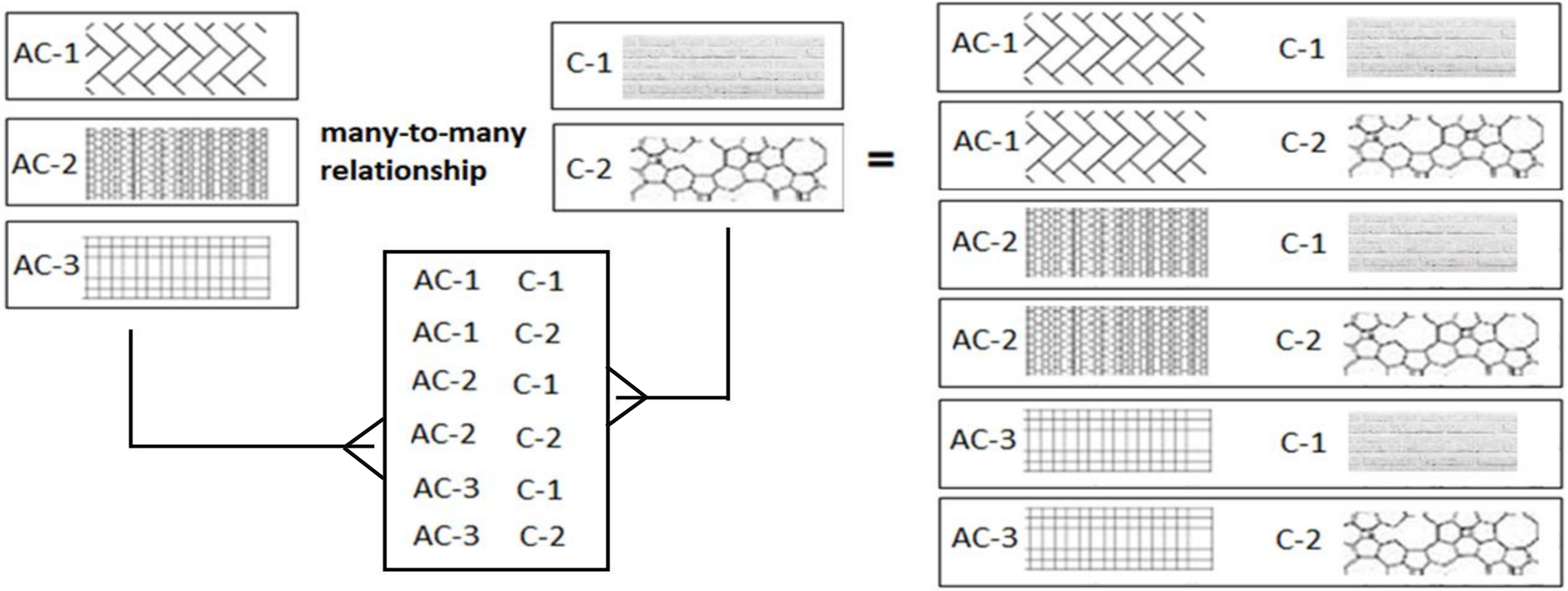


many-to-many relationship

One row in **Table A** can relate to **many rows in Table B**, and one row in **Table B** can also relate to **many rows in Table A**

many-to-many relationship

A *many-to-many* relationship is a type of cardinality that refers to the relationship between two entities ***R*** and ***S*** in which ***R*** may contain a parent instance for which there are many children in ***S*** and vice versa.

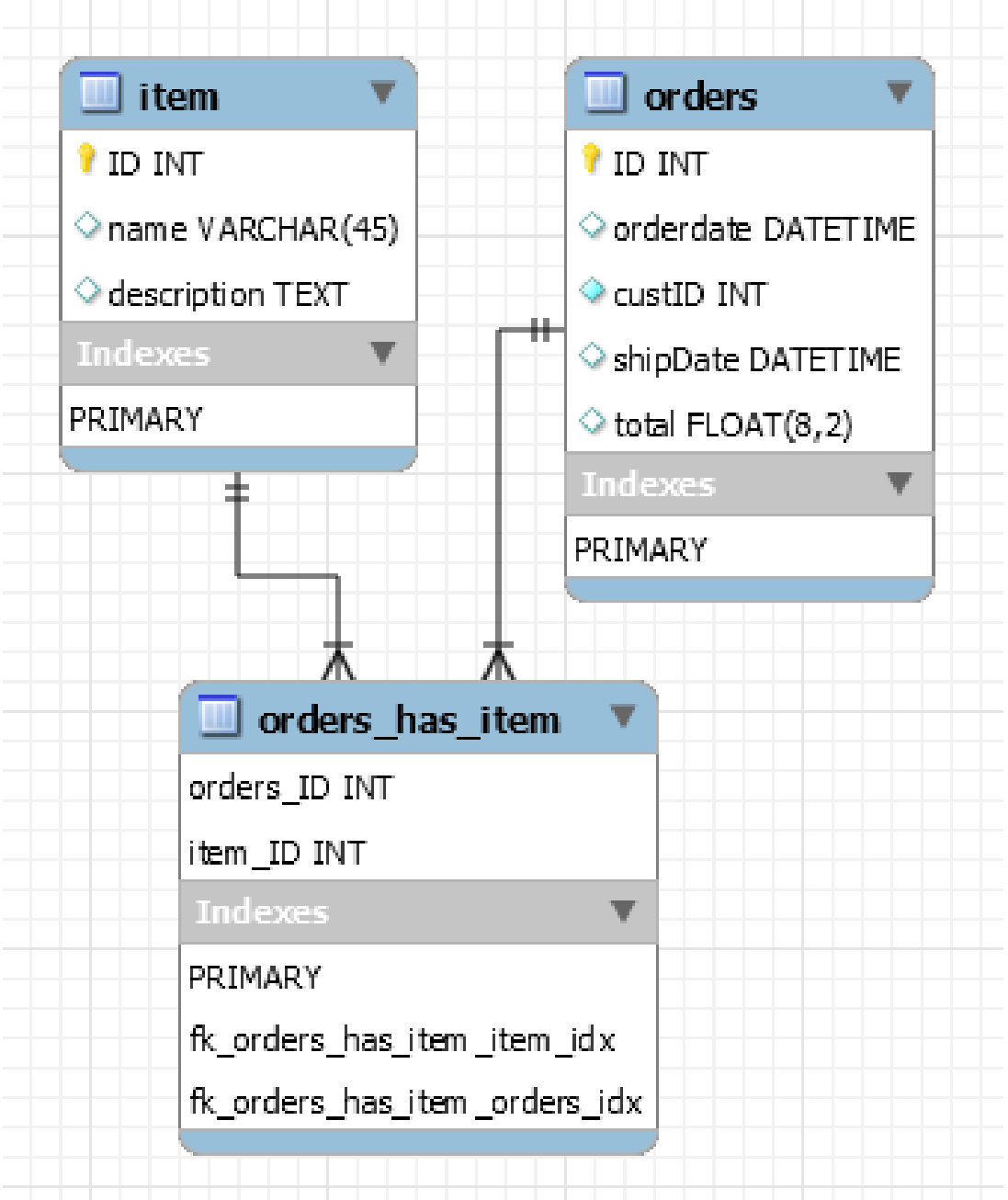


how to create many-to-many relationship

```
CREATE TABLE item (
    ID INT PRIMARY KEY,
    name VARCHAR(45),
    description TEXT
);

CREATE TABLE orders (
    ID INT PRIMARY KEY,
    orderdate DATETIME,
    custID INT NOT NULL,
    shipDate DATETIME,
    total FLOAT(8,2),
    constraint total_greater_zero CHECK(total >= 0)
);

CREATE TABLE orders_has_item (
   orders_ID INT NOT NULL,
   item_ID INT NOT NULL,
   PRIMARY KEY(orders_ID, item_ID),
   constraint fk_orders_has_item_orders FOREIGN KEY(orders_ID)
   REFERENCES orders(ID),
   constraint fk_orders_has_item_item1 FOREIGN KEY(item_ID)
   REFERENCES item(ID)
);
```

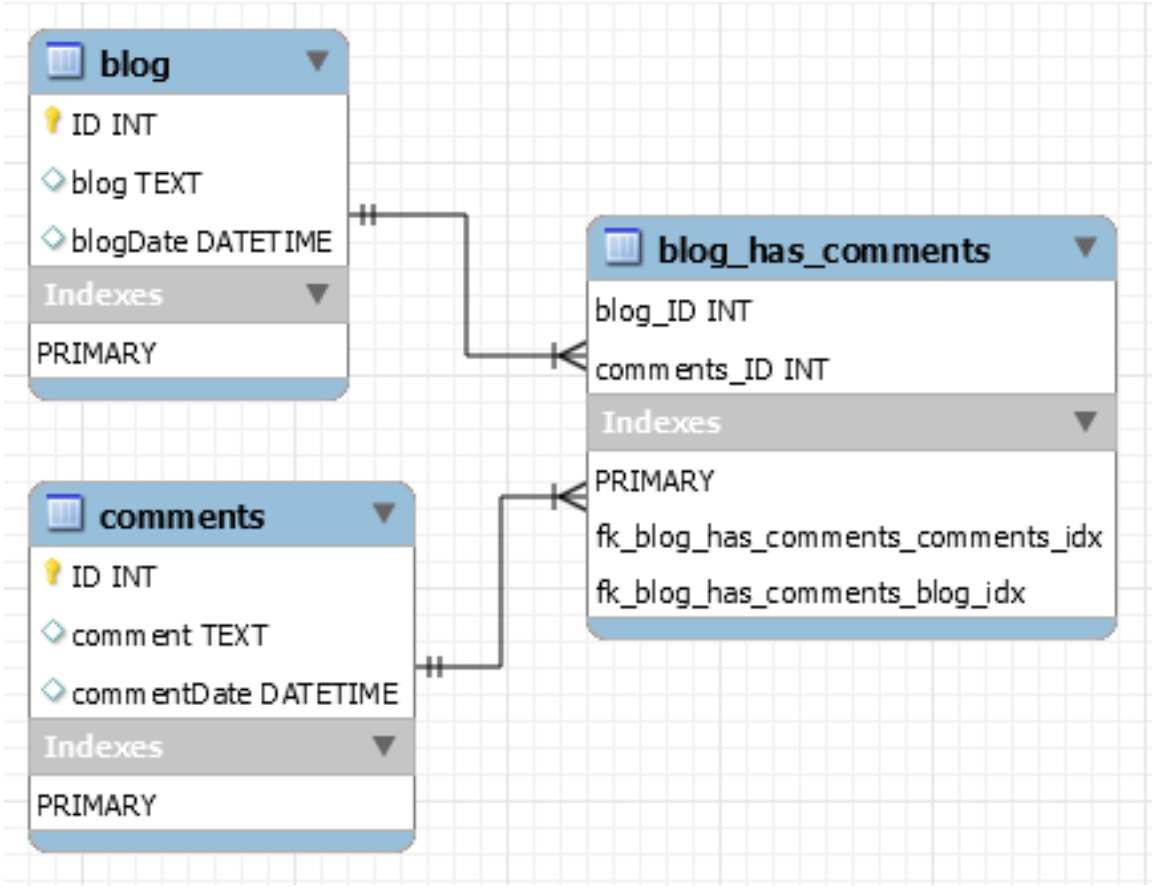


how to create many-to-many relationship

```
CREATE TABLE blog (
  blog_ID INT PRIMARY KEY,
  blog TEXT,
  blogDate DATETIME
);

CREATE TABLE comments (
  comment_ID INT PRIMARY KEY,
  comment TEXT,
  commentDate DATETIME
);

CREATE TABLE comments_has_replies (
   blog_ID INT,
   comment_ID INT,
   PRIMARY KEY(blog_ID, comment_ID),
   constraint fk_blog_has_a_comment FOREIGN KEY(blog_ID) REFERENCES blog(blog_ID ),
   constraint fk_comment_has_a_replies FOREIGN KEY(comment_ID) REFERENCES comments(comment_ID)
);
```

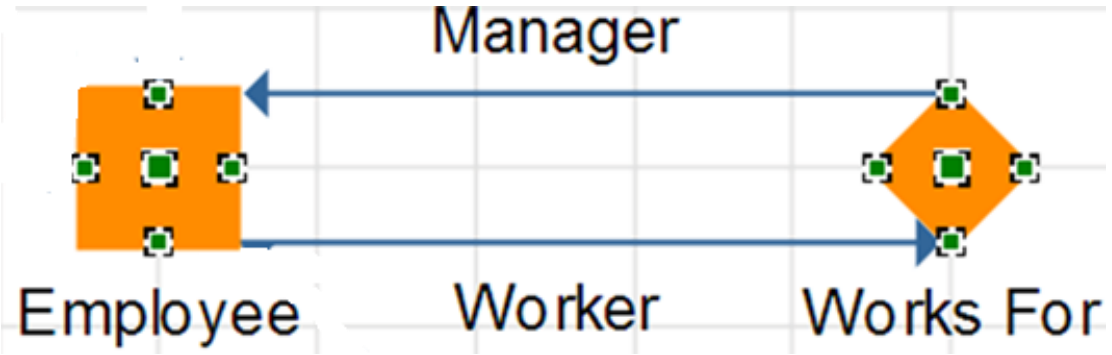


self-referencing relationship

An entity (table) is **related to itself** using a **foreign key** that points to its own primary key.

self-referencing relationship

A *"self-referencing"* or *"recursive"* relationship in databases or data structures means that a record within a table can reference another record in the same table.



Product Categories and Subcategories

CategoryID	CategoryName	ParentCategoryID
1	Electronics	NULL
2	Phones	1
3	Laptops	1
4	Smartphones	2
5	Gaming Laptops	3

self-referencing relationship

```
CREATE TABLE product_category(
  CategoryID INT PRIMARY KEY,
  CategoryName VARCHAR(100),
  ParentCategoryID INT,
  FOREIGN KEY (ParentCategoryID) REFERENCES product_category(CategoryID)
);


CREATE TABLE family(
    PersonID INT PRIMARY KEY,
    RelationName VARCHAR(100),
    ParentID INT,
    FOREIGN KEY (ParentID) REFERENCES family(PersonID)
);
```

- Branch { branch-ID, branch-name, branch-city, assets }
- Customer { customer-id, customer-name, customer-street, customer-only }
- Account { account-number, branch-ID, balance }
- Loan { loan-number, branch-ID, amount }
- Depositor { customer-id, account-number }
- Borrower { customer-id, loan-number }

Relational Algebra

∧ (and), ∨ (or), ¬ (not)

Relational Algebra

- Select (sigma) (σ)
- Project (pi) (Π)
- Union (∪)
- Set different (−)
- Cartesian product (X)
- Rename (ρ)

- and ∧
- or ∨
- not ¬

SELECT

Select (σ) Notation – $\sigma_p(r)$

$\sigma_{condition}(r)$

σ would represent the SELECT command

p would represent the condition for selection.

(r) would represent the Relation or the Table from which we are making a selection of the tuples.

- Select the EMPLOYEE whose employee number is 7, or those whose date of birth is '01-Jan-1980'

$(r)=\sigma_{empno=7}(EMPLOYEE)$
$(r)=\sigma_{dob<='01\text{-}Jan\text{-}1980'}(EMPLOYEE)$
$(r)=\sigma_{empno=7 \vee dob='01\text{-}Jan\text{-}1980'}(EMPLOYEE)$

Returns the result to new relation.

SELECT

Select (σ) Notation – $\sigma_p(r)$

$\sigma_{condition}(r)$

- Select the EMPLOYEE tuples whose employee number is > 7 and department ID is 10

$$(r)=\sigma_{deptno=10}\,[\sigma_{empno>7}\,(EMPLOYEE)]$$

$$(r)=\sigma_{empno>7\ \wedge\ deptno=10}\,(EMPLOYEE)$$

PROJECT

Project (Π) Notation – $\Pi_{A1, A2, An}$ (r)

$\Pi_{<attribute\text{-}list>}(r)$

$\prod$ would represent the ROJECT.

<attribute list> would represent the attributes(columns) we want from a relational.

(r) would represent the Relation or the Table from which we are making a selection of the tuples.

Select Date of Birth (dob) and Employee Number (empno) from the relation EMPLOYE or Select Date of Birth (dob) and Employee Number (empno) from the relation EMPLOYE whose date of birth is before 1980

$(r)=\Pi_{dob, empno}(EMPLOYEE)$

$(r)=\Pi_{dob, empno}[\sigma_{dob<'01\text{-}Jan\text{-}1980'}(EMPLOYEE)]$

UNION

Union (∪) Notation – r U s

$$\Pi_{<attribute\text{-}list>}(r) \cup \Pi_{<attribute\text{-}list>}(s)$$

- ***r***, and ***s*** must have the same number of attributes.
- Attribute domains must be compatible.
- Duplicate tuples are automatically eliminated.

Selects the customer name who have either purchased a book or a computer or both.

$$(r) = \Pi_{cName}(BOOK) \cup \Pi_{cName}(COMPUTER)$$

INTERSECTION

Intersection (∩) Notation – r ∩ s

$$\Pi_{<attribute\text{-}list>}(r) \cap \Pi_{<attribute\text{-}list>}(S)$$

- ***r***, and ***s*** must have the same number of attributes.
- Attribute domains must be compatible.
- Duplicate tuples are automatically eliminated.

Projects the customer name who have purchased a book and a computer both.

$$(r) = \Pi_{cName}(BOOK) \cap \Pi_{cName}(COMPUTER)$$

MINUS

Minus (-) Notation – r - s

$$\Pi_{<attribute\text{-}list>}(r) - \Pi_{<attribute\text{-}list>}(S)$$

- ***r***, and ***s*** must have the same number of attributes.
- Attribute domains must be compatible.
- Duplicate tuples are automatically eliminated.

- Projects the customer name who have purchased a BOOK but not a COMPUTER.
- Projects the customer name who have purchased a COMPUTER but not a BOOK.

$$(r) = \Pi_{cName}(BOOK) - \Pi_{cName}(COMPUTER)$$

$$(r) = \Pi_{cName}(COMPUTER) - \Pi_{cName}(BOOK)$$

CARTESIAN PRODUCT

Cartesian product (X) Notation − r X s

$\Pi_{<attribute\text{-}list>}(r) \times \Pi_{<attribute\text{-}list>}(S)$

Examples Queries

Find all loans of over $1200

$$(r)=\sigma_{amount > 1200}(loan)$$

Find the loan number for each loan of an amount greater than $120

$$(r)=\prod_{loan\text{-}number}(\sigma_{amount > 1200}(loan))$$

Examples Queries

Find the names of all customers who have a loan, an account, or both, from the bank

$$\Pi_{customer\text{-}name}(borrower) \cup \Pi_{customer\text{-}name}(depositor)$$

Find the names of all customers who have a loan and an account at bank.

$$\Pi_{customer\text{-}name}(borrower) \cap \Pi_{customer\text{-}name}(depositor)$$

- Customer (CustomerID, Name, Email, City)
- Product (ProductID, Name, Category, Price)
- Order (OrderID, CustomerID, OrderDate, TotalAmount)
- OrderDetails (OrderID, ProductID, Quantity)
- Payment (PaymentID, OrderID, PaymentStatus)
- Shipment (ShipmentID, OrderID, ShipmentStatus, DeliveryDate)

Relational Algebra Examples

E-Commerce Database Schema

Examples Queries

- Find All Customers from 'New York'

  σ *City='NewYork'(Customer)*

- Show Only Product Names and Prices

  $\pi_{\text{Name, Price}}(\text{Product})$

- Find Customers Who Placed Orders

  $\pi_{\text{Customer.Name, Order.OrderID}}(\text{Customer} \bowtie \text{Order})$

- List All Products Ordered by a Customer Named 'Alice'

  $\pi_{\text{Product.Name, Order.OrderID}}(\sigma\ \text{Customer.Name= 'Alice'}\ (\text{Customer} \bowtie \text{Order} \bowtie \text{OrderDetails} \bowtie \text{Product}))$

Examples Queries

- Find Customers Who Have Never Placed an Order

  $\pi_{Name}(\sigma_{CID=(\pi CID(Customers) - \pi CID(Orders)}))(Customers))$

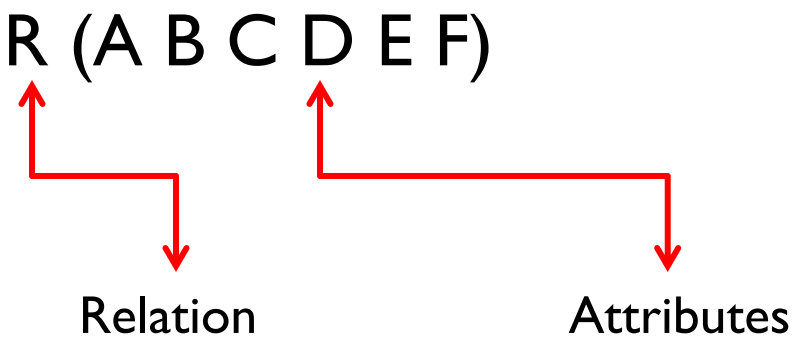
- Find Products Ordered in a Given Category

  $\pi_{Pname}(\sigma_{PID=(\pi PID(OrderDetails)}(\sigma Category='Electronics'(Products)))$



Closure of Attributes

Closure of attributes



Schema: A schema is a collection of database objects (like table, foreign key, primary key, views, columns, data types, stored procedure, etc.) associated with one particular database username. This username is called the schema owner, or the owner of the related group of objects. You may have one or multiple schemas in a database.

What is schema and instance

Instance

- The data stored in database at a particular moment of time is called instance of database.

For example, lets say we have a single table student in the database, today the table has 100 records, so today the instance of the database has 100 records. Lets say we are going to add another 100 records in this table by tomorrow so the instance of database tomorrow will have 200 records in table.

An instance of a relation is a set of tuples, also called records

What is Data Model

Data modeling is the process of creating a data model for the data to be stored in a Database. This data model is a conceptual representation of

- Data objects
- The associations between different data objects
- The rules.

data modeling

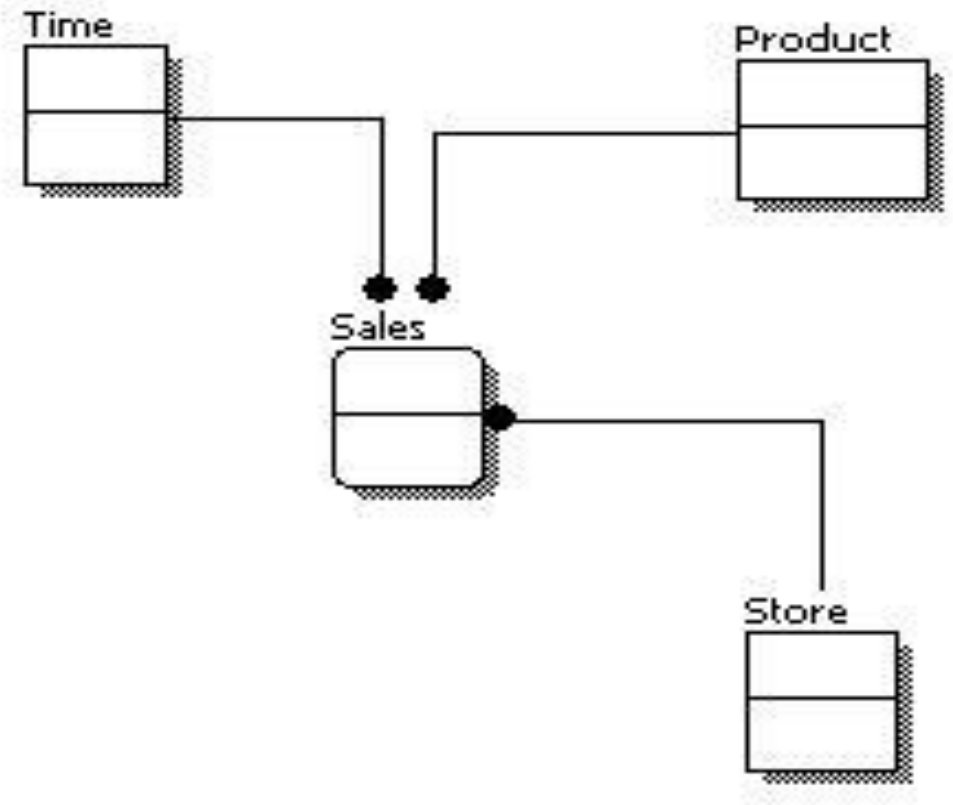
Data models define how data is connected to each other and how they are processed and stored inside the system.

Data modeling involves a progression from conceptual model to logical model to physical **object**.

conceptual data model

Features of conceptual data model include:

- Includes the important entities and the relationships among them.
- No attribute is specified.
- No primary key is specified.



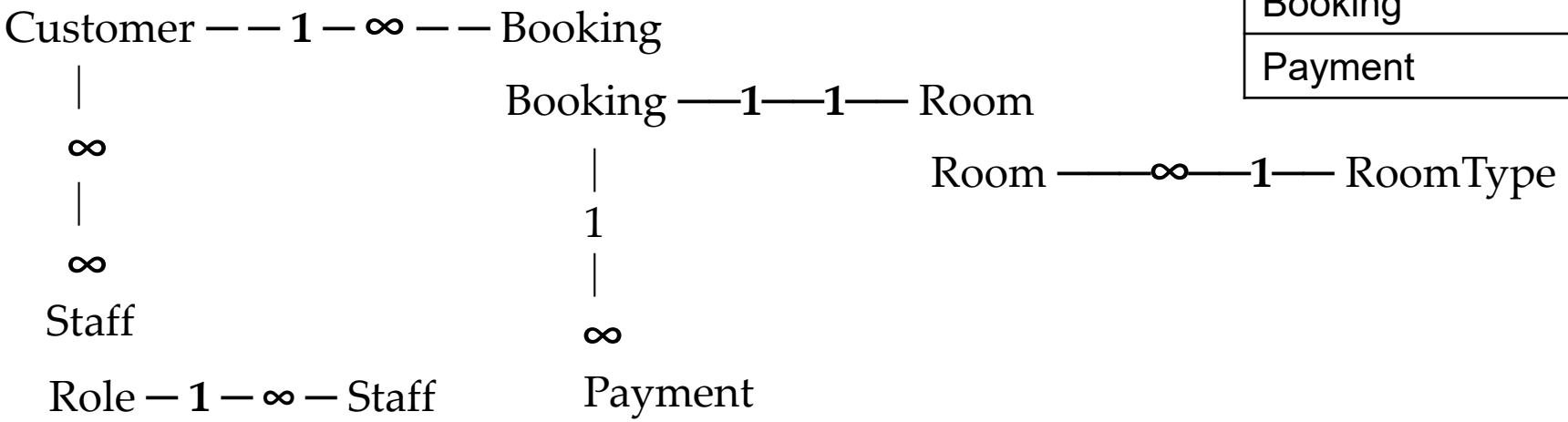
Online Hotel Management System

The system must allow:

1. ***Customers*** to book rooms online.
2. ***Rooms*** are ***categorized/ roomtypes*** (*Standard, Deluxe, Suite*).
3. Each room can have one or more ***bookings*** (but not overlapping dates).
4. Customers can make ***payments*** online.
5. ***Staff*** manage [***Roles***] reservations and guests. (*Receptionist, Housekeeping, Manager, Chef, etc.*)
6. Each booking includes *check-in* and *check-out dates*.
7. Different ***type of services*** are provided like *Room Cleaning, Room Booking, Food Delivery*

conceptual data model

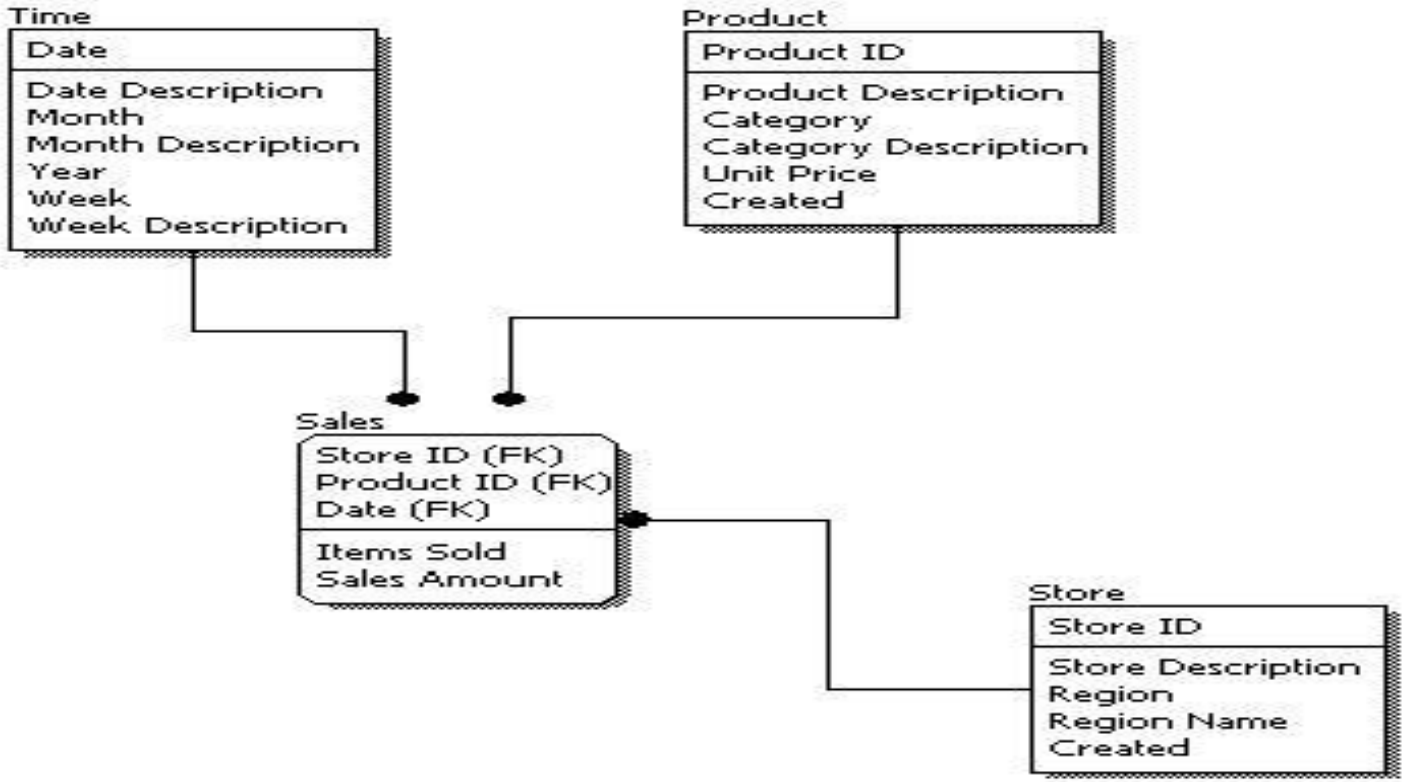
Entity	Description
Customer	A person who books a room
Staff	Employees who manage the hotel
Role	Receptionist, Housekeeping, Manager, Chef, etc.
ServiceType	Room Cleaning, Room Booking, Food Delivery
Customer_Service	CustomerID, StaffID, ServiceID, ...
Room	Hotel room with room type, price, availability, . . .
RoomType	Standard, Deluxe, Suite, **etc.**
Booking	Reservation made by a customer
Payment	Payments made for bookings



logical data model

Features of logical data model include:

- Includes all entities and relationships among them.
- All attributes for each entity are specified.
- The primary key for each entity is specified.
- Foreign keys are specified.



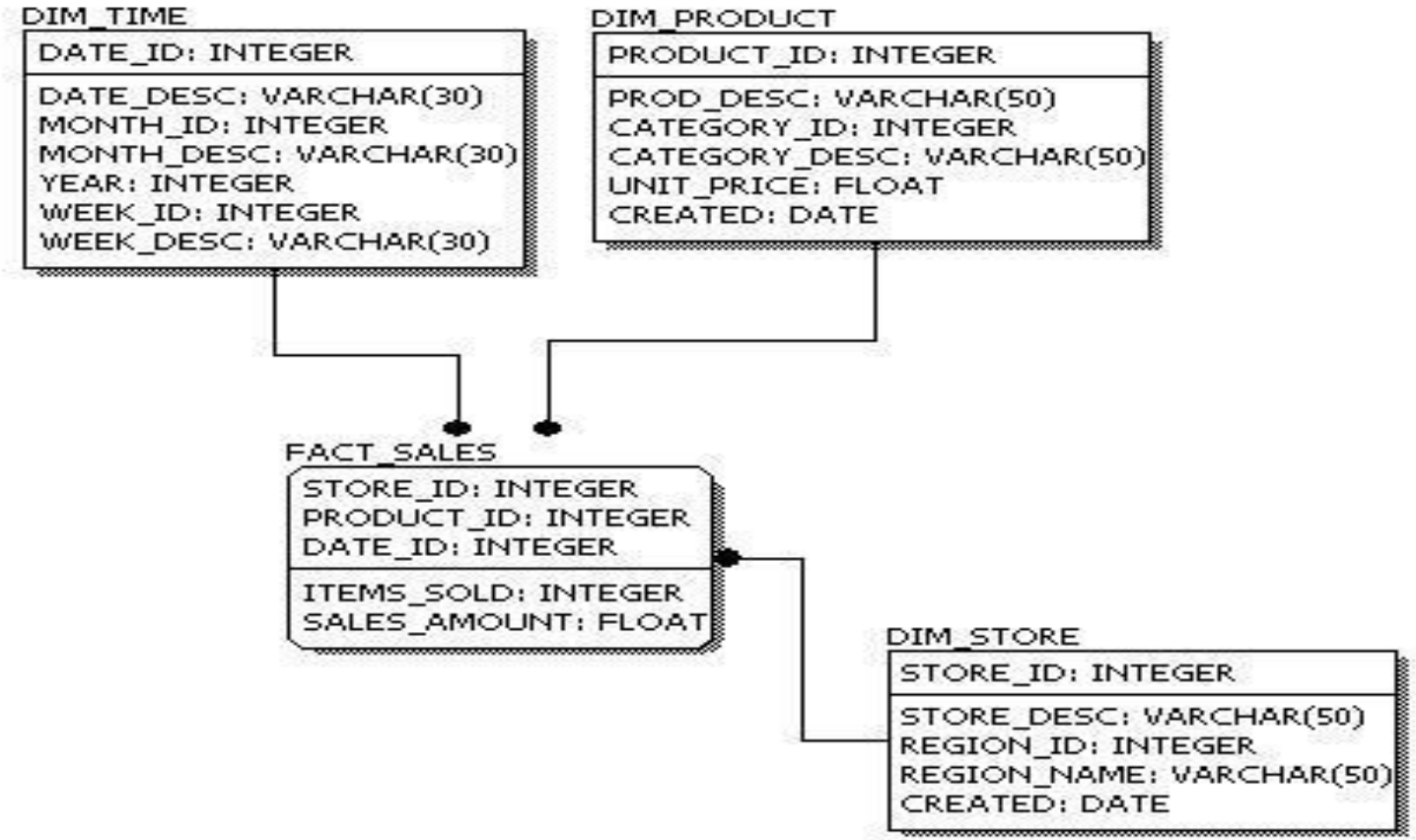
logical data model

1. **Customer** [customer_id (*PK*), name, email, phone, address]
2. **Role** [RoleID (*PK*), RoleName, Description] // Receptionist, Housekeeping, Manager, Chef, etc.
3. **Staff** [staff_id (*PK*), name, role, email, phone, RoleID (*FK*)]
4. **Service** [serviceTypeID (*PK*), ServiceName, Description] // Room Cleaning, Room Booking, Food Delivery
5. **Role_Service [**RoleID (*FK*), serviceTypeID (*FK*), Remarks**]**
6. **Customer_Service** []
7. **Room** [room_id (*PK*), room_number, room_type_id (*FK*), floor, status] // Available, Occupied, Maintenance
8. **RoomType** [room_type_id (*PK*), type_name, price_per_night] // Standard, Deluxe, Suite, **etc.**
9. **Booking** [booking_id (*PK*), customer_id (*FK*), room_id (*FK*), staff_id (FK, optional), check_in_date, check_out_date, booking_status] // Confirmed, Cancelled, Completed
10. **Payment** [payment_id (*PK*), booking_id (*FK*), amount, payment_date, payment_method] // Card, UPI, Cash, Online

physical data model

Features of physical data model include:

- Convert entities into tables.
- Convert relationships into foreign keys.
- Convert attributes into columns.



Feature	Conceptual	Logical	Physical
Entity Names	✓	✓	
Entity Relationships	✓	✓	
Attributes		✓	
Primary Keys		✓	✓
Foreign Keys		✓	✓
Table Names			✓
Column Names			✓
Column Data Types			✓

- Hierarchical databases
- Network databases
- Object-oriented databases
- Relational databases
- NoSQL databases

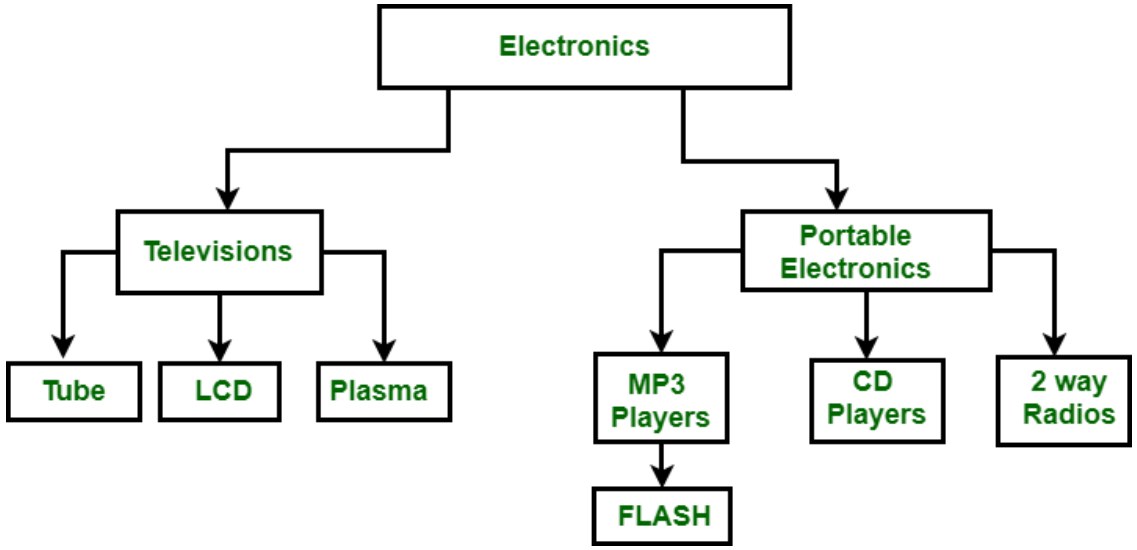
Types of data modeling

There are three main types of DBMS data models: relational, hierarchical and network.

types of data modeling

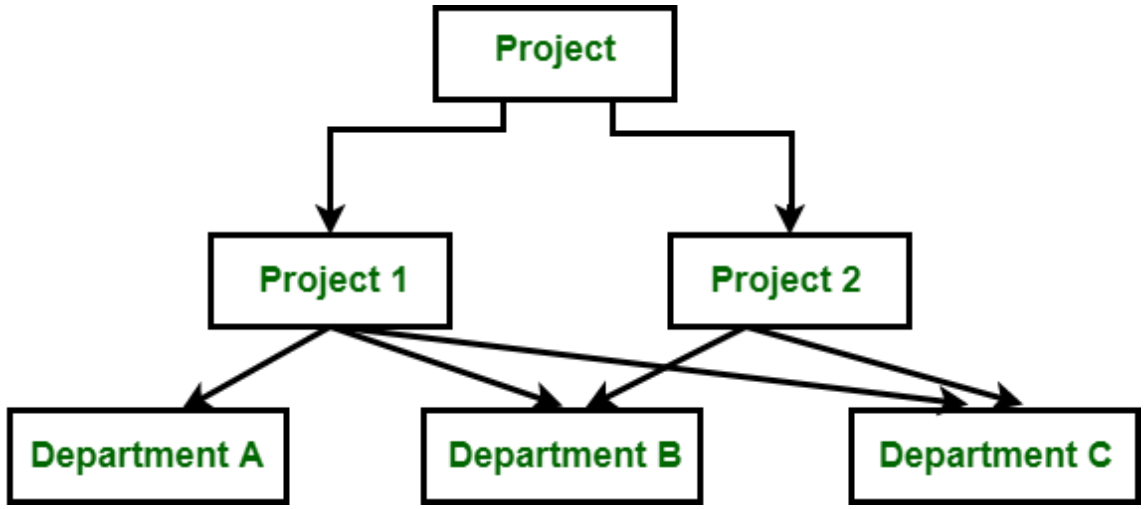
In a hierarchical database, the data is organized in a "**tree structure**", which consists of nodes connected by edges representing the relationships between records.

hierarchical data models



In a network database, the data is organized in a "**graph-like structure**", which consists of multiple parent and child records.

network data model



- Geography :- Country → State → City → ZipCode
- Products :- Category → Sub-category → Product → Variant

Problems that can occure in un-normalized database where all the data is stored in one table.

Anomalies in DBMS

Something that deviates from what is standard, normal, or expected.

There are three types of anomalies that occur when the database is not normalized. These are –

- **Insertion** anomaly
- **Updation** anomaly
- **Deletion** anomaly.

anomalies in dbms - example

student table.

Rollno (PK)	name	branch	hod	office_tel
401	Amit	**CSE**	**Mr. X**	**53337**
402	Rajan	**CSE**	**Mr. X**	**53337**
403	Bhavin	**CSE**	**Mr. X**	**53337**
404	Pankaj	**CSE**	**Mr. X**	**53337**
NULL	NULL	**BCA**	**Mr.Y**	**77127**

insert, update, and delete anomaly

student table.

RollNo (PK)	name	branch	hod	office_tel
401	Amit	**CSE**	**Mr. X**	**53337**
402	Rajan	**CSE**	**Mr. X**	**53337**
403	Bhavin	**CSE**	**Mr. X**	**53337**
404	Pankaj	**CSE**	**Mr. X**	**53337**
NULL	NULL	**BCA**	**Mr.Y**	**77127**

- **Insert anomaly**: What if we want to add new branch in the student table? In this case as the student Rollno is a primary key, it will not allow you to insert new branch. Also, if we have to insert data of 100 students of same branch, then the branch information will be repeated for all those 100 students. This is **Insertion anomalies**.
- **Update anomaly:** What if **Mr. X** telephone number changes? In that case all the student records will have to be updated, and if by mistake we miss any record, it will lead to data inconsistency. This is **Updation anomaly**.
- **Delete anomaly:** In our **Student** table, two different information are kept together, Student information and Branch information. Hence, at the end of the academic year, if student records are deleted, we will also lose the branch information. This is **Deletion anomaly.**

Movie_ID	MovieName	Genres	Film_certificate

If P -> Q and Q -> R is true, then P-> R is a *transitive dependency*.

- **Movie_ID -> Genres**
- **Genres -> Film_certificate**

- Therefore, **Movie_ID -> Film_certificate** has ***transitive dependency***.

Functional & Transitivity Dependency

A functional dependency (FD denoted by an arrow "→") is a relationship between two attributes, typically between the prime attribute (PK) and other non-prime attributes within a table.

If X ->Y , the value of X uniquely determines the value of Y, then Y is functionally dependent on attribute X (usually the PK) .

Note:

The left side is called the **determinant**, and the right side is the **dependent**.

- **Movie_ID -> MovieName**

Therefore, **MovieName** is ***functional dependent*** on attribute **Movie_ID**

Transitivity Dependency

- CREATE TABLE customers (
  customerID INT PRIMARY KEY,
  customerName VARCHAR(100),
  cityID INT,
  cityName VARCHAR(100),
  state VARCHAR(100)
);

customerID → cityID
cityID → cityName, state

customerID → cityName, state (via cityID)

So:- *This is Transitive Dependency*

- CREATE TABLE customers (
  customerID INT PRIMARY KEY,
  customerName VARCHAR(100),
  cityID INT
);

- CREATE TABLE cities (
  cityID INT PRIMARY KEY,
  cityName VARCHAR(100),
  state VARCHAR(100)
);

functional dependency

Movie_ID	MovieName	Genres	Film_certificate
1	X - (2)	Action	U/A
2	P - (1)	Comey	U
3	Q - (1)	Drama	U/A
4	R - **(4)**	Thriller	U/A
5	R - **(4)**	Action	U/A
6	Y - (2)	Action	U/A
7	Z – (3)	Musical	U
8	Z – (3)	Musical	U

In a relation R, a functional dependency X → Y means: If two rows have the same value for attribute X, then they must also have the same value for attribute Y.

Determinant	Dependant
1	a
2	b
Valid F.D.	

Determinant	Dependant
1	a
1	a
Valid F.D.	

Determinant	Dependant
1	a
2	a
Valid F.D.	

Determinant	Dependant
1	a
1	b
Invalid F.D.	

making IT simple

Partial Dependency

A partial dependency would occur whenever a non-prime attribute depends functionally on a part of the given candidate key.

BranchID	AccountNo	holderName	Pin code	Area

R(X, Y, A, B, C)

BranchID	AccountNo	holderName	PincodeID

Pincode	Area

FD: {XY -> A, A -> B, Y -> C }. Find partial dependency.

Where:

- XY is the only candidate key.
- key attributes =X, Y and
- non-key attributes = A, B, C.

Student (studentID, assignmentNo, studentName, assignment, marks)
PRIMARY KEY (studentID, assignmentNo)

{studentID } -> studentName // partial dependency
{assignmentNo } -> assignment // partial dependency
{studentID, assignmentNo } -> marks // Full Dependency

Note:

Key attributes are also called **prime attributes** and **non-key attributes** are also called **non-prime attributes**.

XY -> A is not a partial dependency // it is full dependency.

A -> B is not a partial dependency // // it is a transitive dependency

Y -> C is a partial dependency, since Y is a part of the candidate key and C is a non-key attribute.

Partial Dependency

```
CREATE TABLE orders(
    orderID INT,
    orderDate DATE,
    productID INT,
    productName VARCHAR(100),
    price FLOAT,
    quantity INT,
    PRIMARY KEY (orderID, productID)
);
```

Price inconsistency risk

- Primary Key = (orderID, productID)
- orderID → orderDate
- productID → productName, price
- (orderID, productID) → quantity

```
CREATE TABLE products (
    productID INT PRIMARY KEY,
    productName VARCHAR(100),
    price FLOAT
);
```

```
CREATE TABLE orders (
    orderID INT,
    orderDate DATE);
```

```
CREATE TABLE order_details (
    orderID INT,
    productID INT,
    quantity INT,
    PRIMARY KEY (orderID, productID)
     FOREIGN KEY (productID) REFERENCES products(productID) ,
     FOREIGN KEY (orderID) REFERENCES orders(orderID)
);
```

patient_id	doctor_id	patient_name	doctor_name	treatment

1. Doctor name repeated for every patient
2. Patient name repeated for every doctor visit

Normalization in DBMS

Normalization is the process of removing redundant data from the database by splitting the table into two or more tables and defining relationships between these tables.

Problem

To understand normalization in database with example tables, let's assume that we are supposed to store the details of **courses** and **instructors** in a university.

Course code	Course venue	Instructor Name	Instructor's phone number
CS101	Lecture Hall 20	Prof. Ronald	+1 6514821924
CS152	Lecture Hall 21	Prof. John	+1 6519272918
CS154	CS Auditorium	Prof. Ronald	+1 6514821924

At first, this design seems to be good. However, issues start to develop once we need to modify information. For instance, suppose, if Prof. Ronald changed his mobile number. In such a situation, we will have to make edits in 2 places**.** What if someone just edited the mobile number against **CS101**, but forgot to edit it for **CS154**? This will lead to stale/wrong information in the database.

Solution

This problem, however, can be easily tackled by dividing our table into 2 simpler tables.

Table 1 (Instructor)

Instructor's ID	Instructor's Name	Instructor's Number
1	Prof. Ronald	+1 6514821924
2	Prof. John	+1 6519272918
3	Prof. Smith	+1 6519272919

Table 2 (Course)

Course Code	Course Venue	Instructor ID
CS101	Lecture Hall 20	1
CS152	Lecture Hall 21	2
CS154	CS Auditorium	1

First Normal Form (1NF)

1st NF removes → repeating groups or multi-valued attributes

For a table to be in the First Normal Form, it should follow the following 4 rules:

1. It should only have single(atomic) valued attributes/columns.
2. Values stored in a column should be of the same domain
3. All the columns in a table should have unique names.
4. And the order in which data is stored, does not matter.
5. The table should have no "repeating groups" of fields.

OrderID	CustomerName	Product1	Product2	Product3
1	Rajesh Singh	Apple	Orange	Banana
2	Rajat Kumar	Apple	NULL	NULL
3	Ramesh	Orange	Grape	NULL

First Normal Form (1NF)

The First normal form simply says that each cell of a table should contain exactly one value.

Instructor's Name	Course Code
Prof. Ronald	(CS101, CS154)
Prof. John	(CS152)
Prof. Smith	(CS189, CS107)

Here, the issue is that in the first row, we are storing 2 courses against Prof. Ronald. A better method would be to store the courses separately. This way, if we want to edit some information related to **CS101**, we do not have to touch the data corresponding to **CS154**.

Instructor's Name	Course Code
Prof. Ronald	CS101
Prof. Ronald	CS154
Prof. John	CS152
Prof. Smith	CS189
Prof. Smith	CS107

Second Normal Form (2NF)

2nd NF removes → Partial Dependency

For a table to be in the Second Normal Form

1. It should be in the First Normal form.
2. And, it should not have **Partial Dependency**. (i.e In the second normal form, all **non-key attributes** are fully **functional dependent** on the primary key)

Second Normal Form (2NF)

2nd NF removes → Partial Dependency

For a table to be in second normal form, the following 2 conditions are to be met:

- **The table should be in the first normal form.**
- **The primary key of the table should compose of exactly 1 column (a compound key, if necessary).**

Let us take another example of storing student enrollment in various courses. Each student may enroll in multiple courses. Similarly, each course may have multiple enrollments.
A sample table may look like this (**student name and course code**)

Enrollments

Student Name	Phone	City	Course Code	Course Name
Rahul	●●●●●	Pune	CS152	Computer_Course_1
Rajat	●●●●●	Mumbai	CS101	Computer_Course_2
Rahul	●●●●●	Pune	CS154	Computer_Course_3
Raman	●●●●●	Surat	CS101	Computer_Course_2
Rajat	●●●●●	Mumbai	CS189	Computer_Course_5

Second Normal Form (2NF)

Student

ID(PK)	Student Name	Phone	City
1	Rahul	●●●●●	Pune
2	Rajat	●●●●●	Mumbai
3	Raman	●●●●●	Surat
4	Pankaj	●●●●●	Baroda

Courses

Course Code(PK)	Course Name
CS101	Computer_Course_2
CS152	Computer_Course_1
CS154	Computer_Course_3
CS107	Computer_Course_4
CS189	Computer_Course_5

Student_Enrollment

ID(PK)	StudentID(FK)	Course Code(FK)
1	1	CS152
2	1	CS154
3	2	CS101
4	3	CS101
5	2	CS189

Third Normal Form (3NF)

3rd NF removes → Transitive Dependency

A table is said to be in the Third Normal Form when,

1. It is in the Second Normal form.
2. And, it doesn't have Transitive Dependency. (No Transitive Dependency)

- **Transitive dependency** is expressing the dependency of A on C when A depends on B and B depends on C.
- **A functional dependency** is an association between two attributes of the same relational database table. One of the attributes is called the determinant and the other attribute is called the determined. For each value of the determinant there is associated one and only one value of the determined.

Third Normal Form (3NF)

Column A is said to be functionally dependent on column B if changing the value of A may require a change in the value of B.
Here, the department column is dependent on the professor name column. This is because if in a particular row, we change the name of the professor, we will also have to change the department value.

Course Code	Course venue	Instructor's Name	Department
CS101	Lecture Hall 18	Prof. Ronald	Mathematics Department
CS152	Auditorium building	Prof. John	Electronics Department
CS154	Lecture Hall 19	Prof. Ronald	Mathematics Department
CS189	Auditorium building #1	Prof. Smith	Computer Science Department
CS107	Lecture Hall #4	Prof. Smith	Computer Science Department

Third Normal Form (3NF)

Here, when we changed the name of the professor, we also had to change the department column. This is not desirable since someone who is updating the database may remember to change the name of the professor, but may forget updating the department value. This can cause inconsistency in the database.

Third normal form avoids this by breaking this into separate tables

Course Code	Course Venue	Instructor's ID
CS101	Lecture Hall 18	1
CS152	Auditorium building,	2
CS154	Lecture Hall 19	1
CS189	Auditorium building #1	3
CS107	Lecture Hall #4	3

Here, the first column is the ID of the professor who's taking the course.

Instructor's ID	Instructor's Name	Instructor's Number	Department
1	Prof. Ronald	+1 6514821924	Mathematics Department
2	Prof. John	+1 6519272918	Electronics Department
3	Prof. Smith	+1 6519272919	Computer Science Department

Fourth Normal Form (3NF)

TODO

4th NF removes → Multi-Valued Dependency

A table is said to be in the Fourth Normal Form when,

1. It is in the Third Normal form.
2. And, it has NO Multi-Valued (Multi-Valued attribute) Dependency

Student Name	Hobby	Skill
Amit	Cricket	Java
Amit	Cricket	Python
Amit	Music	Java
Amit	Music	Python
Amit	Cricket	C++

If this is the data then convert in in 3NF

Business Rules.

1. A salesman can sale one or many products.
2. A salesman can sale a product in one or many regions.
3. A product can be sold in one or may regions.

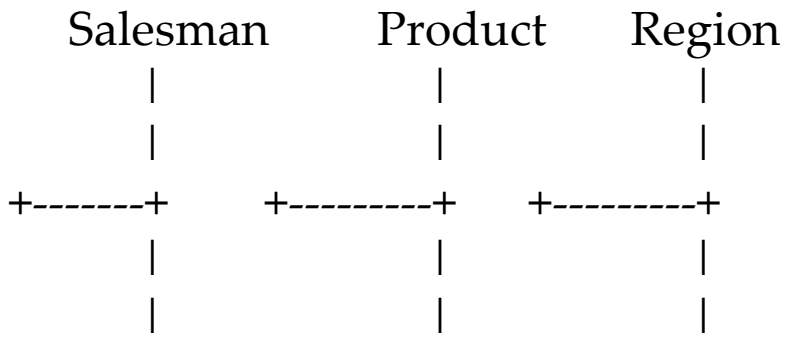
Relational schema

- Salesman(salesmanID, salesmanName)
- Region(regionID, regionName)
- Product(productID, productName)
- Sales(salesmanID, regionID, productID)

Salesman Name	Region1	Region2	Region3	Region4
Saleel	Car, Bus	Car, Bus, Cycle	Cycle, Motorcycle	Car
Rahul	Shirt, Pant, Gloves	Gloves, Book, Pencil	Pencil, Notebook	Keyboard, Mouse
Omkar	Keyboard, Mouse	Pen Drives, CD	Monitors, Scanners	Scanner, CD
Ninad	Photos, Photo Frames	Shirt, Pants, Gloves	Cycle, Motorcycle	Bus, Cycle, Car

Entities

- Salesman
- Product
- Region
- A junction table that connects all 3



Salesman_Product_Region (salesman_id, product_id, region_id)

MySQL is the most popular **Open Source** Relational Database Management System.

MySQL was created by a Swedish company - MySQL AB that was founded in 1995. It was acquired by Sun Microsystems in 2008; Sun was in turn acquired by Oracle Corporation in 2010.

When you use MySQL, you're actually using at least two programmes. One program is the MySQL server (*mysqld.exe*) and other program is MySQL client program (*mysql.exe*) that connects to the database server.



What is SQL?

Remember:

what is sql?

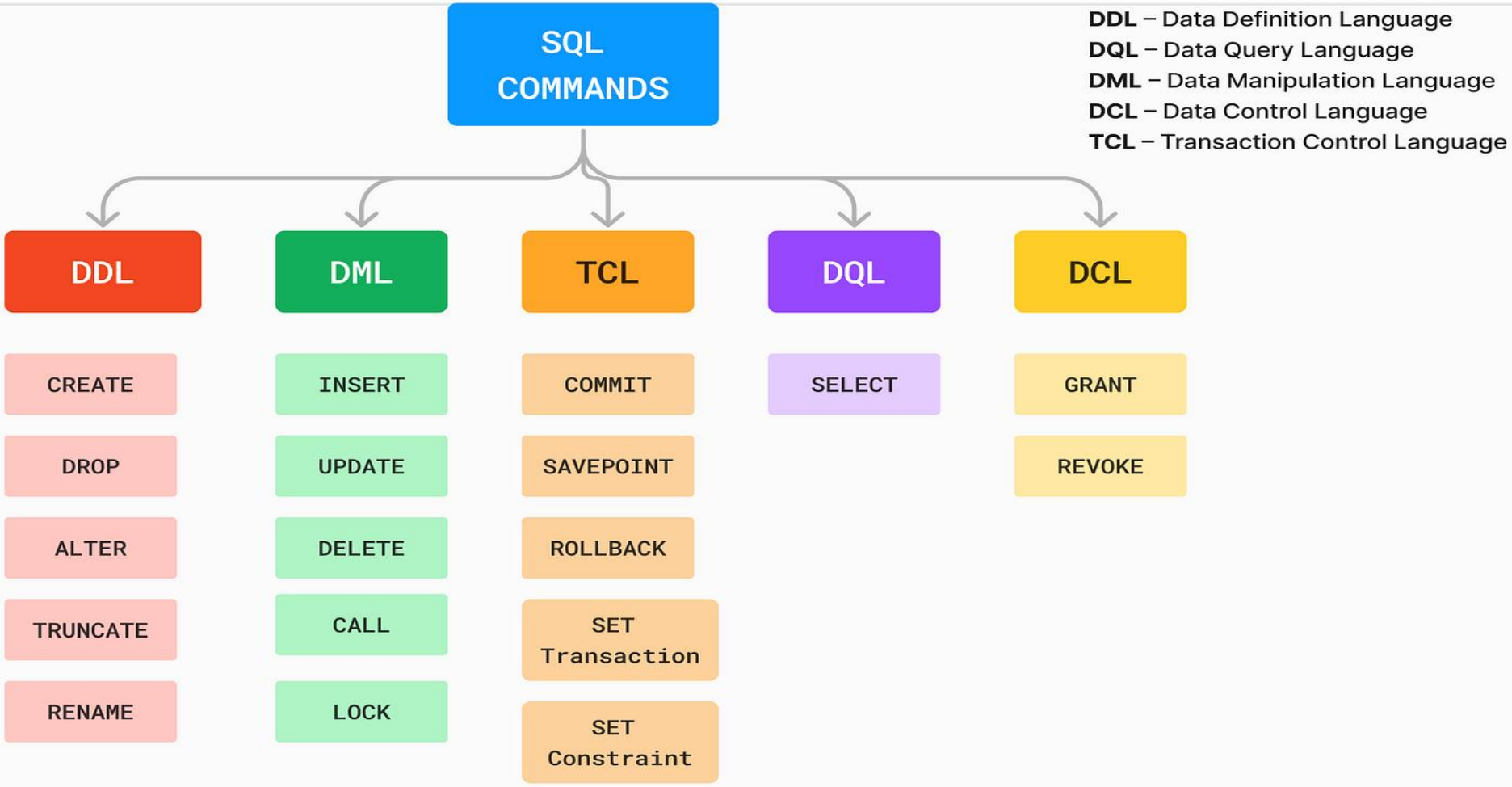
- **EXPLICIT or IMPLICIT commit will commit the data.**

SQL (Structured Query Language) is a database language designed and developed for managing data in relational database management systems (RDBMS). SQL is common language for all Relational Databases.



Remember:

- An ***implicit commit*** occurs ***automatically*** in MySQL ***without the need of COMMIT command***. This means changes made by the SQL statement are immediately saved to the database and ***cannot be rolled back***.
- An ***explicit commit*** is done by the user issuing a ***COMMIT*** command to ***manually save all changes*** made in the current transaction.



comments in mysql

- From a # character to the end of the line.
- From a -- sequence to the end of the line.
- From a /* sequence to the following */ sequence.

Reconnect to the server	**\r**
Execute a system shell command	**\!**
Exit mysql	**\q**
Change your mysql prompt.	**prompt str or \R str**

Tablespace, Control files, Data files

database storage structures

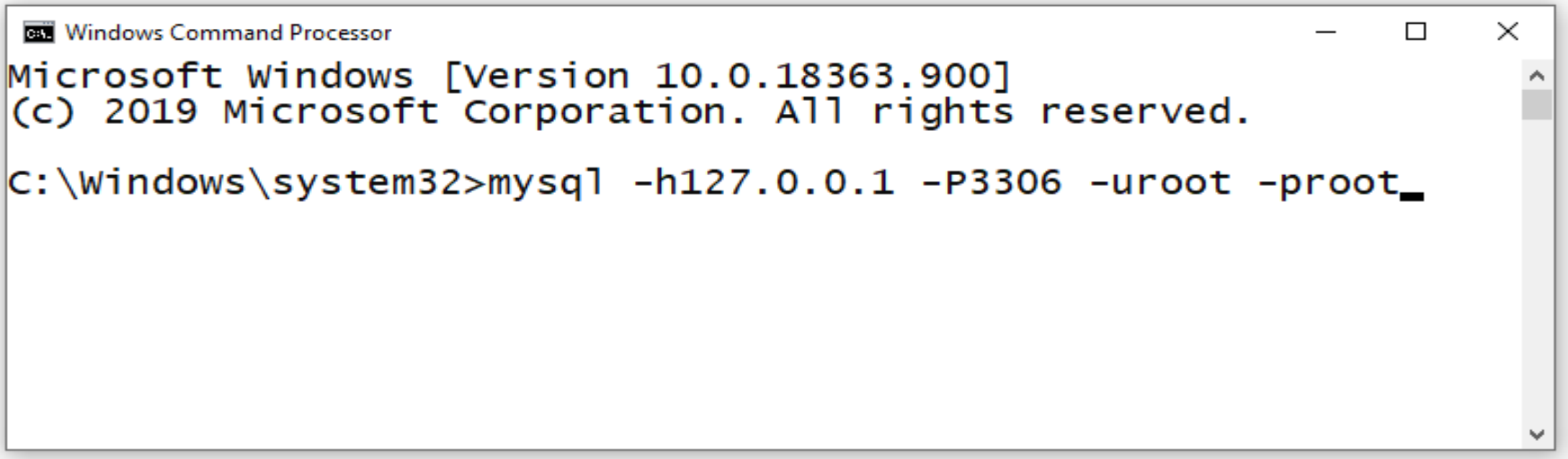
```
SHOW VARIABLES LIKE 'datadir';
```

Login to MySQL

Default port for MySQL Server: 3306

login

- C:\> mysql -hlocalhost -P3307 -uroot -p
- C:\> mysql -h127.0.0.1 -P3307 -uroot -p [*database_name*]
- C:\> mysql -h192.168.100.14 -P3307 -uroot -psaleel [*database_name*]
- C:\> mysql --host localhost --port 3306 --user root --password=ROOT [*database_name*]
- C:\> mysql --host=localhost --port=3306 --user=root --password=ROOT [*database_name*]



- Before MySQL version 5.5, MyISAM is the default storage engine.
- From version 5.5, MySQL uses InnoDB as the default storage engine.

STORAGE ENGINES

A storage engine is a software module that a database management system uses to create, read, update data from a database. There are two types of storage engines in MySQL: **transactional** and **non-transactional.**

Storage Engine	File on disk
MEMORY	Data is not stores on the disk
InnoDB	.idb (data and index)
MyISAM	MYD (data), .MYI (index)
CSV	.CSV (data), CSM (metadata)

SHOW ENGINES Syntax

For MySQL 5.5 and later, the default storage engine is *InnoDB*. The default storage engine for MySQL prior to version 5.5 was *MyISAM*.

SHOW [STORAGE] ENGINES

SHOW ENGINES displays status information about the server's storage engines. This is particularly useful for checking whether a storage engine is supported, or to see what the default engine is.

INFORMATION_SCHEMA.ENGINES

- show engines;
- show STORAGE engines;

INFORMATION_SCHEMA provides access to database metadata, information about the MySQL server such as the name of a database or table, the data type of a column, or access privileges.

ENGINES

- show engines;
- show STORAGE engines;

```
mysql> show engines;
+--------------------+---------+----------------------------------------------------------------+--------------+------+------------+
| Engine             | Support | Comment                                                        | Transactions | XA   | Savepoints |
+--------------------+---------+----------------------------------------------------------------+--------------+------+------------+
| MEMORY             | YES     | Hash based, stored in memory, useful for temporary tables      | NO           | NO   | NO         |
| MRG_MYISAM         | YES     | Collection of identical MyISAM tables                          | NO           | NO   | NO         |
| CSV                | YES     | CSV storage engine                                             | NO           | NO   | NO         |
| FEDERATED          | NO      | Federated MySQL storage engine                                 | NULL         | NULL | NULL       |
| PERFORMANCE_SCHEMA | YES     | Performance Schema                                             | NO           | NO   | NO         |
| MyISAM             | YES     | MyISAM storage engine                                          | NO           | NO   | NO         |
| InnoDB             | DEFAULT | Supports transactions, row-level locking, and foreign keys     | YES          | YES  | YES        |
| BLACKHOLE          | YES     | /dev/null storage engine (anything you write to it disappears) | NO           | NO   | NO         |
| ARCHIVE            | YES     | Archive storage engine                                         | NO           | NO   | NO         |
+--------------------+---------+----------------------------------------------------------------+--------------+------+------------+
9 rows in set (0.00 sec)

mysql>
```

- SET DEFAULT_STORAGE_ENGINE = MyISAM;

- MEMORY tables are visible to another client, but
- TEMPORARY tables are not visible to another client.

ENGINES

- **InnoDB** is the most widely used storage engine with transaction support. It is the only engine which provides foreign key referential integrity constraint. Oracle recommends using InnoDB for tables except for specialized use cases.
- **MyISAM** is the original storage engine. It is a fast storage engine. It does not support transactions. MyISAM provides table-level locking. It is used mostly in Web and data warehousing.
- **Memory** storage engine creates tables in memory. It is the fastest engine. It provides table-level locking. It does not support transactions. Memory storage engine is ideal for creating temporary tables or quick lookups. The data is lost when the database is re-started.
- **CSV** stores data in CSV files. It provides great flexibility because data in this format is easily integrated into other applications.

- MEMORY tables are visible to another client, but
- TEMPORARY tables are not visible to another client.

ENGINES

In the case of InnoDB, it is used to store the tables in tablespace whereas, in the case of MyISAM, it stores each MyISAM table in a separate file.

In my.ini do this changes.

```
[mysqld]
innodb_file_per_table = 1
```

Tablespace: A data file that can hold data for one or more InnoDB tables and associated indexes.

a. System tablespace
b. File per tablespace
c. General tablespace

Note:- By default, InnoDB contains only one tablespace called the System tablespace whose identifier is 0

SHOW DATABASES

SHOW DATABASES Syntax

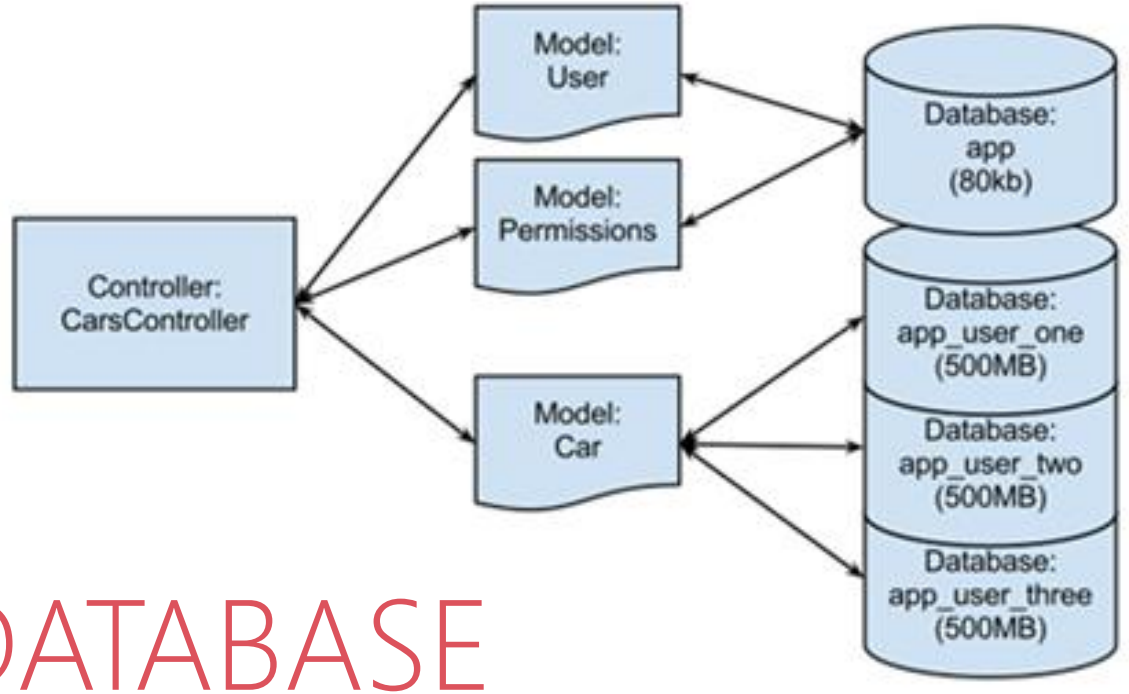
```
SHOW { DATABASES | SCHEMAS } [ LIKE 'pattern' | WHERE expr ]
```

SHOW SCHEMAS is a synonym for SHOW DATABASES.

```
SHOW DATABASES;
SHOW SCHEMAS;
SHOW DATABASES LIKE 'U%';
SHOW SCHEMAS LIKE 'U%';
```

Remember:

NULL means “no database is selected”. Issue the ***USE dbName*** command to select the database.



USE DATABASE

The **USE** ***db_name*** statement tells MySQL to use the db_name database as the default (current) database for subsequent statements. The database remains the default until the end of the session or another **USE** statement is issued.

USE DATABASE Syntax

USE *db_name*
\U *db_name*

Note:

- **USE, does not require a semicolon.**
- **USE must be followed by a database name.**

USE db1

\U db1

CREATE DATABASE
ALTER DATABASE

create / alter database

CREATE DATABASE creates a database with the given name. To use this statement, you need the CREATE privilege for the database.

CREATE { DATABASE | SCHEMA } [IF NOT EXISTS] *db_name*

ALTER { DATABASE | SCHEMA } [*db_name*] READ ONLY [=] { 0 | 1}

CREATE SCHEMA is a synonym for CREATE DATABASE.

- CREATE DATABASE db1;
- CREATE DATABASE IF NOT EXISTS db1;

- ALTER DATABASE db1 READ ONLY = 0; // is in read write mode.
- ALTER DATABASE db1 READ ONLY = 1; // is in read only mode.

Note:

- It is ***not*** possible to Create, Alter, Drop any object, and Write (Insert, Update, and Delete rows) in a read-only database.
- TEMPORARY tables; it is possible to create, alter, drop, and write (Insert, Update, and Delete rows) to TEMPORARY tables in a read-only database.

DROP DATABASE

If the default database is dropped, the default database is unset (the DATABASE() function returns NULL).

drop database

DROP DATABASE drops all tables in the database and deletes the database. Be very careful with this statement! To use DROP DATABASE, you need the DROP privilege on the database.

DROP { DATABASE | SCHEMA } [IF EXISTS] *db_name*

DROP SCHEMA is a synonym for DROP DATABASE.

```
DROP DATABASE db1;
```

```
DROP DATABASE IF EXISTS db1;
```

Information Functions

Use these function with SELECT statement

information function

- **FOUND_ROWS()** - For a SELECT with a LIMIT clause, the number of rows that would be returned were there no LIMIT clause

e.g.

  - SELECT * FORM emp;
  - SELECT **FOUND_ROWS()**;

- **LAST_INSERT_ID()** - Is a function used to retrieve the AUTO_INCREMENT value generated by the most
- recent successful INSERT in the current session.

- **ROW_COUNT()** - Is a function that returns the number of rows affected by the most recent statement.

e.g.

  - SELECT * FORM emp;
  - SELECT **ROW_COUNT();**

- **DATABASE()** –
  1. Is a function that returns the name of the currently selected database for the current session.
  2. If no database is selected then it will return NULL

e.g.

  - SELECT **DATABASE()**;

- **SCHEMA()** - Synonym for DATABASE()

information function

- **CURRENT_USER(), CURRENT_USER** - The authenticated user name and host name
- **USER()** - The user name and host name provided by the client

e.g.

  - SELECT **USER()**;

- **SESSION_USER()** - Synonym for USER()
- **SYSTEM_USER()** - Synonym for USER()
- **VERSION()** - Return a string that indicates the MySQL server version

e.g.

  - SELECT **VERSION()**;

- **CONNECTION_ID()** - Return the connection ID (thread ID) for the connection

mysqlshow [*options*] [*db_name* [*tbl_name* [*col_name*]]]

mysqlshow.exe

The mysqlshow client can be used to quickly see which databases exist, their tables, or a table's columns or indexes

mysqlshow – Get Quick Info On MySQL DB, Table, Column and Indexes

[Display available databases]

- C:\> mysqlshow -hlocalhost -P3306 -uroot –proot

[Display all tables in a database]

- C:\> mysqlshow -hlocalhost -P3306 -uroot -proot db1

[Display tables along with number of columns in a database]

- C:\> mysqlshow -hlocalhost -P3306 -uroot -proot db1 –v

[Display total number of columns and rows of all tables in a database]

- C:\> mysqlshow -hlocalhost -P3306 -uroot -proot db1 –v –v

mysqlshow – Get Quick Info On MySQL DB, Table, Column and Indexes

[Display all columns of a table]

- C:\> mysqlshow -hlocalhost -P3306 -uroot -proot db1 emp

[Display details about a specific column from a table]

- C:\> mysqlshow -hlocalhost -P3306 -uroot -proot db1 emp empno

[Display both indexes and columns of a table]

- C:\> mysqlshow -hlocalhost -P3306 -uroot -proot db1 emp –k

Source Command

source command

You can execute an SQL script file using the source command or \. command

\. file_name
source file_name

- \. D:\mysqldemobld7.sql
- SOURCE D:\mysqldemobld7.sql
- SOURCE //infoserver1/infodomain1/Everyone/DBT/mysqldemobld7.sql

SHOW COLUMNS

SHOW COLUMNS is a statement used to display the structure of columns in a table.

EMP & DEPT Table structure

```
mysql> show columns from emp;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| EMPNO     | int         | NO   | PRI | NULL    |       |
| ENAME     | varchar(12) | YES  |     | NULL    |       |
| GENDER    | char(1)     | YES  |     | NULL    |       |
| JOB       | varchar(20) | YES  | MUL | NULL    |       |
| MGR       | int         | YES  | MUL | NULL    |       |
| HIREDATE  | date        | YES  |     | NULL    |       |
| SAL       | int         | YES  |     | NULL    |       |
| COMM      | int         | YES  |     | NULL    |       |
| DEPTNO    | int         | NO   | MUL | NULL    |       |
| BONUSID   | int         | YES  |     | NULL    |       |
| USER NAME | varchar(20) | YES  |     | NULL    |       |
| PWD       | varchar(20) | YES  |     | NULL    |       |
| PHONE     | varchar(45) | YES  |     | NULL    |       |
| isActive  | tinyint(1)  | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
14 rows in set (0.00 sec)
```

```
mysql> show columns from dept;
+----------+------------+------+-----+---------+-------+
| Field    | Type       | Null | Key | Default | Extra |
+----------+------------+------+-----+---------+-------+
| DEPTNO   | int        | NO   | PRI | NULL    |       |
| DNAME    | varchar(12) | YES  |     | NULL    |       |
| LOC      | varchar(10) | YES  |     | NULL    |       |
| PWD      | varchar(20) | YES  |     | NULL    |       |
| STARTEDON | varchar(10) | YES  |     | NULL    |       |
+----------+------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

SHOW COLUMNS Syntax

SHOW [FULL] { COLUMNS | FIELDS } { FROM | IN } *tbl_name* [{ FROM | IN } *db_name*]
[LIKE '*pattern*' | WHERE *expr*]

- SHOW COLUMNS FROM emp;
- SHOW COLUMNS IN emp;
- SHOW FULL COLUMNS FROM emp; # WITH PRIVILEGES
- SHOW COLUMNS FROM emp FROM dbName;
- SHOW COLUMNS FROM user01.emp;
- SHOW COLUMNS FROM emp LIKE 'E%'; # STARTING WITH E
- SHOW COLUMNS FROM emp WHERE FIELD IN ('ename'); # ONLY ENAME COLUMN

SHOW TABLES

SHOW TABLES is a statement used to display the names of all tables in the currently selected database.

SHOW TABLES Syntax

SHOW [FULL] TABLES [{ FROM | IN } *db_name*] [LIKE '*pattern*' | WHERE *expr*]

- SHOW TABLES;
- SHOW FULL TABLES; // WITH TABLE TYPE
- SHOW TABLES FROM USER01;
- SHOW TABLES WHERE TABLES_IN_USER01 LIKE 'E%' OR TABLES_IN_USER01 LIKE 'B%';
- SHOW TABLES WHERE TABLES_IN_USER01 IN ('EMP');

SHOW TABLES STATUS

SHOW TABLE STATUS is used to display detailed information (metadata) about the tables in a database.

SHOW TABLES STATUS Syntax

SHOW TABLE STATUS [{ FROM | IN } *db_name*] [LIKE '*pattern*' | WHERE *expr*]

- SHOW TABLE STATUS;
- SHOW TABLE STATUS FROM user01;
- SHOW TABLE STATUS IN user01;
- SHOW TABLE STATUS LIKE 'emp';

SHOW VARIABLES

SHOW VARIABLES is a statement used to display the configuration/system variables and their current values for the MySQL server or current session.

SHOW VARIABLES Syntax

```
SHOW [ GLOBAL | SESSION ] VARIABLES [LIKE 'pattern' | WHERE expr]
```

```
SET SQL_SAFE_UPDATES = 0;
SET SQL_SAFE_UPDATES = false;
```

Enable/Disable :- 1 (true) / 0 (false)

explain analyze

SHOW VARIABLES is a statement used to display the configuration/system variables and their current values for the MySQL server or current session.

explain

The DESCRIBE and EXPLAIN statements are synonyms. In practice, the DESCRIBE keyword is more often used to obtain information about table structure, whereas EXPLAIN is used to obtain a query execution plan.

{ EXPLAIN | DESCRIBE | DESC } *tbl_name* [*col_name*]

- DESC emp;
- DESC emp ename;
- DESCRIBE emp;
- EXPLAIN emp;
- EXPLAIN emp empno;

EXPLAIN works with SELECT, DELETE, INSERT, REPLACE, and UPDATE statements.

explain

The DESCRIBE and EXPLAIN statements are synonyms. In practice, the DESCRIBE keyword is more often used to obtain information about table structure, whereas EXPLAIN is used to obtain a query execution plan.

```
{ EXPLAIN | DESCRIBE | DESC }
    {explainable_stmt}

explainable_stmt: {
      SELECT statement
    | DELETE statement
    | INSERT statement
    | REPLACE statement
    | UPDATE statement
}
```

- EXPLAIN ANALYZE SELECT * FROM emp;
- EXPLAIN ANALYZE SELECT * FROM emp where empno = 7788;
- EXPLAIN ANALYZE SELECT * FROM emp INNER JOIN dept USING(deptno);

- The **char** is a fixed-length character data type,
- The **varchar** is a variable-length character data type.

```
1. CREATE TABLE temp (
     c1 CHAR(10),
     c2 VARCHAR(10));
2. INSERT INTO temp
   VALUES('SALEEL', 'SALEEL');
3. SELECT * FROM temp WHERE c1 LIKE 'SALEEL';
```

datatypes

ENAME CHAR (10)	S	A	L	E	E	L				LENGTH -> 10
ENAME VARCHAR2(10)	S	A	L	E	E	L				LENGTH -> 6

In MySQL

ENAME CHAR (10)	S	A	L	E	E	L				LENGTH -> 6
ENAME VARCHAR(10)	S	A	L	E	E	L				LENGTH -> 6

Note:

The BINARY and VARBINARY types are similar to CHAR and VARCHAR, except that they store binary strings rather than nonbinary strings. That is, they store byte strings rather than character strings.

quantity VARCHAR(20) -- values: 20, '20', 'twenty'

datatype - string

Datatypes	Size	Description
CHAR [(length)]	0-255	***fixed-length character data type.***
VARCHAR (length)	0 to 65,535	***variable-length character data type.*** The maximum row size (65,535 bytes, which is shared among all columns.
TINYTEXT [(length)]	$(2^8 - 1)$ bytes	***variable-length string data type***
TEXT [(length)]	$(2^{16} - 1)$ bytes	65,535 bytes ~ 64kb
MEDIUMTEXT [(length)]	$(2^{24} - 1)$ bytes	16,777,215 bytes ~16MB
LONGTEXT [(length)]	$(2^{32} - 1)$ bytes	4,294,967,295 bytes ~4GB
ENUM('value1', 'value2',...)	65,535 distinct values	ENUM is a string data type that allows a column to **store only one value** from a predefined list of values.
SET('value1', 'value2',...)	64 distinct values	SET is a string data type that allows a column to store **zero, one, or multiple values** from a predefined list.
BINARY[(length)]	255	***fixed-length binary string data type***
VARBINARY(length)		***variable-length binary string data type***

PAD_CHAR_TO_FULL_LENGTH

- ***PAD_CHAR_TO_FULL_LENGTH*** is an SQL mode that affects how values stored in a CHAR column are retrieved.
- Normally, MySQL removes trailing spaces from CHAR values when retrieving them.
- When ***PAD_CHAR_TO_FULL_LENGTH*** is enabled, those trailing spaces are preserved.

By default, trailing spaces are trimmed from CHAR column values on retrieval. If ***PAD_CHAR_TO_FULL_LENGTH*** is enabled, trimming does not occur and retrieved CHAR values are padded to their full length.

- *SET sql_mode = '';*
- *SET sql_mode = 'PAD_CHAR_TO_FULL_LENGTH';*

example of char and varchar

Datatypes	Size	Description
CHAR [(length)]	0-255	***fixed-length character data type.***
VARCHAR (length)	0 to 65,535	***variable-length character data type.*** The maximum row size (65,535 bytes, which is shared among all columns.

Try Out

- CREATE TABLE x (x1 CHAR(4), x2 VARCHAR(4));
- INSERT INTO x VALUE('', '');
- INSERT INTO x VALUE('ab', 'ab');
- INSERT INTO x VALUE('abcd', 'abcd');
- SELECT x1, LENGTH(x1), x2, LENGTH(x2) FROM x;
- SET sql_mode = 'PAD_CHAR_TO_FULL_LENGTH';
- SELECT x1, LENGTH(x1), x2, LENGTH(x2) FROM x;
- SET sql_mode = '';
- SELECT x1, LENGTH(x1), x2, LENGTH(x2) FROM x;

* In CHAR, if a table contains value 'a', an attempt to store 'a ' causes a duplicate-key error.

- CREATE TABLE x (x1 CHAR(4) PRIMARY KEY, x2 VARCHAR(4));
- INSERT INTO x VALUE('a', 'a');
- INSERT INTO x VALUE('a ', 'a ');

- CREATE TABLE x (x1 CHAR(4), x2 VARCHAR(4) PRIMARY KEY);
- INSERT INTO x VALUE('a', 'a');
- INSERT INTO x VALUE('a ', 'a ');

example of text, blob, and longblob

Try Out

- `CREATE TABLE movies( _id INT AUTO_INCREMENT PRIMARY KEY,  description  TEXT, img1 BLOB, img2 LONGBLOB);`
- `INSERT INTO movies VALUE(default, load_file("d:/movie1.txt") , load_file("d:/img1.jpg") , load_file("d:/img2.jpg") );`
- `INSERT INTO movies VALUE(default, load_file("d:/movie2.txt") , load_file("d:/img1.jpg") , load_file("d:/img2.jpg") );`
- `INSERT INTO movies VALUE(default, load_file("d:\\movie3.txt") , load_file("d:\\img1.jpg") , load_file("d:\\img2.jpg") );`
- `SELECT * FROM movies;`

- CREATE TABLE z1 (c1 ZEROFILL); // then Insert few records
- CREATE TABLE z1 (c1 INT(4) ZEROFILL);

datatype - numeric

Datatypes	Size	Description
TINYINT	1 byte	-128 to +127 **(The unsigned range is 0 to 255).**
SMALLINT [(length)]	2 bytes	-32768 to 32767. **(The unsigned range is 0 to 65535).**
MEDIUMINT [(length)]	3 bytes	-8388608 to 8388607. **(The unsigned range is 0 to 16777215).**
INT, INTEGER [(length)]	4 bytes	-2147483648 to 2147483647. **(The unsigned range is 0 to 4294967295).**
BIGINT [(length)]	8 bytes	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
FLOAT [(length[,decimals])]	4 bytes	**FLOAT(255,30)**
DOUBLE [PRECISION] [(length[,decimals])], REAL [(length[,decimals])]	8 bytes	**REAL(255,30) / DOUBLE(255,30)** REAL will get converted to DOUBLE
DECIMAL [(length[,decimals])], NUMERIC [(length[,decimals])]		**DECIMAL(65,30) / NUMERIC(65,30)** NUMERIC will get converted in DECIMAL

For: float(M,D), double(M,D) or decimal(M,D), M must be >= D

Here, **(M,D)** means than values can be stored with up to ***M*** digits in total, of which ***D*** digits may be after the decimal point.

TINYINT(1) ["TRUE" = 1 "FALSE" = 0]

UNSIGNED prohibits negative values.

datatype – date and time

Datatypes	**Size**	**Description**
YEAR	1 byte	YYYY ***(1901 to 2155)***
DATE	3 bytes	YYYY-MM-DD ***(1000-01-01 to 9999-12-31)***
TIME	3 bytes	HH:MM:SS
DATETIME	8 bytes	YYYY-MM-DD hh:mm:ss

datatype – boolean

A value of zero is considered false. Nonzero values are considered true.

ColumnName BOOLEAN

```
CREATE TABLE tasks ( id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, completed BOOLEAN);
```

- `INSERT INTO tasks VALUE(default, 'Task1', 0);`
- `INSERT INTO tasks VALUE(default, 'Task2', 1);`
- `INSERT INTO tasks VALUE(default, 'Task3', False);`
- `INSERT INTO tasks VALUE(default, 'Task4', True);`
- `INSERT INTO tasks VALUE(default, 'Task5', NULL);`
- `INSERT INTO tasks VALUE(default, 'Task6', default);`
- `INSERT INTO tasks VALUE(default, 'Task7', 1 > 2);`
- `INSERT INTO tasks VALUE(default, 'Task8', 1 < 2);`
- `INSERT INTO tasks VALUE(default, 'Task9', 12);`
- `INSERT INTO tasks VALUE(default, 'Task10', 58);`
- `INSERT INTO tasks VALUE(default, 'Task11', .75);`
- `INSERT INTO tasks VALUE(default, 'Task12', .15);`
- `INSERT INTO tasks VALUE(default, 'Task13', 'a' = 'a');`

Note:

- BOOL and BOOLEAN are **synonym of TINYINT(1)**

Remember:

MySQL performs type conversion when comparing values of different types.

- 'b' is a string. When compared to a number (0), MySQL converts the string to a number.
- 'b' → numeric conversion = 0 (since it's not a valid number).

datatype – enum

NOTE:

- An ENUM column can have a maximum of **65,535** distinct elements.
- Each ENUM value is stored as a number internally, starting from 1.
- ENUM values are sorted based on their index numbers, which depend on the order in which the enumeration members were listed in the column specification.
- Default value, NULL if the column can be NULL, first enumeration value if NOT NULL

```
CREATE TABLE temp (col1 INT, COL2 ENUM('A','B','C'));
INSERT INTO temp (col1, col2) VALUES(1, 1);
INSERT INTO temp(col1) VALUES (1);  // NULL
```

```
CREATE TABLE temp (col1 INT, col2 ENUM('A','B','C') NOT NULL);
INSERT INTO temp(col1) VALUES (1); // First element from the ENUM datatype
```

```
CREATE TABLE temp (col1 INT, col2 ENUM('') NOT NULL);
INSERT INTO temp (col1, col2) VALUES (1,'This is the test'); // NULL
```

```
CREATE TABLE temp (col1 INT, COL2 ENUM('A','B','C') default 'C' ); // Valid default value for 'COL2'
CREATE TABLE temp (col1 INT, COL2 ENUM('A','B','C') default 'D' ); // Invalid default value for 'COL2'
```

IMP:

- MySQL maps [membership ENUM('Silver', 'Gold', 'Diamond', 'Platinum')] these enumeration member to a numeric index where Silver=1, Gold=2, Diamond=3, Platinum=4 respectively.

datatype – enum

- An ENUM column can have a maximum of **65,535** distinct elements.
- You can use **membership + 0** to get the index value if needed.

```
size ENUM('small', 'medium', 'large', 'x-large')
membership ENUM('Silver', 'Gold', 'Diamond', 'Platinum')
interest ENUM('Movie', 'Music', 'Concert')
zone ENUM('North', 'South', 'East', 'West')
season ENUM('Winter', 'Summer', 'Monsoon', 'Autumn')
sortby ENUM('Popularity', 'Price -- Low to High', 'Price -- High to Low', 'Newest First')
status ENUM('active', 'inactive', 'pending', 'expired', 'shipped', 'in-process', 'resolved', 'on-hold', 'cancelled', 'disputed')
payment_method ENUM('UPI', 'Credit Card', 'Debit Card', 'Net Banking', 'Cash')
```

- `ALTER TABLE product MODIFY COLUMN c2 ENUM('Bread','Butter','Cheese', 'Jam');`

Note:

- You cannot use user variable as an enumeration value. This pair of statements do not work:

```
SET @mysize = 'medium';
CREATE TABLE sizes ( size ENUM('small', @mysize, 'large'));   // error
```

NOTE:

datatype – set

- A SET column can have a maximum of **64** distinct members.
- Prevents invalid or duplicate values from being inserted.
- A SET is a string object that can have zero or more values, each of which must be chosen from a list of permitted values specified when the table is created.
- SET column values that consist of multiple set members are specified with members separated by commas (,) without leaving a spaces.

```
CREATE TABLE clients(id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(10), membership ENUM('Silver', 'Gold', 'Premium', 'Diamond'), interest SET('Movie', 'Music', 'Concert'));

INSERT INTO clients (name, membership, interest) VALUES('Saleel', 'Gold', 'Music');

INSERT INTO clients (name, membership, interest) VALUES('Saleel', 'Premium', 'Movie,Concert');
```

```
FIND_IN_SET(str, { strlist | Field } )

SELECT FIND_IN_SET('Concert', 'Movie,Music,Concert');

SELECT * FROM clients WHERE FIND_IN_SET('Music', interest);
```

MySQL store them as integers (bit values), but when you insert, you must give string values, not numbers (unless you know the internal bit mapping).

IMP:

- The SET data type allows you to specify a list of values to be inserted in the column, like ENUM. But, unlike the ENUM data type, which lets you choose only one value, the SET data type allows you to choose multiple values from the list of specified values.

Using ENUM or SET for Dynamic

Anti-pattern

- Here,
```
CREATE TABLE employees (
    status ENUM('Active', 'Inactive', 'On Leave')
);
```

Why it's bad:

- Adding new values requires altering the table.

Better approach

- Here,
```
CREATE TABLE employee_status (
    status_id INT PRIMARY KEY,
    status_name VARCHAR(50)
);
```

datatype – set

Decimal	Binary	Stored Text
0	0000	(empty set)
1	0001	Red
2	0010	Blue
3	0011	Red,Blue
4	0100	Green
5	0101	Red,Green
6	0110	Blue,Green
7	0111	Red,Blue,Green
8	1000	Yellow
9	1001	Red,Yellow
10	1010	Blue,Yellow
11	1011	Red,Blue,Yellow
12	1100	Green,Yellow
13	1101	Red,Green,Yellow
14	1110	Blue,Green,Yellow
15	1111	Red,Blue,Green,Yellow

- `CREATE TABLE x(c1 INT , c2 SET('Red','Blue','Green', 'Yellow'));`

MySQL store them as integers (bit values), but when you insert, you must give string values, not numbers (unless you know the internal bit mapping).

Use a CREATE TABLE statement to specify the layout of your table.

Remember:

- Max 4096 columns per table provided the row size <= 65,535 Bytes.
- The NULL value is different from values such as 0 for numeric types or the empty string for string types.

create table

Use a **CREATE TABLE** statement to specify the layout of your table.

Note:

- **USER TABLES**: This is a collection of tables created and maintained by the user. Contain USER information.
- **DATA DICTIONARY**: This is a collection of tables created and maintained by the MySQL Server. It contains database information. All data dictionary tables are owned by the SYS user.

create table

Use a **CREATE TABLE** statement to specify the layout of your table.

- `CREATE TABLE `123` (c1 INT, c2 VARCHAR(10));`
- `CREATE TABLE `Order Details` (c1 INT, c2 VARCHAR(10));`

Remember:

- by default, tables are created in the default database, using the InnoDB storage engine.
- table name should not begin with a number or special symbols.
- table name can start with _table_name (underscore) or $table_name (dollar sign)
- table name and column name can have max 64 char.
- multiple words as table_name is invalid, if you want to give multiple words as table_name then give it in `table_name` (backtick)
- error occurs if the table exists.
- error occurs if there is no default database.
- error occurs if the database does not exist.

create table

syntax

CREATE [TEMPORARY] TABLE [IF NOT EXISTS] tbl_name
(*create_defineation*, . . .)
[table_options]
[partition_options]

create_definition:
col_name ***column_definition***

column_definition:
data_type [NOT NULL | NULL] [DEFAULT default_value]
[AUTO_INCREMENT] [UNIQUE [KEY] | [PRIMARY] KEY]
[reference_definition]
| data_type [GENERATED ALWAYS] AS (expression) [VIRTUAL]
[VISIBLE | INVISIBLE]

table_options:
AUTO_INCREMENT = <number> // must be used with AUTO_INCREMENT definition
ENGINE [=] engine_name

e.g.

- CREATE TABLE student(
  ID INT,
  firstName VARCHAR(45),
  lastName VARCHAR(45),
  DoB DATE,
  emailID VARCHAR(128)
  );

show engines;
set default_storage_engine = memory;

Using NULLs Everywhere

Anti-pattern

- Here,
```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    name VARCHAR(50),
    phone VARCHAR(15),
    alt_phone VARCHAR(15) NULL,
    remarks TEXT NULL
);
```

Why it's bad:

- Excessive NULL values waste space and complicate queries.
- You have to use IS NULL or COALESCE() everywhere.
- Confusing logic for "missing", "unknown", or "not applicable."

Better approach:

- Use DEFAULT values where possible.
- Split optional data into separate tables.
- Here,
```
CREATE TABLE employee_contacts (
    emp_id INT PRIMARY KEY,
    phone VARCHAR(15) NOT NULL,
    alt_phone VARCHAR(15) DEFAULT ''
);
```

- Literals, built-in functions (both deterministic and nondeterministic), and operators are permitted.
- Subqueries, parameters, variables, and stored functions are not permitted.
- An expression default value cannot depend on a column that has the AUTO_INCREMENT attribute.

default value

The DEFAULT specifies a default value for the column.

- `CREATE TABLE temp (c1 INT PRIMARY KEY AUTO_INCREMENT, c2 INT DEFAULT(c1 + c2));  // Error`
- `CREATE TABLE temp (c1 INT, c2 INT DEFAULT(c1 < c2)); // Error`
- `CREATE TABLE temp (c1 INT, c2 INT , c3 INT DEFAULT(c1 < c2)); // OK`

default value

col_name data_type DEFAULT value

The **DEFAULT** specifies a **default** value for the column.

- CREATE TABLE posts(
  postID INT,
  postTitle VARCHAR(255),
  postDate DATETIME DEFAULT NOW(),
  deleted INT
  );

	Field	Type	Null	Key	Default	Extra
▶	postID	int	YES		NULL	
	postTitle	varchar(255)	YES		NULL	
	postDate	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
	deleted	int	YES		NULL	

version 8.0 and above.

- CREATE TABLE empl(
  ID INT PRIMARY KEY,
  firstName VARCHAR(45),
  phone INT,
  city VARCHAR(10) DEFAULT 'PUNE',
  salary INT,
  comm INT,
  total INT DEFAULT(salary + comm)
  );

	Field	Type	Null	Key	Default	Extra
▶	ID	int	NO	PRI	NULL	
	firstName	varchar(45)	YES		NULL	
	phone	int	YES		NULL	
	city	varchar(10)	YES		PUNE	
	salary	int	YES		NULL	
	comm	int	YES		NULL	
	total	int	YES		(`salary` + `comm`)	DEFAULT_GENERATED

default value - insert

The **DEFAULT** example.

- CREATE TABLE t(
  c1 INT,
  c2 INT DEFAULT 1,
  c3 INT DEFAULT 3,
  );

	Field	Type	Null	Key	Default	Extra
▶	c1	int	YES		NULL	
	c2	int	YES		1	
	c3	int	YES		3	

  - INSERT INTO t VALUES();
  - INSERT INTO t VALUES(-1, DEFAULT, DEFAULT);
  - INSERT INTO t VALUES(-2, DEFAULT(c2), DEFAULT(c3));
  - INSERT INTO t VALUES(-3, DEFAULT(c3), DEFAULT(c2));

default value - update

The **DEFAULT** example.

- ```
  CREATE TABLE temp(
    c1 INT,
    c2 INT,
    c3 INT DEFAULT(c1 + c2),
    c4 INT DEFAULT(c1 * c2)
  );
  ```

  - `INSERT INTO temp (c1, c2, c3, c4) VALUES(1, 1, 1, 1);`
  - `INSERT INTO temp (c1, c2, c3, c4) VALUES(2, 2, 2, 2);`
  - `UPDATE temp SET c3 = DEFAULT;`
  - `UPDATE temp SET c4 = DEFAULT;`
  ```

insert rows

INSERT is used to add a single or multiple tuple to a relation. We must specify the relation name and a list of values for the tuple. **The values should be listed in the same order in which the corresponding attributes were specified in the CREATE TABLE command.**

You can insert data using following methods:

- INSERT ... VALUES
- INSERT ... SET
- INSERT ... SELECT

INSERT can violate for any of the four types of constraints.

Important:

- If an attribute value is not of the appropriate data type.
- Entity integrity can be violated if a key value in the new tuple *t* already exists in another tuple in the relation r(R).
- Entity integrity can be violated if any part of the primary key of the new tuple *t* is NULL.
- Referential integrity can be violated if the value of any foreign key in *t* refers to a tuple that does not exist in the referenced relation.

INSERT will also fail in following cases.

Important :

- Your database table has **X** columns, Where as the **VALUES** you are passing are for (**X-1**) or (**X+1**). This mismatch of column-values will giving you the error.
- Inserting a string into a string column that exceeds the column maximum length. Data too long for column error will be raise.
- Inserting data into a column than does not exists, then Unknown column error will raise.
- INSERT INTO *tbl_name* (col1,col2) VALUES(15,col1*2); // is legal.
- INSERT INTO *tbl_name* (col1,col2) VALUES(col2*2,15); // is not legal, because the value for col1 refers to col2, which is assigned after col1.

- **INSERT** is used to add a single or multiple tuple to a relation. We must specify the relation name and a list of values for the tuple. **The values should be listed in the same order in which the corresponding attributes were specified in the CREATE TABLE command.**
- A second form of the **INSERT** statement allows the user to specify explicit attribute names that correspond to the values provided in the **INSERT** command. This is useful if a relation has many attributes but only a few of those attributes are assigned values in the new tuple. However, the values must include all attributes with **NOT NULL** specification and no default value. Attributes with **NULL** allowed or **DEFAULT** values are the ones that can be left out.

insert rows using values

ON DUPLICATE KEY UPDATE clause is used in INSERT statements to deal with situations where a duplicate key (like a **PRIMARY KEY** or **UNIQUE KEY**) would lead to a conflict. Instead of throwing an error, MySQL updates the existing record.

dml- insert … values

INSERT inserts new row(s) into an existing table. The INSERT ... VALUES

INSERT [IGNORE] [INTO] tbl_name [PARTITION (*partition_name* [, *partition_name*] ...)] [(*field_name*, . . .)]
{ VALUES | VALUE } [ROW] ({ *expr* | *DEFAULT* }, . . .), [ROW] (. . .), [ROW] . . . [ON DUPLICATE KEY UPDATE
assignment_list]

The affected-rows value for an INSERT can be obtained using the ROW_COUNT() function.

```
INSERT INTO DEPT VALUES (1, 'HRD', 'Pune')
```

Column Values

```
INSERT INTO DEPT(ID, NAME, LOC) VALUES (1, 'HRD', 'Pune')
```

Column List

```
INSERT INTO DEPT(ID, NAME, LOC) VALUES (1, 'HRD', 'Baroda'),
 (2,'Sales','Surat'), (3,'Purchase','Pune'), (4,'Account','Mumbai')
```

Inserting multiple rows

dml- insert … values

INSERT inserts new row(s) into an existing table. The INSERT ... VALUES

INSERT [IGNORE] [INTO] tbl_name [PARTITION (*partition_name* [, *partition_name*] ...)] [(*field_name*, . . .)]
{ VALUES | VALUE } [ROW] ({ *expr* | *DEFAULT* }, . . .), [ROW] (. . .), [ROW] . . . [ON DUPLICATE KEY UPDATE
assignment_list]

```
CREATE TABLE student (
   ID INT PRIMARY KEY,
   nameFirst VARCHAR(45),
   nameLast VARCHAR(45),
   DoB DATE ,
   emailID VARCHAR(128)
);
```

- INSERT INTO a(c1, c2) VALUES(2, elt(c1, 'cod', 'cash', 'credit', 'debit'));

e.g.

- INSERT INTO student VALUES (29, 'sharmin', 'patil', '1999-11-10', 'sharmin.patil@gmail.com');
- INSERT INTO student (ID, nameFirst, nameLast, DOB, emailID) VALUES (30, 'john', 'thomas', '1983-11-10', 'john.thomas@gmail.com’);
- INSERT INTO student (ID, nameFirst, emailID) VALUES (31, 'jack', 'jack.thorn@gmail.com');
- INSERT INTO student (ID, nameFirst) VALUES (32, 'james'), (33, 'jr. james'), (34, 'sr. james');

insert rows using values with on duplicate key UPDATE

dml- insert … with on duplicate key UPDATE

INSERT inserts new rows into an existing table. The INSERT ... VALUES

```
INSERT [IGNORE] [INTO] tbl_name { VALUES | VALUE } [ROW] ( { expr | DEFAULT }, . . .), [ROW]  (. . .), [ROW]
. . . [ ON DUPLICATE KEY UPDATE assignment_list ]
```

- ```
  CREATE TABLE `login_count`(`userID` INT NOT NULL,
  `loginCount` INT DEFAULT 1,
  `loginDate` DATETIME DEFAULT NULL,
  PRIMARY KEY (`userID`));
  ```

- ```
  INSERT INTO login_count(userID, loginDate) VALUES(1, NOW()) ON DUPLICATE KEY UPDATE loginCount = loginCount + 1,
  loginDate = NOW();
  ```
- ```
  INSERT INTO login_count(userID, loginDate) VALUES(1, NOW()) ON DUPLICATE KEY UPDATE loginCount = loginCount + 1,
  loginDate = NOW();
  ```
- ```
  INSERT INTO login_count(userID, loginDate) VALUES(1, NOW()) ON DUPLICATE KEY UPDATE loginCount = loginCount + 1,
  loginDate = NOW();
  ```

insert multiple rows

To insert multiple rows in a single INSERT statement, provide multiple sets of values separated by commas.

dml- insert … values

INSERT inserts new rows into an existing table. The INSERT ... VALUES

```
INSERT [INTO] tbl_name { VALUES | VALUE } [ROW] ( { expr | DEFAULT }, . . .), [ROW]  (. . .), [ROW] (. . .)
```

```
CREATE TABLE student(
   ID INT PRIMARY KEY,
   nameFirst VARCHAR(45),
   nameLast VARCHAR(45),
   DoB DATE ,
   emailID VARCHAR(128)
);
```

e.g.

```
INSERT INTO student (ID, nameFirst)
   VALUES
      (32, 'james'),
      (33, 'jr. james'),
      (34, 'sr. james');
```

e.g.

```
INSERT INTO student (ID, nameFirst)
   VALUES
      ROW(32, 'james'),
      ROW(33, 'jr. james'),
      ROW(34, 'sr. james');
```

Inserting Multiple Rows Using Multiple INSERT

Anti-pattern

- Here,
```
INSERT INTO emp(ename, sal) VALUES('A', 10000);
INSERT INTO emp(ename, sal) VALUES('B', 15000);
INSERT INTO emp(ename, sal) VALUES('C', 12000);
```

Why it's bad:

- Generates high network overhead and slow performance.

Better approach: Bulk Insert

- Here,
```
INSERT INTO emp(ename, sal)
VALUES
  ('A', 10000),
  ('B', 15000),
  ('C', 12000);
```

INSERT to Fetch Auto-Increment ID Incorrectly

Anti-pattern

- Here, `INSERT INTO dept(deptno) VALUES(1001);`
  `SELECT MAX(deptno) FROM dept;  -- WRONG`

Why it's bad:

- `MAX(deptno)` is not safe if multiple users insert simultaneously.

Better approach:

- Here, `SELECT LAST_INSERT_ID();`

INSERT Without Using Transactions

Anti-pattern

- Here,
```
INSERT INTO invoice VALUES(. . .);
INSERT INTO invoice_details VALUES(. . .);   -- WRONG
```

Why it's bad:

- If the ***second*** INSERT command **fails**, the ***first*** INSERT statement will be executed and a record will be created.

Better approach:

- Here,
```
START TRANSACTION READ WRITE;
INSERT INTO invoice VALUES(. . .);
INSERT INTO invoice_details VALUES(. . .);
COMMIT;
```

INSERTing Sensitive Data Without Encryption

Anti-pattern

- Here, `INSERT INTO user(password) VALUES('Saleel');`

Why it's bad:

- Sensitive data will be displayed

Better approach:

- Here, password `VARBINARY(60)`

Point	Replace Into	Insert ... On Duplicate Key Update
on duplicate	Deletes old row, inserts new one	Updates existing row
Auto-increment field changes?	Yes	No (same row is updated)
Triggers (BEFORE/AFTER DELETE)	Fires	Not fired

replace statement

REPLACE works exactly like INSERT, except that if an old row in the table has the same value as a new row for a PRIMARY KEY or a UNIQUE index, **the old row is deleted before the new row is inserted / it deletes the old row and inserts the new one..**

Note:

- REPLACE makes sense only if a **table has a PRIMARY KEY** or **UNIQUE index**. Otherwise, it behaves like to INSERT statement.

replace statement

TODO

```
REPLACE INTO tbl_name [(field_name,...)] { VALUES | VALUE } ( { value_list | expr | DEFAULT }, . . .)
```

```
CREATE TABLE student(
  ID INT PRIMARY KEY,
  nameFirst VARCHAR(45),
  emailID VARCHAR(128)
);
```

e.g.

- `REPLACE INTO student(ID, nameFirst, emailID) VALUES (1, 'james', 'james@gmail.com'), (2, 'jr. james', 'james@redifmail.com'), (3, 'sr. james', 'james@mail.com');`

- `REPLACE INTO student(ID, nameFirst, emailID) VALUES (1, 'jack', 'jack@gmail.com'), (2, 'jr. jack', 'jack@redifmail.com'), (3, 'sr. jack', 'jack@mail.com');`

encoding/decoding

aes_encrypt / aes_decrypt

AES_ENCRYPT() and AES_DECRYPT() implement encryption and decryption of data using the official AES (Advanced Encryption Standard) algorithm, previously known as "Rijndael".

- AES_ENCRYPT(str, key_str)
- AES_DECRYPT(str, key_str)

```
CREATE TABLE pwd (
  ID INT PRIMARY KEY,
  name VARCHAR(45),
  pwd VARBINARY(45)
);
```

```
INSERT INTO pwd VALUES (1, 'sharmin', AES_ENCRYPT('sharmin', 'sharmin1'));
INSERT INTO pwd VALUES (2, 'saleel', AES_ENCRYPT('saleel', 'sharmin1'));
INSERT INTO pwd VALUES (3, 'kaushal', AES_ENCRYPT('kaushal', 'kaushal1'));
INSERT INTO pwd VALUES (4, 'ruhan', AES_ENCRYPT('ruhan', 'ruhan1'));
```

```
SELECT * FROM pwd;
SELECT ID, name, AES_DECRYPT(pwd, 'sharmin1') FROM pwd;
SELECT ID, name,  CAST(AES_DECRYPT(pwd, 'sharmin1') AS CHAR) FROM pwd;
```

md5()

- MD5() is a built-in function that generates a 128-bit hash value from a string.
- MD5 stands for Message-Digest Algorithm 5.
- It converts a string into a fixed-length 128-bit (16-byte) hash, normally represented as 32 hexadecimal characters.

MD5(*string*)

```
CREATE TABLE pwd (
    ID INT PRIMARY KEY,
    user_name VARCHAR(45),
    pwd VARCHAR(45)
);
```

```
INSERT INTO pwd(ID, user_name, password_hash) VALUES (1, 'saleel', MD5('p1'));
INSERT INTO pwd(ID, user_name, password_hash) VALUES (2, 'sharmin', MD5('p1'));
INSERT INTO pwd(ID, user_name, password_hash) VALUES (3, 'ruhan', MD5('p2'));
INSERT INTO pwd(ID, user_name, password_hash) VALUES (4, 'natasha', MD5('p2’));
```

```
SELECT * FROM pwd WHERE password_hash= MD5('p1');
```

```
SELECT MD5('S');
SELECT MD5('s');
SELECT MD5('saleel');
```

SHA2(*string, hash_length*)

str → text/value to hash
hash_length → 224, 256, 384, 512, or 0
0 means SHA-512

sha2

SHA2() is a function used to generate a SHA-2 family cryptographic hash from a string.

*

sha2()

INSERTing Sensitive Data Encryption

```
CREATE TABLE users
(
   user_id INT PRIMARY KEY,
   user_name VARCHAR(10),
   password_hash VARCHAR(255) GENERATED ALWAYS AS (SHA2(CONCAT(user_id, user_name), 256)) STORED
);
```

- ```
  INSERT INTO users(user_id, user_name)
  VALUES
    (1, 'saleel’),
    (2, 'sharmin’),
    (3, 'Ruhan’),
    (4, 'deep’),
    (5, 'neel’),
    (6, 'gau');
  ```

```
SELECT * FROM users WHERE password_hash = SHA2(CONCAT(user_id, user_name), 256);

SELECT * FROM users WHERE user_id = 1 AND password_hash = SHA2(CONCAT(user_id, user_name), 256);
```
```

insert rows using select

insert ... select

With INSERT ... SELECT, you can quickly insert many rows into a table from one or many tables.

INSERT [INTO] tbl_name [(*col_name*, . . .)] SELECT . . .

- INSERT INTO dept(deptno) SELECT 1 + 1;
- INSERT INTO dept(deptno) SELECT deptno FROM dept;
- INSERT INTO dept SELECT * FROM dept;
- INSERT INTO dept(SELECT MAX(deptno) + 1, 'HRD', 'BARODA', 'r57px33px' FROM dept);

- set @x := 40;
- INSERT INTO dept VALUES (@x := @x + 1 , 'HRD', 'BARODA', 'r57px33px');

- set @x := (SELECT MAX(deptno) FROM dept);
- INSERT INTO dept VALUES (@x := @x + 1 , 'HRD', 'BARODA', 'r57px33px');

Do not use the * operator in your SELECT statements. Instead, use column names. Reason is that in MySQL Server scans for all column names and replaces the * with all the column names of the table(s) in the SELECT statement. Providing column names avoids this search-and-replace, and enhances performance.

SELECT statement...

```
SELECT what_to_select
FROM which_table
WHERE conditions_to_satisfy;
```

SELECT CLAUSE

The **SELECT** statement retrieves or extracts data from tables in the database.

- You can use one or more tables separated by comma to extract data.
- You can fetch one or more fields/columns in a single SELECT command.
- You can specify star (*) in place of fields. In this case, SELECT will return all the fields.
- SELECT can also be used to retrieve rows computed without reference to any table e.g. SELECT 1 + 2;

SELECT * means:

- retrieve all rows
- retrieve all columns

Capabilities of SELECT Statement

1. SELECTION
2. PROJECTION
3. JOINING

Capabilities of SELECT Statement

➢ SELECTION

Selection capability in SQL is to choose the record's/row's/tuple's in a table that you want to return by a query.

R

EMPNO	ENAME	JOB	HIREDATE	DEPTNO
1	Saleel	Manager	1995-01-01	10
2	Janhavi	Sales	1994-12-20	20
3	Snehal	Manager	1997-05-21	10
4	Rahul	Account	1997-07-30	10
5	Ketan	Sales	1994-01-01	30

Capabilities of SELECT Statement

➢ PROJECTION

Projection capability in SQL to choose the column's/attribute's/field's in a table that you want to return by your query.

R

EMPNO	ENAME	JOB	HIREDATE	DEPTNO
1	Saleel	Manager	1995-01-01	10
2	Janhavi	Sales	1994-12-20	20
3	Snehal	Manager	1997-05-21	10
4	Rahul	Account	1997-07-30	10
5	Ketan	Sales	1994-01-01	30

Table DEPARTMENTS

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
60	IT	103	1400
90	Executive	100	1700



Table EMPLOYEES

EMPLOYEE_ID	LAST_NAME	EMAIL	HIRE_DATE	JOB_ID	MANAGER_ID	DEPARTMENT_ID
100	King	SKING		AD_PRES		90
101	Kochhar	NKOCHHAR	21-SEP-89	AD_VP	100	90
102	De Hann	LDEHANN	13-JAN-93	AD_VP	100	90
103	Hunold	AHUNOLD		IT_PROG	102	60

Capabilities of SELECT Statement

➢ JOINING

Join capability in SQL to bring together data that is stored in different tables by creating a link between them.

R

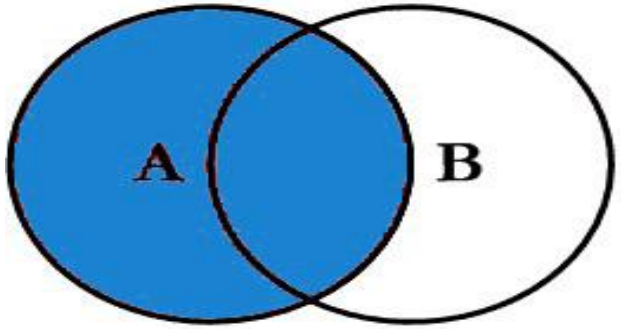
EMPNO	ENAME	JOB	HIREDATE	DEPTNO
1	Saleel	Manager	1995-01-01	20
2	Janhavi	Sales	1994-12-20	10
3	Snehal	Manager	1997-05-21	10
4	Rahul	Account	1997-07-30	20
5	Ketan	Sales	1994-01-01	30



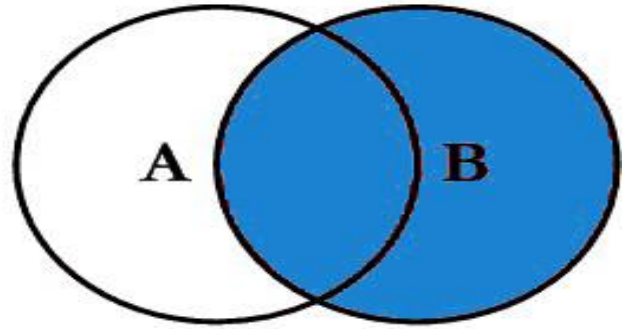
S

DEPTNO	DNAME	LOC
10	HRD	PUNE
20	SALES	BARODA
40	PURCHASE	SURAT

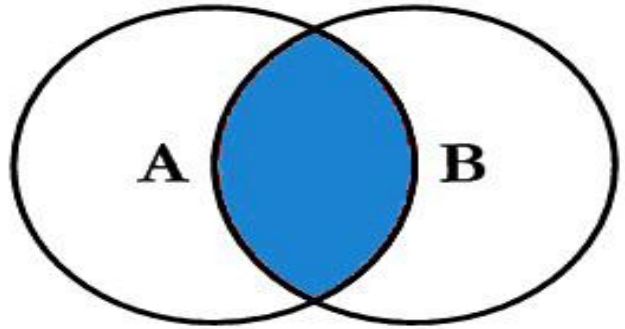
SQL JOINS



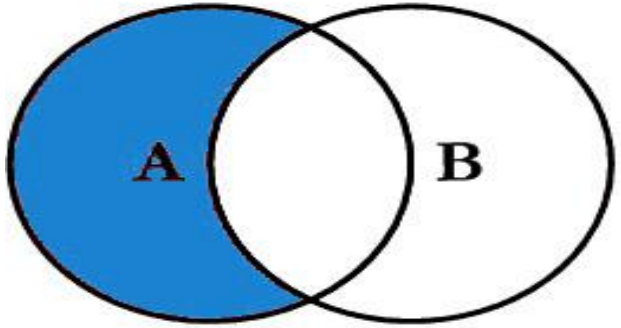

```
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key
```



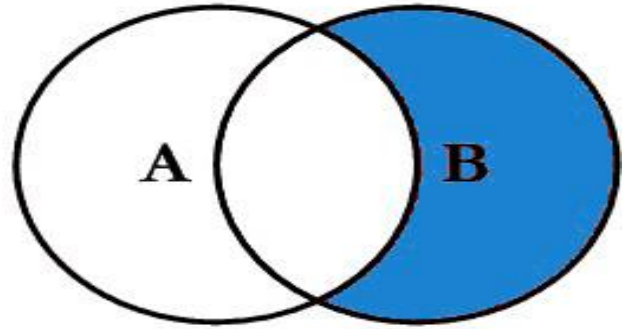

```
SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key
```



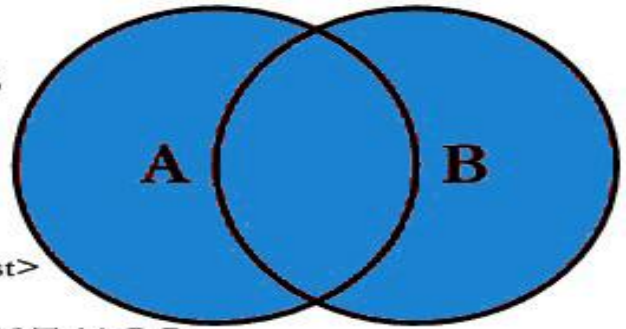

```
SELECT <select_list>
FROM TableA A
INNER JOIN TableB B
ON A.Key = B.Key
```



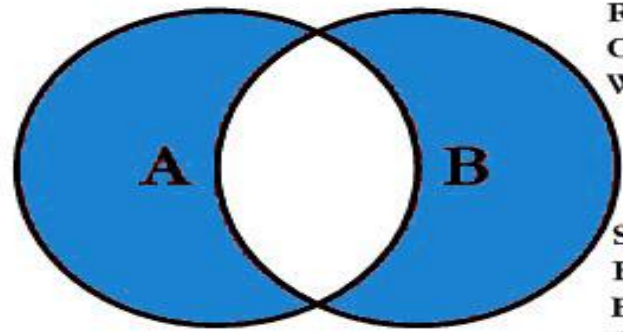

```
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key
WHERE B.Key IS NULL
```




```
SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL
```




```
SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key
```




```
SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL
OR B.Key IS NULL
```

select statement

SELECTION Process

SELECT * FROM <table_references>

selection-list | field-list | column-list

Remember:

- Here, " * " is known as metacharacter (all columns)

PROJECTION Process

SELECT column-list FROM <table_references>

selection-list | field-list | column-list

Remember:

- Position of columns in SELECT statement will determine the position of columns in the output (as per user requirements)

ORDER BY in UPDATE: if the table contains two values 1 and 2 in the id column and 1 is updated to 2 before 2 is updated to 3, an error occurs. To avoid this problem, add an ORDER BY clause to cause the rows with larger id values to be updated before those with smaller values.

In a **SET** statement, = is treated identically to :=

Note:

Here c1 column is a Primary Key

- UPDATE temp SET c1 = c1 - 1 ORDER BY c1 ASC; # In case of decrement
- UPDATE temp SET c1 = c1 + 1 ORDER BY c1 DESC; # In case of increment

```
SELECT ename, job, sal, sal * 1.1, sal * 1.25 FROM emp;
```

Salary increased by **10%**

Salary increased by **25%**

single-table update

UPDATE is used to change/modify the values of some attributes of one or more selected tuples.

- SET @x := 0;
- UPDATE emp SET id = @x := @x + 1;
- UPDATE t, (SELECT isactive, COUNT(isactive) r1 FROM emp GROUP BY isactive) a SET t.c2 = a.r1 WHERE t.c1 = a.isactive;

```
mysql> SELECT * FROM t;
+------+--------+      +------+------+
| c1   | c2     |      | c1   | c2   |
+------+--------+      +------+------+
|   0  | NULL   |      |   0  |   6  |
|   1  | NULL   |      |   1  |  14  |
+------+--------+      +------+------+
```

e.g.

1. Update top 2 rows.
2. Update UnitPrice for the top 5 most expensive products.

UPDATE can violate.

Important:

- TODO

single-table update

The UPDATE statement updates columns of existing rows in the named table with new values. The SET clause indicates which columns to modify and the values they should be given. The **WHERE** clause, if given, specifies the conditions that identify which rows to update. With **no WHERE** clause, all rows are updated. If the **ORDER BY** clause is specified, the rows are updated in the order that is specified. The **LIMIT** clause places a limit on the number of rows that can be updated.

```
UPDATE tbl_name SET col_name1 = CASE WHEN [condition] THEN result [WHEN [condition] THEN result . . .]
[ELSE result] END
```

```
UPDATE tbl_name SET col_name1 = { expr1 | DEFAULT | query } [, col_name2 = { expr2 | DEFAULT | query } ] . . .
    [WHERE where_condition]  [ORDER BY . . .]  [LIMIT row_count]
```

single-table update

- `UPDATE temp SET dname = 'new_value' LIMIT 2;`
- `UPDATE temp SET c1 = 'new_value' ORDER BY loc LIMIT 2;`
- `UPDATE temp SET c1 := 'new_value' WHERE deptno < 50;`
- `UPDATE temp SET c1 := 'new_value' WHERE deptno < 50 LIMIT 2;`
- `ALTER TABLE dept ADD SUMSALARY INT;`
- `UPDATE dept SET sumsalary = (SELECT SUM(sal) FROM emp WHERE emp.deptno = dept.deptno GROUP BY emp.deptno);`
- `UPDATE candidate SET totalvotes = (SELECT COUNT(*) FROM votes WHERE candidate.id = votes.candidateID GROUP BY votes.candidateID);`

Important:

- `UPDATE duplicate SET id = ( SELECT @cnt := @cnt + 1 );`
- `UPDATE subscriptions SET status = CASE WHEN expiry_date < CURDATE() THEN 'Expired' ELSE 'Active' END;`

- `UPDATE subscription SET start_date = '2026-06-10', expiration_date = '2026-07-10' WHERE subscription_id = (SELECT subscription_id FROM (SELECT MAX(subscription_id) FROM subscription) t);`

single-table delete

DELETE is used to delete tuples from a relation.

delete can violate only in referential integrity.

Important:

- The **DELETE** operation can violate only referential integrity. This occurs if the tuple *t* being deleted is referenced by foreign keys from other tuple *t* in the database.

single-table delete

The DELETE statement deletes rows from tbl_name and returns the number of deleted rows. To check the number of deleted rows, call the *ROW_COUNT()* function. The optional WHERE clause identify which rows to delete. With no WHERE clause, all rows are deleted. If the ORDER BY clause is specified, the rows are deleted in the order that is specified. The LIMIT clause places a limit on the number of rows that can be deleted.

```
DELETE FROM tbl_name
   [PARTITION (partition_name [, partition_name] . . .)]
   [WHERE where_condition]
   [ORDER BY . . .]
   [LIMIT row_count]
```

Note:

- LIMIT clauses apply to single-table deletes, but not multi-table deletes.

- `DELETE FROM temp;`
- `DELETE FROM temp ORDER BY loc LIMIT 2;`
- `DELETE FROM temp WHERE deptno < 50;`
- `DELETE FROM temp WHERE deptno < 50 LIMIT 2;`

auto_increment column

The **AUTO_INCREMENT** attribute can be used to generate a unique number/identity for new rows.

Gap occurs when :- INSERT fails or is rolled back

auto_increment

col_name data_type AUTO_INCREMENT [UNIQUE [KEY] | [PRIMARY] KEY]

IDENTITY is a synonym to the ***LAST_INSERT_ID*** variable.

Remember:

- There can be **only one** AUTO_INCREMENT column per table.
- The column defined with AUTO_INCREMENT **must be indexed**.
- it cannot have a DEFAULT value.
- it contains only **positive values**.
- AUTO_INCREMENT works only with **integer types** (TINYINT, SMALLINT, INT, BIGINT . . .)
- when you insert a value of NULL or 0 into AUTO_INCREMENT column, it generates **next value**.
- If a row is deleted or a transaction is rolled back, the auto_increment number is not reused.
- Truncating a table **resets** AUTO_INCREMENT back to 1.
- If you insert multiple rows in one INSERT, each row gets a **different value**.
- *use LAST_INSERT_ID*() function to find the row that contains the most recent AUTO_INCREMENT value.
- you cannot use AUTO_INCREMENT directly with a GENERATED ALWAYS AS column.

- CREATE TABLE animals(id INT NOT NULL AUTO_INCREMENT, breed INT, PRIMARY KEY (id, breed)); //valid
- CREATE TABLE animals(id NT NOT NULL, breed INT AUTO_INCREMENT, PRIMARY KEY (id, breed)); //invalid

- SELECT @@IDENTITY
- SELECT LAST_INSERT_ID()
- SET INSERT_ID = 7

```
CREATE TABLE posts (
  c1 INT UNIQUE KEY AUTO_INCREMENT,
  c2 VARCHAR(20)
) AUTO_INCREMENT = 2;      // auto_number will start with value 2.
```

auto_increment

The **auto_increment** specifies a **auto_increment** value for the column.

- CREATE TABLE posts(
  postID INT AUTO_INCREMENT UNIQUE KEY,
  postTitle VARCHAR(255),
  postDate DATETIME DEFAULT NOW(),
  deleted INT
  );

Field	Type	Null	Key	Default	Extra
postID	int	NO	PRI	NULL	auto_increment
postTitle	varchar(255)	YES		NULL	
postDate	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
deleted	int	YES		NULL	

- CREATE TABLE comments(
  commentID INT AUTO_INCREMENT PRIMARY KEY,
  comment TEXT,
  commentDate DATETIME DEFAULT NOW(),
  deleted INT
  );

Field	Type	Null	Key	Default	Extra
commentID	int	NO	PRI	NULL	auto_increment
comment	text	YES		NULL	
commentDate	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
deleted	int	YES		NULL	

- CREATE TABLE animals(id INT NOT NULL AUTO_INCREMENT, breed INT, PRIMARY KEY (id, breed)); //valid
- CREATE TABLE animals(id NT NOT NULL, breed INT AUTO_INCREMENT, PRIMARY KEY (id, breed)); //invalid

auto_increment

- **auto_increment_increment** – is the incremental value, controls the interval between successive column values.
- **auto_increment_offset** – determines the starting value for the AUTO_INCREMENT field; this value is used for the first record inserted into the table.

```
SET @@auto_increment_offset=5;
SET @@auto_increment_increment=10;
ALTER TABLE temp AUTO_INCREMENT = 0;
```

Remember:

- If value of auto_increment_offset set is greater than that of auto_increment_increment, the value of auto_increment_offset is ignored.

auto_increment

NO_AUTO_VALUE_ON_ZERO affects handling of *AUTO_INCREMENT* columns. Normally, you generate the next sequence number for the column by inserting either NULL or 0 into it. *NO_AUTO_VALUE_ON_ZERO* suppresses this behaviour for 0 so that only NULL generates the next sequence number.

- *SET sql_mode = '';*
- *SET sql_mode = 'NO_AUTO_VALUE_ON_ZERO';*

- **CREATE TABLE . . . LIKE . . .**, the destination table *preserves generated column information* from the original table.
- **CREATE TABLE . . . SELECT . . .**, the destination table *does not preserves generated column information* from the original table.

generated column

A SQL generated column is a type of column that stores values calculated from an expression applied to data in other columns of the same table. The value of a generated column cannot be altered manually and is automatically updated whenever the data it depends on changes.

Remember:

- Stored functions and user-defined functions are not permitted.
- Stored procedure and function with parameters are not permitted.
- Variables (system variables, user-defined variables, and stored program local variables) are not permitted.
- Subqueries are not permitted.
- You cannot use a generated column in a DEFAULT clause.
- You cannot use AUTO_INCREMENT directly with a GENERATED ALWAYS AS column.
- The AUTO_INCREMENT attribute cannot be used in a generated column definition.
- You **cannot insert/update** values into a generated column directly.
- Triggers cannot use NEW.COL_NAME or use OLD.COL_NAME to refer to generated columns.
- Stored column cannot be converted to virtual column and virtual column cannot be converted to stored column.
- Generated column can be made as invisible column.

Note:

- The expression can contain literals, built-in functions with no parameters, operators, or references to any column within the same table. If you use a function, it must be scalar and deterministic.

virtual column - generated always

col_name data_type [GENERATED ALWAYS] AS (*expression*) | (CASE WHEN . . . THEN . . . ELSE . . . END)
[VIRTUAL | STORED]

- **VIRTUAL**: Column values are not stored, but are evaluated when rows are read, immediately after any BEFORE triggers. A virtual column takes no storage.
- **STORED**: Column values are evaluated and stored when rows are inserted or updated. A stored column does require storage space and can be indexed.

Note:

- The default is **VIRTUAL** if neither keyword is specified.

- ```
  CREATE TABLE product(
    productCode INT AUTO_INCREMENT PRIMARY KEY,
    productName VARCHAR(45),
    productVendor VARCHAR(45),
    productDescription TEXT,
    quantityInStock INT,
    buyPrice FLOAT,
    stockValue FLOAT GENERATED ALWAYS AS(quantityInStock * buyPrice) VIRTUAL
  );
  ```

Field	Type	Null	Key	Default	Extra
productCode	int	NO	PRI	NULL	auto_increment
productName	varchar(45)	YES		NULL	
productVendor	varchar(45)	YES		NULL	
productDescription	text	YES		NULL	
quantityInStock	int	YES		NULL	
buyPrice	float	YES		NULL	
stockValue	float	YES		NULL	VIRTUAL GENERATED
  ```

generated always as column with case when . . . end

col_name data_type [GENERATED ALWAYS] AS (*expression*) | (CASE WHEN . . . THEN . . . ELSE . . . END)

- ```
  CREATE TABLE loan(
    loanID BIGINT PRIMARY KEY AUTO_INCREMENT,
    bankName VARCHAR(50),
    accountNumber VARCHAR(30),
    amount INT,
    riskStatus VARCHAR(30) GENERATED ALWAYS AS (
    CASE
       WHEN amount < 1000 THEN 'Low Risk'
       WHEN amount BETWEEN 1000 AND 2000 THEN 'Above Average Risk'
       ELSE 'High Risk’
    END));
  ```
  ```

generated always as column with ENUM and case when . . . end

col_name data_type [GENERATED ALWAYS] AS (*expression*) | (CASE WHEN . . . THEN . . . ELSE . . . END)

- ```
  CREATE TABLE patientAdmission(
    admissionID INT PRIMARY KEY,
    patientName VARCHAR(10),
    referred_by ENUM('ref1', 'ref2', 'ref3', 'ref4', 'ref5', 'ref9'),
    admissionType VARCHAR(40) GENERATED ALWAYS AS (
    CASE
       WHEN referred_by = 'ref1' THEN 'Emergency'
       WHEN referred_by = 'ref2' THEN 'Urgent'
       WHEN referred_by = 'ref3' THEN 'Elective'
       WHEN referred_by = 'ref4' THEN 'Newborn'
       WHEN referred_by = 'ref5' THEN 'Trauma Center'
       WHEN referred_by = 'ref9' THEN 'Information Not Available'
    END));
  ```
  ```

generated always as column with ENUM and case when . . . end

col_name data_type [GENERATED ALWAYS] AS (*expression*) | (CASE WHEN . . . THEN . . . ELSE . . . END)

- ```
  CREATE TABLE ticketReservation (
    ticketID INT PRIMARY KEY AUTO_INCREMENT,
    reservationType ENUM('GN', 'TQ','LD','FT', 'HP', 'OQ'),
    description VARCHAR(100) GENERATED ALWAYS AS (
    CASE
       WHEN reservationType = 'GN' THEN 'General Quota'
       WHEN reservationType = 'TQ' THEN 'Tatkal Quota'
       WHEN reservationType = 'LD' THEN 'Ladies Quota'
       WHEN reservationType = 'FT' THEN 'Foreign Tourists Quota'
       WHEN reservationType = 'HP' THEN 'Physically Handicapped'
       WHEN reservationType = 'OQ' THEN 'Headquarters/High Official, Parliament House, Defence, Duty Pass Quota'
   END));
  ```
  ```

generated always as column with case when . . . end

col_name data_type [GENERATED ALWAYS] AS (*expression*) | (CASE WHEN . . . THEN . . . ELSE . . . END)

- CREATE TABLE hostelBilling(
  billingID INT AUTO_INCREMENT PRIMARY KEY,
  studentID INT,
  roomRent INT,
  messCharges INT,
  laundryCharges INT,
  discountPercent INT,
  amount INT GENERATED ALWAYS AS ((roomRent + messCharges + laundryCharges) * discountPercent / 100,
  amenities varchar(150) GENERATED ALWAYS AS (
  CASE
  WHEN (roomRent + messCharges + laundryCharges) BETWEEN 5000 AND 6000 THEN 'Wi-Fi & Bed-sheets'
  WHEN (roomRent + messCharges + laundryCharges) BETWEEN 6001 AND 8000 THEN 'Wi-Fi, Bed-sheets, Reading Room & Lockers'
  WHEN (roomRent + messCharges + laundryCharges) BETWEEN 8001 AND 10000 THEN 'Wi-Fi, Bed-sheets, Reading Room, Lockers and Ac, Fridge, Induction stove'
  END));

generated always as column with JSON object

col_name data_type [GENERATED ALWAYS] AS (JSON *expression*)

- CREATE TABLE product(
  id INT PRIMARY KEY AUTO_INCREMENT,
  document JSON,
  brand VARCHAR(10),
  total INT GENERATED ALWAYS AS (document->>'$.rate' * document->>'$.qty'));

Note:

- MySQL doesn't allow you to insert a value into GENERATED column — not even `DEFAULT`

generated always as column with JSON object

col_name data_type [GENERATED ALWAYS] AS (JSON *expression*)

SELECT

- SELECT id, brand, document->> '$.rate' rate, document->> '$.qty' quantity, total FROM product;
- SELECT JSON_LENGTH(document, '$.size') FROM product; // return the number of elements in array

generated always as column with JSON object

col_name data_type [GENERATED ALWAYS] AS (JSON *expression*)

*INSERT in **Array, Binary***

- INSERT INTO product(id, document, brand) VALUES (DEFAULT, JSON_OBJECT("productName", "shirt", "rate", 1250, "qty", 9, "size", JSON_ARRAY('S', 'M')), 'Raymonds');
- INSERT INTO product(document) VALUES(JSON_OBJECT('userName', 'saleel', 'password', binary 'saleel'));

UPDATE

- UPDATE product SET document = JSON_SET(document, '$.qty', 100) WHERE id = 1; // update the existing value
- UPDATE product SET document = JSON_SET(document, '$.discount', 10.34) WHERE id = 1001; // add new key
- UPDATE product SET document = JSON_ARRAY_APPEND(document, '$.size', 'L'); // append a new item to the array
- UPDATE product SET document = JSON_ARRAY_APPEND(document, '$.size', 'XS', '$.size', 'XL'); // append multiple new item to the array
- UPDATE product SET document = JSON_REMOVE(document, '$.size[0]'); // remove element from array

PROBLEMS *virtual column – MODIFY with ALTER command*

col_name data_type [GENERATED ALWAYS] AS (*expression*) | (CASE WHEN . . . THEN . . . ELSE . . . END)

Note:

- **Virtual column cannot be converted to stored column**.

- ```
  CREATE TABLE product1(
    productCode INT,
    quantityInStock INT,
    buyPrice INT,
    stockValue FLOAT GENERATED ALWAYS AS (quantityInStock * buyPrice) VIRTUAL
  );
  ```

- ```
  ALTER TABLE product1 MODIFY stockValue FLOAT (12, 2);   // error: We are trying to convert virtual column  [ GENERATED ALWAYS AS  ] to stored column by not giving GENERATED ALWAYS AS
  ```

Then What-→ give

- ```
  ALTER TABLE product1 MODIFY stockValue FLOAT (12, 2) GENERATED ALWAYS AS (quantityInStock * buyPrice) VIRTUAL ;
  ```
  ```

PROBLEMS *virtual column – MODIFY with ALTER command*

col_name data_type [GENERATED ALWAYS] AS (*expression*) | (CASE WHEN . . . THEN . . . ELSE . . . END)

Note:

- **Stored column cannot be converted to virtual column**.

- ```
  CREATE TABLE product1(
    productCode INT,
    quantityInStock INT,
    buyPrice INT,
    stockValue FLOAT
  );
  ```

- ALTER TABLE product1 MODIFY stockValue FLOAT GENERATED ALWAYS AS (quantityInStock * buyPrice) VIRTUAL; // error: We are trying to convert stored column to virtual column by giving GENERATED ALWAYS AS
  ```

visible / invisible columns

Columns are visible by default. To explicitly specify visibility for a new column, use a VISIBLE or INVISIBLE keyword as part of the column definition for CREATE TABLE or ALTER TABLE.

Note:

- An invisible column is normally hidden to queries, but can be accessed if explicitly referenced. Prior to MySQL 8.0.23, all columns are visible.
- A table must have at least one visible column. Attempting to make all columns invisible produces an error.
- SELECT * queries by default does not include invisible columns.
- Invisible columns can be indexed and used in constraints (PK, UNIQUE, FK).
- You can change visibility with ALTER TABLE.
- Both STORED and VIRTUAL generated columns can also be declared VISIBLE or INVISIBLE.

invisible column

col_name data_type INVISIBLE

```
CREATE TABLE employee(
  ID INT AUTO_INCREMENT PRIMARY KEY,
  firstName VARCHAR(40),
  salary INT,
 commission INT,
 total INT DEFAULT(salary + commission) INVISIBLE
 tax INT GENERATED ALWAYS AS (total * .25) VIRTUAL INVISIBLE
) ;
```

```
CREATE TABLE employee(
  ID INT PRIMARY KEY AUTO_INCREMENT INVISIBLE ,
  firstName VARCHAR(40)
) ;
```

```
INSERT INTO employee(firstName, salary, commission) VALUES('ram', 4700, -700);
INSERT INTO employee(firstName, salary, commission) VALUES('pankaj', 3400, NULL);
INSERT INTO employee(firstName, salary, commission) VALUES('rajan', 3200, 250);
INSERT INTO employee(firstName, salary, commission) VALUES('ninad', 2600, 0);
INSERT INTO employee(firstName, salary, commission) VALUES('omkar', 4500, 300);
SELECT * FROM employee;
```

```
ALTER TABLE employee MODIFY total INT VISIBLE;
ALTER TABLE employee MODIFY total INT INVISIBLE;
```

varbinary column

VARBINARY is a variable-length binary string column. It's similar to VARCHAR, but it stores binary data, not text.

Note:

- VARBINARY stores binary strings, not character strings.
- Comparisons are based on numeric byte values, not text collation.
- You can index VARBINARY columns.
- You can assign default values but must be valid binary literals.

varbinary column

col_name VARBINARY

```
CREATE TABLE login (
  ID INT AUTO_INCREMENT PRIMARY KEY,
  userName VARCHAR(40),
  password VARBINARY(40) INVISIBLE
) ;
```

```
INSERT INTO login(userName, password) VALUES('ram', 'ram@123');
INSERT INTO login(userName, password) VALUES('pankaj', 'pankaj');
INSERT INTO login(userName, password) VALUES('rajan', 'rajan');
INSERT INTO login(userName, password) VALUES('ninad', 'ninad');
INSERT INTO login(userName, password) VALUES('omkar', 'omkar');
```

```
SELECT * FROM login;
SELECT username, CAST(password as CHAR) FROM login;
```

```
CREATE TABLE temp(
  c1 INT PRIMARY KEY,
  c2 VARBINARY(4) DEFAULT 0x89504E47);
```

a1 INT(10) zerofill	0	0	0	0	0	0	1	5	0	0	length(ID1) -> 10
a2 INT(10)							**3**	**6**	**7**	**5**	length(ID2) -> 4

zerofill value

When you select a column with the type ZEROFILL it pads the displayed value of the field with zeros up to the display width specified in the column definition.

Note:

- ZEROFILL attribute is use for numeric datatype.
- If you specify ZEROFILL for a column, MySQL automatically adds the ***UNSIGNED*** attribute to the column.

zerofill column

col_name data_type ZEROFILL

```
CREATE TABLE employee (
  ID INT ZEROFILL AUTO_INCREMENT PRIMARY KEY,
  firstName VARCHAR(40),
  salary INT,
  commission INT,
  total INT DEFAULT(salary + commission)
) ;
```

- INSERT INTO employee VALUES(NULL, 'ram', 4700, NULL, default);
- INSERT INTO employee VALUES(0, 'pankaj', 3400, 400 , default);
- INSERT INTO employee VALUES(100, 'rajan', 3200, NULL , default);
- INSERT INTO employee VALUES(NULL, 'ninad', 2600, 0, default);
- INSERT INTO employee VALUES(0, 'omkar', 4500, 300, default);
- INSERT INTO employee VALUES (-200, 'rahul', 3000, 300 , default);

- SELECT * FROM employee;
- SELECT ID, LENGTH(ID), salary, LENGTH(salary) FROM employee;

zerofill column

```
CREATE TABLE account (
  accountNumber INT ZEROFILL PRIMARY KEY,
  balance FLOAT,
  openingBalance FLOAT,
  accountName VARCHAR(45),
  custonerID INT,
  openingDate DATE
 );
```

Field	Type	Null	Key	Default	Extra
accountNumber	int(10) unsigned zerofill	NO	PRI	NULL	
balance	float	YES		NULL	
openingBalance	float	YES		NULL	
accountName	varchar(45)	YES		NULL	
custonerID	int	YES		NULL	
openingDate	date	YES		NULL	

int

Note:

- If you specify ZEROFILL for a numeric column, MySQL automatically adds the ***UNSIGNED*** attribute to the column.

Constraints define specific rules to the column(s) data in a database table. While ***inserting***, ***updating***, or ***deleting*** the data rows, if the rules of the constraint are **not followed**, the system will display an **error** message and ***the action will be terminated***. The SQL Constraints are defined while creating a new table. We can also **ALTER** the table and **ADD new** SQL Constraints. The MySQL Constraints are mainly used to maintain **data integrity**.

constraints

CONSTRAINT is used to define rules to allow or restrict what values can be stored in columns. The purpose of inducing constraints is to enforce the integrity of a database.

CONSTRAINTS can be classified into two types –

- *Column Level*
- *Table Level*

The column level constraints can apply only to one column where as table level constraints are applied to the entire table.

In MySQL, when you run SHOW INDEXES or SHOW CREATE TABLE, you may see MUL under the Key column.

Remember:

- PRI => primary key
- UNI => unique key
- MUL=> is basically an index that is neither a **primary key** nor a **unique key**. The name comes from "multiple" because multiple occurrences of the same value are allowed (Since it's not unique, the column can contain **duplicate values**.).

constraints

To limit or to restrict or to check or to control.

Note:

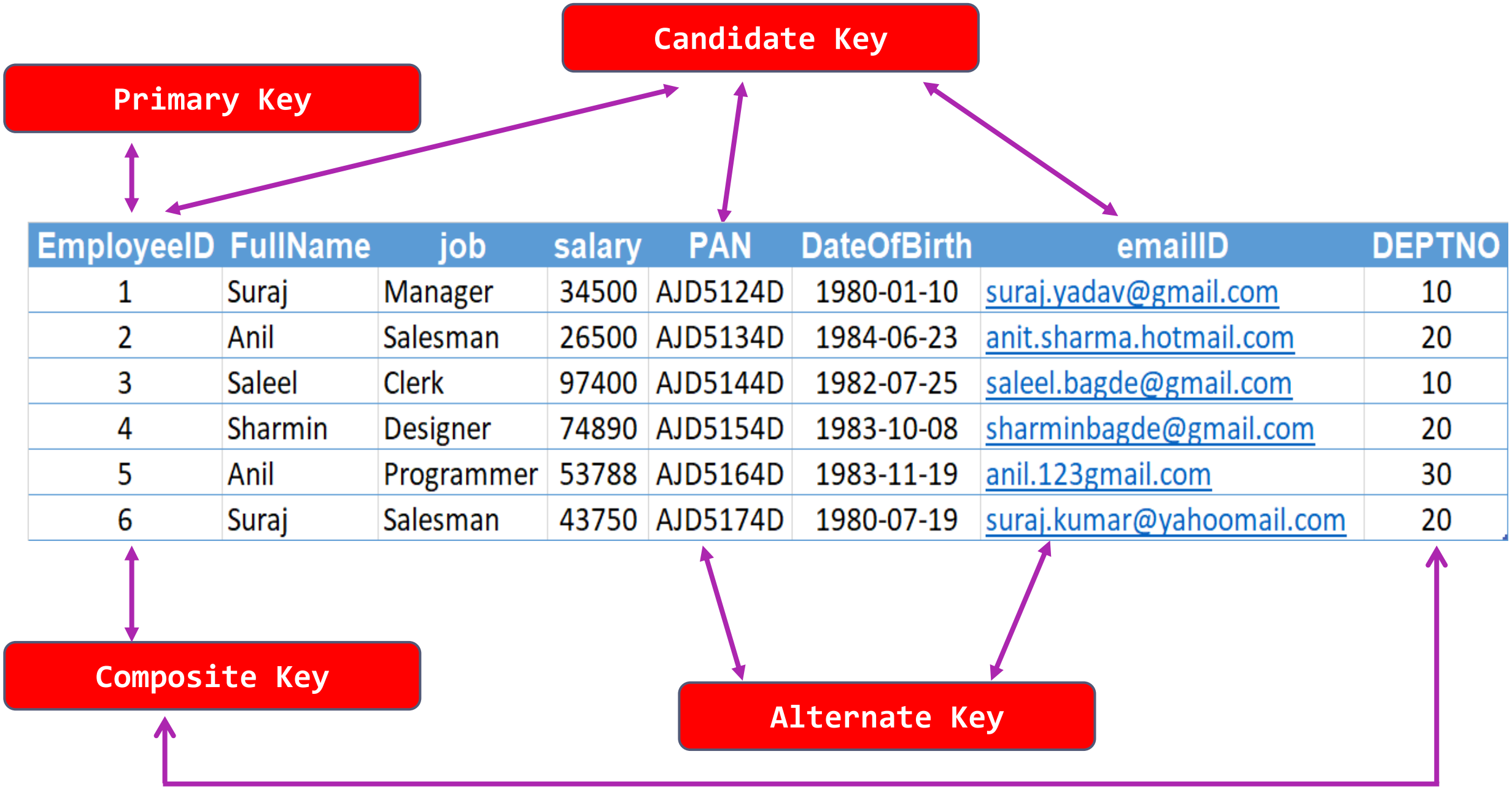
- a table with a foreign key that references another table's primary key is **MUL**.
- If more than one of the Key values applies to a given column of a table, Key displays the one with the highest priority, in the order **PRI**, **UNI**, and **MUL**.
- If a table has a PRIMARY KEY or UNIQUE NOT NULL index that consists of a single column that has an integer type, you can use **_rowid** to refer to the indexed column in SELECT statements.

Keys are used to establish relationships between tables and also to uniquely identify any record in the table.

types of Keys?

r = Employee(EmployeeID, FullName, job, salary, PAN, DateOfBirth, emailID, deptno)

- **Candidate Key:** are individual columns in a table that qualifies for uniqueness of all the rows. Here in Employee table EmployeeID, PAN or emailID are Candidate keys.
- **Primary Key**: is the columns you choose to maintain uniqueness in a table. Here in Employee table you can choose either EmployeeID, PAN or emailID columns, EmployeeID is preferable choice.
- **Alternate Key**: Candidate column other the primary key column, like if EmployeeID is primary key then , PAN or emailID columns would be the Alternate key.
- **Super Key**: If you add any other column to a primary key then it become a super key, like EmployeeID + FullName is a Super Key.
- **Composite Key**: If a table do not have any single column that qualifies for a Candidate key, then you have to select 2 or more columns to make a row unique. Like if there is no EmployeeID, PAN or emailID columns, then you can make FullName + DateOfBirth as Composite key. But still there can be a narrow chance of duplicate row.


Candidate Key
Primary Key
EmployeeID FullName job salary PAN DateOfBirth emailID DEPTNO
1 Suraj Manager 34500 AJD5124D 1980-01-10 suraj.yadav@gmail.com 10
2 Anil Salesman 26500 AJD5134D 1984-06-23 anit.sharma.hotmail.com 20
3 Saleel Clerk 97400 AJD5144D 1982-07-25 saleel.bagde@gmail.com 10
4 Sharmin Designer 74890 AJD5154D 1983-10-08 sharminbagde@gmail.com 20
5 Anil Programmer 53788 AJD5164D 1983-11-19 anil.123gmail.com 30
6 Suraj Salesman 43750 AJD5174D 1980-07-19 suraj.kumar@yahoomail.com 20
Composite Key
Alternate Key

No Primary Key in a Table

Anti-pattern

- Here,
```
CREATE TABLE orders (
    order_id INT,
    customer_name VARCHAR(100)
);
```

Why it's bad:

- Rows cannot be uniquely identified.
- Makes updates/deletes unreliable.

Better approach

- Here,
```
CREATE TABLE orders (
    order_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_name VARCHAR(100)
);
```

Remember:

- A primary key cannot be NULL (absence of a value).
- A primary key value must be unique.
- A table has only one primary key.
- The primary key values cannot be changed, if it is referred by some other column.
- The primary key must be given a value when a new record is inserted.
- **An index can consist of 16 columns, at maximum. Since a PRIMARY KEY constraint automatically adds an index, it can't have more than 16 columns.**

Database	Max Columns in Primary Key
MySQL	16
PostgreSQL	32
Oracle	32
SQL Server	16
DB2	16
MariaDB	16

PRIMARY KEY constraint

A primary key is a special column (or set of combined columns) in a relational database table, that is used to uniquely identify each record. Each database table needs a primary key.

Note:

- Primary key in a relation is always associated with an **INDEX** object.
- If, we give on a column a combination of **NOT NULL & UNIQUE** key then it behaves like a PRIMARY key.
- If, we give on a column a combination of **UNIQUE key & AUTO_INCREMENT** then also it behaves like a PRIMARY key.
- Stability: The value of the primary key should be stable over time and not change frequently.

clustered and non-clustered index

Indexing in MySQL is a process that helps us to return the requested data from the table very fast. If the table does not have an index, it scans the whole table for the requested data.

MySQL allows two different types of Indexing:

- Clustered Index
- Non-Clustered Index

Clustered Index:- The InnoDB table uses a clustered index for optimizing the speed of most common lookups (SELECT statement) and DML operations like INSERT, UPDATE, and DELETE command. Clustered indexes sort and store the data rows in the table based on their key values that can be sorted in only one way. If the table column contains a primary key or unique key, MySQL creates a clustered index named PRIMARY based on that specific column.

Non-Clustered Index:- The indexes other than PRIMARY indexes (i.e. clustered indexes) called a non-clustered index. The non-clustered indexes are also known as secondary indexes. The non-clustered index and table data are both stored in different places. It is not sorted (ordering) the table data.

- CREATE TABLE test(c1 INT, c2 INT, c3 INT, c4 INT,c5 INT, c6 INT, c7 INT, c8 INT, c9 INT, c10 INT, c11 INT, c12 INT, c13 INT, c14 INT, c15 INT, c16 INT, c17 INT, c18 INT, c19 INT, c20 INT, PRIMARY KEY (c1, c2, c3, c4, c5, c6, c7, c8, c9, c10, c11, c12, c13, c14, c15, c16, c17, c18)); // error

constraints – add primary key

col_name data_type PRIMARY KEY

The following example creates tables with **PRIMARY KEY** column.

- ```
  CREATE TABLE users(
    ID INT PRIMARY KEY,
    userName VARCHAR(25),
    password VARCHAR(25),
    email VARCHAR(255)
  ) ;
  ```

- ```
  CREATE TABLE purchase_orders(
    po_number INT,
    vendor_id INT NOT NULL,
    po_status INT NOT NULL,
    PRIMARY KEY(po_number)
  );
  ```

- ```
  CREATE TABLE person(
    ID INT NOT NULL UNIQUE,
    lastName VARCHAR(45),
    firstName VARCHAR(45),
    age INT,
    email VARCHAR(255)
  );
  ```

- ```
  CREATE TABLE supplier(
   supplier_id INT,
   supplier_name VARCHAR(50),
   contact_name VARCHAR(50),
   constraint pk_supplier_id PRIMARY KEY(supplier_id)
  );
  ```

constraints – add composite primary key

The following example creates tables with **COMPOSITE PRIMARY KEY** column.

- CREATE TABLE salesDetails (
  customerID INT,
  productID INT,
  timeID INT,
  qty INT,
  salesDate DATE,
  salesAmount INT,
  PRIMARY KEY(customerID , productID, timeID)
) ;

customerID	productID	timeID	quantity	salesDate	salesAmount
Cust-001	PRD-1	D1-T1	100	●●●●●●	25,00,000
Cust-001	PRD-1	D2-T1	100	●●●●●●	25,00,000
Cust-001	PRD-2	D1-T1	200	●●●●●●	50,00,000
Cust-002	PRD-1	D1-T1	100	●●●●●●	25,00,000
Cust-004	PRD-1	D1-T1	100	●●●●●●	25,00,000
Cust-004	PRD-2	D3-T1	200	●●●●●●	50,00,000

constraints – add composite primary key

The following example creates tables with **COMPOSITE PRIMARY KEY** column.

- ```
  CREATE TABLE payments (
    paymentID INT,
    orderID INT,
    amount INT,
    bankDetails VARCHAR(255),
    PRIMARY KEY(paymentID , orderID)
  ) ;
  ```

- ```
  CREATE TABLE order_product (
    orderID INT,
    productID INT,
    qty INT,
    rate INT,
    constraint pk_orderID_productID PRIMARY KEY(orderID, productID)
  ) ;
  ```

Try It

```
CREATE TABLE try(c1 INT, c2 INT, c3 INT, c4 INT, c5 INT, c6 INT
,c7 INT, c8 INT, c9 INT, c10 INT,c11 INT,c12 INT, c13 INT, c14 INT,
c15 INT, c16 INT,c17 INT, c18 INT, c19 INT, c20 INT, c21 INT, c22
INT, c23 INT, c24 INT, c25 INT, c26 INT, c27 INT, c28 INT, c29 INT ,
c30 INT, c31 INT, c32 INT, c33 INT, PRIMARY KEY(c1, c2, c3, c4,
c5, c6, c7, c8, c9, c10, c11, c12, c13, c14, c15, c16, c17));
```

Note:

- ERROR 1070 (42000): Too many key parts specified; max 16 parts allowed

```
ALTER TABLE table_name
  ADD [ CONSTRAINT constraint_name ]
    PRIMARY KEY (column1, column2, . . . column_n)
```

Add Primary Key using Alter

constraints – add primary key using alter

You can use the **ALTER TABLE** statement to **ADD PRIMARY KEY** on existing column.

```
ALTER TABLE table_name
 ADD [ CONSTRAINT constraint_name ]
  PRIMARY KEY (column1, column2, . . . column_n)
```

- ```
  CREATE TABLE vendors (
    vendor_id INT,
    vendor_name VARCHAR(25),
    address VARCHAR(255)
  );
  ```

- ```
  ALTER TABLE vendors ADD PRIMARY KEY(vendor_id);
  ```

- ```
  ALTER TABLE vendors ADD constraint pk_vendor_id PRIMARY KEY(vendor_id);
  ```
  ```

```
ALTER TABLE table_name
  DROP PRIMARY KEY
```

Drop Primary Key

constraints – drop primary key

You can use the **ALTER TABLE** statement to **DROP PRIMARY KEY**.

```
ALTER TABLE table_name
 DROP PRIMARY KEY
```

- ```
  CREATE TABLE vendors (
      vendor_id INT,
      vendor_name VARCHAR(25),
      address VARCHAR(255)
  );
  ```

```
ALTER TABLE vendors DROP PRIMARY KEY;
```
```

Remember:

- A unique key can be NULL (absence of a value).
- A unique key value must be unique.
- A table can have multiple unique key.
- A column can have unique key as well as a primary key.

UNIQUE KEY constraint

A **UNIQUE key** constraint is a set of one or more than one fields/columns of a table that uniquely identify a record in a database table.

Note:

- Unique key in a relation is always associated with an ***INDEX*** object.

constraints – add unique key

col_name data_type UNIQUE KEY

The following example creates table with **UNIQUE KEY** column.

- ```
  CREATE TABLE clients (
    client_id INT,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    company_name VARCHAR(255),
    email VARCHAR(255) UNIQUE
  );
  ```

- ```
  CREATE TABLE brands (
    ID INT,
    brandName VARCHAR(30),
    constraint uni_brandName UNIQUE(brandName)
  );
  ```

- ```
  SHOW  INDEX FROM clients;
  ```

- ```
  CREATE TABLE contacts (
    ID INT,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    phone VARCHAR(15),
    UNIQUE(phone)
  );
  ```

```
ALTER TABLE table_name
  ADD [ CONSTRAINT constraint_name ]
    UNIQUE (column1, column2, . . . column_n)
```

Add Unique Key using Alter

constraints – add unique key using alter

You can use the **ALTER TABLE** statement to **ADD UNIQUE KEY** on existing column.

```
ALTER TABLE table_name
 ADD [ CONSTRAINT constraint_name ]
  UNIQUE (column1, column2, . . . column_n)
```

- ```
  CREATE TABLE shop (
    ID INT,
    shop_name VARCHAR(30)
  );
  ```

- ```
  ALTER TABLE shop ADD UNIQUE(shop_name);
  ```
- ```
  ALTER TABLE shop ADD constraint uni_shop_name UNIQUE(shop_name);
  ```
  ```

```
ALTER TABLE table_name
  DROP INDEX constraint_name;
```

Drop Unique Key

constraints – drop unique key

You can use the **ALTER TABLE** statement to **DROP UNIQUE KEY**.

```
ALTER TABLE table_name
 DROP INDEX constraint_name;
```

- SELECT table_name, constraint_name, constraint_type FROM information_schema.table_constraints WHERE constraint_schema = 'z' AND table_name IN ('A', 'B');

- ALTER TABLE users DROP INDEX <COLUMN_NAME>;
- ALTER TABLE users DROP INDEX U_USER_ID; #CONSTRAINT NAME

-
```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255) UNIQUE KEY
) ;
```

-
```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255),
  constraint uni_email UNIQUE KEY(email)
) ;
```

- ALTER TABLE users DROP INDEX email;

- ALTER TABLE users DROP INDEX uni_email;

[CONSTRAINT [*symbol*]] FOREIGN KEY (*col_name*, . . .) REFERENCES *tbl_name* (*col_name*, . . .)

[ON DELETE CASCADE | SET NULL]
[ON UPDATE CASCADE | SET NULL]

FOREIGN KEY constraint

A **FOREIGN KEY** is a **key** used to link two tables together. A **FOREIGN KEY** is a field (or collection of fields) in one table that refers to the PRIMARY **KEY** in another table. The table containing the **foreign key** is called the child table, and the table containing the candidate **key** is called the referenced or parent table.

constraints – foreign key

Remember:

- A foreign key can have a different column name from its primary key.
- DataType of primary key and foreign key column must be same.
- It ensures rows in one table have corresponding rows in another.
- Unlike the Primary key, they do not have to be unique.
- Foreign keys can be null even though primary keys can not.

Note:

- The table containing the FOREIGN KEY is referred to as the child table, and the table containing the PRIMARY KEY (referenced key) is the parent table.
- PARENT and CHILD tables must use the same storage engine,
- and they cannot be defined as temporary tables.

insert, update, & delete – (primary key/foreign key)

A referential constraint could be violated in following cases.

- An **INSERT** attempt to add a row to a child table that has a value in its foreign key columns that does not match a value in the corresponding parent table's column.
- An **UPDATE** attempt to change the value in a child table's foreign key columns to a value that has no matching value in the corresponding parent table's parent key.
- An **UPDATE** attempt to change the value in a parent table's parent key to a value that does not have a matching value in a child table's foreign key columns.
- A **DELETE** attempt to remove a record from a parent table that has a matching value in a child table's foreign key columns.

Note:

- PARENT and CHILD tables must use the same storage engine,
- and they cannot be defined as temporary tables.
- If we don't give constraint name. System will automatically generated the constraint name and will assign to foreign key constraint. **e.g. login_ibfk_1, login_ibfk_2, …..**

Remember:

anomaly – (primary key/foreign key)

Student (parent) Table

RollNo	Name	Mobile	City	State	isActive
1	Ramesh	●●●●	Pune	MH	1
2	Amit	●●●●	Baroda	GJ	1
3	Rajan	●●●●	Surat	GJ	1
4	Bhavin	●●●●	Baroda	GJ	1
5	Pankaj	●●●●	Surat	GJ	1

student_course (child) Table

RollNo	CourceDuration	CourceName
1	1.5 month	RDBMS
2	1.2 month	NoSQL
3	2 month	Networking
1	2 month	Java
2	2 month	.NET

Insertion anomaly:

- If we try to insert a record in Student_Course (child) table with RollNo = 7, it will not allow.

Updation and Deletion anomaly:

- If you try to chance the RollNo from Student (parent) table with RollNo = 6 whose RollNo = 1 , it will not allow.
- If you try to chance the RollNo from Student_Course (child) table with RollNo = 9 whose RollNo = 3 , it will not allow.
- If we try to delete a record from Student (parent) table with RollNo = 1, it will not allow.

alter, drop – (primary key/foreign key)

Remember:

Parent Table

```
student  = {
   rollno INT,  * (PK)
   name VARCHAR(10),
   mobile VARCHAR(10),
   city VARCHAR(10),
   state VARCHAR(10),
   isActive BOOL
}
```

Child Table

```
student_course = {
   rollno INT,   * (FK)
   courceduration VARCHAR(10),
   courcename VARCHAR(10)
}
```

DDL command could be violated in following cases.

Alter command:

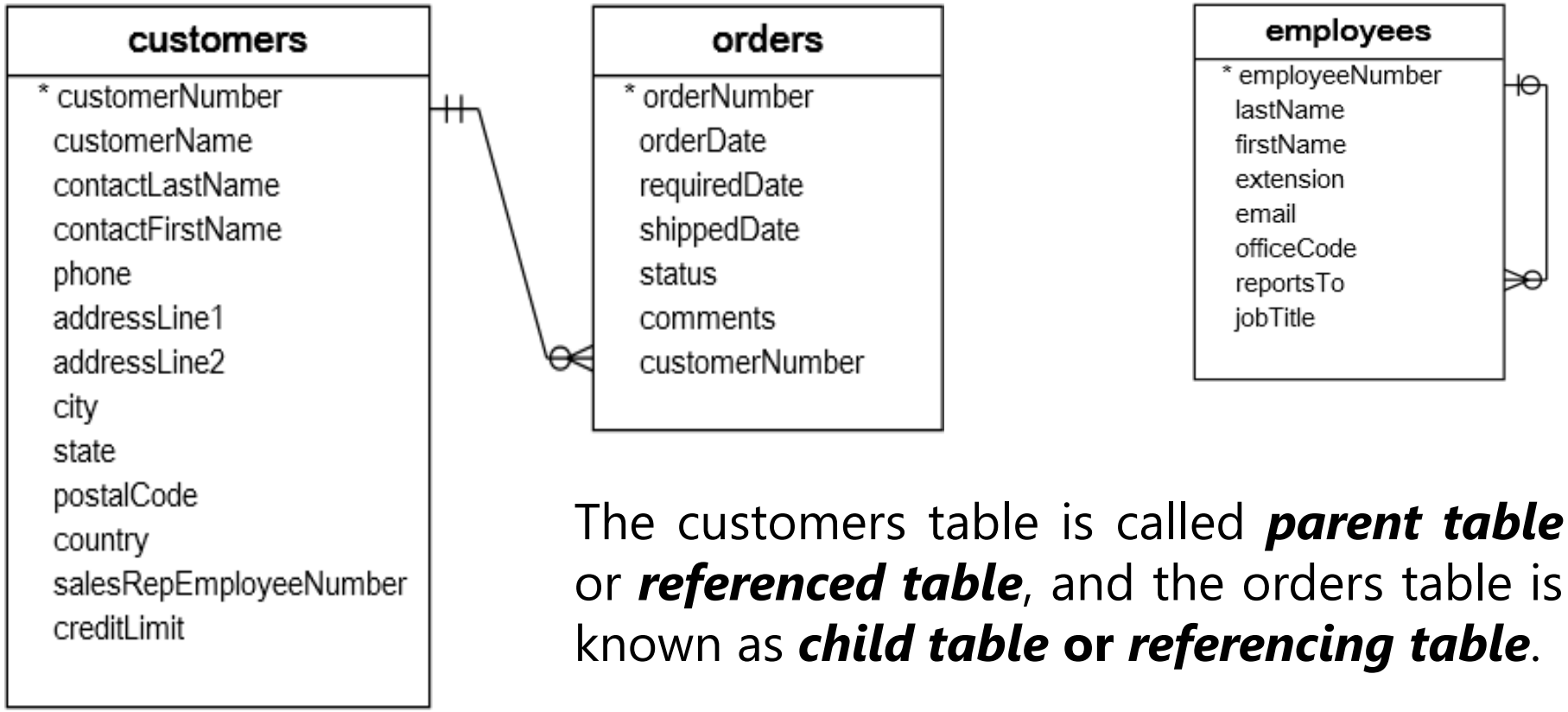
- If we try to modify datatype of RollNo in Student or Student_Course table with VARCHAR, it will not allow.
- If we try to apply auto_increment to RollNo in Student table, it will not allow
- If we try to drop RollNo column from Student table , it will not allow.

Drop command:

- If we try to drop Student (parent) table, it will not allow.

constraints – foreign key

A foreign key is a field in a table that matches another field of another table. A foreign key places constraints on data in the related tables, which enables MySQL to maintain referential integrity.



The customers table is called ***parent table*** or ***referenced table***, and the orders table is known as ***child table*** **or** ***referencing table***.

constraints – foreign key

- ```
  CREATE TABLE department(
    department_id INT UNSIGNED  PRIMARY KEY
  );
  ```

```
CREATE TABLE employee(
  department_id INT,
  FOREIGN KEY (department_id)  REFERENCES
   department(department_id)
);
```

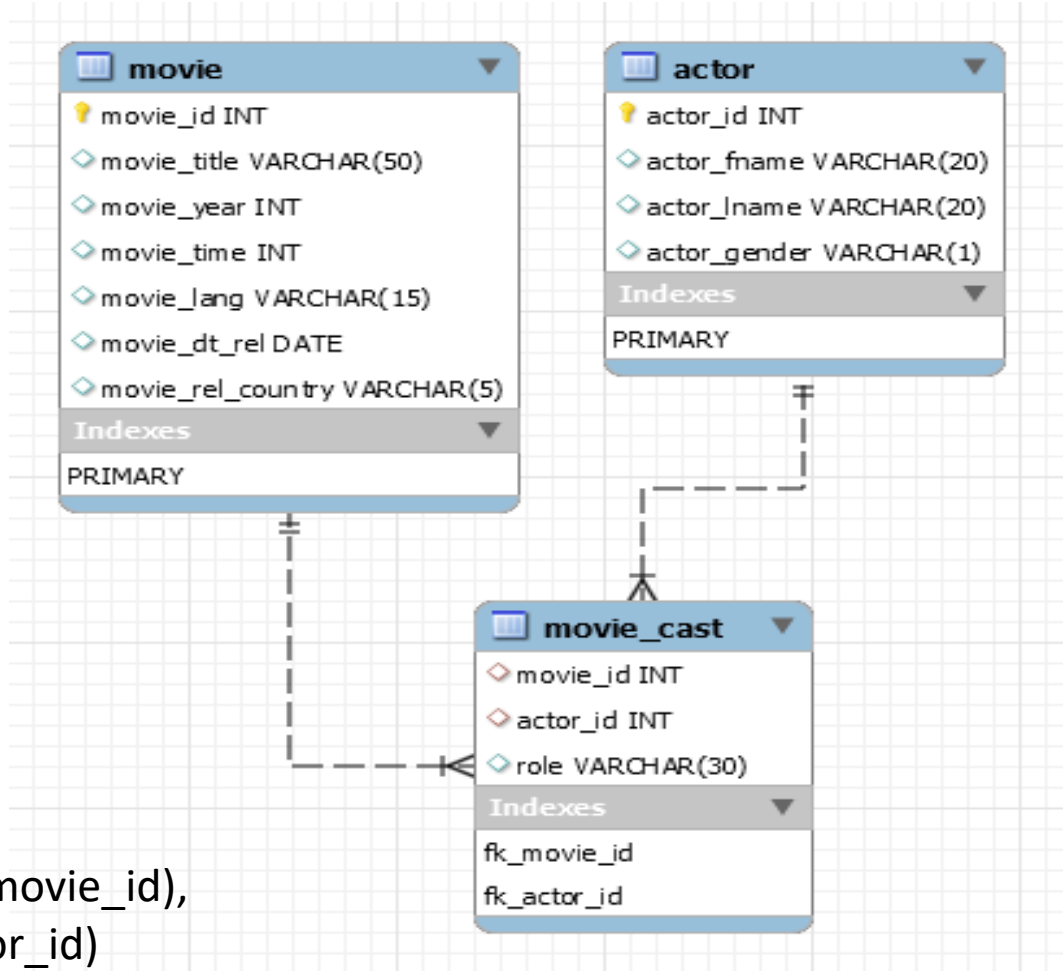
```
CREATE TABLE employee(
  department_id INT UNSIGNED,
  FOREIGN KEY (department_id)  REFERENCES
   department(department_id)
);
```
```

constraints – foreign key

A foreign key is a field in a table that matches another field of another table. A foreign key places constraints on data in the related tables, which enables MySQL to maintain referential integrity.

- CREATE TABLE movie (
  movie_id INT PRIMARY KEY,
  movie_title VARCHAR(50),
  movie_year INT,
  movie_time INT,
  movie_lang VARCHAR(15),
  movie_dt_rel DATE,
  movie_rel_country VARCHAR(5)
  );

- CREATE TABLE actor (
  actor_id INT PRIMARY KEY,
  actor_fname VARCHAR(20),
  actor_lname VARCHAR(20),
  actor_gender VARCHAR(1)
  );



- CREATE TABLE movie_cast (
  movie_id INT,
  actor_id INT,
  role VARCHAR(30),
  constraint fk_movie_id FOREIGN KEY(movie_id) REFERENCES movie(movie_id),
  constraint fk_actor_id FOREIGN KEY(actor_id) REFERENCES actor(actor_id)
  );

QUESTION – find foreign key columns

The following example find ***Foreign Key*** columns.

- ```
  CREATE TABLE owner (
    owner_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(255)
  );
  ```

- ```
  CREATE TABLE shop (
    shop_id INT,
    owner_id INT,
    shop_name VARCHAR(30)
  );
  ```

- ```
  CREATE TABLE brands (
    brand_id INT PRIMARY KEY,
    brand_name VARCHAR(30) UNIQUE
   );
  ```

- ```
  CREATE TABLE contacts (
    contact_id INT PRIMARY KEY,
    owner_id INT,
    contact_number VARCHAR(15)
  );
  ```

- ```
  CREATE TABLE shop_brand (
    ID INT PRIMARY KEY,
    shop_id INT,
    brand_id INT
  );
  ```
  ```

```
ALTER TABLE table_name
  ADD [ CONSTRAINT constraint_name ]
    FOREIGN KEY (child_col1, child_col2, . . . child_col_n)
    REFERENCES parent_table (parent_col1, parent_col2, . . . parent_col_n);
```

Add Foreign Key Constraint using Alter

constraints – add foreign key using alter

You can use the **ALTER TABLE** statement to **ADD FOREIGN KEY** on existing column.

```
ALTER TABLE table_name
 ADD [ CONSTRAINT constraint_name ]
   FOREIGN KEY (child_col1, child_col2, . . . child_col_n)
   REFERENCES parent_table (parent_col1, parent_col2, . . . parent_col_n);
```

```
CREATE TABLE users (
   ID INT PRIMARY KEY,
   userName VARCHAR(40),
   password VARCHAR(255),
   email VARCHAR(255) UNIQUE KEY
) ;
```

```
CREATE TABLE login (
   ID INT PRIMARY KEY,
   userID INT,
   loginDate DATE,
   loginTime TIME
) ;
```

- ALTER TABLE login ADD FOREIGN KEY(userID) REFERENCES users(ID);
- ALTER TABLE login ADD constraint fk_userID FOREIGN KEY(userID) REFERENCES users(ID);

```
ALTER TABLE table_name
  DROP  FOREIGN KEY constraint_name
```

Drop Foreign Key Constraint using Alter

constraints – drop foreign key

You can use the **ALTER TABLE** statement to **DROP FOREIGN KEY**.

- ```
  CREATE TABLE users (
    ID INT PRIMARY KEY ,
    userName VARCHAR(40),
    password VARCHAR(255),
    email VARCHAR(255)
  ) ;
  ```

- ```
  CREATE TABLE login (
    ID INT PRIMARY KEY,
    userID INT,
    loginDate DATE,
    loginTime TIME,
    FOREIGN KEY(userID) REFERENCES users(ID)
  ) ;
  ```

- ```
  CREATE TABLE login (
    ID INT PRIMARY KEY,
    userID INT,
    loginDate DATE,
    loginTime TIME,
    constraint fk_userID FOREIGN KEY(userID)  REFERENCES users(ID)
  ) ;
  ```

- ```
  ALTER TABLE login DROP FOREIGN KEY fk_userID;
  ```

- ```
  ALTER TABLE login DROP FOREIGN KEY login_ibfk_1; // login_ibfk_1 is the default constraint name.
  ```

- ```
  SELECT table_name, constraint_name, constraint_type FROM information_schema.table_constraints WHERE
  table_schema = 'DB2';
  ```

constraints – foreign key

- [ON DELETE *reference_option*]
- [ON UPDATE *reference_option*]

reference_option: RESTRICT | CASCADE | SET NULL | NO ACTION | SET DEFAULT

Cascaded FOREIGN KEY actions do not activate triggers.

Remember:

- When an UPDATE or DELETE operation affects a key value in the parent table that has matching rows in the child table, the result depends on the referential action specified using ON UPDATE and ON DELETE sub clauses of the FOREIGN KEY clause.

- ON DELETE or ON UPDATE that is not specified, the default action is always RESTRICT.

constraints – foreign key

Remember:

- CASCADE: DELETE or UPDATE the row from the parent table, and automatically DELETE or UPDATE the matching rows in the child table. Both ON DELETE CASCADE and ON UPDATE CASCADE are supported.
- SET NULL: DELETE or UPDATE the row from the parent table, and set the foreign key column or columns in the child table to NULL. Both ON DELETE SET NULL and ON UPDATE SET NULL clauses are supported.

```
CREATE TABLE users (
  ID INT PRIMARY KEY ,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255)
);
```

```
CREATE TABLE login (
  ID INT PRIMARY KEY,
  userID INT,
  loginDate DATE,
  loginTime TIME,
  FOREIGN KEY(userID)  REFERENCES users(ID) ON DELETE CASCADE ON UPDATE CASCADE
);
```

```
CREATE TABLE login (
  ID INT PRIMARY KEY,
  userID INT,
  loginDate DATE,
  loginTime TIME,
  FOREIGN KEY(userID)  REFERENCES users(ID) ON DELETE SET NULL ON UPDATE SET NULL
);
```

on delete / on update – foreign key

SET NULL: Delete or update the row from the parent table and set the foreign key column or columns in the child table to NULL. If you specify a SET NULL action, make sure that you have not declared the columns in the child table as NOT NULL.

RESTRICT: Rejects the delete or update operation for the parent table. Specifying RESTRICT (or NO ACTION) is the same as omitting the ON DELETE or ON UPDATE clause.

NO ACTION: Is equivalent to RESTRICT. The MySQL Server rejects the delete or update operation for the parent table if there is a related foreign key value in the referenced table.

SET DEFAULT: This action is recognized by the MySQL parser, but both InnoDB and NDB reject table definitions containing ON DELETE SET DEFAULT or ON UPDATE SET DEFAULT clauses.

- `CREATE TABLE test (c1 INT, c2 INT, c3 INT, check (c3 = SUM(c1)));`

```
// ERROR
SUM(SAL) MIN(SAL) COUNT(*)
AVG(SAL) MAX(SAL) COUNT(JOB)
```

Check Constraint

operators like =, >, <, BETWEEN, IN, LIKE. Can use.

constraints – check

CHECK condition expressions must follow some rules.

- Literals, deterministic built-in functions, and operators are permitted.
- Non-generated and generated columns are permitted, except columns with the AUTO_INCREMENT attribute.
- Sub-queries are not permitted.
- Environmental variables (such as CURRENT_USER, CURRENT_DATE, …) are not permitted.
- Non-Deterministic built-in functions (such as AVG, COUNT, RAND, LAST_INSERT_ID, FIRST_VALUE, LAST_VALUE, ...) are not permitted.
- Variables (system variables, user-defined variables, and stored program local variables) are not permitted.
- Stored functions and user-defined functions are not permitted.

Note:

Prior to MySQL 8.0.16, CREATE TABLE permits only the following limited version of table CHECK constraint syntax, which is parsed and ignored.

Remember:

If you omit the constraint name, MySQL automatically generates a name with the following convention:

- table_name_chk_n

constraints – check

col_name data_type CHECK(*expr*)

The following example creates **USERS** table with **CHECK** column.

```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255),
  ratings INT CHECK(ratings > 50)
);
```

```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255),
  ratings INT,
  CHECK(ratings > 50)
);
```

```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255),
  ratings INT,
  constraint chk_ratings CHECK(ratings > 50)
);
```

```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255),
  ratings INT,
  constraint chk_ratings  CHECK(ratings > 50),
  constraint chk_email CHECK(LENGTH(email) > 12)
);
```

constraints – check

col_name data_type CHECK(*expr*)

The following example creates **USERS** table with **CHECK** column.

```
CREATE TABLE users (
   ID INT PRIMARY KEY,
   startDate DATE,
   endDate DATE,
  constraint chk_endDate CHECK(endDate > startDate +  INTERVAL 7 day)
);
```

- SELECT * FROM check_constraints WHERE CONSTRAINT_SCHEMA = 'z';

```
ALTER TABLE table_name
  ADD   [ CONSTRAINT constraint_name ]
    CHECK (conidiation)
```

Add Check Constraint using Alter

constraints – add check using alter

You can use the **ALTER TABLE** statement to **ADD CHECK KEY** on existing column.

```
ALTER TABLE table_name
 ADD  CONSTRAINT [ constraint_name ]
  CHECK (conidiation)
```

```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255),
  ratings INT
) ;
```

- ALTER TABLE users ADD CHECK(ratings > 50);
- ALTER TABLE users ADD constraint chk_ratings CHECK(ratings > 50);

```
ALTER TABLE table_name
  DROP { CHECK | CONSTRAINT } constraint_name
```

drop check constraint

constraints – drop check key

You can use the **ALTER TABLE** statement to **DROP CHECK KEY**.

```
ALTER TABLE table_name
 DROP { CHECK | CONSTRAINT } constraint_name
```

```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255),
  ratings INT,
  constraint chk_ratings CHECK(ratings > 50)
) ;
```

- ALTER TABLE users DROP CHECK chk_ratings;
- ALTER TABLE users DROP constraint chk_ratings;
- ALTER TABLE users DROP CHECK users_chk_1;

```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255),
  email VARCHAR(255),
  ratings INT,
  CHECK(ratings > 50)
) ;
```

- SELECT table_name, constraint_name, constraint_type FROM information_schema.table_constraints WHERE table_schema = 'DB2' AND (table_name LIKE 'U%' OR table_name LIKE 'L%');

The check constraint defined on a table must refer to only columns in that table. It can not refer to columns in other tables.

- `CREATE TABLE test (CHECK(c3 > (c1 + c2)), c1 INT, c2 INT, c3 INT);`
- `CREATE TABLE test (c1 INT, c2 INT, c3 INT, CHECK(c3 > (c1 + c2)));`
- `CREATE TABLE test (CHECK(c3 > (c1 + c2)), PRIMARY KEY(c1), c1 INT, c2 INT, c3 INT);`
- `CREATE TABLE test (a INT CHECK (a >= 0),b INT  CHECK (b >= 0), CHECK ( a + b <= 10));`

- `ALTER TABLE test ADD constraint chk_id CHECK(ID > 10);`
- `ALTER TABLE test DROP CHECK chk_id;`

- ```
  CREATE TABLE payments(
    paymentID INT PRIMARY KEY,
    paymentMode VARCHAR(20),
   CHECK (paymentMode IN ('Cash', 'Card', 'UPI', 'Insurance'))
  );
  ```
  ```

check with (in, like, and between)

The **CHECK** constraint using IN, LIKE, and BETWEEN.

```
CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(255) CHECK(LENGTH(password) > 5),
  email VARCHAR(255),
  country VARCHAR(255) CHECK(country LIKE ('I%') OR country LIKE ('U%')),
  ratings INT CHECK(ratings BETWEEN 1 AND 5 OR ratings BETWEEN 12 AND 25),
  isActive BOOL CHECK(isActive IN (1, 0)),
  startDate DATE,
  endDate DATE,
  constraint chk_endDate CHECK(endDate > startDate +  INTERVAL 7 day)
) ;
```

alter table

ALTER TABLE changes the structure of a table.

Note:

- you can add or delete columns,
- create or destroy indexes,
- change the type of existing columns, or
- rename columns or the table itself.
- You cannot change the position of columns in table structure. If not, then what? create a new table with SELECT statement.

syntax

alter table

ALTER TABLE tbl_name

[*alter_specification* [, *alter_specification*] . . .

- | ADD [COLUMN] *col_name column_definition* [FIRST | AFTER *col_name*]
- | ADD [COLUMN] (*col_name column_definition*, . . .)
- | ADD {INDEX|KEY} [index_name] (*index_col_name*, . . .)
- | ADD [CONSTRAINT [*symbol*]] PRIMARY KEY
- | ADD [CONSTRAINT [*symbol*]] UNIQUE KEY
- | ADD [CONSTRAINT [*symbol*]] FOREIGN KEY *reference_definition*
- | CHANGE [COLUMN] *old_col_name new_col_name* column_definition [FIRST|AFTER *col_name*]
- | MODIFY [COLUMN] *col_name column_definition* [FIRST | AFTER *col_name*]
- | DROP [COLUMN] *col_name*
- | DROP PRIMARY KEY
- | DROP {INDEX|KEY} *index_name*
- | DROP FOREIGN KEY *fk_symbol*
- | RENAME [TO|AS] *new_tbl_name*
- | RENAME COLUMN *old_col_name* TO *new_col_name*
- | ALTER [COLUMN] *col_name* { SET DEFAULT {*literal* | (*expr*)} | DROP DEFAULT } | SET { VISIBLE | INVISIBLE }

Remember:

- **Change Columns** :- You can rename a column using a CHANGE old_col_name new_col_name column_definition clause. To do so, specify the old and new column names and the definition that the column currently has.
- **Modify Columns :-** You can also use MODIFY to change a column's type without renaming it.
- **Dropping Columns :-** If a table contains only one column, the column cannot be dropped. If columns are dropped from a table, the columns are also removed from any index of which they are a part. If all columns that make up an index are dropped, the index is dropped as well.

Note:

- To convert a table from one storage engine to another, use an ALTER TABLE statement that indicates the new engine:

```
ALTER TABLE tbl_name ENGINE = InnoDB;

ALTER TABLE tbl_name ADD col1 INT, ADD col2 INT;

ALTER TABLE tbl_name DROP COLUMN col1, DROP COLUMN col2 , ADD col3 INT;
```

add column

ALTER TABLE tbl_name [alter_specification [, alter_specification] . . .]

alter_specification

- ADD [COLUMN] col_name column_definition [FIRST | AFTER col_name]
- ADD [COLUMN] (col_name column_definition, . . .)

add column

```
CREATE TABLE vehicles(
    vehicleID INT PRIMARY KEY ,
    year INT,
    make VARCHAR(100)
);
```

Field	Type	Null	Key	Default	Extra
vehicleID	int	NO	PRI	NULL	
year	int	YES		NULL	
make	varchar(100)	YES		NULL	

```
INSERT INTO vehicles VALUES (111, 2000, 'Honda');
INSERT INTO vehicles VALUES (112, 2002, 'Hyundai');
INSERT INTO vehicles VALUES (113, 2000, 'Jeep');
INSERT INTO vehicles VALUES (114, 2005, 'Toyota');
```

```
ALTER TABLE vehicles
    ADD ID INT UNIQUE auto_increment first,
    ADD model VARCHAR(100) NOT NULL,
    ADD color VARCHAR(50),
    ADD note VARCHAR(255);
```

Field	Type	Null	Key	Default	Extra
ID	int	NO	UNI	NULL	auto_increment
vehicleID	int	NO	PRI	NULL	
year	int	YES		NULL	
make	varchar(100)	YES		NULL	
model	varchar(100)	NO		NULL	
color	varchar(50)	YES		NULL	
note	varchar(255)	YES		NULL	

ID	vehicleID	year	make	model	color	note
1	111	2000	Honda		NULL	NULL
2	112	2002	Hyundai		NULL	NULL
3	113	2000	Jeep		NULL	NULL
4	114	2005	Toyota		NULL	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

modify column

ALTER TABLE tbl_name [alter_specification [, alter_specification] ...]

alter_specification

- MODIFY [COLUMN] col_name column_definition [FIRST | AFTER col_name]

modify column

```
CREATE TABLE vehicles(
    vehicleID INT PRIMARY KEY ,
    year INT,
    make VARCHAR(100),
    model VARCHAR(100) NOT NULL,
    color VARCHAR(50),
    note VARCHAR(255)
);
```

Field	Type	Null	Key	Default	Extra
vehicleID	int	NO	PRI	NULL	
year	int	YES		NULL	
make	varchar(100)	YES		NULL	
model	varchar(100)	NO		NULL	
color	varchar(50)	YES		NULL	
note	varchar(255)	YES		NULL	

```
ALTER TABLE vehicles
    MODIFY year SMALLINT NOT NULL,
    MODIFY make VARCHAR(150) NOT NULL,
    MODIFY color VARCHAR(20) NOT NULL;
```

Field	Type	Null	Key	Default	Extra
vehicleID	int	NO	PRI	NULL	
year	smallint	NO		NULL	
make	varchar(150)	NO		NULL	
model	varchar(100)	NO		NULL	
color	varchar(20)	NO		NULL	
note	varchar(255)	YES		NULL	

```
INSERT INTO vehicles VALUES (111, 2000, 'Honda', 'A1', 'silver', ' Honda was the first Japanese automobile manufacturer to release a dedicated luxury brand, Acura, in 1986.');
```

```
INSERT INTO vehicles VALUES (112, 2002, 'Hyundai', 'AC1', 'white', ' Hyundai operates the world's largest integrated automobile manufacturing facility in Ulsan, South Korea which has an annual production capacity of 1.6 million units.');
```

```
INSERT INTO vehicles VALUES (113, 2000, 'Jeep', 'D2', 'black', ' Fiat Chrysler Automobiles has owned Jeep since 2014. Previous owners include the Kaiser Jeep Corporation and American Motors Corporation. Most Jeeps are American-made, except for a select few models. The Toledo Assembly Complex in Ohio manufactures the Jeep Wrangler.');
```

rename column

ALTER TABLE tbl_name [alter_specification [, alter_specification] ...]

alter_specification

- RENAME COLUMN old_col_name TO new_col_name

rename column

- CREATE TABLE vehicles (
  vehicleID INT,
  year SMALLINT,
  make VARCHAR(150),
  model VARCHAR(100),
  color VARCHAR(20),
  note VARCHAR(255)
  );

Field	Type	Null	Key	Default	Extra
vehicleID	int	YES		NULL	
year	smallint	YES		NULL	
make	varchar(150)	YES		NULL	
model	varchar(100)	YES		NULL	
color	varchar(20)	YES		NULL	
note	varchar(255)	YES		NULL	

- ALTER TABLE vehicles
  RENAME COLUMN year TO model_year

Field	Type	Null	Key	Default	Extra
vehicleID	int	YES		NULL	
model_year	int	NO		NULL	
make	varchar(150)	YES		NULL	
model	varchar(100)	YES		NULL	
model_color	varchar(20)	YES		NULL	
vehicleCondition	varchar(150)	YES		NULL	

change column

ALTER TABLE tbl_name [alter_specification [, alter_specification] ...]

alter_specification

- CHANGE [COLUMN] old_col_name new_col_name column_definition [FIRST | AFTER col_name]

change column

- CREATE TABLE vehicles (
  vehicleID INT,
  year SMALLINT,
  make VARCHAR(150),
  model VARCHAR(100),
  color VARCHAR(20),
  note VARCHAR(255)
  );

Field	Type	Null	Key	Default	Extra
vehicleID	int	YES		NULL	
year	smallint	YES		NULL	
make	varchar(150)	YES		NULL	
model	varchar(100)	YES		NULL	
color	varchar(20)	YES		NULL	
note	varchar(255)	YES		NULL	

- ALTER TABLE vehicles
  CHANGE year model_year INT,
  CHANGE color model_color VARCHAR(20),
  CHANGE note vehicleCondition VARCHAR(150);

Field	Type	Null	Key	Default	Extra
vehicleID	int	YES		NULL	
model_year	int	NO		NULL	
make	varchar(150)	YES		NULL	
model	varchar(100)	YES		NULL	
model_color	varchar(20)	YES		NULL	
vehicleCondition	varchar(150)	YES		NULL	

change column

- CREATE TABLE users (
  ID INT PRIMARY KEY,
  userName VARCHAR(40),
  password VARCHAR(25),
  email VARCHAR(255)
  );

- CREATE TABLE login (
  ID INT PRIMARY KEY,
  userID INT,
  loginDate DATE,
  loginTime TIME,
  constraint fk_userID FOREIGN KEY(userID) REFERENCES users(ID)
) ;

- INSERT INTO users VALUES (1, 'rajan', 'ranaj123', 'rajan447.gmail.com');
- INSERT INTO users VALUES (2, 'raj', 'raj', 'raj.gmail.com');

- INSERT INTO login VALUES (1, 1, curdate(), curtime());
- INSERT INTO login VALUES (2, 1, curdate(), curtime());
- INSERT INTO login VALUES (3, 2, curdate(), curtime());
- INSERT INTO login VALUES (4, NULL, curdate(), curtime());

- ALTER TABLE users CHANGE ID userID INT;
- ALTER TABLE login CHANGE userID UID INT;

- INSERT INTO login VALUES (5, NULL, curdate(), curtime());

drop column

ALTER TABLE tbl_name [*alter_specification* [, *alter_specification*] . . .]

alter_specification

- DROP [COLUMN] *col_name*

drop column

- CREATE TABLE vehicles(
  vehicleID INT,
  model_year SMALLINT,
  make VARCHAR(150),
  model VARCHAR(100),
  model_color VARCHAR(20),
  vehicleCondition VARCHAR(150)
  );

Field	Type	Null	Key	Default	Extra
vehicleID	int	YES		NULL	
model_year	smallint	YES		NULL	
make	varchar(150)	YES		NULL	
model	varchar(100)	YES		NULL	
model_color	varchar(20)	YES		NULL	
vehicleCondition	varchar(150)	YES		NULL	

- ALTER TABLE vehicles
  CHANGE model_year year INT NOT NULL,
  DROP model,
  DROP model_color,
  DROP vehicleCondition;

Field	Type	Null	Key	Default	Extra
vehicleID	int	YES		NULL	
year	int	NO		NULL	
make	varchar(150)	YES		NULL	

alter table

Sample table

```
CREATE TABLE vehicles(
vehicleID INT PRIMARY KEY ,
year INT,
make VARCHAR(100)
);
```

Add new columns to a table

```
ALTER TABLE vehicles
ADD model VARCHAR(100) NOT NULL,
ADD color VARCHAR(50),
ADD note VARCHAR(255);
```

Modify columns

```
ALTER TABLE vehicles
MODIFY year SMALLINT NOT NULL,
MODIFY color VARCHAR(20) NOT NULL,
MODIFY make VARCHAR(150) NOT NULL;
```

Rename columns

```
ALTER TABLE vehicles
CHANGE year model_year SMALLINT NOT NULL,
CHANGE color model_color VARCHAR(20),
CHANGE note vehicleCondition VARCHAR(150);
```

DROP columns

```
ALTER TABLE vehicles
CHANGE model_year year INT NOT NULL,
DROP model,
DROP model_color,
DROP vehicleCondition;
```

drop table

Remember:

- DROP and TRUNCATE are DDL commands, whereas DELETE is a DML command.
- DELETE operations can be rolled back (undone), while DROP and TRUNCATE operations cannot be rolled back (DDL statements are auto committed).
- Dropping a TABLE also drops any TRIGGERS for the table.
- Dropping a TABLE also drops any INDEX for the table.
- Dropping a TABLE will not drops any VIEW for the table.
- If you try to drop a PARENT/MASTER TABLE, it will not get dropped.

drop table

DROP [TEMPORARY] TABLE [IF EXISTS] tbl_name [, tbl_name] ...

Note:

- All table data and the table definition are removed/dropped.
- If it is desired to delete only the records but to leave the table definition for future use, then the ***DELETE*** command should be used instead of ***DROP TABLE***.

- DROP login;
- DROP TABLE users;
- DROP TABLE login, users;

create table using different engines

```
show engines;
set default_storage_engine = memory;
```

create table with memory engine

- **MEMORY** storage engine tables are visible to another client/user.
- Structure is stored and rows will be removed, after re-starting mysql server (MySQL80) from Services.
- Provides in-memory tables, formerly known as HEAP.
- It sores all data in RAM for faster access than storing data on disks.
- Operations involving non-critical data such as session management or caching.

e.g.
```
CREATE TABLE temp(c1 INT,  c2 INT) ENGINE = MEMORY;
```

-
```
INSERT INTO temp VALUES(10, 10);
```
-
```
SELECT * FROM temp;
```

re-start mysql server.

-
```
SELECT * FROM temp;
```

create table with csv engine

```
show engines;
set default_storage_engine = csv;
```

- **CSV** storage engine tables are visible to another client.
- The CSV storage engine stores data in text/csv files using comma-separated values format.
- The storage engine for the table doesn't support nullable (NULL) columns.
- Doesn't support AUTO_INCREMENT columns.
- Doesn't support PRIMARY KEY and UNIQUE KEY constraints.
- CHECK constraint with NOT NULL is allowed.

e.g.
```
CREATE TABLE x(
    ID INT NOT NULL,
    ename VARCHAR(10) NOT NULL,
    job VARCHAR(10) NOT NULL,
    sal INT NOT NULL) ENGINE = CSV;
```

- `INSERT INTO x VALUES(1, 'saleel', 'manager', 3400);`
- `SELECT * FROM x;`

Note:

- ERROR 1194 (HY000): Table 'x' is marked as crashed and should be repaired.
- mysql> REPAIR TABLE x;

create table with blackhole engine

```
show engines;
set default_storage_engine = blackhole;
```

- **BLACKHOLE** tables are visible to another client.
- storage engine acts as a “black hole” that accepts data but throws it away and does not store it.
- Triggers can be written on this type of tables

e.g. CREATE TABLE temp(c1 INT PRIMARY KEY AUTO_INCREMENT, c2 INT UNIQUE, c3 INT NOT NULL, c4 INT CHECK(c4 >= 100)) ENGINE = BLACKHOLE;

- INSERT INTO temp(c2, c3, c4) VALUES(100, 200, 300);
- SELECT * FROM temp;

- DROP TRIGGER IF EXISTS triggername;

```
delimiter $$
CREATE TRIGGER triggername BEFRE INSERT ON temp FOR EACH ROW
begin
   INSERT INTO temp1 VALUES (NEW.c1, NEW.c2);
end $$
delimiter ;
```

create temporary table

- **TEMPORARY** tables are not visible to another client.
- Structure and rows is removed, after exit.

e.g. CREATE TEMPORARY TABLE temp(c1 INT, c2 INT);

- INSERT INTO temp VALUES(10, 10);
- SELECT * FROM temp;
- EXIT

table partitioning

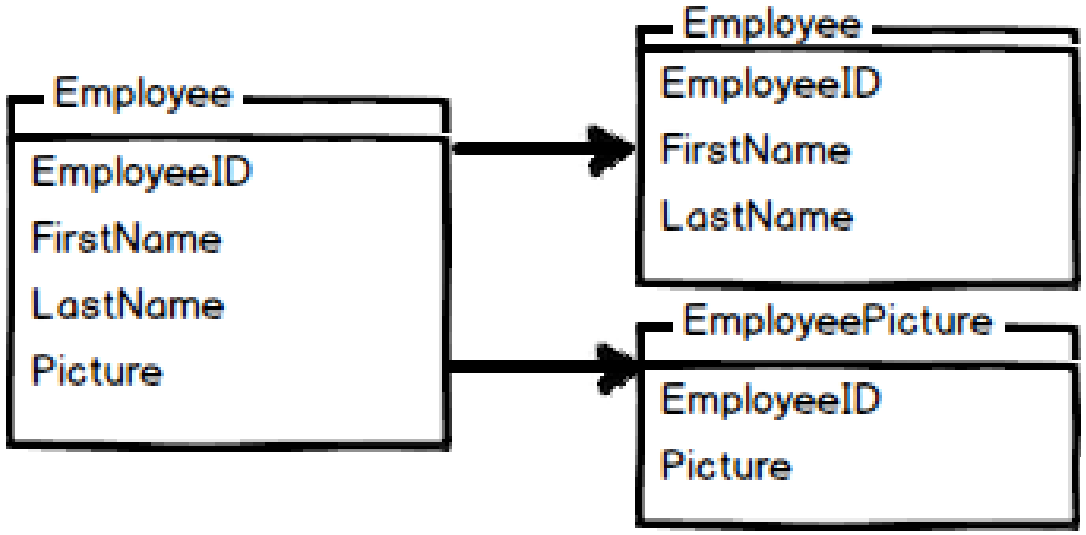
Partitioning separates data into logical units.

table partitioning

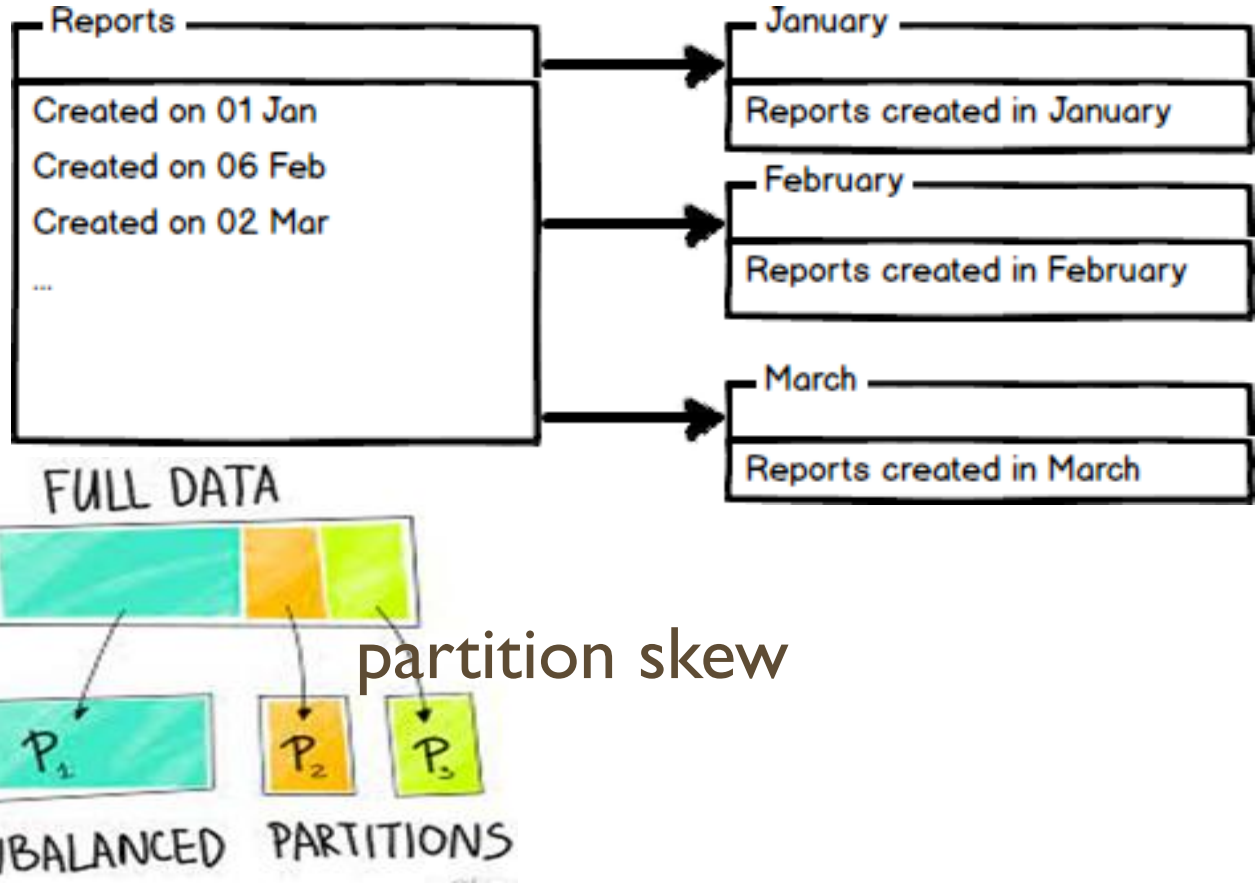
What is a database table partitioning?

Partitioning is the database process where very large tables are divided into multiple smaller parts. By splitting a large table into smaller, individual tables. The main of goal of partitioning is to aid in maintenance of large tables and to reduce the overall response time to read and load data for particular SQL operations.

Vertical Partitioning



Horizontal Partitioning



partition skew

MySQL has mainly six types of partitioning, which are given below:

- RANGE Partitioning
- LIST Partitioning
- COLUMNS Partitioning
- HASH Partitioning
- KEY Partitioning
- Subpartitioning

table partitioning

Original Table

CUSTOMER ID	FIRST NAME	LAST NAME	FAVORITE COLOR
1	TAEKO	OHNUKI	BLUE
2	O.V.	WRIGHT	GREEN
3	SELDA	BAĞCAN	PURPLE
4	JIM	PEPPER	AUBERGINE

Vertical Partitions

VP1

CUSTOMER ID	FIRST NAME	LAST NAME
1	TAEKO	OHNUKI
2	O.V.	WRIGHT
3	SELDA	BAĞCAN
4	JIM	PEPPER

VP2

CUSTOMER ID	FAVORITE COLOR
1	BLUE
2	GREEN
3	PURPLE
4	AUBERGINE

Horizontal Partitions

HP1

CUSTOMER ID	FIRST NAME	LAST NAME	FAVORITE COLOR
1	TAEKO	OHNUKI	BLUE
2	O.V.	WRIGHT	GREEN

HP2

CUSTOMER ID	FIRST NAME	LAST NAME	FAVORITE COLOR
3	SELDA	BAĞCAN	PURPLE
4	JIM	PEPPER	AUBERGINE

partitioning by range / list

RANGE Partitioning

```
PARTITION BY RANGE (COLUMNS)
  (
    PARTITION part_name1 VALUES LESS THAN (int_value),
    PARTITION part_name2 VALUES LESS THAN (int_value),
    PARTITION part_name3 VALUES LESS THAN MAXVALUE
  )
```

LIST Partitioning

```
PARTITION BY LIST (COLUMNS)
  (
     PARTITION part_name1 VALUES IN (int_value_list),
     PARTITION part_name2 VALUES IN (int_value_list),
     PARTITION part_name3 VALUES IN (int_value_list)
  )
```

RANGE Partitioning

```
e.g. CREATE TABLE employee (
        empno  INT,
        ename  VARCHAR(10),
        salary INT
     )
     PARTITION BY RANGE (salary) (
       PARTITION p0 VALUES LESS THAN (2000),
       PARTITION p1 VALUES LESS THAN (4000),
       PARTITION p2 VALUES LESS THAN (6000),
       PARTITION p3 VALUES LESS THAN MAXVALUE
     );
```

- INSERT INTO employee PARTITION(p0) VALUES(1, 'saleel', 1500);
- SELECT * FROM employee PARTITION(p0);
- UPDATE employee PARTITION(p0) set salary = 1500;
- UPDATE employee PARTITION(p0) set salary = 3000 WHERE empno = 1; // Invalid statement
- DELETE FROM employee PARTITION(p0);

RANGE Partitioning

```
e.g. CREATE TABLE bookings (
       booking_id INT,
       guest_name VARCHAR(100),
       checkin_date DATE
     )
     PARTITION BY RANGE(MONTH(checkin_date)) (
       PARTITION January VALUES LESS THAN (2),
       PARTITION February VALUES LESS THAN (3),
       PARTITION March VALUES LESS THAN (4),
       PARTITION April VALUES LESS THAN (5),
       PARTITION May VALUES LESS THAN (6),
       PARTITION June VALUES LESS THAN (7),
       PARTITION July VALUES LESS THAN (8),
       PARTITION August VALUES LESS THAN (9),
       PARTITION September VALUES LESS THAN (10),
       PARTITION October VALUES LESS THAN (11),
       PARTITION November VALUES LESS THAN (12),
       PARTITION December VALUES LESS THAN (13)
     );
```

LIST Partitioning

```
e.g. CREATE TABLE item (
        itemID  INT,
        itemDesc  VARCHAR(10),
        storeID INT
     )
    PARTITION BY LIST(storeID) (
       PARTITION p0 VALUES IN(1, 3, 5, 7),
       PARTITION p1 VALUES IN(2, 4, 6, 8),
       PARTITION p2 VALUES IN(10, 12, 14, 16),
       PARTITION p3 VALUES IN(11, 13, 15, 17)
    );
```

Warehouse	storeID
AC Warehouse	1, 3, 5, 7
National	2, 4, 6, 8
Global	10, 12, 14, 16
Migrant System	11, 13, 15, 17



```
CREATE TABLE customers (
   id INT,
   name VARCHAR(50),
   region VARCHAR(20)
 )
 PARTITION BY LIST COLUMNS(region) (
   PARTITION east VALUES IN ('East', 'Northeast'),
   PARTITION west VALUES IN ('West', 'Southwest'),
   PARTITION central VALUES IN ('Central'),
   PARTITION other VALUES IN ('North', 'South')
);
```

alter / drop partitioning by range / list

Alter Partitioning

RANGE Partitioning

```
ALTER TABLE a ADD PARTITION (PARTITION p3 VALUES LESS THAN(130));
```

- MAXVALUE can only be used in last partition definition

LIST Partitioning

```
ALTER TABLE a ADD PARTITION (PARTITION p3 VALUES IN (10, 11));
```

DROP Partitioning

```
ALTER TABLE a ADD PARTITION p3;
```

create temporary table

Note:

- it is possible to create, alter, drop, and write (Insert, Update, and Delete rows) to TEMPORARY tables.
- You can use all constraints (PRIMARY KEY, UNIQUE, DEFAULT, etc.).
- You can create generated column in temporary table.

temporary table

Remember:

- You can use the *TEMPORARY* keyword when creating a table.
- A *TEMPORARY* table is visible only to the current session, and is dropped automatically when the session is closed.
- Use *TEMPORARY* table with the same name as the original can be useful when you want to try some statements that modify the contents of the table, without changing the original table.
- The permanent (original) table becomes hidden (inaccessible) to the client who creates the *TEMPORARY* table with same name as the original.
- If you issue a DROP TABLE statement, the *TEMPORARY* table is removed and the original table reappears, it is possible, only when then original *tbl_name* and temporary *tbl_name* are same.
- The original table also reappears if you rename the *TEMPORARY* table.

  e.g. ALTER TABLE **dept** RENAME TO d;

  Temporary table_name

temporary table

e.g.

```
CREATE TEMPORARY TABLE student (
    ID INT PRIMARY KEY,
    namefirst VARCHAR(45),
    namelast VARCHAR(45),
    DOB DATE,
    emailID VARCHAR(128)
) ;

CREATE TEMPORARY TABLE temp (
    ID INT PRIMARY KEY,
    firstName VARCHAR(45),
    phone INT,
    city VARCHAR(10) DEFAULT 'PUNE',
    salary INT,
    comm INT,
    total INT GENERATED ALWAYS AS(salary + comm) VIRTUAL
) ;
```

create temporary table ... like

Use CREATE TABLE ... LIKE to create an empty table based on the definition of another table.

```
CREATE TEMPORARY TABLE [IF NOT EXISTS] new_tbl LIKE orig_tbl;
```

- `CREATE TEMPORARY TABLE tempEmployee LIKE employee;`

Remember:

- LIKE works only for base tables, not for VIEWS.
- You can use the TEMPORARY keyword when creating a table. A TEMPORARY table is visible only to the current session, and is dropped automatically when the session is closed.
- Use TEMPORARY table with the same name as the original can be useful when you want to try some statements that modify the contents of the table, without changing the original table.

- `CREATE TEMPORARY TABLE new_tbl SELECT * FROM orig_tbl LIMIT 0;`

Do not use the * operator in your SELECT statements. Instead, use column names. Reason is that in MySQL Server scans for all column names and replaces the * with all the column names of the table(s) in the SELECT statement. Providing column names avoids this search-and-replace, and enhances performance.

continue with SELECT statement…

```
SELECT what_to_select
FROM which_table
WHERE conditions_to_satisfy;
```

The asterisk symbol " * " can be used in the SELECT clause to denote "all attributes."

SELECT CLAUSE

The **SELECT** statement retrieves or extracts data from tables in the database.

- You can use one or more tables separated by comma to extract data.
- You can fetch one or more fields/columns in a single SELECT command.
- You can specify star (*) in place of fields. In this case, SELECT will return all the fields.
- SELECT can also be used to retrieve rows computed without reference to any table e.g. SELECT 1 + 2;

Capabilities of SELECT Statement

1. SELECTION
2. PROJECTION
3. JOINING

Capabilities of SELECT Statement

➢ SELECTION

Selection capability in SQL is to choose the rows in a table that you want to return by a query.

R

EMPNO	ENAME	JOB	HIREDATE	DEPTNO
1	Saleel	Manager	1995-01-01	10
2	Janhavi	Sales	1994-12-20	20
3	Snehal	Manager	1997-05-21	10
4	Rahul	Account	1997-07-30	10
5	Ketan	Sales	1994-01-01	30

Capabilities of SELECT Statement

➢ PROJECTION

Projection capability in SQL to choose the columns in a table that you want to return by your query.

R

EMPNO	ENAME	JOB	HIREDATE	DEPTNO
1	Saleel	Manager	1995-01-01	10
2	Janhavi	Sales	1994-12-20	20
3	Snehal	Manager	1997-05-21	10
4	Rahul	Account	1997-07-30	10
5	Ketan	Sales	1994-01-01	30

Capabilities of SELECT Statement

➢ JOINING

Join capability in SQL to bring together data that is stored in different tables by creating a link between them.

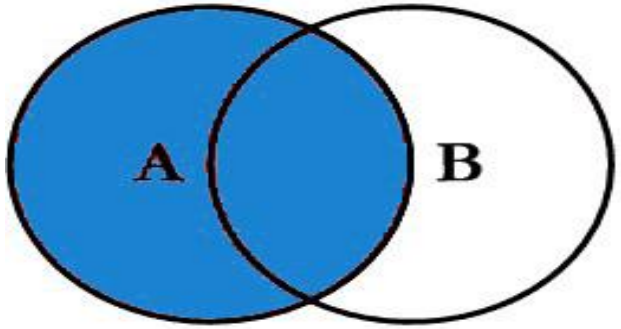
R

EMPNO	ENAME	JOB	HIREDATE	DEPTNO
1	Saleel	Manager	1995-01-01	20
2	Janhavi	Sales	1994-12-20	10
3	Snehal	Manager	1997-05-21	10
4	Rahul	Account	1997-07-30	20
5	Ketan	Sales	1994-01-01	30

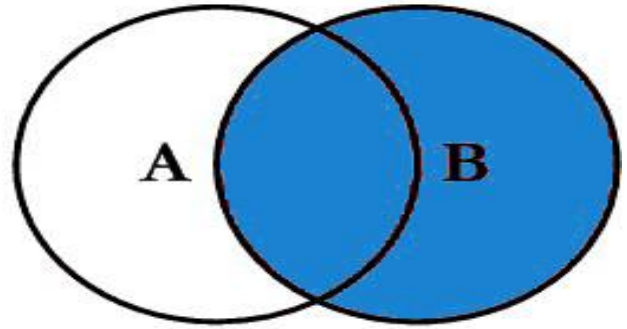
S

DEPTNO	DNAME	LOC
10	HRD	PUNE
20	SALES	BARODA
40	PURCHASE	SURAT

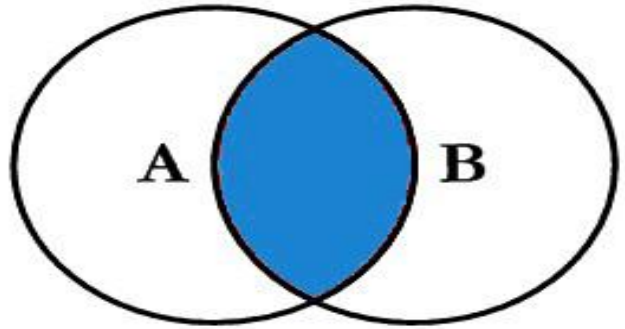
SQL JOINS



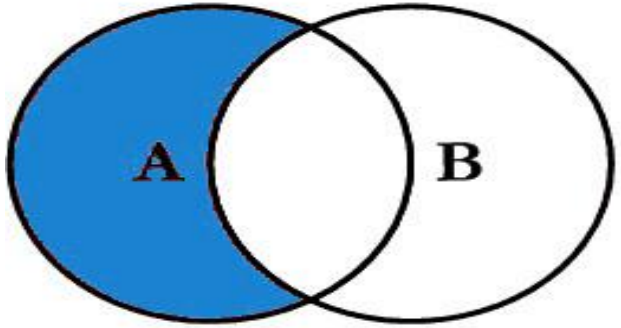

```
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key
```



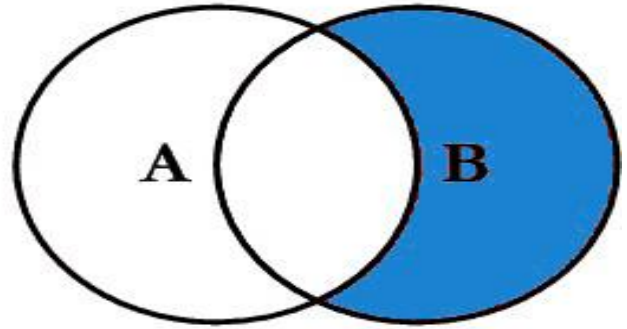

```
SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key
```



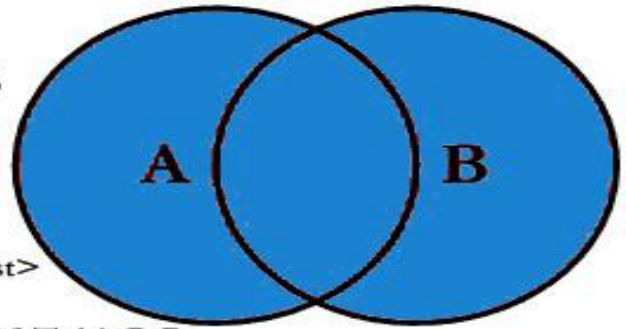

```
SELECT <select_list>
FROM TableA A
INNER JOIN TableB B
ON A.Key = B.Key
```



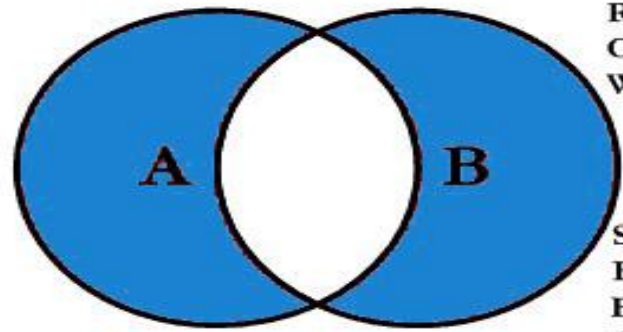

```
SELECT <select_list>
FROM TableA A
LEFT JOIN TableB B
ON A.Key = B.Key
WHERE B.Key IS NULL
```




```
SELECT <select_list>
FROM TableA A
RIGHT JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL
```




```
SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key
```




```
SELECT <select_list>
FROM TableA A
FULL OUTER JOIN TableB B
ON A.Key = B.Key
WHERE A.Key IS NULL
OR B.Key IS NULL
```

select statement

SELECTION Process

SELECT * FROM <table_references>

selection-list | field-list | column-list

When you write a SELECT query, the list of columns or expressions you mention after the SELECT keyword is called the SELECT-LIST (also referred to as FIELD-LIST).

Remember:

- Here, " * " is known as metacharacter (all columns)

PROJECTION Process

SELECT column-list FROM <table_references>

selection-list | field-list | column-list

Remember:

- Position of columns in SELECT statement will determine the position of columns in the output (as per user requirements)

anti-pattern in select statement

Using SELECT *** (SELECT all columns)

SELECT * FROM <table_references>

selection-list | field-list | column-list

Anti-pattern

- Here, SELECT * FROM employees;

Why it's bad:

- Fetches unnecessary columns → wastes bandwidth and memory.
- Breaks code if schema changes (e.g., new columns added).

PROJECTION Process

SELECT column-list FROM <table_references>

selection-list | field-list | column-list

Better approach

- Here, SELECT emp_id, emp_name, salary FROM employees;

- SELECT 'HELLO' ' WORLD';
- SELECT 'HELLO' AS 'WORLD';
- SELECT ename `EmployeeName` FROM emp;
- SELECT ename AS `EmployeeName` FROM emp;

column - alias

A programmer can use an alias to temporarily assign another name to a **column** or **table** for the duration of a ***SELECT*** query**.**

In the selection-list, a quoted column alias can be specified using identifier **(`)** or string quote **(' or ")** characters.

Note:

- Assigning an alias_name does not actually rename the column or table.
- You cannot use alias in an expression.

select statement - alias

SELECT A_1 [[AS] *alias_name*], A_2 [[AS] *alias_name*], . . ., A_N FROM ***r*** [[AS*] alias_name*]

column-name as new-name

table-name as new-name

Remember:

- A select_expr can be given an alias using **AS alias_name**. The alias is used as the expression's column name and can be used in **GROUP BY, HAVING,** or **ORDER BY** clauses.
- The **AS** keyword is optional when aliasing a select_expr with an identifier.
- Standard SQL **disallows** references to column aliases in a ***WHERE*** clause.
- A table reference can be aliased using **tbl_name alias_name** or **tbl_name AS alias_name**
- If the column alias contains spaces, **put it in quotes**.
- Alias name is **max 256 characters**.

- SELECT empno AS *EmployeeID*, ename *EmployeeName* FROM emp;
- SELECT ID AS *'Employee ID'*, ename *"Employee Name"* FROM emp;
- SELECT * FROM emp *employee*;

comparison functions and operator

Comparison operations result in a value of 1 **(TRUE),** 0 **(FALSE)**, or **NULL**.

assignment _operator

= (assignment), :=

- The value on the right hand side may be a literal value, another variable storing a value, or any legal expression that yields a scalar value, including the result of a query (provided that this value is a scalar value). You can perform multiple assignments in the same SET statement. You can perform multiple assignments in the same statement.
- Unlike **=**, the **:=** operator is never interpreted as a comparison operator. This means you can use := in any valid SQL statement (not just in SET statements) to assign a value to a variable.

comparison functions and operator

1. ***arithmetic_operators:***
   * | / | DIV | % |MOD | - | +
2. ***comparison_operator:***
   = | <=> | >= | > | <= | < | <> | !=
3. ***boolean_ predicate:***
   IS [NOT] *NULL* | IS [BOOLEAN]
   | *expr* <=> *null*
4. ***predicate*:**
   expr [NOT] LIKE *expr* [ESCAPE *char*]
   | *expr* [NOT] IN (*expr1, expr2, . . .*)
   | *expr* [NOT] IN (*subquery*)
   | *expr* [NOT] BETWEEN *expr1* AND *expr2*
5. ***logical_operators***
   { AND | && } | { OR | || }
6. ***assignment _operator***
   = (assignment), :=

operand meaning: the quantity on which an operation is to be done.

e.g.

1. *operand1 * operand2*
2. *operand1 = operand2*
3. *operand* IS [NOT] *NULL*
4. *operand* [NOT] LIKE 'pattern'
5. *expr* AND *expr*
6. *Operand := 1001*

- SELECT 23 DIV 6 ; #3
- SELECT 23 / 6 ; #3 .8333

Note:

- NOT has higher precedence than AND **and** AND has higher precedence than OR

- `WHERE col * 4 < 16`
- `WHERE col < 16 / 4`

- `SELECT CONCAT(1, "saleel");`

column - expressions

"Strings are automatically converted to numbers (this behavior is known as **implicit type conversion (type coercion)** in MySQL.)."

- If a string starts with a number, MySQL extracts the number and uses it.
- If a string does not start with a number, MySQL converts it to 0.

select statement - expressions

Column EXPRESSIONS

SELECT A_1, A_2, A_3, A_4, expressions, . . . FROM *r*

- SELECT 1001 + 1;
- SELECT 1001 + '1';
- SELECT '1' + '1' ;
- SELECT '1' + 'a1';
- SELECT '1' + '1a';
- SELECT 'a1' + 1;
- SELECT '1a' + 1;
- SELECT 1 + -1;
- SELECT 1 + -2;
- SELECT -1 + -1;
- SELECT -1 - 1;
- SELECT -1 - -1;
- SELECT 123 * 1;
- SELECT -123 * 1;
- SELECT 123 * -1 ;
- SELECT -123 * -1;
- SELECT 2 * 0;
- SELECT 2435 / 1;
- SELECT 2 / 0;
- SELECT '2435Saleel' / 1;
- SELECT NULL <=> NULL;
- SELECT '20' > 15;
- SELECT 'abc' > 0;
- SELECT 'abc' = 0;
- SELECT '123abc' = 123;
- SELECT 'abc123' = 0;
- SELECT sal, sal + 1000 AS *'New Salary'* FROM emp;
- SELECT sal, comm, sal + comm FROM emp;
- SELECT sal, comm, sal + IFNULL(comm, 0) FROM emp;
- SELECT ename, ename = ename FROM emp;
- SELECT ename, ename = 'smith' FROM emp;
- SELECT c1, c1 / 1 *R1* FROM numberString;

Note:

If any expression evaluated with NULL, returns NULL.

- SELECT 2 + NULL ;
- SELECT 2 - NULL ;
- SELECT 2 * NULL ;
- SELECT 2 / NULL ;
- SELECT 2 % NULL ;

coalesce()

Return the first non-NULL argument

coalesce()

TODO

- SELECT primaryphone, bphone, cphone, hphone, IFNULL(COALESCE(bphone, cphone, hphone), 'Contact customer care') 'Active Phone' FROM coalesce;

Note:

between ... and ...

If any expression is NULL, the BETWEEN operator returns a NULL value.

Check whether a value is within a range of values

Comparison BETWEEN ... AND ...

All three expressions: expr, begin_expr, and end_expr must have the same data type.

```
SELECT 5 BETWEEN 0 AND 10 // Returns 1

SELECT 5 BETWEEN 6 AND 10 // Returns 0
```

Note:

- If any expression is NULL, the BETWEEN operator returns a NULL value.

Tests a value against a boolean value, where boolean_value can be TRUE, FALSE, or UNKNOWN.

Comparison IS

```
SELECT 0 is false //return 1
SELECT 0 is true //return 0
SELECT 1 is true //return 1
SELECT 1 is false //return 0
SELECT 0 is unknown //return 0
SELECT 1 is unknown //return 0
SELECT null is unknown //return 1
```

Note:

In SQL, the same name can be used for two (or more) attributes as long as the attributes are in different relations. If this is the case, and a multi-table query refers to two or more attributes with the same name, we must ***qualify*** the attribute name with the relation name to prevent ambiguity. This is done by prefixing the relation name to the attribute name and separating the two by a period (**.**).

identifiers

Certain objects within MySQL, including database, table, index, column, alias, view, stored procedure, stored functions, triggers, partition, tablespace, and other object names are known as **identifiers**.

identifiers

The maximum length for each type of identifiers like (Database, Table, Column, Index, Constraint, View, Stored Program, Compound Statement Label, User-Defined Variable, Tablespace) is 64 characters, whereas for Alias is 256 characters.

- You can refer to a table within the default database as
  1. tbl_name
  2. db_name.tbl_name.

- You can refer to a column as
  1. col_name
  2. tbl_name.col_name
  3. db_name.tbl_name.col_name.

Note:

- You need not specify a ***tbl_name*** or ***db_name.tbl_name*** prefix for a column reference unless the reference would be ambiguous.
- The identifier quote character is the backtick (`)

- NULL value does not occupy space in memory.
- NULL value is independent of data type.
- NULL can be written in any lettercase.
- NULL represents a missing or unknown value.
- It is not a number (0) or an empty string ("")
- A missing / unknown value cannot be compared as equal or not-equal
- SQL uses three-valued logic: TRUE, FALSE and UNKNOWN.

null-to-null comparisons

- SELECT NULL = NULL, NULL IS NULL; // Output will be NULL and 1
- SELECT NULL = NULL, NULL IS NOT NULL; // Output will be NULL and 0

- SELECT FALSE AND NULL;
- SELECT FALSE OR NULL;
- SELECT TRUE AND NULL;
- SELECT TRUE OR NULL;

- SELECT NULL = NULL;
- SELECT NULL <> NULL;
- SELECT NULL is NULL;
- SELECT NULL is NOT NULL;

the null value

The **NULL** (absence of a value) **value** is special. It means **"*NO VALUE*"** or **"*UNKNOWN VALUE*".**

Remember:

- You cannot be compared two **"*UNKNOWN VALUE*"** the way you compere two **"*KNOWN VALUE*"** to each other.
- If you attempt to use **NULL** with the usual arithmetic comparison operators, the result is **NULL** (*undefined*).
- Instead of using =, <>, or != to test for equality or inequality with **NULL** *values, use* ***IS NULL*** *or* ***IS NOT NULL***
- MySQL specific **<=>** comparison operator is used for NULL-to-NULL comparisons.

```
SELECT * FROM EMP WHERE COMM <=> NULL
```

Note:

The result of (FALSE AND UNKNOWN) is FALSE, whereas the result of (FALSE OR UNKNOWN) is UNKNOWN.

control flow functions

control flow functions - ifnull

IFNULL function

MySQL IFNULL() takes two expressions, if the first expression is not NULL, it returns the first expression. Otherwise, it returns the second expression, **it returns either numeric or string value.**

IFNULL(*expression1*, *expression2*)

- SELECT IFNULL (1, 2) AS *R1*;
- SELECT IFNULL (NULL, 2) AS *R1*;
- SELECT IFNULL (1/0, 2) AS *R1*;
- SELECT IFNULL (1/0, 'Yes') AS *R1*;
- SELECT comm, IFNULL(comm + comm*.25, 1000) FROM emp;

control flow functions - if

IF function

If **expr1 is TRUE or expr1 <> 0 or expr1 <> NULL**, then IF() returns expr2, otherwise it returns expr3, **it returns either numeric or string value.**

IF(*condition, value_if_true, value_if_false*)

A year is a leap year if:
- Divisible by **400** → leap year
- Divisible by **4** but **not by 100** → leap year
- Otherwise → **not a leap year**

- SELECT IF(1 > 2, 2, 3) as *R1*;
- SELECT sal, IF(sal = 3000, 'Ok', 'Not Bad') *R1* FROM emp;
- SELECT ename, sal, IF(sal = 3000 AND ename = 'FORD', 'Y', 'N') *R1* FROM emp;
- SELECT ename, sal, comm, IF(comm IS NULL && ename = 'FORD', 'Y', 'N') *R1* FROM emp;
- SELECT deptno, IF(deptno = 10, 'Sales', IF(deptno = 20 , 'Purchase' ,'N/A')) *R1* FROM emp;
- SELECT productid, productname, unitprice, unitsinstock, reorderlevel, IF(unitsinstock < reorderlevel, 'Stock is less', 'Good Stock') as 'Stock Report' FROM products;
- SELECT hiredate, IF((YEAR(hiredate) % 4 = 0 AND YEAR(hiredate) % 100 <> 0) OR YEAR(hiredate) % 400 = 0 ,'Leap Year', 'Not A Leap Year') FROM emp;

control flow functions - nullif

NULLIF function

Returns **NULL** if expr1 = expr2 is true, otherwise returns expr1.

NULLIF(*expr1*, *expr2*)

- SELECT NULLIF(1, 1) as *R1*;
- SELECT NULLIF(1, 2) as *R1*;

- shipped and actualShipDate must be same.

control flow functions - nullif

todo

NULLIF(*expr1*, *expr2*)

Table:- ord

ordid	. . .	shipdate	actualShipDate	. . .
601	. . .	1986-05-30	1986-06-02	. . .
602	. . .	1986-06-20	NULL	. . .
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
634	. . .	1987-02-22	1987-02-24	. . .
635	. . .	1987-02-23	1987-02-23	. . .

Output

ordid	. . .	shipdate	actualShipDate	. . .
606	. . .	1986-07-30	1986-07-30	. . .
608	. . .	1986-07-25	1986-07-25	. . .
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
632	. . .	1987-03-14	1987-03-14	. . .
635	. . .	1987-02-23	1987-02-23	. . .

- SELECT * FROM ord WHERE NULLIF(shipdate, actualshipdate) IS NULL;
- SELECT * FROM ord WHERE NULLIF(shipdate, actualshipdate) IS NOT NULL;

- SELECT * FROM ord WHERE shipdate = actualshipdate;

- email and userName must be same.

control flow functions - nullif

todo

NULLIF(*expr1*, *expr2*)

Table:- student

studentID	. . .	emailID	userName
1001	. . .	saleel.songs@gmail.com	saleel.songs@gmail.com
1002	. . .	sharmin@gmail.com	sharmin@gmail.com
1003	. . .	vrusha@hotmail.com	vrushali@hotmail.com
. .	. .	. .	. .
1013	. . .	NULL	NULL

- SELECT student.*, NULLIF(emailID, userName) FROM student WHERE NULLIF(emailID, userName) IS NULL;
- SELECT student.*, NULLIF(emailID, userName) FROM student WHERE NULLIF(emailID, userName) IS NOT NULL;

Output

studentID	. . .	emailID	userName
1001	. . .	saleel.songs@gmail.com	saleel.songs@gmail.com
1002	. . .	sharmin@gmail.com	sharmin@gmail.com
. .	. .	. .	. .

see difference

- SELECT student.*, NULLIF(emailID, userName) FROM student WHERE emailID = userName;

control flow functions - case

CASE function

Returns the result for the first condition that is true. If there was no matching result value, the result after ELSE is returned, or NULL if there is no ELSE part.

```
CASE value WHEN [compare_value] THEN result [WHEN [compare_value] THEN result . . .] [ELSE result] END
```

- `SELECT deptno, CASE deptno WHEN 10 THEN 'Accounts' WHEN 20 THEN 'Sales' ELSE 'N/A' END R1 FROM emp;`
- `SELECT deptno, CASE deptno WHEN 10 THEN 'Accounts' ELSE 'N/A' END CASE FROM emp; # error`
- `SELECT custId, type, amount, CASE type WHEN 'd' THEN amount WHEN 'c' THEN amount * -1 END amount FROM transactions;`

- `SELECT job, SUM(CASE job WHEN 'manager1' THEN 1 ELSE 0 END) R1 FROM emp; # returns 0`
- `SELECT job, SUM(CASE job WHEN 'manager1' THEN 1 END) R1 FROM emp; # returns NULL`

control flow functions - case

CASE function

Returns the result for the first condition that is true. If there was no matching result value, the result after ELSE is returned, or NULL if there is no ELSE part.

CASE WHEN [*condition*] THEN result [WHEN [*condition*] THEN result . . .] [ELSE *result*] END

CASE WHEN EXISTS (SELECT true FROM R WHERE $R.a_1 = S.a_1$) THEN result [WHEN [*condition*] THEN result . . .] [ELSE *result*] END

- SELECT deptno, CASE WHEN deptno = 10 THEN 'Sales' WHEN deptno = 20 THEN 'Purchase' ELSE 'N/A' END *R1* FROM emp;
- SELECT companyName, CASE WHEN country IN ('USA', 'Canada') THEN 'North America' WHEN country = 'Brazil' THEN 'South America' WHEN country IN ('Japan', 'Singapore') THEN 'Asia' WHEN country = 'Australia' THEN 'Australia' ELSE 'Europe' END as *Continent* FROM suppliers ORDER BY companyName;
- SELECT hiredate, CASE WHEN (YEAR(hiredate) % 4 = 0 AND YEAR(hiredate) % 100 <> 0) OR YEAR(hiredate) % 400 = 0 THEN 'LEAP YEAR' END *R1* FROM emp;

control flow functions - case

CASE function

Returns the result for the first condition that is true. If there was no matching result value, the result after ELSE is returned, or NULL if there is no ELSE part.

CASE WHEN [*condition*] THEN result [WHEN [*condition*] THEN result . . .] [ELSE *result*] END

CASE WHEN EXISTS (SELECT true FROM R WHERE $R.a_1 = S.a_1$) THEN result [WHEN [*condition*] THEN result . . .] [ELSE *result*] END

```
* Count(custID)
  ORDER BY CASE orderCount
  WHEN 1 THEN 'One-time Customer'
  WHEN 2 THEN 'Repeated Customer'
  WHEN 3 THEN 'Frequent Customer'
  ELSE 'Loyal Customer' END customerType
```

```
* ORDER BY CASE
  WHEN filter = 'Debit' THEN 1
  WHEN filter = 'Credit' THEN 2
  WHEN filter = 'Total' THEN 3
  END transactionType;
```

```
* ORDER BY FIELD(status, 'In Process',
  'On Hold', 'Cancelled', 'Resolved',
  'Disputed', 'Shipped');
```

```
* ORDER BY CASE status
  WHEN 'active' THEN 1
  WHEN 'approved' THEN 2
  WHEN 'rejected' THEN 3
  WHEN 'submitted' THEN 4
  ELSE 5 END statusType
```

control flow functions - case

CASE function

Returns the result for the first condition that is true. If there was no matching result value, the result after ELSE is returned, or NULL if there is no ELSE part.

CASE WHEN [*condition*] THEN result [WHEN [*condition*] THEN result . . .] [ELSE *result*] END

CASE WHEN EXISTS (SELECT true FROM R WHERE $R.a_1 = S.a_1$) THEN result [WHEN [*condition*] THEN result . . .] [ELSE *result*] END

- ```
  SELECT cnum, COUNT(*),   CASE
      WHEN COUNT(*) = 1 THEN 'one-time-customer'
      WHEN COUNT(*) = 2 THEN 'repeated-customer'
      WHEN COUNT(*) = 3 THEN 'frequent-customer'
      WHEN COUNT(*) >= 4 THEN 'loyal-customer'
   END "Customer Report"
  FROM orders GROUP BY cnum ORDER BY 2;
  ```
  ```

- DATEDIFF(CURDATE(), hiredate) / 365.25

datetime functions

sysdate(), now(), curdate(), curtime()

In MySQL, the **NOW()** function returns a default value for a **DATETIME**.
MySQL inserts the current **date and time** into the column whose default value is NOW().

In MySQL, the **CURDATE()** returns the current date in 'YYYY-MM-DD'. **CURRENT_DATE()** and **CURRENT_DATE** are the **synonym of CURDATE().**

In MySQL, the **CURTIME()** returns the value of current time in 'HH:MM:SS'. **CURRENT_TIME()** and **CURRENT_TIME** are the **synonym of CURTIME().**

sysdate(), now(), curdate(), curtime()

NOW() returns a constant time that indicates the time at which the statement began to execute. (Within a stored function or trigger, NOW() returns the time at which the function or triggering statement began to execute.) This differs from the behavior for SYSDATE(), which returns the exact time at which it executes.

- SELECT SYSDATE()
- SELECT NOW()
- SELECT CURDATE()
- SELECT CURTIME()

Result in something like this:

SYSDATE()	**NOW()**	**CURDATE()**	**CURTIME()**
2017-02-11 10:22:31	2017-02-11 10:22:31	2017-02-11	10:22:31

```
mysql> SELECT NOW(), SLEEP(7), NOW();
mysql> SELECT SYSDATE(), SLEEP(7), SYSDATE();
```

date()

The DATE() function extracts the date value from a date or datetime expression.

DATE(*expr*)

- SELECT DATE("2017-06-15 09:34:21");
- SELECT NOW(), DATE(NOW());

Note:

- Returns NULL if expression is not a date or a datetime value.

- SELECT hiredate, DAYNAME(hiredate), if(DAYNAME(hiredate)='Saturday', hiredate + INTERVAL 2 DAY, IF(DAYNAME(hiredate)='Sunday', hiredate + INTERVAL 1 DAY, IF(DAYNAME(hiredate)='Monday', hiredate + INTERVAL 7 DAY, IF(DAYNAME(hiredate)='Tuesday', hiredate + INTERVAL 6 DAY, IF(DAYNAME(hiredate)='Wednesday', hiredate + INTERVAL 5 DAY, IF(DAYNAME(hiredate)='Thursday', hiredate + INTERVAL 4 DAY, IF(DAYNAME(hiredate)='Friday', hiredate + INTERVAL 3 DAY, 'TODO')))))) R1 FROM emp;

+ or - operator

Date arithmetic also can be performed using INTERVAL together with the + or - operator

date + INTERVAL *expr* unit + INTERVAL *expr* unit + INTERVAL *expr* unit + . . .
date - INTERVAL *expr* unit - INTERVAL *expr* unit - INTERVAL *expr* unit - . . .

- SELECT NOW(), NOW() + INTERVAL 1 DAY;
- SELECT NOW(), NOW() + INTERVAL '1-3' YEAR_MONTH;

unit Value	expr	unit Value	expr
SECOND	SECONDS	DAY_HOUR	'DAYS HOURS' e.g. '1 1'
MINUTE	MINUTES	DAY_MINUTE	'DAYS HOURS:MINUTES' e.g. '1 3:34'
HOUR	HOURS	DAY_SECOND	'DAYS HOURS:MINUTES:SECONDS'
DAY	DAYS	HOUR_MINUTE	'HOURS:MINUTES' e.g. '3:34'
WEEK	WEEKS	HOUR_SECOND	'HOURS:MINUTES:SECONDS'
MONTH	MONTHS	MINUTE_SECOND	'MINUTES:SECONDS' e.g. '27:34'
QUARTER	QUARTERS	YEAR_MONTH	'YEARS-MONTHS' e.g. '1-3'
YEAR	YEARS		

+ or - operator

TODO

date + INTERVAL *expr* unit + INTERVAL *expr* unit + INTERVAL *expr* unit + . . .
date - INTERVAL *expr* unit - INTERVAL *expr* unit - INTERVAL *expr* unit - . . .

1. Delivery + Return Window
2. Subscription with Grace Period
3. Loan Repayment
4. Event Planning
5. Software License Expiry
6. Hospital Follow-up
7. Online Course Access
8. Job Offer Validity
9. Restaurant Reservation
10. Insurance Policy Expiry
11. ……..

+ or - operator

Date arithmetic also can be performed using INTERVAL together with the + or - operator

date + INTERVAL *expr* unit + INTERVAL *expr* unit + INTERVAL *expr* unit + . . .
date - INTERVAL *expr* unit - INTERVAL *expr* unit - INTERVAL *expr* unit - . . .

- SELECT NOW(), NOW() + INTERVAL 1 DAY;
- SELECT hiredate, DAYNAME(hiredate), IF(DAYNAME(hiredate) = 'Saturday', hiredate + INTERVAL 2 DAY, IF(DAYNAME(hiredate) = 'Sunday', hiredate + INTERVAL 1 DAY, IF(DAYNAME(hiredate) = 'Monday', hiredate + INTERVAL 7 DAY, IF(DAYNAME(hiredate) = 'Tuesday', hiredate + INTERVAL 6 DAY, IF(DAYNAME(hiredate) = 'Wednesday', hiredate + INTERVAL 5 DAY, IF(DAYNAME(hiredate) = 'Thursday', hiredate + INTERVAL 4 DAY, IF(DAYNAME(hiredate) = 'Friday', hiredate + INTERVAL 3 DAY, 'TODO'))))))) R1 FROM emp;

adddate()

ADDDATE() is a synonym for **DATE_ADD()**

ADDDATE(date, INTERVAL *expr* unit) / DATE_ADD (date, INTERVAL *expr* unit)

- SELECT NOW(), ADDDATE(NOW(), INTERVAL 1 DAY);
- SELECT NOW(), ADDDATE(NOW(), 1);

unit Value	ExpectedexprFormat
SECOND	SECONDS
MINUTE	MINUTES
HOUR	HOURS
DAY	DAYS
WEEK	WEEKS
MONTH	MONTHS
QUARTER	QUARTERS
YEAR	YEARS

SUBDATE() is a synonym for **DATE_SUB()**

SUBDATE(date, INTERVAL *expr* unit) / DATE_SUB (date, INTERVAL *expr* unit)

- SELECT NOW(), SUBDATE(NOW(), INTERVAL 1 DAY);
- SELECT NOW(), SUBDATE(NOW(), 1);

unit Value	ExpectedexprFormat
SECOND	SECONDS
MINUTE	MINUTES
HOUR	HOURS
DAY	DAYS
WEEK	WEEKS
MONTH	MONTHS
QUARTER	QUARTERS
YEAR	YEARS

addtime()

ADDTIME() adds expr2 to expr1 and returns the result. expr1 is a time or datetime expression, and expr2 is a time expression.

```
ADDTIME(expr1, expr2)
```

- `SELECT NOW(), ADDTIME(NOW(), '1:1:1')  # 'HH:MM:SS'`
- `SELECT NOW(), ADDTIME(NOW(), '2 2:10:5')  # 'DAY HH:MM:SS'`

- Adding 15 seconds.
  ```
  SELECT NOW(), ADDTIME(NOW(), "15") AS Updated_time;
  ```
- Adding 10 minutes.
  ```
  SELECT NOW(), ADDTIME(NOW(), "00:10:00") AS Updated_time;
  ```
- Adding 2 hours.
  ```
  SELECT NOW(), ADDTIME(NOW(), "2:00:00") AS Updated_time;
  ```
- Adding 10 hours 30 minute 25 second and 100000 Microseconds.
  ```
  SELECT NOW(), ADDTIME(NOW(), "10:30:25.100000") AS Updated_time;
  ```
- Adding 1 day 2 hours 30 minute 25 second.
  ```
  SELECT NOW(), ADDTIME(NOW(), "1 2:30:25") AS Updated_time;
  ```

subtime()

SUBTIME() returns expr1 − expr2 expressed as a value in the same format as expr1. expr1 is a time or datetime expression, and expr2 is a time expression.

SUBTIME(*expr1*, *expr2*)

- `SELECT NOW(), SUBTIME(NOW(), '1:1:1');` `# 'HH:MM:SS'`
- `SELECT NOW(), SUBTIME(NOW(), '2 1:1:1');` `# 'DAY HH:MM:SS'`

- Subtract 15 seconds.
  `SELECT NOW(), SUBTIME(NOW(), "15") AS Updated_time;`
- Subtract 10 minutes.
  `SELECT NOW(), SUBTIME(NOW(), "00:10:00") AS Updated_time;`
- Subtract 2 hours.
  `SELECT NOW(), SUBTIME(NOW(), "2:00:00") AS Updated_time;`
- Subtract 10 hours 30 minute 25 second and 100000 Microseconds.
  `SELECT NOW(), SUBTIME(NOW(), "10:30:25.100000") AS Updated_time;`
- Subtract 1 day 2 hours 30 minute 25 second.
  `SELECT NOW(), SUBTIME(NOW(), "1 2:30:25") AS Updated_time;`

extract

The EXTRACT() function is used to return a single part of a date/time, such as year, month, day, hour, minute, etc.

EXTRACT(*unit* FROM *date*)

Unit Value				
MICROSECOND	SECOND	MINUTE	HOUR	DAY
WEEK	MONTH	QUARTER	YEAR	
MINUTE_SECOND	HOUR_SECOND	DAY_SECOND	DAY_HOUR	
HOUR_MINUTE	DAY_MINUTE	YEAR_MONTH		

- SELECT EXTRACT(MONTH FROM NOW());
- SELECT EXTRACT(YEAR_MONTH FROM NOW()) ;

Note:

- There must no space between extract function and ().

e.g.

SELECT EXTRACT (MONTH FROM NOW()); # error

period_diff()

MySQL PERIOD_DIFF() returns the difference between two periods. Periods should be in the same format i.e. YYYYMM or YYMM.

```
PERIOD_DIFF(Period1, Period2);
```

- `SELECT PERIOD_DIFF(201701, 201601);`
- `SELECT PERIOD_DIFF(EXTRACT (YEAR_MONTH FROM NOW()), EXTRACT(YEAR_MONTH FROM hiredate)) AS R1 FROM emp;`

datetime functions

Syntax	Result
DAY(*date*)	DAY() is a **synonym for DAYOFMONTH().**
DAYNAME(*date*)	Returns the name of the weekday for date.
DAYOFMONTH(*date*)	Returns the day of the month for date, in the range 1 to 31
DAYOFWEEK(*date*)	Returns the weekday index for date (1 = Sunday, 2 = Monday, …, 7 = Saturday).
DAYOFYEAR(*date*)	Returns the day of the year for date, in the range 1 to 366
LAST_DAY(*date*)	Takes a date or datetime value and returns the corresponding value for the last day of the month. Returns NULL if the argument is invalid.
MONTH(*date*)	Returns the month for date, in the range 1 to 12 for January to December
MONTHNAME(*date*)	Returns the full name of the month for date.
YEAR(*date*)	Returns the year in 4 digit

- SELECT DAYOFWEEK(NOW()), WEEKDAY(NOW());
- SELECT DAYOFWEEK(ADDDATE(NOW(), INTERVAL 1 DAY)), WEEKDAY(ADDDATE(NOW(), INTERVAL 1 DAY));

datetime functions

Syntax	Result
WEEKDAY(*date*)	Returns the weekday index for date (0 = Monday, 1 = Tuesday, … 6 = Sunday).
WEEKOFYEAR(*date*)	Returns the calendar week of the date as a number in the range from 1 to 53.
QUARTER(*date*)	Returns the quarter of the year for date, in the range 1 to 4.
HOUR(*time*)	Returns the hour for time. The range of the return value is 0 to 23 for time-of-day values.
MINUTE(*time*)	Returns the minute for time, in the range 0 to 59.
SECOND(*time*)	Returns the second for time, in the range 0 to 59.
DATEDIFF(*expr1, expr2*)	Returns the number of days between two dates or datetimes.
STR_TO_DATE(*str, format*)	Convert a string to a date.

- `SELECT NOW(), NOW() + INTERVAL 1 DAY, WEEKDAY(NOW() + INTERVAL 1 DAY);`
- `SELECT * FROM emp WHERE DAY(hiredate) = 17;`
- `SELECT YEAR(hiredate), ( YEAR(hiredate) % 4 = 0 AND YEAR(hiredate) % 100 != 0 ) OR YEAR(hiredate) % 400 = 0 R1 FROM emp ;`
- `SELECT STR_TO_DATE('24/05/2022', '%d/%m/%Y');`

datetime formats

datetime formats

Formats	Description
%a	Abbreviated weekday name (Sun-Sat)
%b	Abbreviated month name (Jan-Dec)
%c	Month, numeric (1-12)
%D	Day of month with English suffix (0th, 1st, 2nd, 3rd, □)
%d	Day of month, numeric (01-31)
%e	Day of month, numeric (1-31)
%f	Microseconds (000000-999999)
%H	Hour (00-23)
%h	Hour (01-12)

- SELECT DATE_FORMAT(NOW(), '%a');
- SELECT DATE_FORMAT(CURDATE(), '%Y-%m-01') AS *"First Day"*, LAST_DAY(CURDATE()) AS *"Last Day"*;
- SELECT hiredate, DATE_FORMAT(hiredate, '%Y-%m-01') AS *"First Day"*, LAST_DAY(hiredate) AS *"Last Day"* FROM emp;

datetime formats

Formats	Description
%I	Hour (01-12)
%i	Minutes, numeric (00-59)
%j	Day of year (001-366)
%k	Hour (0-23)
%l	Hour (1-12)
%M	Month name (January-December)
%m	Month, numeric (01-12)
%p	AM or PM
%r	Time, 12-hour (hh:mm:ss followed by AM or PM)
%S	Seconds (00-59)
%s	Seconds (00-59)

- `SELECT DATE_FORMAT(NOW(), '%j');`

datetime formats

Formats	Description
%T	Time, 24-hour (hh:mm:ss)
%U	Week (00-53) where Sunday is the first day of week
%u	Week (00-53) where Monday is the first day of week
%V	Week (01-53) where Sunday is the first day of week, used with %X
%v	Week (01-53) where Monday is the first day of week, used with %x
%W	Weekday name (Sunday-Saturday)
%w	Day of the week (0=Sunday, 6=Saturday)
%X	Year for the week where Sunday is the first day of week, four digits, used with %V
%x	Year for the week where Monday is the first day of week, four digits, used with %v
%Y	Year, numeric, four digits
%y	Year, numeric, two digits

- `SELECT DATE_FORMAT(NOW(), '%Y');`

- Print the **first** and the **last** day of every month.

datetime formats

Input

curdate()
2025-06-30

Output

First Date	Last Date
2025-06-01	2025-06-30

- SELECT DATE_FORMAT(CURDATE(), '%Y-%m-01') AS *"First Day"*, LAST_DAY(CURDATE()) AS *"Last Day"*;
- SELECT hiredate, DATE_FORMAT(hiredate, '%Y-%m-01') AS *"First Day"*, LAST_DAY(hiredate) AS *"Last Day"* FROM emp;

- ```
  CREATE TABLE hobbies(
    hobbyid ENUM('1','2','3','4'),
    hobbyName VARCHAR(20) GENERATED ALWAYS AS (ELT(hobbyid, 'Windsurfing', 'Fishing', 'Swimming', 'Shopping'))
  );
  ```

string functions
  ```

string functions

Syntax	Result
ASCII(*str*)	Returns the numeric value of the leftmost character of the string str. Returns 0 if str is the empty string. Returns NULL if str is NULL. e.g. • SELECT ASCII(ename) FROM emp;
CHAR(*N*, , . . .)	CHAR() interprets each argument N as an integer and returns a string consisting of the characters given by the code values of those integers. **NULL values are skipped**. e.g. • SELECT CHAR(65, 66, 67); / SELECT CAST(CHAR(65 66, 67) AS CHAR);
CONCAT(*str1, str2*, . . .)	Returns the string that results from concatenating the arguments. CONCAT() **returns NULL if any argument is NULL**. e.g. • SELECT CONCAT('Mr. ' , ename) FROM emp; • SELECT CONCAT('My', NULL, 'SQL'); #output will be NULL
ELT(*N, str1, str2, str3,* . . .)	ELT() returns the Nth element of the list of strings: str1 if N = 1, str2 if N = 2, and so on. Returns NULL if N is less than 1 or greater than the number of arguments. • SELECT hiredate, ELT(MONTH(hiredate),'Winter', 'Winter', 'Spring', 'Spring', 'Spring', 'Summer', 'Summer', 'Summer', 'Autumn', 'Autumn', 'Autumn', 'Winter') R1 FROM emp;

- SELECT Name, rnk, ELT(rnk, 'A+', 'A', 'B', 'C', 'D') *grade* FROM student;
- SELECT bankName, accountNumber, riskStatus, CASE WHEN riskStatus = 0 THEN 'No risk' ELSE ELT(riskStatus, 'Low', 'Moderate', 'High', 'Critical') END *riskStatus* FROM bankLoan;

string functions

Syntax	Result
STRCMP(*expr1, expr2*)	STRCMP() returns 0 if the strings are the same, -1 if the first argument is smaller than the second according to the current sort order, and 1 otherwise.
LCASE(*str*)	Returns lower case string. LCASE() **is a synonym for** LOWER().
UCASE(*str*)	Returns upper case string. UCASE() **is a synonym for** UPPER().
LENGTH(*str*)	Returns the length of the string.
LPAD(*str, len, padstr*)	Returns the string str, left-padded with the string padstr to a length of len characters.
RPAD(*str, len, padstr*)	Returns the string str, right-padded with the string padstr to a length of len characters.
REPEAT(*str, count*)	Returns a string consisting of the string str repeated count times. If count is less than 1, returns an empty string. Returns NULL if str or count are NULL.

- SELECT UCASE(ename) FROM emp;
- SELECT LPAD(id, 7, '#'), bookName, Type, Cost FROM books;
- SELECT LPAD(prodid, 9, 'x'), Descrip, Price FROM product;
- SELECT CONCAT('TKT', LPAD(id, 7, 0)) *'Ticket ID'* FROM hall_ticket;
- SELECT COUNT(*) *R1* FROM emp WHERE BINARY ename = BINARY UPPER(ename);
- SELECT CASE WHEN status = 'In Process' THEN LPAD(ordid, LENGTH(ordid) + 3, '(I)') WHEN status = 'Delivered' THEN LPAD(ordid, LENGTH(ordid) + 3, '(D)') ELSE 'TODO' END ordid, orderDate, custID, Status, Total FROM ord;

string functions

Syntax	Result
LEFT(*str, len*)	Returns the leftmost len characters from the string str, or NULL if any argument is NULL.
RIGHT(*str, len*)	Returns the rightmost len characters from the string str, or NULL if any argument is NULL.
LTRIM(*str*)	Returns the string str with leading space characters removed.
RTRIM(*str*)	Returns the string str with trailing space characters removed.
TRIM(*str*)	Returns the string str with leading and trailing space characters removed.
BINARY *value*	Convert a value to a binary string.
INSTR(*str, substr*)	• *str,* The string that will be searched • *substr,* The string to search for in string1. If string2 is not found, this function returns 0
FIND_IN_SET(*str*, *string_list*)	• *str,* The string to search for • *string_list,* The list of string values to be searched (separated by commas)

FIND_IN_SET:- This function of MySQL is used to find the position of a string within a comma-separated list of strings.

- If *string* is not found in *string_list*, this function returns 0
- If *string* or *string_list* is NULL, this function returns NULL
- If *string_list* is an empty string (""), this function returns 0

- SELECT ename, BINARY ename FROM emp;
- SELECT CONCAT(UCASE(LEFT(ename, 1)), LCASE(SUBSTRING(ename, 2))) *"Title Case"* FROM emp;

string functions

Syntax	Result
CONCAT_WS(*separator*, *string1*, *string2*, *string3*, . . .)	• separator → Character/string placed between each value. • string1, string2, ... → Values that you want to concatenate. • NULL values are skipped by CONCAT_WS().
REGEXP_REPLACE(*expr*, *pattern*, *replacement*)	

```
SELECT patientId,   CONCAT_WS(' ', firstName, middleName, lastName) AS 'Full Name' FROM patient;
```

- Find the position of the given name.

string functions - examples

FIND_IN_SET(*str*, *string_list*)

Table:- teams

TeamID	Members
1	Saleel, Sharmin, Vrushali, Gau
2	Kaushal, Natsha, Ruhan
3	Lalu, Brooke, Boy, Naru, Bhavin
4	Bhuru, Baba, Anoop, Meera, Boy
5	Dilu, Fredy
6	Daigo, Harshil, Boy, Raju, Bothyo
7	Bandish, Supriya, Sangita
8	Monika, Karishma, William, Varsha, Jack, Grace
9	Vasant, Neel
10	Nitish, Bipin, Mai, Didi

Output

TeamID	Members	Position
1	Saleel, Sharmin, Vrushali, Gau	0
2	Kaushal, Natsha, Ruhan	0
3	Lalu, Brooke, Boy, Naru, Bhavin	3
4	Bhuru, Baba, Anoop, Meera, Boy	5
5	Dilu, Fredy	0
6	Daigo, Harshil, Boy, Raju, Bothyo	3
7	Bandish, Supriya, Sangita	0
8	Monika, Karishma, William, Varsha, Jack, Grace	0
9	Vasant, Neel	0
10	Nitish, Bipin, Mai, Didi	0

Output →

- SELECT members, FIND_IN_SET('Boy', REPLACE(members, ' ', '')) *Position* FROM teams;

SUBSTRING_INDEX(*str*, *delimiter*, *count*) **// count can be either +ve or -ve**

string functions

Syntax	Result
REPLACE(*str, from_str, to_str*)	Returns the string str with all occurrences of the string from_str replaced by the string to_str. REPLACE() performs a case-sensitive match when searching for from_str. e.g. • SELECT REPLACE('Hello', 'l', 'x');
REVERSE(*str*)	Returns the string str with the order of the characters reversed.
SUBSTR(*str, pos, len*)	**SUBSTR() is a synonym for SUBSTRING().** e.g. • SELECT SUBSTR ('This is the test by IWAY', 6); • SELECT SUBSTR ('This is the test by IWAY', -4, 4);
MID(*str, pos, len*)	MID function **is a synonym for** SUBSTRING**.**

- SELECT ename, job, IF(ISNULL(phone), '**********', RPAD(LEFT(phone, 4), 10, '*')) FROM emp;
- SELECT ename, job, phone, IF(ISNULL(phone), REPEAT('*',10), RPAD(LEFT(phone, 4), 10, '*')) FROM emp;
- SELECT `user name`, IF(LENGTH(SUBSTR(`user name`, INSTR(`user name`," "))) = 0,"Weak User", `user name`) *R1* FROM emp;
- UPDATE emp SET job = REPLACE(job, job, LOWER(job));

string functions - examples

- SELECT sal, REPEAT('$', sal/100) FROM emp;
- SELECT emailid, REPEAT('*', LENGTH(emailid)) FROM emp;
- SELECT pwd, REPEAT('*', LENGTH(pwd)) *password* FROM emp;
- SELECT c1, CONCAT(REPEAT('0', 10 - LENGTH(c1)) , c1) FROM leading_zeroes;
- SELECT ename, job, IF(ISNULL(phone), '**********', RPAD(LEFT(phone, 4), 10, '*')) FROM emp;
- SELECT 9850884228, LPAD(SUBSTR(9850884228, -4),10,'*') *R1*;
- SELECT `user name`, IF(LENGTH(SUBSTR(`user name`, INSTR(`user name`," "))) = 0,"Weak User", `user name`) *R1* FROM emp;
- SELECT LENGTH('saleel') - LENGTH(REPLACE('saleel', 'e', ""));
- SELECT empno, datePresent, LENGTH(datePresent) - LENGTH(REPLACE(datePresent, ",", "")) + 1 *"Days Present"* FROM emp_attendance;
- SELECT c1, c1 / 1, SUBSTR(c1, LENGTH(c1 / 1) + 1) FROM numberString;
- SELECT c1, REVERSE(c1) / 1, LENGTH(REVERSE(c1) / 1), REVERSE(SUBSTR(REVERSE(c1), LENGTH(REVERSE(c1) / 1) + 1)) FROM Stringnumber;
- UPDATE emp SET job := REPLACE(job, 'officers', 'Officers');

- Count total ***y*** votes.

string functions - examples

Table:- vote_response

CandidateID	response
1	n,n,n,y,y,y,y,n,y,n,y,y,n,n,y,n,n,n,n,n,n,n,n,y,y,y,y,y,y,y,n,y,n,y,y,y,n,n,n,y,y,y,y,n,n,n,n,n,y
2	y,n,n,y,y,y,y,n,y,n,y,y,n,n,y,n,n,y,n,y,n,y,n,y,y,y,y,y,y,y,n,y,n,y,y,y,n,n,n,y,y,y,y,y,y,y,n,y,n,y,n,y,n,n,n,y,y,y,n
3	y,y,y,y,y,y,y,n,y,y,y,y,n,y,y,y,y,y,n,y,y,y,y,y,y,y,y,y,y,y,y,y,y,n,y,y,y,y,y,y,y,n,y,y,y,y,y,n,y,y,y,y,y,y,y,n,y,n
4	n,n,n,y,y,n,n,n,y,n,y,y,n,n,y,n,y,n,n,y,n,y,n,y,n,y,n,y,n,n,n,y,y,n,n

Output

CandidateID	Yes Votes	Total Votes
1	yyyyyyyyyyyyyyyyyyyyyyy	24
2	yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy	36
3	yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy	40
4	yyyyyyyyyyyyyyyyyy	19

Output →

- SELECT candidateID, REPLACE(REPLACE(response, ',', ''), 'n', '') *'Yes Votes'*, LENGTH(REPLACE(REPLACE(response, ',', '') ,'n', '')) *'Total Votes'* FROM vote_response;

string functions - examples

- `SELECT * FROM emp1 WHERE ename = BINARY "sherlock";`
- `SELECT * FROM emp1 WHERE ename = BINARY "Sherlock";`
- `SELECT * FROM emp1 WHERE ename = BINARY UPPER(ename);`
- `SELECT * FROM emp1 WHERE ename = BINARY LOWER(ename);`
- `SELECT CONCAT(UCASE(LEFT(ename, 1)), LCASE(SUBSTRING(ename, 2))) "Title Case" FROM emp;`
- `SELECT * FROM emp1 WHERE ename = BINARY CONCAT(UCASE(LEFT(ename, 1)), LCASE(SUBSTRING(ename, 2)));`

-
```
SELECT   SUBSTRING_INDEX('State Bank of India', ' ', 1) AS part1,                    -- 'State'
SUBSTRING_INDEX(SUBSTRING_INDEX('State Bank of India', ' ', 2), ' ', -1) AS part2,   -- 'Bank'
SUBSTRING_INDEX(SUBSTRING_INDEX('State Bank of India', ' ', 3), ' ', -1) AS part3,   -- 'of'
SUBSTRING_INDEX('State Bank of India', ' ', -1) AS part4;                            -- 'India'
```

- Count elements in **enum** data-type

string functions - examples

Table:- teams

TeamID	Members
1	Saleel, Sharmin, Vrushali, Gau
2	Kaushal, Natsha, Ruhan
3	Lalu, Brooke, Boy, Naru, Bhavin
4	Bhuru, Baba, Anoop, Meera, Boy
5	Dilu, Fredy
6	Daigo, Harshil, Boy, Raju, Bothyo
7	Bandish, Supriya, Sangita
8	Monika, Karishma, William, Varsha, Jack, Grace
9	Vasant, Neel
10	Nitish, Bipin, Mai, Didi

Output →

Output

TeamID	Members	**Total Members**
1	Saleel, Sharmin, Vrushali, Gau	4
2	Kaushal, Natsha, Ruhan	3
3	Lalu, Brooke, Boy, Naru, Bhavin	5
4	Bhuru, Baba, Anoop, Meera, Boy	5
5	Dilu, Fredy	2
6	Daigo, Harshil, Boy, Raju, Bothyo	5
7	Bandish, Supriya, Sangita	3
8	Monika, Karishma, William, Varsha, Jack, Grace	6
9	Vasant, Neel	2
10	Nitish, Bipin, Mai, Didi	4

- `SELECT TeamID, members, LENGTH(members) - LENGTH(REPLACE(members,",", "")) + 1 `Total Members` FROM teams;`

string functions

- SELECT * FROM emp WHERE LEFT(ename, 1) IN ('a', 'e', 'o', 'i', 'u');
- SELECT * FROM emp WHERE REGEXP_LIKE(ename, '(a|e|i|o|u)');
- SELECT * FROM emp WHERE REGEXP_LIKE(ename, '^(a|m)'); //starts with
- SELECT * FROM emp WHERE REGEXP_LIKE(ename, '(n|r)$');   //ends with

- SELECT "abc,,abc,,,,,bc,,,,,,abc" ; / SELECT REPLACE(REPLACE(REPLACE("abc,,abc,,,,,bc,,,,,,abc", ",", ".,"),",.",""),".,",",")
  R1 ;

mathematical functions

mathematical functions

Syntax	Result
ABS(x)	Returns the absolute value of X.
CEIL(x)	CEIL() is a synonym for CEILING().
CEILING(x)	Returns CEIL value.
FLOOR(x)	Returns FLOOR value.
MOD(n, m), n % m, n MOD m	Returns the remainder of N divided by M. MOD(N,0) returns NULL.
POWER(x, y)	This is a synonym for POW().
RAND()	Returns a random floating-point value
ROUND(x) ROUND(x, d)	Rounds the argument X to D decimal places. The rounding algorithm depends on the data type of X. D defaults to 0 if not specified. D can be negative to cause D digits left of the decimal point of the value X to become zero.
TRUNCATE(x, d)	Returns the number X, truncated to D decimal places. If D is 0, the result has no decimal point or fractional part. D can be negative to cause D digits left of the decimal point of the value X to become zero.

mathematical functions

e.g.

- SELECT CEIL(1.23);
- SELECT CEIL(-1.23);
- SELECT FLOOR(1.23);
- SELECT FLOOR(-1.23);
- SELECT ROUND(-1.23);
- SELECT ROUND(-1.58);
- SELECT FLOOR(1 + RAND() * 100); // between 1 to 100
- SELECT FLOOR(20 + RAND() * 56); // between 20 to 75
- SELECT FLOOR(899999 + RAND() * 100000) *OTP*; // 6 digit
- SELECT weight, TRUNCATE(weight, 0) AS *kg*, MID(weight, INSTR(weight, ".") + 1) AS *gms* FROM mass_table;
- SELECT weight, TRUNCATE(weight, 0) AS *kg*, RIGHT(MOD(weight , 1), 2) AS *gms* FROM mass_table;

random characters / number

FLOOR(*start_value* + RAND() * difference between (*start_value* – *end_value*) + 1);

- CAST(CHAR(FLOOR(65 + RAND() * 26)) AS CHAR) → Random uppercase letter (A–Z)
- CAST(CHAR(FLOOR(97 + RAND() * 26)) AS CHAR) → Random lowercase letter (a–z)
- FLOOR(0 + RAND() * 10)) → Random digit (0–9)

random numbers between

- 1 to 100 → FLOOR(1 + RAND() * 100);
- 1 to 14 → FLOOR(1 + RAND() * 14);
- 20 to 75 → FLOOR(20 + RAND() * 56);
- 6 digit OPT number → FLOOR(899999 + RAND() * 100000);

Table:- cNumbers

ID	num
1	1
2	1
3	1
4	2
5	1
6	2
7	2
8	7
9	7
10	7
11	5
12	5
13	2
14	4
15	4
16	4

Output →

OUTPUT

ID	R1	R2
1	1	1
2	2	1
3	3	1
4	4	1
5	5	1
6	1	2
7	2	2
8	3	2
9	4	2
10	5	2
11	1	3
12	2	3
13	3	3
14	4	3
15	5	3
16	1	4

TODO

TODO

- SELECT id, CASE WHEN id%5 = 0 THEN 5 ELSE id%5 END *R1*, CEIL(id/5) *R2* FROM cnumbers;

random characters / number

FLOOR(*start_value* + RAND() * difference between (*start_value* – *end_value*) + 1);

- CAST(CHAR(FLOOR(65 + RAND() * 26)) AS CHAR) → Random uppercase letter (A–Z)
- CAST(CHAR(FLOOR(97 + RAND() * 26)) AS CHAR) → Random lowercase letter (a–z)
- FLOOR(0 + RAND() * 10)) → Random digit (0–9)

- WITH *RECURSIVE* ***cte*** AS (SELECT 1 *n* UNION SELECT *n* + 1 FROM ***cte*** WHERE *n* <= 10) SELECT CAST(CHAR(FLOOR(65 + RAND() * 26)) AS CHAR) *R1* FROM ***cte***; // letter (A–Z)
- WITH *RECURSIVE* ***cte*** AS (SELECT 1 *n* UNION SELECT *n* + 1 FROM ***cte*** WHERE *n* <= 10) SELECT CAST(CHAR(FLOOR(97 + RAND() * 26)) AS CHAR) *R1* FROM ***cte***; // letter (a–z)
- WITH *RECURSIVE* ***cte*** AS (SELECT 1 *n* UNION SELECT n + 1 FROM ***cte*** WHERE *n* <= 10) SELECT FLOOR(0 + RAND() * 10) *R1* FROM ***cte***; // digit (0–9)

- SELECT ELT(FLOOR(1 + RAND() * 14), 'Pending', 'Shipped', 'Delivered', 'Cancelled', 'Processing', 'In Transit', 'Out for Delivery', 'Failed Attempt', 'Returned to Origin', 'Returned', 'Lost', 'Damaged', 'On Hold', 'Ready for Pickup') *Status*;

reset password – using NodeJS, Express

User requests a password reset every after 10 minutes

R = { userID INT PRIMARY KEY AUTO_INCREMENT, userName VARCHAR(20) NOT NULL, password VARCHAR(50) NOT NULL, passwordExpiry DATE NOT NULL }

- Generate 10-character password and updating user record

JSON_OBJECT

It creates a JSON object from a list of key–value pairs

JSON_OBJECT with select

JSON_OBJECT("key1", value1, "key2", value2 , "key3", value3, JSON_ARRAY("value1", "value2" , . . .) , . . .)

OR

'{ "key1" : value1, "key2" : value2 , "key3" : ["value1", "value2" , . . .] , . . . }'

- SELECT JSON_OBJECT('orderID', onum, 'orderDate', DATE_FORMAT(odate, '%Y-%m-%d %h:%m:%s'), 'customerID', cnum, 'Type', type, 'paymentType', paymentType, 'amount', ROUND(amt, 2)) *R1* FROM orders;
- WITH ***cte*** AS (SELECT JSON_OBJECT('orderID', onum, 'orderDate', DATE_FORMAT(odate, '%Y-%m-%d %h:%m:%s'), 'customerID', cnum, 'Type', type, 'paymentType', paymentType, 'amount', ROUND(amt, 2)) *R1* FROM orders) SELECT * FROM ***cte***;
- WITH ***cte*** AS (SELECT JSON_OBJECT('orderID', onum, 'orderDate', DATE_FORMAT(odate, '%Y-%m-%d %h:%m:%s'), 'customerID', cnum, 'Type', type, 'paymentType', paymentType, 'amount', ROUND(amt, 2)) *R1* FROM orders) SELECT JSON_ARRAY(*R1*) FROM ***cte***;

JSON_OBJECT with insert

TODO

INSERT [INTO] tbl_name (*json_column*) VALUES (JSON_OBJECT("key1", value1, "key2", value2 , "key3", value3, JSON_ARRAY("value1", "value2" , . . .)

- INSERT INTO product VALUES(default, JSON_OBJECT("product_id", 1001, "productName", "Jam", "stock", 120, "rol", 24, "rate", 73));
- INSERT INTO product VALUES(default, JSON_OBJECT("product_id", 1001, "productName", "shirt", "stock", 120, "rol", 24, "rate", 73, "size", JSON_ARRAY('s', 'm', 'l')));
- INSERT INTO product VALUES(default, JSON_OBJECT("product_id", 1002, "productName", "Bread", "stock", 25, "rol", 7, "rate", 34, "size", JSON_ARRAY('small', 'medium', 'large'), "type", JSON_ARRAY('white', 'brown')));

-> / ->> … (select)

MySQL supports a native JSON data type that enables efficient access to data in JSON (JavaScript Object Notation) documents.

- SELECT id, document->'$.ename', document->'$.sal' FROM empJ;
- SELECT id, document->'$.ename', document->'$.sal', document->'$.comm' FROM empJ;
- SELECT id, document->'$.ename', document->'$.sal', document->'$.comm', document->'$.sal' + document->'$.comm' FROM empJ;
- SELECT id, document->'$.address.coord[0]' FROM empJ;
- SELECT id, document->'$.phone[0]' FROM empJ;
- SELECT author_details->>'$. authorName', book_details->'$[1].*' FROM author_book;
- SELECT id, document->"$.phone", JSON_LENGTH(document, "$.phone") FROM empJ;

Note:

- TABLE statement always displays all columns of the table.
- TABLE statement does not support any WHERE clause.
- TABLE statement can be used with temporary tables.

table statement…

TABLE is a DML statement introduced in MySQL 8.0.19 which returns rows and columns of the named table.

- WITH *RECURSIVE* ***a*** AS (SELECT 1 AS *n* UNION SELECT *n* + 1 FROM ***a*** WHERE *n* < 10)
  SELECT *n* FROM ***a***;

table statement

The TABLE statement in some ways acts like SELECT. You can order and limit the number of rows produced by TABLE using ORDER BY and LIMIT clauses, respectively.

TABLE tbl_name [ORDER BY *col_name*] [LIMIT *number* [OFFSET *number*]]

- TABLE emp;
- TABLE emp ORDER BY 2;
- TABLE emp ORDER BY 2 LIMIT 1, 2;
- TABLE t1 UNION TABLE t2;

Remember:

- Here, "*" is known as metacharacter (all columns)
- You can print data in JSON format using built-in JSON functions like JSON_OBJECT(), JSON_ARRAYAGG(), and JSON_ARRAY().

e.g

- SELECT JSON_OBJECT('empno', empno, 'ename', ename, 'job', job, 'sal', sal) AS emp_json FROM emp;
- SELECT JSON_ARRAYAGG(JSON_OBJECT('empno', empno, 'ename', ename, 'job', job, 'sal', sal)) AS emp_data FROM emp;

select statement... syntax

SELECT is used to retrieve rows selected from one or more tables (using JOINS), and can include UNION statements and SUBQUERIES.



syntax

select statement

modifiers

SELECT [ALL / DISTINCT / DISTINCTROW] identifier.* / identifier.A_1 [[as] *alias_name*], identifier.A_2 [[as] *alias_name*], identifier.A_3 [[as] *alias_name*], expression1 [[as] *alias_name*], expression2 [[as] *alias_name*] . . .

- [FROM <identifier.r_1> [as] *alias_name*], <identifier.r_2> [as] *alias_name*], . . .]
- [WHERE < *where_condition1* > { and | or } < *where_condition2* > . . .]
- [GROUP BY < { *col_name* | *expr* | *position* }, . . . [WITH ROLLUP] >]
- [HAVING < *having_condition1* > { and | or } < *having_condition2* > . . .]
- [ORDER BY < { *col_name* | *expr* | *position* } [ASC | DESC], . . . >]
- [LIMIT < { [*offset*,] *row_count* | *row_count* OFFSET *offset* } >]
- [FOR { UPDATE }]
- [{ INTO OUTFILE '*file_name*' | INTO DUMPFILE '*file_name*' | INTO *var_name* [, *var_name*], . . . }]

found_rows() (returns the number of rows)

Remember:

- **ALL** (**modifier is default**) specifies that all matching rows should be returned, including duplicates.
- **DISTINCT** (**modifier**) specifies removal of duplicate rows from the result set.
- **DISTINCTROW** (**modifier**) is a synonym for **DISTINCT**.
- It is an error to specify both modifiers.
- Whenever you use **DISTINCT**, sorting takes place in server.

If <*condition1*> is TRUE, then MySQL does not check <*condition2*>.

If <*condition1*> is FALSE, then MySQL must check <*condition2*>.

*Why is SELECT * often undesirable?*

SELECT * can……

1. Read unnecessary data
2. Increase network traffic
3. Prevent use of a covering index
4. Increase in memory consumption
5. Make application dependent on table structure
6. Cause ambiguity in JOINs
7. Fetches unnecessary columns
8. Expose newly added scnsitive columns
9. Creates problem when the table structure is changed

*Why is SELECT * often undesirable?*

1. SELECT * unnecessarily

2. Functions on indexed columns in WHERE

e.g. WHERE YEAR(join_date) = 2026;

3. Leading wildcard with LIKE in WHERE

e.g. WHERE employeeName LIKE '%aleel';

4. Using OR unnecessarily

e.g. WHERE department_id = 1 OR

department_id = 2

5. NOT IN with nullable columns

e.g. WHERE id NOT IN (SELECT id FROM B);

*Why is SELECT * often undesirable?*

6. Missing WHERE in UPDATE/DELETE

7. Using HAVING instead of WHERE

e.g. HAVING department_id = 10;

8. Comparing with NULL using =

e.g. WHERE bonus = NULL;

9. Implicit data-type conversion

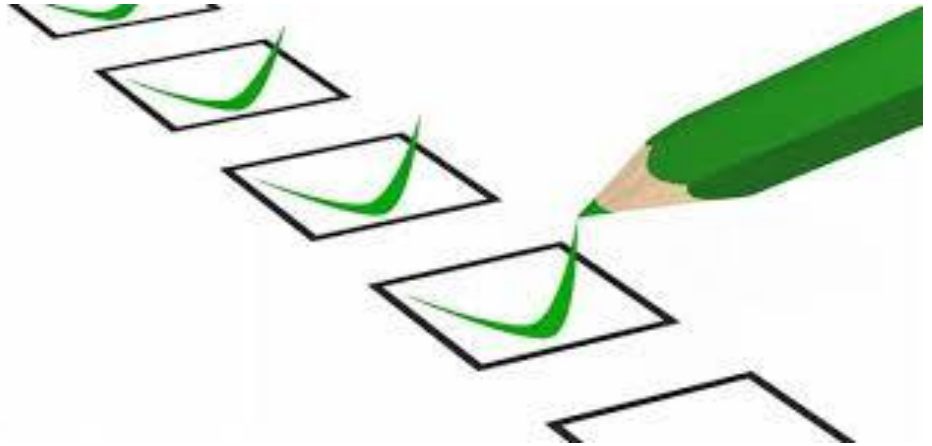
e.g. WHERE employee_id = '101'

10. No ORDER BY when order matters

e.g. FROM employee LIMIT 10;

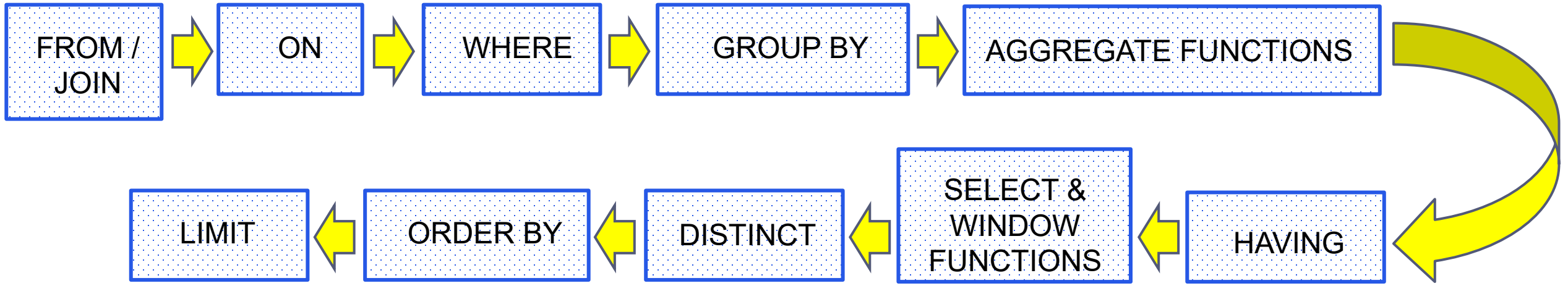
sequence of clauses





(The "Waterfall" of SQL Processing)

select statement... execution



select statement... execution

1. FROM & JOIN (Identify tables & Cartesian product)
2. ON (Filter rows during join conditions)
3. WHERE (Filter rows before grouping)
4. GROUP BY (Arrange rows into groups/buckets)
5. AGGREGATE FUNCTIONS (Compute SUM, AVG, COUNT, etc. for HAVING)
6. HAVING (Filter groups based on aggregates)
7. SELECT & WINDOW FUNCTIONS ◄– These are computed together (Compute expressions, aliases, and OVER() clauses)
8. DISTINCT (Remove duplicate rows from the resultset)
9. ORDER BY (Sort the final dataset)
10. LIMIT / OFFSET (Truncate to desired number of rows)

select statement... (is checks for)

Syntax Check

MySQL Database must check each SQL statement for syntactic validity.

```
mysql> SELECT *  FORM emp;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'FORM emp' at line 1
```

Semantic Check

A semantic check determines whether a statement is meaningful, for example, whether the objects and columns in the statement exist.

```
mysql> SELECT *  FROM nonexistent_table;

ERROR 1146 (42S02): Table 'db1.nonexistent_table' doesn't exist
```

UUID()/ UUID_SHORT()

A UUID is a Universal Unique Identifier and 128-bit long value.

Remember:

- UUID values in MySQL are **unique** across tables, databases, and servers..

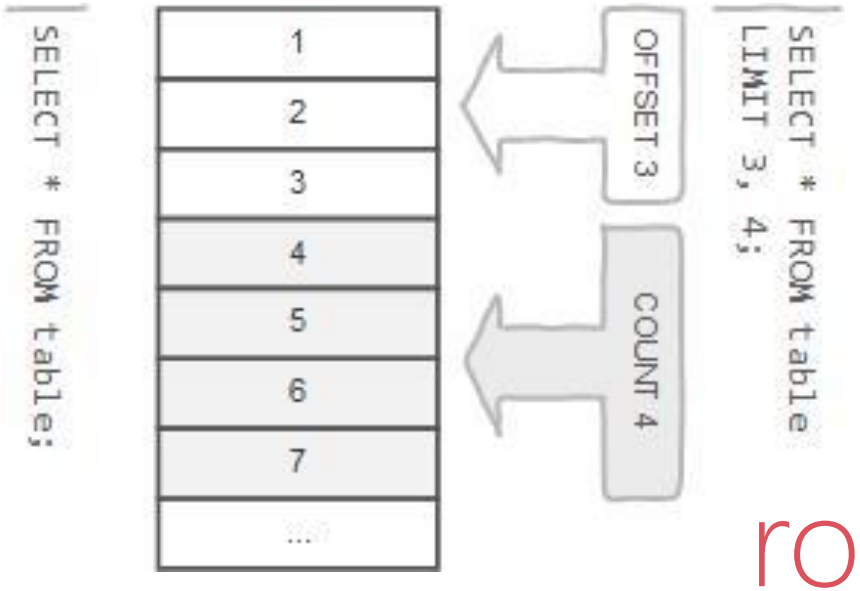
- SELECT UUID() AS R1, UUID_SHORT() AS R2 FROM *tbl_name*;

found_rows()

TODO

Note:

- **FOUND_ROWS()** only works for the immediately previous SELECT query.
- Session-Specific



row limiting clause

LIMIT is applied after HAVING

Remember:

- LIMIT enables you to pull a section of rows from the middle of a result set. Specify two values: The number of rows to skip at the beginning of the result set, and the number of rows to return.
- MySQL does not accept user-defined session variables directly in LIMIT cause inside a **stored procedure**.
  e.g. SELECT * FROM ccatStudents LIMIT @x, @y; // This statement is invalid.

Note:

- Limit value are not to be given within **(. . .)**
- Limit takes one or two numeric arguments, which must both be **non-negative** integer value.

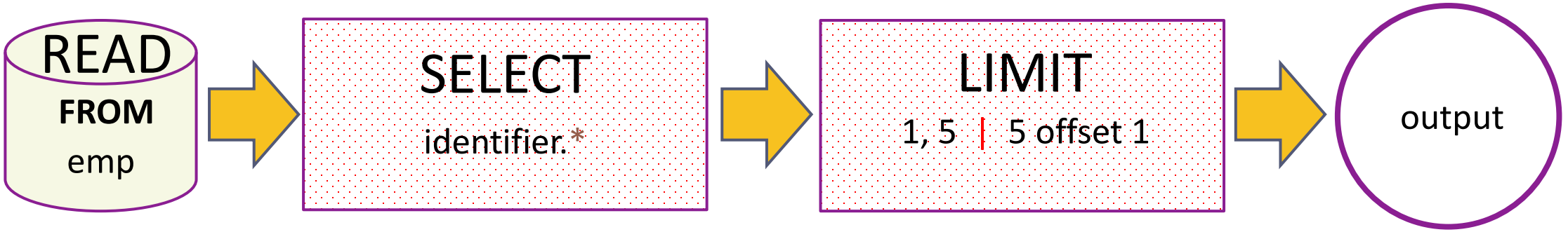
select - limit

SELECT $A_1, A_2, A_3, \ldots$ FROM *r*

[LIMIT { [*offset*,] *row_count* | *row_count* OFFSET *offset* }]

You can specify an offset using OFFSET from where SELECT will start returning records. By default ***offset is zero.***

- SELECT * FROM emp LIMIT 5 OFFSET 1;



- SELECT * FROM student LIMIT 5;
- SELECT * FROM student LIMIT 1, 5;
- SELECT * FROM student LIMIT 5 offset 1;
- SELECT RAND(), student.* FROM student ORDER BY 1 LIMIT 1;
- SELECT student.* FROM student ORDER BY RAND() LIMIT 1;

select - limit

This variable is used to return the maximum number of rows from SELECT statements. Its default value is unlimited. But if you changed the limit then SELECT statement returns the rows equals to the value.

SQL_SELECT_LIMIT = { *value* | *DEFAULT* }

This variable does not apply to SELECT statement that executed in the stored procedures or functions.

- SET *SQL_SELECT_LIMIT* = 2;
- SET *SQL_SELECT_LIMIT* = DEFAULT;

- SELECT * FROM emp;

anti-pattern in order by

Storing Multiple Values in One Column

Anti-pattern

- Here,
```
CREATE TABLE products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(50),
    tags VARCHAR(100)   -- 'Electronics,Home,Smart'
);
```

Why it's bad:

- Hard to query, sort, or filter.
- Violates first normal form (1NF).

Better approach

- Here,
```
CREATE TABLE tags (
    tag_id INT PRIMARY KEY,
    tag_name VARCHAR(50)
);

CREATE TABLE product_tags (
    product_id INT,
    tag_id INT,
    FOREIGN KEY (product_id) REFERENCES products(product_id),
    FOREIGN KEY (tag_id) REFERENCES tags(tag_id)
);
```

Nulls by default occur at the top, but you can use *IsNull* to assign default values, that will put it in the position you require. . The *ISNULL()* function tests whether an expression is NULL. If expression is a NULL value, the ISNULL() function returns 1. Otherwise, it returns 0.

- SELECT id AS 'a' FROM *tbl_name* ORDER BY `a`;
- SELECT id AS 'a' FROM *tbl_name* ORDER BY 'a';

order by clause

SQL allows the user to order the tuples in the result of a query by the values of one or more of the attributes that appear in the query result, by using the ORDER BY clause.

Remember:

- The default sort order is ascending **ASC**, with smallest values first. To sort in descending (reverse) order, add the **DESC** keyword to the name of the column you are sorting by.
- You can sort on multiple columns, and you can sort different columns in different directions.
- If the **ASC** or **DESC** modifier is not provided in the ORDER BY clause, the results will be sorted by expression in **ASC** (ascending) order. This is equivalent to ORDER BY expression ASC.

select - order by

When doing an ORDER BY, NULL values are placed **first** if you do ORDER BY ... ASC and **last** if you do ORDER BY ... DESC.

SELECT A_1, A_2, A_3, A_n FROM *r*

[ORDER BY { A_1, A_2, A_3, . . . | *expr* | *position* } [ASC | DESC] , . . .]

"Ordered by attributes A_1, A_2, A_3 …"

- Tuples are sorted by specified attributes
- Results are sorted by A_1 first
- Within each value of A_1, results are sorted by A_2 then within each value of A_2, results are sorted by A_3

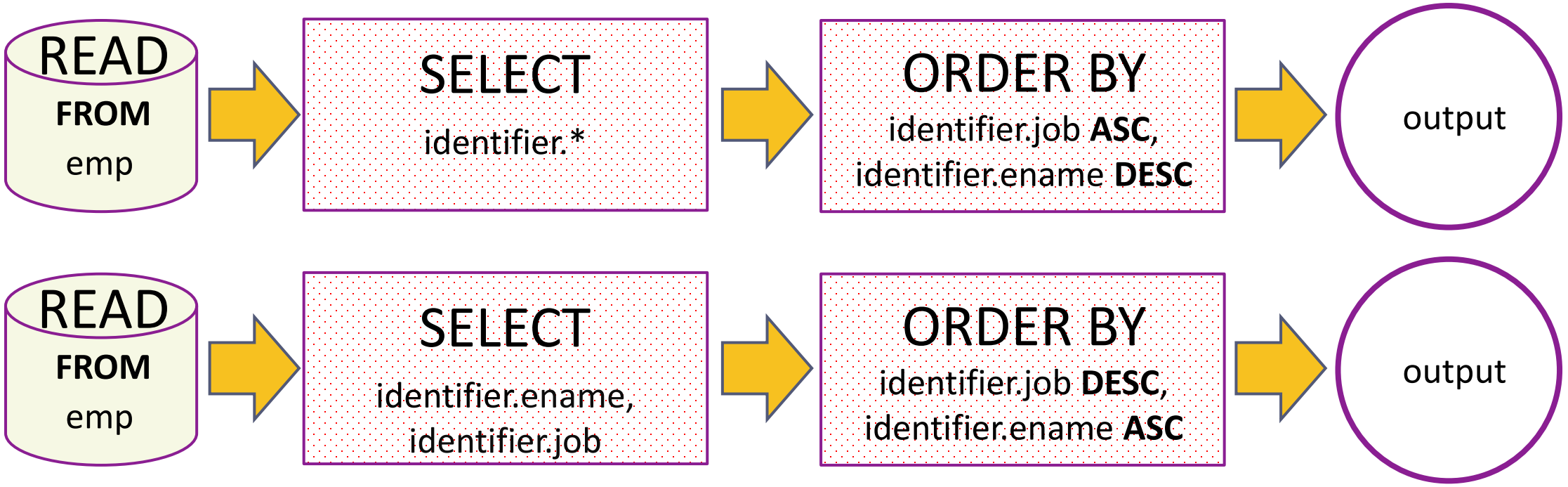
- SELECT * FROM ***r*** ORDER BY *key_part1, key_part2;* // optimizer does not use the index.
- SELECT *key_part1, key_part2* FROM ***r*** ORDER BY *key_part1, key_part2;* // optimizer uses the index.

select - order by

The ORDER BY clause is used to sort the records in your result set.

SELECT A_1, A_2, A_3, A_n FROM *r*

[ORDER BY { A_1, A_2, A_3, . . . | *expr* | *position* } [ASC | DESC] , . . .]



select - order by

When doing an ORDER BY, NULL values are presented **first** if you do ORDER BY ... ASC

- SELECT * FROM emp ORDER BY comm ASC;

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO	BONUSID	USER NAME	PWD	isActive
▶	7369	SMITH	CLERK	7902	1980-12-17 00:00:00	800.00	NULL	20	2	NULL	NULL	0
	7415	AARAV	CLERK	7902	1981-12-31 00:00:00	3350.00	NULL	10	NULL	NULL	NULL	0
	7421	THOMAS	CLERK	7920	1981-07-19 00:00:00	1750.00	NULL	10	1	NULL	NULL	0
	7566	JONES	MANAGER	7839	1981-04-02 00:00:00	2975.00	NULL	20	4	NULL	NULL	1
	7698	BLAKE	MANAGER	7839	1981-05-01 00:00:00	2850.00	NULL	30	1	NULL	NULL	1
	7782	CLARK	MANAGER	7839	1981-06-09 00:00:00	2450.00	NULL	10	3	NULL	NULL	1
	7788	SCOTT	ANALYST	7566	1982-12-09 00:00:00	3000.00	NULL	20	3	NULL	NULL	1
	7839	KING	PRESIDENT	NULL	1981-11-17 00:00:00	5000.00	NULL	10	1	NULL	NULL	1
	7876	ADAMS	CLERK	7788	1983-01-12 00:00:00	1100.00	NULL	20	1	NULL	NULL	1
	7900	JAMES	CLERK	7698	1981-12-03 00:00:00	950.00	NULL	30	2	NULL	NULL	1
	7902	FORD	ANALYST	7566	1981-12-03 00:00:00	3000.00	NULL	20	4	NULL	NULL	0
	7919	HOFFMAN	MANAGER	7566	1982-03-24 00:00:00	4150.00	NULL	30	3	NULL	NULL	1
	7934	MILLER	CLERK	7782	1982-01-23 00:00:00	1300.00	NULL	10	2	NULL	NULL	0
	7844	TURNER	SALESMAN	7698	1981-09-08 00:00:00	1500.00	0.00	30	5	NULL	NULL	1
	7499	ALLEN	SALESMAN	7698	1981-02-20 00:00:00	1600.00	300.00	30	4	NULL	NULL	1
	7521	WARD	SALESMAN	7698	1981-02-22 00:00:00	1250.00	500.00	30	1	NULL	NULL	1
	7949	ALEX	MANAGER	7698	1982-01-24 00:00:00	1250.00	500.00	30	5	NULL	NULL	1
	7654	MARTIN	SALESMAN	7698	1981-09-28 00:00:00	1250.00	1400.00	30	6	NULL	NULL	1
	7920	GRASS	SALESMAN	7919	1980-02-14 00:00:00	2575.00	2700.00	30	5	NULL	NULL	1
	7945	AARUSH	SALESMAN	7902	1980-02-14 00:00:00	1350.00	2700.00	30	NULL	NULL	NULL	0
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

select - order by

When doing an ORDER BY, NULL values are presented **last** if you do ORDER BY ... DESC.

- SELECT * FROM emp ORDER BY comm DESC;

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO	BONUSID	USER NAME	PWD	isActive
▶	7920	GRASS	SALESMAN	7919	1980-02-14 00:00:00	2575.00	2700.00	30	5	NULL	NULL	1
	7945	AARUSH	SALESMAN	7902	1980-02-14 00:00:00	1350.00	2700.00	30	NULL	NULL	NULL	0
	7654	MARTIN	SALESMAN	7698	1981-09-28 00:00:00	1250.00	1400.00	30	6	NULL	NULL	1
	7521	WARD	SALESMAN	7698	1981-02-22 00:00:00	1250.00	500.00	30	1	NULL	NULL	1
	7949	ALEX	MANAGER	7698	1982-01-24 00:00:00	1250.00	500.00	30	5	NULL	NULL	1
	7499	ALLEN	SALESMAN	7698	1981-02-20 00:00:00	1600.00	300.00	30	4	NULL	NULL	1
	7844	TURNER	SALESMAN	7698	1981-09-08 00:00:00	1500.00	0.00	30	5	NULL	NULL	1
	7369	SMITH	CLERK	7902	1980-12-17 00:00:00	800.00	NULL	20	2	NULL	NULL	0
	7415	AARAV	CLERK	7902	1981-12-31 00:00:00	3350.00	NULL	10	NULL	NULL	NULL	0
	7421	THOMAS	CLERK	7920	1981-07-19 00:00:00	1750.00	NULL	10	1	NULL	NULL	0
	7566	JONES	MANAGER	7839	1981-04-02 00:00:00	2975.00	NULL	20	4	NULL	NULL	1
	7698	BLAKE	MANAGER	7839	1981-05-01 00:00:00	2850.00	NULL	30	1	NULL	NULL	1
	7782	CLARK	MANAGER	7839	1981-06-09 00:00:00	2450.00	NULL	10	3	NULL	NULL	1
	7788	SCOTT	ANALYST	7566	1982-12-09 00:00:00	3000.00	NULL	20	3	NULL	NULL	1
	7839	KING	PRESIDENT	NULL	1981-11-17 00:00:00	5000.00	NULL	10	1	NULL	NULL	1
	7876	ADAMS	CLERK	7788	1983-01-12 00:00:00	1100.00	NULL	20	1	NULL	NULL	1
	7900	JAMES	CLERK	7698	1981-12-03 00:00:00	950.00	NULL	30	2	NULL	NULL	1
	7902	FORD	ANALYST	7566	1981-12-03 00:00:00	3000.00	NULL	20	4	NULL	NULL	0
	7919	HOFFMAN	MANAGER	7566	1982-03-24 00:00:00	4150.00	NULL	30	3	NULL	NULL	1
	7934	MILLER	CLERK	7782	1982-01-23 00:00:00	1300.00	NULL	10	2	NULL	NULL	0
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

SELECT A_1, A_2, A_3, A_n FROM *r*

select - order by

[ORDER BY { $A_1, A_2, A_3, \ldots$ | *expr* | *position* | CASE WHEN *condition* END]) [*over_clause*] } [ASC | DESC] , . . .]

- SELECT * FROM emp ORDER BY comm;
- SELECT * FROM emp ORDER BY comm IS NULL ;
- SELECT * FROM emp ORDER BY comm IS NOT NULL ;
- SELECT * FROM emp ORDER BY 1 + 1;
- SELECT * FROM emp ORDER BY True;
- SELECT sal FROM emp ORDER BY -sal;
- SELECT ename, LENGTH(ename) FROM emp ORDER BY LENGTH(ename), ename DESC ;
- SELECT * FROM emp ORDER BY IF(job = 'manager', 3, IF(job = 'salesman', 2, NULL)) ;
- SELECT onum, paymentType, amt FROM orders ORDER BY FIELD(paymentType, 'Credit Cards', 'Debit Cards', 'Wallets', 'Paytm', 'PhonePe', 'GPay', 'EMI Options', 'COD');
- SELECT * FROM emp ORDER BY ISNULL(comm), comm ;
- SELECT ename `e` FROM emp ORDER BY `e` ;
- SELECT ename `e` FROM emp ORDER BY e ;
- SELECT ename 'e' FROM emp ORDER BY 'e' ;
- SELECT * FROM emp ORDER BY CASE WHEN ename='sharmin' THEN 0 ELSE 1 END, *ename*;
- SELECT ename, job, sal, sal > 2500 FROM emp ORDER BY sal > 2500, sal;
- SELECT * FROM emp ORDER BY NULL;
- SELECT * FROM emp ORDER BY (SELECT NULL);

- swap every pair of rows (1↔2, 3↔4, 5↔6, etc.)

select - order by

swap every pair of row

Table:- swapCity

ID	ename	cityName
1	Saleel	Baroda
2	Sharmin	Rajkot
3	Vrushali	Surat
4	Ruhan	Bharuch
5	Nitish	Jamnagar
6	Deep	Anand
7	Neel	Bhuj

Output →

Output

ID	ename	cityName
2	Sharmin	Rajkot
1	Saleel	Baroda
4	Ruhan	Bharuch
3	Vrushali	Surat
6	Deep	Anand
5	Nitish	Jamnagar
7	Neel	Bhuj

- SELECT id, CASE WHEN id % 2 = 1 THEN id + 1 ELSE id - 1 END *R1*, ename, cityName FROM swapcity ORDER BY *R1*;
- SELECT id, ename, cityName FROM swapcity ORDER BY CASE WHEN id % 2 = 1 THEN id + 1 ELSE id - 1 END;

- swap every pair of rows (1↔2, 3↔4, 5↔6, etc.)

select - order by

swap city with original sequence of city names

Table:- swapCity

ID	ename	cityName
1	Saleel	Baroda
2	Sharmin	Rajkot
3	Vrushali	Surat
4	Ruhan	Bharuch
5	Nitish	Jamnagar
6	Deep	Anand
7	Neel	Bhuj

Output →

Output

Swap City Names	Original City Names
Rajkot	Baroda
Baroda	Rajkot
Bharuch	Surat
Surat	Bharuch
Anand	Jamnagar
Jamnagar	Anand

- WITH ***cte1*** AS (SELECT id, cityName FROM swapcity), ***cte2*** AS (SELECT cityName, CASE WHEN id%2 = 0 THEN id - 1 ELSE id + 1 END *R2* FROM swapcity) SELECT ***cte1***.cityName 'Swap City Names' , ***cte2***.cityName 'Original City Names' FROM ***cte1***, ***cte2*** WHERE ***cte1***.id = ***cte2***.*R2*;

- swap every 3rd row (1↔3, 4↔6, 7↔9, etc.)

select - order by

swap every third city

Table:- swapCity

ID	ename	cityName
1	Saleel	Baroda
2	Sharmin	Rajkot
3	Vrushali	Surat
4	Ruhan	Bharuch
5	Nitish	Jamnagar
6	Deep	Anand
7	Neel	Bhuj

Output →

Output

ID	ename	Swap City Names
3	Vrushali	Surat
1	Saleel	Baroda
2	Sharmin	Rajkot
6	Deep	Anand
4	Ruhan	Bharuch
5	Nitish	Jamnagar
7	Neel	Bhuj

- SELECT id, ename, cityName FROM swapcity ORDER BY CASE WHEN id%3 = 1 THEN id + 1 WHEN id%3 = 2 THEN id + 1 ELSE id - 2 end;

Remember:

In **WHERE** clause operations can be performed using…

- *CONSTANTS*
- *TABLE columns*
- *(SELECT clause)*
- *FUNCTION calls (PRE-DEFINED / UDF)*

```
SELECT patientID, patientName FROM patient WHERE (SELECT COUNT(*) FROM doctorvisits
WHERE patient.patientID = doctorVisits.patientid GROUP BY doctorvisits.patientID) > 8;
```

* In SQL, a logical expression is often called a *predicate*.

where clause

The WHERE Clause is used when you want to retrieve specific information from a table excluding other irrelevant data.

Note:

Expressions in WHERE clause can use.

- *Arithmetic operators*
- *Comparison operators*
- *Logical operators*

with OR

- If <*condition1*> is **TRUE**, then MySQL **does not check** <*condition2*>.
- If <*condition1*> is **FALSE**, then MySQL **must check** <*condition2*>.

with AND

- If <*condition1*> is **FALSE**, then MySQL **does not check** <*condition2*>.
- If <*condition1*> is **TRUE**, then MySQL **must check** <*condition2*>.

Note:

- All comparisons return FALSE when either argument is NULL, so no rows are ever selected.

select - where

We can use a conditional clause called WHERE clause to filter out results. Using WHERE clause, we can specify a selection criteria to select required records from a table.

SELECT $A_1, A_2, A_3, \ldots$ FROM $r_1, r_2, r_3, \ldots$ [WHERE P]

- r_i are the relations (tables)
- A_i are attributes (columns)
- P is the selection predicate

SQL permits us to use the notation $(v_1, v_2,\ldots,v_n)$ to denote a tuple of arity (attribute) n containing values $v_1, v_2,\ldots,v_n$.

```
WHERE (a1, a2) <= (b1, b2)
```

```
WHERE (emp.deptno, dname) = (dept.deptno, 'sales');
```

Remember:

- A predicate is a condition expression that evaluates to a boolean value, either **true** or **false**.
- Predicates can be used as follows: In a SELECT statement's ***WHERE*** clause or ***HAVING*** clause to determine which rows are relevant to a particular query.

A value of zero is considered false. Nonzero values are considered true.

- `SELECT true, false, TRUE, FALSE, True, False;`

select - where

SELECT $A_1, A_2, A_3, \ldots$ FROM $r_1, r_2, r_3, \ldots$ [WHERE *P*] CASE WHEN *condition* END

2. *comparison_operator:*

= | <=> | >= | > | <= | < | <> | !=

5. *logical_operators*

{ AND | && } | { OR | || }

What will be the result of the query below?

- SELECT 1 = 1;
- SELECT True = 1;
- SELECT True = 2;
- SELECT True = True;
- SELECT 0 = 0;
- SELECT False = False;
- SELECT False = 1;
- SELECT 'a' = 1;
- SELECT 'a' = 0;
- SELECT * FROM emp WHERE ename = 0;
- SELECT * FROM emp WHERE ename = 1;
- SELECT * FROM emp WHERE ename = False;
- SELECT * FROM emp WHERE ename = True;
- SELECT * FROM emp WHERE True AND False;
- SELECT * FROM emp WHERE True OR False;
- SELECT * FROM emp WHERE True AND 1;
- SELECT * FROM emp WHERE True OR 0;

Note:

AND has higher precedence than **OR.**

- EXPLAIN ANALYZE SELECT * FROM emp WHERE job = 'salesman' OR job = 'manager' AND sal > 2000;

select - where

SELECT $A_1, A_2, A_3, \ldots$ FROM $r_1, r_2, r_3, \ldots$ [WHERE *P*]
CASE WHEN *condition* END

WHERE state = 'NY' OR 'CA' --Illegal
WHERE salary > 20000 AND < 30000 –Illegal
WHERE state NOT = 'CA' --Illegal

Logical Operators		
AND, &&	Logical AND	e.g. SELECT 1 AND 1; / SELECT 1 AND 0; SELECT 0 AND NULL; / SELECT NULL AND 0; SELECT 1 AND NULL; / SELECT NULL AND 1;
OR, ||	Logical OR	e.g. SELECT 1 OR 1; / SELECT 1 OR 0; SELECT 0 OR NULL; / SELECT NULL OR 0; SELECT 1 OR NULL; / SELECT NULL OR 1;
NOT, !	Negates value	e.g. SELECT NOT 1;

- **Logical AND.** Evaluates to 1 if all operands are non-zero and not NULL, to 0 if one or more operands are 0, otherwise NULL is returned.
- **Logical OR.** When both operands are non-NULL, the result is 1 if any operand is nonzero, and 0 otherwise. With a NULL operand, the result is 1 if the other operand is nonzero, and NULL otherwise. If both operands are NULL, the result is NULL.
- **Logical NOT.** Evaluates to 1 if the operand is 0, to 0 if the operand is nonzero, and NOT NULL returns NULL.

select - where

SELECT $A_1, A_2, A_3, \ldots$ FROM $r_1, r_2, r_3, \ldots$ [WHERE *P*] CASE WHEN *condition* END

Comparison Functions and Operators	
LEAST(value1, value2, . . .)	With two or more arguments, returns the smallest argument.
GREATEST(value1, value2, . . .)	With two or more arguments, returns the largest argument.
(expr, [expr] . . .)	Multiple columns in expr. (sub-query returning multiple columns to compare)
COALESCE(value, . . .)	Returns the first **non-NULL** value in the list, or NULL if there are no non-NULL values.

- ```
  SELECT GREATEST(10, 20, 30),      # 30
         LEAST(10, 20, 30);         # 10
  ```

- ```
  SELECT GREATEST(10, null, 30),    # null
         LEAST(10, NULL , 30);      # null
  ```

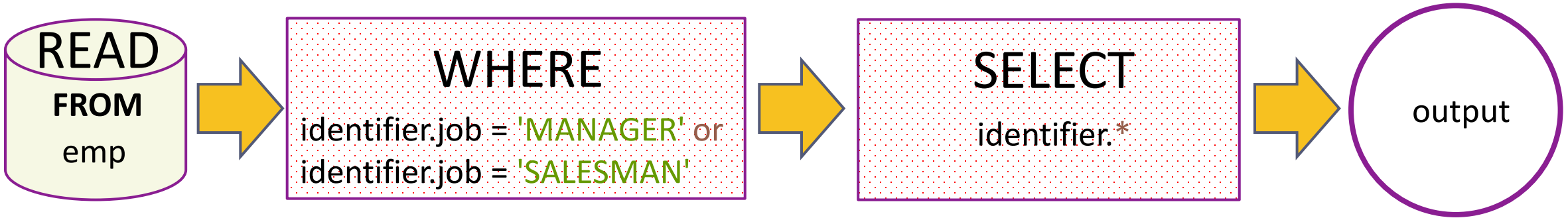
- ```
  SELECT * FROM emp WHERE (deptno, pwd) = (SELECT deptno, pwd FROM dept WHERE deptno = 30);
  ```

Remember:

- If any argument is NULL, the both functions return NULLs immediately without doing any comparison..
  ```

select - where

- SELECT * FROM emp WHERE job = 'MANAGER' OR job = 'SALESMAN';



	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO	BONUSID	USER NAME	PWD	phone	isActive
▶	7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30	4	ALWAYS TESTE	sales@2017	7032300096	1
	7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30	1	WARD	sales@2017	7132300034	1
	7566	JONES	MANAGER	7839	1981-04-02	2975	NULL	20	4	HONEYCOMB	a12recmpm	7132300039	1
	7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30	6	LIFE RACER	sales@2017	7132300050	1
	7698	BLAKE	MANAGER	7839	1981-05-01	2850	NULL	30	1	BIG BEN	sales@2017	7132300027	1
	7782	CLARK	MANAGER	7839	1981-06-09	2450	NULL	10	3	CLARK	r50mpm	7032300001	1
	7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30	5	SAND DUST	sales@2017	NULL	1
	7919	HOFFMAN	MANAGER	7566	1982-03-24	4150	NULL	30	3	INTERVAL	sales@2017	NULL	1

combining and & or - where

Note:

AND has higher precedence than **OR.**

- `SELECT * FROM andor WHERE ename = 'saleel' AND city = 'pune' OR city = 'baroda';`
- `SELECT * FROM andor WHERE ename = 'saleel' AND (city = 'pune' OR city = 'baroda');`
- `SELECT ename, job, comm FROM emp WHERE comm = 0 OR comm IS NULL AND job = 'CLERK';`
- `SELECT ename, job, comm FROM emp WHERE (comm = 0 OR comm IS NULL) AND job = 'CLERK';`

- `EXPLAIN ANALYZE  SELECT * FROM emp WHERE job = 'salesman' OR job = 'manager' AND sal > 2000;`

select - where

SELECT $A_1, A_2, A_3, \ldots$ FROM $r_1, r_2, r_3, \ldots$ [WHERE *P*] CASE WHEN *condition* END

What will be the output of the following statement?

- SELECT "Hello" # "World ";
- SELECT 10 + 10 as *Result* WHERE False;
- SELECT 10 + 10 as *Result* WHERE True;
- SELECT 10 + 10 as *Result* WHERE 10 - 10;
- SELECT 10 + 10 as *Result* WHERE 10 - 0;
- SELECT 10 + 10 as *Result* WHERE 10 - 30;
- SELECT '5' * '5' as *Result*;
- SELECT 5 * 5 - '-5' as *Result*;

- SELECT * FROM orders WHERE amt > CASE WHEN registered = 1 THEN 2000 ELSE 5000 END;
- SELECT * FROM orders WHERE CASE WHEN paymentType = 'Wallets' THEN amt > 1500 WHEN paymentType = 'COD' THEN amt > 3000 END;
- SELECT * FROM orders WHERE CASE WHEN paymentType IN ('GPay', 'Paytm', 'PhonePe') THEN amt > 1000 WHEN paymentType IN ('Credit Cards', 'Debit Cards', 'Wallets') THEN amt > 2000 END ORDER BY paymentType;

- Bitwise (**&**)

select - bitwise search

Table:- status

ID	statusID	status
1	1	New
2	2	Draft
3	4	Approved
4	8	Rejected
5	16	Cancel
6	32	In Process
7	64	Send
8	128	Hold
9	256	Selected
10	512	Received
11	1024	More

Output →

Query

- SELECT * FROM status WHERE statusid & 5;

ID	statusID	status
1	1	New
3	4	Approved

- SELECT * FROM status WHERE statusid & 36;

ID	statusID	status
3	4	Approved
6	32	In Process

- SELECT * FROM status WHERE statusid & 38;

ID	statusID	status
2	2	Draft
3	4	Approved
6	32	In Process

- Bitwise (**&**) example.

bitwise tables

Table:- bitwise

Feature	Decimal Value
Window seat	1
Aisle seat	2
Exit row	4
Extra legroom	8
Near restroom	16
Power outlet available	32
USB charging	64
Reclining seat	128
Frist class seat	256
Business class seat	512
Economy Class	1024

Table:- airline_seats

seat_id	flight_id	seat_number	features
101	456	'12A'	73
102	431	'01F'	521
103	346	'17G'	50
104	101	'15H'	521
105	204	'09L'	336
106	876	'24J'	141
107	683	'34K'	1034
108	456	'13A'	73

A seat with **Window + Extra Legroom + USB Charging**:

- ***Window = 1***
- ***Extra Legroom = 8***
- ***USB Charging = 64***

→ Total = 1 + 8 + 64 = 73

- INSERT INTO airline_seats (seat_id, flight_id, seat_number, features) VALUES (101, 456, '12A', 73);

- Bitwise (**&**) insert example.

insert - bitwise

Table:- bitwise

Feature	Decimal Value
Window seat	1
Aisle seat	2
Exit row	4
Extra legroom	8
Near restroom	16
Power outlet available	32
USB charging	64
Reclining seat	128
Frist class seat	256
Business class seat	512
Economy Class	1024

- INSERT INTO airline_seats (seat_id, flight_id, seat_number, features) VALUES (101, 456, '12A', 73);
- INSERT INTO airline_seats (seat_id, flight_id, seat_number, features) VALUES (102, 456, '01F', 521);
- INSERT INTO airline_seats (seat_id, flight_id, seat_number, features) VALUES (103, 346, '17G', 50);
- INSERT INTO airline_seats (seat_id, flight_id, seat_number, features) VALUES (104, 101, '15H', 521);
- INSERT INTO airline_seats (seat_id, flight_id, seat_number, features) VALUES (105, 204, '09L', 336);
- INSERT INTO airline_seats (seat_id, flight_id, seat_number, features) VALUES (106, 876, '24J', 141);
- INSERT INTO airline_seats (seat_id, flight_id, seat_number, features) VALUES (107, 683, '34K', 1034);
- INSERT INTO airline_seats (seat_id, flight_id, seat_number, features) VALUES (108, 456, '13A', 73);
- . . .

select - bitwise search

- **Bitwise (&) Select Example**

```
SELECT * FROM airline_seats a, bitwise b WHERE a.features & b.bitvalue & 16;
SELECT * FROM airline_seats a, bitwise b WHERE a.features & b.bitvalue & 1;
SELECT * FROM airline_seats a, bitwise b WHERE a.features & b.bitvalue & (1 | 128);
SELECT * FROM airline_seats a, bitwise b WHERE (a.features & b.bitvalue) & (1 | 128);
SELECT * FROM airline_seats a, bitwise b WHERE a.features & b.bitvalue & (2 | 16 | 128);
```

- SELECT * FROM emp WHERE comm IS UNKNOWN;
- SELECT * FROM emp WHERE comm IS NOT UNKNOWN;

- *operand* IS [NOT] *NULL*

3. *boolean_ predicate:*

IS [NOT] NULL | *expr* <=> NULL

is null / is not null

- "*IS NULL*" is the keyword that performs the Boolean comparison. It returns true if the supplied value is NULL and false if the supplied value is not NULL.
- *"IS NOT NULL*" is the keyword that performs the Boolean comparison. It returns true if the supplied value is not NULL and false if the supplied value is null.
- SQL uses a three-valued logic: besides true and false, the result of logical expressions can also be unknown. SQL's three valued logic is a consequence of supporting null to mark absent data.

Note:

- IS UNKNOWN is synonym of *IS NULL*.
- IS NOT UNKNOWN is synonym of *IS NOT NULL*.

Remember:

is null / is not null

SELECT * FROM emp WHERE comm = NULL; # will return Empty set

- SELECT empno, ename, job, sal, comm FROM emp WHERE comm IS NOT NULL;

- SELECT empno, ename, job, sal, comm FROM emp WHERE comm IS NOT UNKNOWN;

- SELECT empno, ename, job, sal, comm FROM emp WHERE comm is TRUE;

empno	ename	job	sal	comm
7499	ALLEN	SALESMAN	1600.00	300.00
7521	WARD	SALESMAN	1250.00	500.00
7654	MARTIN	SALESMAN	1250.00	1400.00
7844	TURNER	SALESMAN	1500.00	0.00
7920	GRASS	SALESMAN	2575.00	2700.00
7945	AARUSH	SALESMAN	1350.00	2700.00
7949	ALEX	MANAGER	1250.00	500.00

empno	ename	job	sal	comm
7499	ALLEN	SALESMAN	1600.00	300.00
7521	WARD	SALESMAN	1250.00	500.00
7654	MARTIN	SALESMAN	1250.00	1400.00
7920	GRASS	SALESMAN	2575.00	2700.00
7945	AARUSH	SALESMAN	1350.00	2700.00
7949	ALEX	MANAGER	1250.00	500.00

Remember:

1. NULL represents an **unknown** OR **missing** value.
2. NULL value does not occupy space in memory.
3. NULL value is independent of data type.
4. NULL can be written in any case.

```
SELECT * FROM emp WHERE comm = NULL;  # will return Empty set
```

select – boolean

- BOOL and BOOLEAN are synonym of TINYINT(1)

A value of zero is considered false. Nonzero values are considered true.

```
SELECT true, false, TRUE, FALSE, True, False;
```

- SELECT * FROM tasks WHERE completed;
- SELECT * FROM tasks WHERE completed IS True;
- SELECT * FROM tasks WHERE completed = 1;
- SELECT * FROM tasks WHERE completed = True;

id	title	completed
2	Task2	1
4	Task4	1
8	Task8	1
9	Task9	12
10	Task10	58
11	Task11	1
13	Task13	1
NULL	NULL	NULL

id	title	completed
2	Task2	1
4	Task4	1
8	Task8	1
11	Task11	1
13	Task13	1
NULL	NULL	NULL

- SELECT * FROM tasks WHERE NOT completed;
- SELECT * FROM tasks WHERE completed IS False;
- SELECT * FROM tasks WHERE completed = 0;
- SELECT * FROM tasks WHERE completed = False;

id	title	completed
1	Task1	0
3	Task3	0
7	Task7	0
12	Task12	0
NULL	NULL	NULL

select – boolean

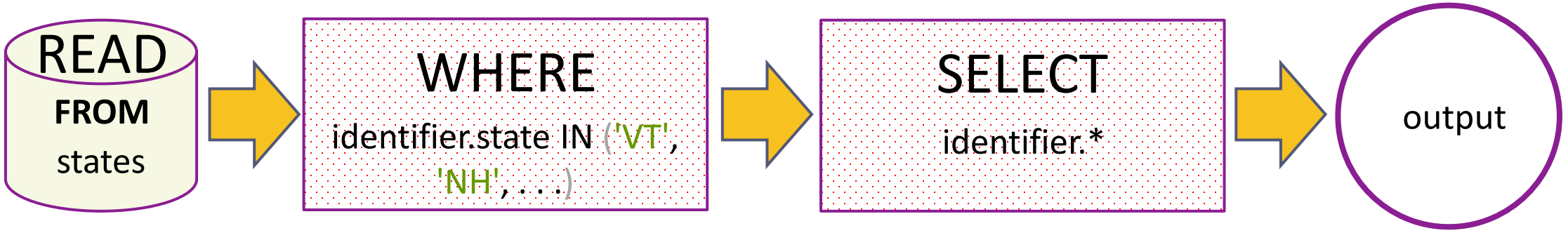
- BOOL and BOOLEAN are synonym of TINYINT(1)

A value of zero is considered false. Nonzero values are considered true.

```
SELECT true, false, TRUE, FALSE, True, False;
```

What will be the result of the query below?

- SELECT * FROM emp WHERE 1;
- SELECT * FROM emp WHERE True;
- SELECT * FROM emp WHERE 0;
- SELECT * FROM emp WHERE False;
- SELECT * FROM emp WHERE ename = '' OR 0;
- SELECT * FROM emp WHERE ename = '' OR 1;
- SELECT * FROM emp WHERE ename = '' OR 1 = 1;
- SELECT * FROM emp WHERE ename = 'smith' OR True;
- SELECT * FROM emp WHERE ename = 'smith' AND True;
- SELECT * FROM emp WHERE ename IN('smith', True);
- SELECT * FROM emp WHERE ename = 'smith' OR False;
- SELECT * FROM emp WHERE ename = 'smith' AND False;
- SELECT * FROM emp WHERE ename IN('smith', False);



4. ***predicate***:

expr [NOT] IN (*expr1, expr2, . . .*)
| *expr* [NOT] IN (*subquery*)

in

The IN statement is used in a WHERE clause to choose items from a set. The IN operator allows you to determine if a specified value matches any value in a set of values or value returned by a subquery.

```
SELECT . . . FROM r1 WHERE (
    state = 'VT'  OR
    state = 'NH' OR
    state = 'ME' OR
    state = 'MA' OR
    state = 'CT'   OR
    state = 'RI'
);
```

IN compares a value against a list (or the result of a subquery).

- SELECT . . . FROM r_1 WHERE state IN ('VT', 'NH', 'ME', 'MA', 'CT', 'RI');
- SELECT . . . FROM r_1 WHERE state IN (SELECT . . .);

A IN (***B1, B2, B3,*** etc.) A is found in the list (B1, B2, etc.)

in with values

syntax

column | expression IN (*v1, v2, v3, . . .*)
column | expression IN (*subquery*)

IN → "Is my value IN this list?"
EXISTS → "Does, at least ONE matching row EXIST?"

Remember:

- If a value in the column or the expression is equal to any value in the list, the result of the IN operator is TRUE.
- The IN operator is equivalent to multiple OR operators.
- To negate the IN operator, you use the NOT IN operator.

- SELECT empno, ename, job, hiredate, sal, comm, deptno, isactive FROM emp WHERE job IN ('salesman', 'manager');

empno	ename	job	hiredate	sal	comm	deptno	isactive
7499	ALLEN	SALESMAN	1981-02-20	1600.00	300.00	30	1
7521	WARD	SALESMAN	1981-02-22	1250.00	500.00	30	1
7566	JONES	MANAGER	1981-04-02	2975.00	NULL	20	1
7654	MARTIN	SALESMAN	1981-09-28	1250.00	1400.00	30	1
7698	BLAKE	MANAGER	1981-05-01	2850.00	NULL	30	1
7782	CLARK	MANAGER	1981-06-09	2450.00	NULL	10	1
7844	TURNER	SALESMAN	1981-09-08	1500.00	0.00	30	1
7919	HOFFMAN	MANAGER	1982-03-24	4150.00	NULL	30	1

in with subquery

syntax

column | expression IN (*v1, v2, v3, . . .*)
column | expression IN (*subquery*)

- SELECT deptno, ename FROM emp WHERE deptno IN (SELECT deptno FROM dept WHERE dname = 'accounting' OR dname = 'RESEARCH');

How it works:

- The subquery returns a list of **deptno**.
- SQL checks whether each employee's **deptno** is in that list.

Think of it as:
department_id IN (10, 20)

column | expression IN (*subquery*)

in VS exists

Point	IN Operator	EXISTS Operator
Meaning	Checks whether a value exists in a list/result set	Checks whether the subquery returns at least one row
Main purpose	Compare a value with values returned by a **subquery**	Check for the existence of matching rows
Works with	Usually compares one column/value with a list	Checks a condition between outer and inner query
Result	Returns TRUE if the value matches any value in the result	Returns TRUE if the subquery returns one or more rows
Subquery dependency	Subquery can be independent of the outer query	Usually a correlated subquery, though it can also be uncorrelated
NULL behavior	NULL can cause unexpected results, especially with NOT IN	Generally safer with NULL, especially NOT EXISTS
Performance	Can be efficient when the result set is small or optimized well	Often preferred when checking existence in a large related table
What is important?	The values returned by the subquery	Whether any row is returned
Stops searching?	Conceptually evaluates membership in the result	Can stop once a matching row is found
Common use	"Find employees whose department is in these departments"	"Find employees who have at least one matching record"

column | expression NOT IN (*subquery*)

not in VS not exists

Point	NOT IN	NOT EXISTS
Purpose	Checks that a value is not present in a list/subquery result	Checks that no matching row exists in a subquery
Syntax	WHERE column NOT IN ***(subquery)***	WHERE NOT EXISTS ***(subquery)***
Type of check	Value-based	Row/existence-based
NULL handling	Can cause unexpected results if subquery returns NULL	Generally handles NULL safely
Correlated subquery	Usually not required	Commonly uses a correlated subquery
Subquery relationship	Compares outer value with all returned values	Checks for a matching row for each outer row
Best suited for	Checking against a list of values	Checking whether a related record does not exist
Example	WHERE dept_id NOT IN (SELECT dept_id FROM dept)	WHERE NOT EXISTS (SELECT 1 FROM dept d WHERE d.dept_id = e.dept_id)
Main problem	NULL in the subquery can make the condition UNKNOWN	No equivalent NULL problem in the matching condition
Common recommendation	Use when you know the subquery column cannot contain NULL	Often preferred when NULL values are possible

Problem with NOT IN:

not in

employee {ID, employee_name, department_ID);

- SELECT * FROM a WHERE c1 NOT IN(1, 2, NULL);
- SELECT * FROM a WHERE c1 NOT IN(SELECT c1 FROM b);
  Empty set (0.00 sec)

Problem with NOT IN:

not in

a

c1	c2
1	1
2	1
3	1
4	1
5	1

b

c1	c2
1	7
NULL	7
3	7

- SELECT * FROM a WHERE c1 NOT IN(1, 2, NULL);
- SELECT * FROM a WHERE c1 NOT IN(SELECT c1 FROM b);
  Empty set (0.00 sec)

"color NOT IN (Red, Blue, NULL)" This is equivalent to: NOT(color = 'Red' OR color = 'Blue' OR color = NULL)

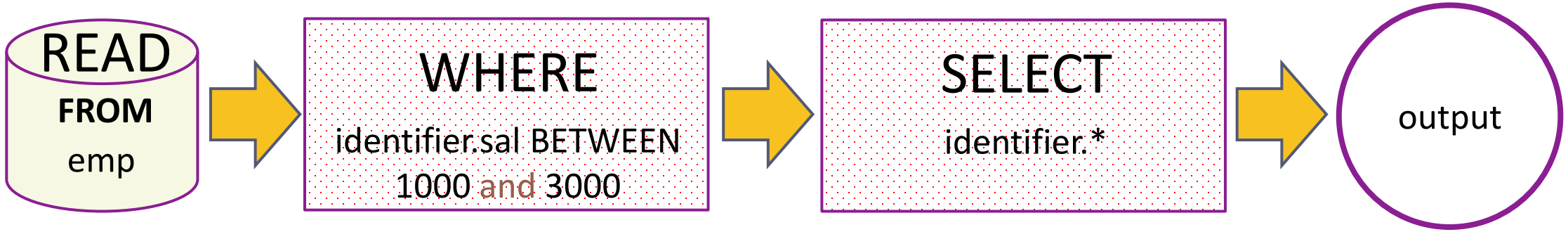
Remember:

in

- On the left side of the IN() predicate, the row constructor contains only column references.
- On the right side of the IN() predicate, there is more than one row constructor.

What will be the result of the query below?

- `SELECT * FROM emp WHERE deptno IN (10);`
- `SELECT * FROM emp WHERE deptno IN (10, 20);`
- `SELECT * FROM emp WHERE False IN (10, 20, 0);`
- `SELECT * FROM emp WHERE True IN (10, 20, 1);`
- `SELECT * FROM emp WHERE 10 IN (10, 20);`
- `SELECT * FROM emp WHERE 7788 IN (empno, mgr);` ←
- `SELECT * FROM emp WHERE 1 IN (10, 20, True, False);`
- `SELECT * FROM emp WHERE deptno IN (10, 20) OR True;`
- `SELECT * FROM emp WHERE deptno IN (10, 20) AND True;`
- `SELECT * FROM emp WHERE deptno IN (SELECT deptno FROM dept);`
- `SELECT * FROM emp WHERE deptno IN (SELECT deptno FROM dept WHERE dname='accounting');`
- `SELECT * FROM emp WHERE deptno IN (TABLE deptno);` # ERROR 1241 (21000): Operand should contain 1 column(s)
- `SELECT * FROM emp WHERE 1 NOT IN (NULL);`



4. ***predicate***:

expr [NOT] BETWEEN *expr1* AND *expr2*

between

The BETWEEN operator is a logical operator that allows you to specify a range to test.

A BETWEEN ***B*** AND ***C*** A is between B and C

Table:- betweenSalary

id	sal
1	2000

Output

R1	R2
1	1

- SELECT COUNT(CASE WHEN sal BETWEEN 1000 AND 2000 THEN 1 END) *R1*, COUNT(CASE WHEN sal BETWEEN 2000 AND 3000 THEN 1 END) *R2* FROM betweenSalary;

between

syntax

WHERE salary BETWEEN (20000 AND 30000) – Illegal

column | expression BETWEEN *start_expression* AND *end_expression*

Remember:

- The BETWEEN operator returns TRUE if the expression to test is greater than or equal to the value of the start_expression and less than or equal to the value of the end_expression.
- You can use the greater than or equal to (**>=**) and less than or equal to (**<=**) to substitute the BETWEEN operator.

Note:

- if any input to the BETWEEN or NOT BETWEEN is NULL, then the result is UNKNOWN.

e.g.

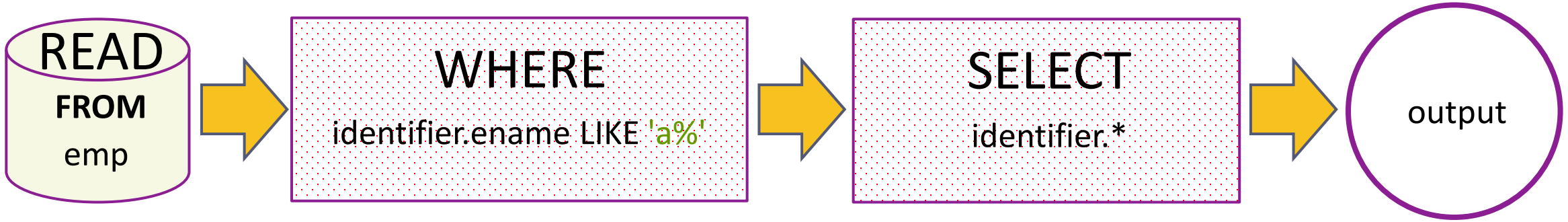
```
SELET empno, ename, job, hiredate, sal, comm, deptno, isactive FROM emp WHERE sal BETWEEN 1000 AND NULL;
```

- SELECT * FROM salespeople WHERE FORMAT(comm, 2) > 0.1 AND FORMAT(comm, 2) < 0.26;

between

- SELECT empno, ename, job, hiredate, sal, comm, deptno, isactive FROM emp WHERE sal BETWEEN 1000 AND 3000;

	empno	ename	job	hiredate	sal	comm	deptno	isactive
▶	7421	THOMAS	CLERK	1981-07-19	1750.00	NULL	10	0
	7499	ALLEN	SALESMAN	1981-02-20	1600.00	300.00	30	1
	7521	WARD	SALESMAN	1981-02-22	1250.00	500.00	30	1
	7566	JONES	MANAGER	1981-04-02	2975.00	NULL	20	1
	7654	MARTIN	SALESMAN	1981-09-28	1250.00	1400.00	30	1
	7698	BLAKE	MANAGER	1981-05-01	2850.00	NULL	30	1
	7782	CLARK	MANAGER	1981-06-09	2450.00	NULL	10	1
	7788	SCOTT	ANALYST	1982-12-09	3000.00	NULL	20	1
	7844	TURNER	SALESMAN	1981-09-08	1500.00	0.00	30	1
	7876	ADAMS	CLERK	1983-01-12	1100.00	NULL	20	1
	7902	FORD	ANALYST	1981-12-03	3000.00	NULL	20	0
	7920	GRASS	SALESMAN	1980-02-14	2575.00	2700.00	30	1



4. ***predicate***:

expr [NOT] LIKE *expr* [ESCAPE *char*]

REGEXP_LIKE(*expr*, *pattern*[, match_type])

like

The LIKE operator is a logical operator that tests whether a string contains a specified pattern or not.

like - string comparison functions

syntax

column | expression LIKE 'pattern' [ESCAPE *escape_character*]
REGEXP_LIKE(*expr*, *pattern*[, match_type])

Remember:

- **%** matches any number of characters, even zero characters. ("%" represents any sequence of characters.)
- **_** matches exactly one character. ("_" represents a single character.)
- If we use **default escape character '\'**, then don't use ESCAPE keyword.

REGEXP_LIKE

- ^ [start]
- $ [end]

- SELECT ename FROM emp WHERE regexp_like(ename, "^['sa']");
- SELECT ename FROM emp WHERE regexp_like(ename, "['sa']$");

Note:

- The ESCAPE keyword is used to escape pattern matching characters such as the (%) percentage and underscore (_) if they form part of the data.
- If you do not specify the ESCAPE character, \ is assumed.

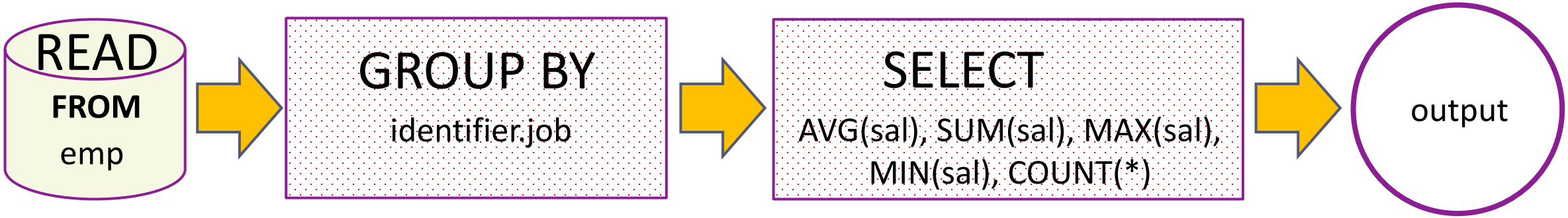
like

- SELECT empno, ename, job, hiredate, sal, comm, deptno, isactive FROM emp WHERE ename LIKE 'a%';

empno	ename	job	hiredate	sal	comm	deptno	isactive
7415	AARAV	CLERK	1981-12-31	3350.00	NULL	10	0
7499	ALLEN	SALESMAN	1981-02-20	1600.00	300.00	30	1
7876	ADAMS	CLERK	1983-01-12	1100.00	NULL	20	1
7945	AARUSH	SALESMAN	1980-02-14	1350.00	2700.00	30	0
7949	ALEX	MANAGER	1982-01-24	1250.00	500.00	30	1

What will be the result of the query below?

- `SELECT * FROM emp WHERE ename LIKE 's%';`
- `SELECT * FROM emp WHERE 'saleel' LIKE 's%';`
- `SELECT * FROM emp WHERE True LIKE '1';`
- `SELECT * FROM emp WHERE True LIKE '1%';`
- `SELECT * FROM emp WHERE True LIKE 001;`
- `SELECT * FROM emp WHERE True LIKE 100;`
- `SELECT * FROM emp WHERE False LIKE 100 OR 0;`
- `SELECT * FROM emp WHERE False LIKE 0 AND 1;`



aggregate functions

SUM, AVG, MAX, MIN, COUNT, GROUP_CONCAT, JSON_OBJECTAGG, and JSON_ARRAYAGG

SELECT . . . . . FROM table_name WHERE <condition> / GROUP BY column_name

this is invalid

SUM(colNM) / AVG(colNM) / MAX(colNM)
MIN(colNM) / COUNT(colNM) / COUNT(*)

- SET *SQL_MODE* = ' ';
- SET *SQL_MODE* = IGNORE_SPACE;

Remember:

None of the below two queries get executed unsuccessfully. The reason is that a condition in a WHERE clause cannot contain any aggregate function (or group function) without a subquery!

- SELECT empno, ename, sal, deptno FROM emp WHERE sal = MAX(sal); #error
- SELECT empno, ename, sal, deptno FROM emp WHERE MAX(sal) = sal; #error

aggregate functions

Remember:

There are 3 places where aggregate functions can appear in a query.

- in the SELECT-LIST/FIELD-LIST (the items before the FROM clause).
- in the ORDER BY clause.
- in the HAVING clause.

Note:

- The aggregate functions allow you to perform the calculation of a set of values and **return a *single* value**.
- The WHERE clause cannot refer to aggregate functions. e.g. WHERE SUM(sal) = 5000 # Invalid, Error
- The HAVING clause can refer to aggregate functions. e.g. HAVING SUM(sal) = 5000 # Valid, No Error
- Nesting of aggregate functions are not allowed.
  e.g.
  ```
  SELECT MAX(COUNT(*)) FROM emp GROUP BY deptno;
  ```
- Blank space between aggregate functions like (SUM, MIN, MAX, COUNT) are not allowed.
  e.g.
  ```
  SELECT SUM (sal) FROM emp;
  ```
- The GROUP BY clause is often used with an aggregate function to perform calculation and **return a single value for each subgroup**.
- To eliminate duplicates before applying the aggregate function is available by including the keyword DISTINCT.

Things to… Remember:

aggregate function - AVG

TODO

AVG([DISTINCT] *expr* | [*condition* | CASE WHEN *condition* END]) [*over_clause*]

- If there are no matching rows, AVG() **returns NULL.**
- AVG() may take a numeric argument, and it returns a average of non-NULL values.

e.g.

- SELECT AVG(1) *"R1"*;
- SELECT AVG (NULL) *"R1"*;
- SELECT AVG (1) *"R1"* WHERE True;
- SELECT AVG(1) *"R1"* WHERE False;
- SELECT AVG(1) *"R1"* FROM emp;
- SELECT AVG(sal) *"R1"* FROM emp WHERE empno = -1;
- SELECT AVG(sal) *"Avg Salary"* FROM emp;
- SELECT job, AVG(sal) *"Avg Salary"* FROM emp GROUP BY job;

- SELECT AVG('a' IS NOT NULL) *"R1"*;
- SELECT AVG('a' IS NOT NULL) + AVG('b' IS NOT NULL) *"R1"*;
- SELECT AVG(CASE WHEN job = 'manager' THEN sal END) *"R1"* FROM emp;

Things to… Remember:

aggregate function - SUM

TODO

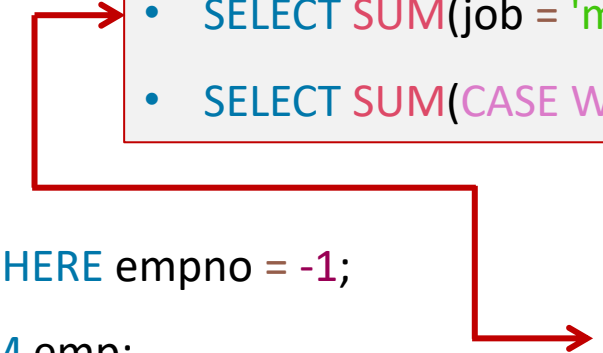
SUM([DISTINCT] *expr* | [*condition* | CASE WHEN *condition* END]) [*over_clause*]

- If there are no matching rows, SUM() **returns NULL.**
- SUM() may take a numeric argument , and it returns a sum of non-NULL values.

e.g.

- SELECT SUM(1) *"R1"*;
- SELECT SUM(NULL) *"R1"*;
- SELECT SUM(2 + 2 * 2) *"R1"*;;
- SELECT SUM(1) *"R1"* WHERE True;
- SELECT SUM(1) *"R1"* WHERE False;
- SELECT SUM(1) *"R1"* FROM emp;
- SELECT SUM(sal) *"R1"* FROM emp WHERE empno = -1;
- SELECT SUM(sal) *"Total Salary"* FROM emp;
- SELECT job, SUM(sal) *"Total Salary"* FROM emp GROUP BY job;

- SELECT COUNT(*) total_employees, SUM(job = 'manager') total_managers FROM emp;
- SELECT SUM('a' IS NOT NULL) *"R1"*;
- SELECT SUM('a' IS NOT NULL) + SUM('b' IS NOT NULL) *"R1"*;
- SELECT SUM(job = 'manager') FROM emp;
- SELECT SUM(CASE WHEN job = 'manager' THEN 1 END) *"R1"* FROM emp;



IMP:

Here, job = 'manager' returns 1 (TRUE) or 0 (FALSE), and SUM adds up the 1s.

Things to… Remember:

aggregate function - SUM

TODO

SUM([DISTINCT] *expr* | [*condition* | CASE WHEN *condition* END]) [*over_clause*]

- If there are no matching rows, SUM() **returns NULL.**
- SUM() may take a numeric argument , and it returns a sum of non-NULL values.

r = { -2, 1, 2, -1, 3, -2, 1, 2, 1 }

- SELECT SUM(c1) *"R1"* FROM r;
- SELECT SUM(IF(c1 >= 0, c1, NULL)) *"R1"* FROM r;
- SELECT SUM(IF(c1 < 0, c1, NULL)) *"R1"* FROM r;

- SELECT custId, type, amount, CASE type WHEN 'd' THEN amount WHEN 'c' THEN -amount END *amount* FROM transactions;
- SELECT SUM(job = 'manager') managers, SUM(job = 'clerk') clerks, SUM(job = 'salesman') salesman FROM emp;
- SELECT SUM(sal > 3000) *R1*, SUM(sal <= 3000) *R2* FROM emp;
- SELECT SUM(gender = 'M') *R1*, SUM(gender = 'F') *R2* FROM emp;
- SELECT SUM(comm IS NOT NULL) *R1*, SUM(comm IS NULL) *R2* FROM emp;

- Running total

aggregate function - SUM

SUM with OVER clause

➢ SUM([DISTINCT] *expr* | [*condition* | CASE WHEN *condition* END]) [OVER([PARTITION BY *expr1, expr2,* . . .] ORDER BY *expr1* [ASC|DESC], . . .)]

Table:- yearly_revenue

id	year	amount
1	2014	7.75
2	2015	9.23
3	2016	5.44
4	2017	7.77
.	.	.
.	.	.
.	.	.
10	2022	6.83
11	2023	2.90

Output →

Output

id	year	amount	running total of amount per year
1	2014	7.75	7.75
2	2015	9.23	16.98
.	.		.
.	.		.
.	.		.
11	2023	2.90	55.43

- SELECT id, year, amount, SUM(amount) OVER(ORDER BY id) *"running total of amount per year"* FROM yearly_revenue;

- Count range-wise salary including total rows

sum with case when . . . end

todo

Table:- emp

Empno	Ename	. . .	sal	isActive
6001	GITA	. . .	1750	. . .
6129	VRUSHALI	. . .	1750	. . .
6473	SHARMIN	. . .	1750	. . .
6781	BANDISH	. . .	1750	. . .
. . .	. . .	. . .	. . .	. . .
9400	SANGITA	. . .	1750	. . .
9473	SUPRIYA	. . .	1750	. . .

Output

total	low_salary < 2000	mid_salary between 2000 and 4000	high_salary > 4000
28	18	7	3

Output →

- SELECT COUNT(*) *total*, COUNT(CASE WHEN sal < 2000 THEN 1 END) *low_salary*, COUNT(CASE WHEN sal BETWEEN 2000 AND 4000 THEN 1 END) *mid_salary*, COUNT(CASE WHEN sal > 4000 THEN 1 END) AS *high_salary* FROM emp;

- Count active and in-active staff.

sum with case when . . . end

todo

Table:- emp

Empno	Ename	. . .	sal	isActive
6001	GITA	. . .	1750	1
6129	VRUSHALI	. . .	1750	1
6473	SHARMIN	. . .	1750	1
6781	BANDISH	. . .	1750	1
. . .	. . .	. . .	. . .	. . .
9400	SANGITA	. . .	1750	1
9473	SUPRIYA	. . .	1750	1

Output →

Output

active_staff	inactive_staff
22	6

- SELECT COUNT(CASE WHEN isActive = 1 THEN 1 END) active_staff, COUNT(CASE WHEN isActive = 0 THEN 1 END) inactive_staff FROM emp;

- TODO

sum function- examples

todo

Table:- stages

CenterID	Case_ID	Stage1	Stage2	Stage3	Stage4	Stage5
C1	1	2024-01-01	NULL	NULL	2024-01-13	NULL
C1	2	2024-01-05	2024-01-10	NULL	NULL	NULL
C2	3	NULL	2024-01-10	NULL	NULL	2024-01-20
C3	4	2024-01-05	2024-01-12	2024-01-12	2024-01-14	2024-01-20
C3	5	2024-01-10	2024-01-15	NULL	NULL	NULL
C3	6	NULL	NULL	NULL	2024-01-15	NULL

Output →

Output

CenterID	**Stage1**	**Stage2**	**Stage3**	**Stage4**	**Stage5**
C1	2	1	0	1	0
C1	0	1	0	0	1
C2	2	2	1	2	1

- SELECT centerID, SUM(Stage1 IS NOT NULL) AS *stage1*, SUM(Stage2 IS NOT NULL) AS *stage2*, SUM(Stage3 IS NOT NULL) AS *stage3*, SUM(Stage4 IS NOT NULL) AS *stage4*, SUM(Stage5 IS NOT NULL) AS *stage5* FROM stages GROUP BY centerID;

Things to… Remember:

aggregate function - MAX

TODO

MAX([DISTINCT] *expr* | [CASE WHEN *condition* END]) [*over_clause*]

- If there are no matching rows, MAX() **returns NULL**.
- MAX() may take a string, number, and date argument, and it returns a maximum of non-NULL values.

e.g.

- SELECT MAX(1) *"R1"*;
- SELECT MAX(NULL) *"R1"*;
- SELECT MAX('VIKAS') *"R1"* ;
- SELECT MAX(1) *"R1"* WHERE True;
- SELECT MAX(1) *"R1"* WHERE False;
- SELECT MAX(1) *"R1"* FROM emp;
- SELECT MAX(sal) *"R1"* FROM emp WHERE empno = -1;
- SELECT MAX(sal) *"Maximum Salary"* FROM emp;
- SELECT job, MAX(sal) *"Maximum Salary"* FROM emp GROUP BY job;

- SELECT MAX('a' IS NOT NULL) *"R1"*;
- SELECT MAX('a' IS NOT NULL) + MAX('b' IS NOT NULL) *"R1"*;
- SELECT MAX(CASE WHEN job = 'manager' THEN sal END) *"R1"* FROM emp;
- SELECT MAX(CASE WHEN job = 'manager' THEN sal END) - MIN(CASE WHEN job = 'manager' THEN sal END) FROM emp;

- Running maximum.

aggregate function - MAX

todo

MAX with OVER clause

➢ MAX([DISTINCT] *expr*) [OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)]

day	companyName	stockPrice
1	Reliance Industries	1429.70
2	Reliance Industries	1429.70
. .	Reliance Industries	. .
8	Only Vimal	3494.93
9	Only Vimal	2833.22
. .	Only Vimal	. .
17	Raymonds	1254.00
18	Raymonds	2833.22

- Stock prices: Track highest price so far.
- Test scores: Get highest score so far by date.
- Sales: Running max revenue per day.

day	companyName	stockPrice	**Running Max Stock Price**
8	Only Vimal	3494.93	3494.93
9	Only Vimal	2833.22	3494.93
10	Only Vimal	1904.43	3494.93
11	Only Vimal	4303.98	4303.98
12	Only Vimal	2357.97	4303.98

- SELECT day, companyName, stockPrice, MAX(stockprice) OVER(PARTITION BY companyName ORDER BY day) as *'Running Max Stock Price'* FROM stocks;

- TODO

max function- examples

todo

Table:- activity

machine_id	process_id	activity_type	timestamp
0	0	Start	0.712
0	0	End	1.520
0	1	Start	3.140
0	1	End	4.120
1	0	Start	0.550
1	0	End	1.550
1	1	Start	0.430
1	1	End	1.420
2	0	Start	4.100
2	0	End	4.512
2	1	Start	2.500
2	1	End	5.000

Output →

machine_id	process_id	Start	End	Difference
0	0	0.712	1.520	0.808
0	1	3.140	4.120	0.980
1	0	0.550	1.550	1.000
1	1	0.430	1.420	0.990
2	0	4.100	4.512	0.412
2	1	2.500	5.000	2.500

- SELECT machine_id, process_id, MAX(CASE WHEN activity_type = 'start' THEN timestamp END) *Start*, MAX(CASE WHEN activity_type = 'end' THEN timestamp END) *End*, MAX(CASE WHEN activity_type = 'end' THEN timestamp END) - MAX(CASE WHEN activity_type = 'start' THEN timestamp END) *Difference* FROM activity GROUP BY machine_id, process_id;

Things to… Remember:

aggregate function - MIN

todo

TODO

MIN([DISTINCT] *expr* | [CASE WHEN *condition* END]) [*over_clause*]

- If there are no matching rows, MIN() **returns NULL.**
- MIN() may take a string, number, and date argument, and it returns a minimum of non-NULL values.

e.g.

- SELECT MIN(1) *"R1"*;
- SELECT MIN(NULL) *"R1"*;
- SELECT MIN(1) *"R1"* WHERE True;
- SELECT MIN(1) *"R1"* WHERE False;
- SELECT MIN(1) *"R1"* FROM emp;
- SELECT MIN(sal) *"R1"* FROM emp WHERE empno = -1;
- SELECT MIN(sal) *"Minimum Salary"* FROM emp;
- SELECT job, MIN(sal) *"Minimum Salary"* FROM emp GROUP BY job;

- SELECT MIN('a' IS NOT NULL) *"R1"*;
- SELECT MIN('a' IS NOT NULL) + MIN('b' IS NOT NULL) *"R1"*;
- SELECT MIN(CASE WHEN job = 'manager' THEN sal END) *"R1"* FROM emp;
- SELECT MIN(CASE WHEN job = 'manager' THEN sal END) - MAX(CASE WHEN job = 'manager' THEN sal END) FROM emp;

- Running minimum.

aggregate function - MIN

todo

MIN with OVER clause

➢ MIN([DISTINCT] *expr* | [CASE WHEN *condition* END]) [OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)]

- Track lowest temperature so far in a weather table.
- Show lowest score a student has received up to a given test.
- Show stock price dips in financial analysis.

Things to… Remember:

aggregate function - COUNT

In MySQL, COUNT(*), COUNT(*column*), and COUNT(DISTINCT *column*) are used to count rows/values, but they behave differently, especially with NULL and duplicate values.

COUNT([DISTINCT] *expr* | [CASE WHEN *condition* END]) [*over_clause*]

- If there are no matching rows, COUNT() **returns 0.**
- Returns a count of the number of non-NULL values.
- COUNT(*) is somewhat different in that it returns a count of the number of rows retrieved, whether or not they contain NULL values.
- COUNT (*) is a special implementation of the COUNT function that returns the count of all the rows in a specified table.
- COUNT (*) also considers Nulls and duplicates.
- SQL does not allow the use of DISTINCT with COUNT (*)

Note:

- **COUNT (*):** Returns a number of rows in a table including duplicates rows and rows containing null values in any of the columns.
- **COUNT (EXP):** Returns the number of non-null values in the column identified by expression.
- **COUNT (DISTINCT EXP):** Returns the number of unique, non-null values in the column identified by expression.
- **COUNT (DISTINCT *):** is illegal.

- Count NULLs in each columns.

aggregate function - COUNT

todo

➢ COUNT([DISTINCT] *expr*) [OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)]

Table:- countNulls

id	ename	city	phone
1	saleel	baroda	NULL
2	sharmin	pune	NULL
3	ruhan	NULL	1111
4	vrushali	NULL	NULL
5	neel	NULL	2222

Output

Total Rows	NULLs in city	NULLs in phone
5	3	2

- SELECT COUNT(*) 'Total Rows', COUNT(*) - COUNT(city) *'NULLs in city'*, COUNT(*) - COUNT(phone) *'NULLs in phone'* FROM countnulls;

- SELECT COUNT(*) 'Total Rows', SUM(CASE WHEN city IS NULL THEN 1 END) *'NULLs in city'*, SUM(CASE WHEN phone IS NULL THEN 1 END) *'NULLs in phone'* FROM countnulls;

- Same number for NULL data.

aggregate function - COUNT

todo

COUNT with OVER clause

➢ COUNT([DISTINCT] *expr*) [OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)]

Table:- countNumbers

id	c1
1	1
2	NULL
3	NULL
.	.
.	.
.	.
6	2
	.
	.
	.
19	NULL
20	6

Output

id	c1	Output
1	1	1
2	NULL	1
3	NULL	1
.	.	.
.	.	.
.	.	.
6	2	2
	.	.
	.	.
	.	.
19	NULL	5
20	6	6

- SELECT id, c1, COUNT(c1) OVER(ORDER BY id) *Output* FROM countNumbers;

- WITH ***cte*** AS (SELECT id, c1, COUNT(c1) OVER(ORDER BY id) *R1* FROM countnumbers) SELECT id, c1, FIRST_VALUE(c1) OVER(PARTITION BY *R1*) *R2* FROM ***cte***;

- Same number for NULL data.

aggregate function - COUNT

COUNT with OVER clause

➢ COUNT([DISTINCT] *expr*) [OVER([PARTITION BY *expr1, expr2,* . . .] ORDER BY *expr1* [ASC|DESC], . . .)]

Table:- stocks

day	companyName	stockPrice	Since
1	Reliance Industries Ltd	1429.70	1966
. .	. .	. .	NULL .
8	Only Vimal	3494.93	1975
. .	. .	. .	NULL .
17	Raymonds	1254.00	1925
. .	. .	. .	NULL .

Output

day	companyName	stockPrice	Since	**Output**
1	Reliance Industries Ltd	1429.70	1966	1
. .	. .	. .	NULL .	1 .
8	Only Vimal	3494.93	1975	2
. .	. .	. .	NULL .	2 .
17	Raymonds	1254.00	1925	3
. .	. .	. .	NULL .	3 .

- SELECT day, companyName, stockPrice, COUNT(Since) OVER(ORDER BY day) *Output* FROM stocks;

Things to… Remember:

aggregate function - COUNT

TODO

COUNT([DISTINCT] *expr* | [CASE WHEN *condition* END]) [*over_clause*]

e.g.

- SELECT COUNT(*) *"R1"*;
- SELECT COUNT(NULL) *"R1"*;
- SELECT COUNT(*) *"R1"* WHERE True;
- SELECT COUNT(*) *"R1"* WHERE False;
- SELECT COUNT(0) FROM emp;
- SELECT COUNT(1) FROM emp;
- SELECT COUNT(*) FROM emp WHERE empno = -1;
- SELECT COUNT(comm) *"R1"* FROM emp;
- SELECT job, COUNT(*) *"R1"* FROM emp GROUP BY job;
- SELECT CASE WHEN sal <= 1500 THEN 'low' WHEN sal > 1501 AND sal < 3000 THEN 'medium' WHEN sal >= 3000 THEN 'high' END *"R1"*, COUNT(*) FROM emp GROUP BY *R1*;

- SELECT COUNT('a' IS NOT NULL) *"R1"*;
- SELECT COUNT('a' IS NOT NULL) + COUNT('b' IS NOT NULL) *"R1"*;
- SELECT COUNT(CASE WHEN job = 'manager' THEN 1 END) *"R1"* FROM emp;

IMP:

- SELECT COUNT(job = 'manager') *"R1"* FROM emp;

1. COUNT(job = 'manager') returns a Boolean value (TRUE or FALSE).
2. When used COUNT(job = 'manager') , MySQL **doesn't count** how many are TRUE. Instead:
   - **It counts** how many values are **non-NULL**.
   - Since job = 'manager' is always **non-NULL** (TRUE or FALSE), it counts all rows.

Things to… Remember: *aggregate function – GROUP_CONCAT*

Function that **joins static string and multiple row values into a single comma-separated string.**

```
GROUP_CONCAT([DISTINCT] expr OR CASE WHEN condition END
                    [ORDER BY { unsigned_integer | col_name | expr }  [ASC | DESC] [, col_name . . .]]
                    [SEPARATOR str_val])
```

e.g.

- SELECT job, GROUP_CONCAT(ename) FROM emp GROUP BY job;

- SELECT job, CONCAT(GROUP_CONCAT(ename),' (', COUNT(*), ')') *"R1"* FROM emp GROUP BY job;
- SELECT job, CONCAT(GROUP_CONCAT(sal),' (', MAX(sal), ')') *"R1"* FROM emp GROUP BY job;
- SELECT job, CONCAT(GROUP_CONCAT(sal),' (', SUM(sal), ')') *"R1"* FROM emp GROUP BY job;
- SELECT custid, GROUP_CONCAT(MONTHNAME(orderdate)) *"R1"* FROM ord GROUP BY custid;

- SELECT GROUP_CONCAT(CONCAT('`', COLUMN_NAME, '`')) FROM information_schema.columns WHERE table_schema = 'db1' AND table_name = 'emp';

- TODO

group_concat with case when . . .end

➢ GROUP_CONCAT(CASE WHEN *condition* THEN *value* ELSE *value* END)

Table:- emp

ename	job	gender
.	.	.
.	.	.
.	.	.
.	.	.
.	.	.
.	.	.
.	.	.
.	.	.
.	.	.

Output

job	male_employees	female_employees
analyst	scott,ford	supriya
clerk	smith,aarav,thomas,adams,james ,miller	natasha
.	.	.
.	.	.
.	.	.
salesman	allen,ward,martin,turner,grass,aarush	null
vice president	fred	null

```
SELECT job , GROUP_CONCAT(CASE WHEN gender = 'M' THEN ename END)  male_employees, GROUP_CONCAT(CASE WHEN
gender = 'F' THEN ename END)  female_employees FROM emp GROUP BY job;
```

Things to… Remember:

aggregate function examples

TODO

- ```
  SELECT productname,
  SUM(CASE WHEN storelocation = 'North' THEN totalsales END) North,
  SUM(CASE WHEN storelocation = 'South' THEN totalsales END) South,
  SUM(CASE WHEN storelocation = 'East' THEN totalsales END) East,
  SUM(CASE WHEN storelocation = 'West' THEN totalsales END) West,
  SUM(CASE WHEN storelocation = 'Central' THEN totalsales END) Central,
  SUM(totalsales) TotalSales FROM pivot_table GROUP BY productname;
  ```
- ```
  SELECT productName, COUNT(CASE WHEN storelocation = 'North' THEN productName END) North, COUNT(CASE WHEN storelocation = 'South' THEN productName END) South, COUNT(CASE WHEN storelocation = 'East' THEN productName END) East, COUNT(CASE WHEN storelocation = 'West' THEN productName END) West  FROM pivot_table GROUP BY productName;
  ```

- ```
  SELECT itemname,
  COUNT(CASE WHEN color = 'white' AND size = 'medium' THEN 1 END) White,
  COUNT(CASE WHEN color = 'dark' AND size = 'medium' THEN 1 END) Dark,
  COUNT(CASE WHEN color = 'pastel' AND size = 'medium' THEN 1 END) Pastel FROM shop GROUP BY itemname;
  ```
  ```

aggregate function – JSON_OBJECTAGG

TODO

JSON_OBJECTAGG (key, value) [OVER([PARTITION BY *expr1*, *expr2,* . . .] ORDER BY *expr1* [ASC|DESC], . . .)]

- **key** → Column or expression used as the JSON key (must be non-NULL and unique, otherwise duplicates are ignored except last one wins).
- **value** → Column or expression used as the JSON value.

aggregate function – JSON_ARRAYAGG

It's like GROUP_CONCAT(), but instead of returning a comma-separated string, it returns a **JSON array**.

JSON_ARRAYAGG(col_or_expr) [OVER([PARTITION BY *expr1, expr2, . . .*] ORDER BY *expr1* [ASC|DESC], . . .)]

Takes two column names or expressions as arguments, the first of these being used as a key and the second as a value, and returns a JSON object containing key-value pairs. Returns NULL if the result contains no rows.

```
SELECT DISTINCT COUNT(JOB) FROM EMP
SELECT COUNT(DISTINCT JOB) FROM EMP
```



- SET *SQL_MODE* = '';
- SET *SQL_MODE* = 'ONLY_FULL_GROUP_BY';

$$G_{A_1, A_2, \ldots, A_n,\ G_{F_1(A_1), F_2(A_2), \ldots, F_m(A_m)}}(r)$$

Group Attributes

Group Function Attributes

group by clause

Remember:

- Standard SQL does not allow you to use an ALIAS in the GROUP BY clause, however, MySQL supports this.
- GROUP BY is used in conjunction with aggregating functions to group the results by the unaggregated columns.

Note:

- DISTINCT (if used outside an aggregation function) that is superfluous.

e.g.

```
SELECT DISTINCT COUNT(ename) FROM emp;
```

select - group by

- Columns selected for output can be referred to in ORDER BY and GROUP BY clauses using column names, column aliases, or column positions. Column positions are integers and begin with 1

- If you use GROUP BY, output rows are sorted according to the GROUP BY columns as if you had an ORDER BY for the same columns. To avoid the overhead of sorting that GROUP BY produces, add ORDER BY NULL.

- If a query includes GROUP BY but you want to avoid the overhead of sorting the result, you can suppress sorting by specifying ORDER BY NULL.

For example:

- `SELECT job, COUNT(*) FROM emp GROUP BY job ORDER BY NULL;`
- `SELECT * FROM emp ORDER BY FIELD (job, 'MANAGER', 'SALESMAN');`

This function's will produce a single value for an entire group or a table.

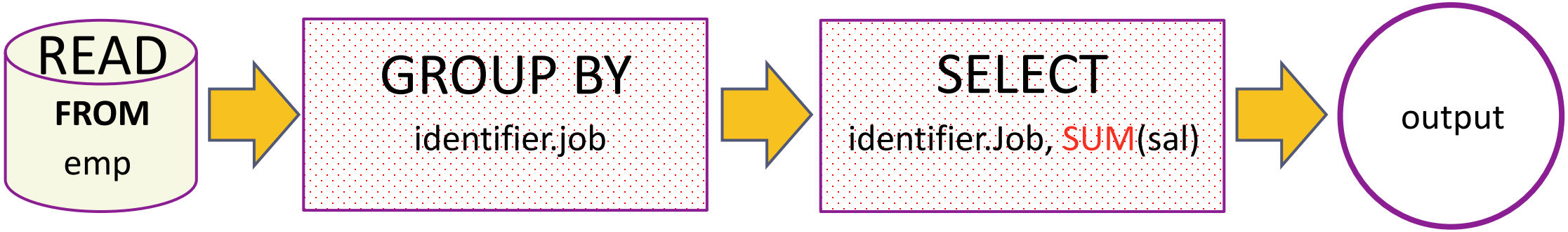
select - group by

You can use GROUP BY to group values from a column, and, if you wish, perform calculations on that column.

SELECT G_1, G_2, . . . , $F_1(A_1)$, $F_2(A_2)$, . . . FROM r_1, r_2, . . .

[GROUP BY {G_1, G_2, . . . | *expr* | *position*}, . . . [WITH ROLLUP]]

- SELECT job, SUM(sal) FROM emp GROUP BY job;
- SELECT job, SUM(sal) FROM emp GROUP BY job WITH ROLLUP;



job	sum(sal)
CLERK	9250
SALESMAN	9525
MANAGER	13675
ANALYST	6000
PRESIDENT	5000

job	sum(sal)
ANALYST	6000
CLERK	9250
MANAGER	13675
PRESIDENT	5000
SALESMAN	9525
NULL	43450

select - group by

- SET *SQL_MODE* = '';
- SET *SQL_MODE* = 'ONLY_FULL_GROUP_BY';

Examples:

- SELECT job, sal + 1001 FROM emp GROUP BY job;
- SELECT job, COUNT(job) FROM emp GROUP BY COUNT(job); # error
- SELECT job, sal + 1001 FROM emp GROUP BY sal + 1001;
- SELECT LENGTH(ename) *R1* FROM emp GROUP BY *R1*;
- SELECT job, SUM(sal) FROM emp GROUP BY job WITH ROLLUP;
- SELECT COALESCE (job, 'Total'), SUM(sal) FROM emp GROUP BY job WITH ROLLUP;

- SELECT CASE WHEN deptno = 10 THEN deptno WHEN deptno = 20 THEN deptno END *R1*, COUNT(*) FROM emp GROUP BY *R1*;

Remember:

- The WHERE clause **cannot refer** to aggregate functions. [WHERE SUM(sal) = 5000 # Error]
- The HAVING clause **can refer** to aggregate functions. [HAVING SUM(sal) = 5000 # No Error]

having clause

The MySQL **HAVING clause** is used in the SELECT statement to specify filter conditions for a group of rows. **HAVING clause** is often used with the GROUP BY clause. When using with the GROUP BY clause, we can apply a filter condition to the columns that appear in the GROUP BY clause.

Note:

- Columns given in **HAVING** clause must be present in selection-list.

e.g.

1. `SELECT COUNT(*) FROM emp HAVING deptno=10;` *
2. `SELECT deptno, COUNT(*) FROM emp GROUP BY deptno HAVING job='manager';` *

* ERROR: Unknown column '. . .' in 'having clause'

- **HAVING** is merged with **WHERE** if you do not use GROUP BY or Aggregate Functions (COUNT(), . . .)

select - having

SELECT G_1, G_2, . . . , $F_1(A_1)$, $F_2(A_2)$, . . . FROM r_1, r_2, . . .

[GROUP BY {G_1, G_2, . . . | *expr* | *position*}, . . . [WITH ROLLUP]]

[HAVING *having_condition*]

- SELECT COUNT(*), job FROM emp GROUP BY job HAVING COUNT(*) > 2;



count(*)	job
6	CLERK
6	SALESMAN
5	MANAGER

having example

SELECT G_1, G_2, . . . , $F_1(A_1)$, $F_2(A_2)$, . . . FROM r_1, r_2, . . .

[GROUP BY {G_1, G_2, . . . | *expr* | *position*}, . . . [WITH ROLLUP]]

[HAVING *having_condition*]

- SELECT descrip, COUNT(*) FROM product GROUP BY descrip HAVING COUNT(DISTINCT price) > 1; // retrieves products with the same name but different prices.

When WHERE and HAVING clause are used together in a SELECT query with aggregate function, WHERE clause is applied first on individual rows and only rows which pass the condition is included for creating groups. Once group is created, HAVING clause is used to filter groups based upon condition specified.

difference between where and having clause

where and having clause

Remember:

- ***WHERE*** clause can be used with - SELECT, UPDATE, and DELETE statements, where as ***HAVING*** clause can only be used with the SELECT statement.
- ***WHERE*** clause filters rows before aggregation (GROUPING), where as, ***HAVING*** clause filters groups, after the aggregations are performed.
- ***WHERE*** is used before the 'GROUP BY' clause if required and ***HAVING*** is used after the 'GROUP BY' clause.
- Aggregate functions (SUM***,*** MIN***,*** MAX***,*** AVG and COUNT) cannot be used in the ***WHERE*** clause, unless it is in a sub query contained in a ***WHERE*** or ***HAVING*** clause, whereas, aggregate functions can be used in ***HAVING*** clause.

Note:

- The WHERE clause acts as a pre-filter where as HAVING clause acts as a post-filter.

WHERE Vs HAVING

where vs having

WHERE	HAVING
Implemented in row operations.	Implemented in column operations.
Single row	Summarized row or group or rows.
It only fetches the data from particular rows or table according to the condition.	It only fetches the data from grouped data according to the condition.
Aggregate Functions **c**cannot appear in WHERE clause.	Aggregate Functions Can appear in HAVING clause.
Used with SELECT and other statements such as UPDATE, DELETE or either one of them.	Used with SELECT statement only.
Pre-filter	Post-filter
GROUP BY Comes after WHERE.	GROUP BY Comes before HAVING.

window function

Note:

MySQL does not support these window function features.

- DISTINCT syntax for aggregate functions.
- Nested window functions
- Window function cannot be the part of WHERE condition
- **OVER() function apply aggregate functions row by row without collapsing rows like GROUP BY does.**

window function

Use ORDER BY *expr* with PARTITION BY *expr* to see the effect of PARTITION BY *expr*.

- RANK() OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)
- DENSE_RANK() OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)
- ROW_NUMBER() OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)
- LAG(*expr* [, *N* [, *default*]]) OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)
- LEAD(*expr* [, *N* [, *default*]]) OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)
- FIRST_VALUE(expr) OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . . [ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING])
- LAST_VALUE(expr) OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . . [ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING])
- NTH_VALUE(*expr*, *N*) OVER([PARTITION BY *expr1*, *expr2*, . . .] ORDER BY *expr1* [ASC|DESC], . . .)

Note:

The N and default argument in the function is optional.

- ***expr***: It can be a column or any built-in function.
- ***N***: It is a positive value which determine number of rows preceding/succeeding the current row. If it is omitted in query then its default value is 1.
- ***default***: It is the default value return by function in-case no row precedes/succeeds the current row by N rows. If it is missing then it is by default NULL.

- ***UNBOUNDED PRECEDING***: Start from the first row in the partition.
- ***UNBOUNDED FOLLOWING***: End at the last row in the partition.

- Repeating numbers 1 to 5 cyclically

window function - row_number

```
ROW_NUMBER() OVER (
    PARTITION BY expr1, expr2,  . . .     -- Optional
    ORDER BY expr1 [ ASC|DESC ],  . . . ])
) AS repeating numbers 1 to 5 cyclically
```

This pattern is useful for **grouping into buckets** of 5 (like food groupings, shift rotations, etc.).

Output1	Output2	item_name	item_unit
1	1	biscuit	packet
2	1	cakes	Pcs
3	1	cheese	packet
4	1	butter	packet
5	1	bread	packet
1	2	eggs	Pcs
2	2	salt	packet
3	2	jam	bottle
4	2	pizza	Pcs
5	2	burger	Pcs
1	1	juice	bottle
2	1	milk	bottle
3	1	ice-cream	packet

- WITH ***cte*** AS (SELECT ROW_NUMBER() OVER() *R1*, ename FROM emp) SELECT CEIL(*R1*/5) *a*, CASE WHEN *R1*%5 = 0 THEN 5 ELSE *R1*%5 END *b* FROM ***cte***;

- SELECT CASE WHEN *R1*%5 <> 0 THEN *R1*%5 ELSE 5 END *Output1*, CEIL(*R1*/5) *Output2*, item_name, item_unit FROM (SELECT ROW_NUMBER() OVER(ORDER BY NULL) *R1*, item_name, item_unit FROM food) t1;

- Get temperature between (1->2, 3->4, 5->6)

window function - row_number

Table:- temperature

id	date	temperature
1	2025-12-01	10
2	2025-12-02	8
3	2025-12-03	7
4	2025-12-04	43
5	2025-12-05	21
6	2025-12-06	10
.	.	.
.	.	.
.	.	.
30	2025-12-30	14
31	2025-12-31	38

Output →

output

id	date	temperature
1	2025-12-01	10
2	2025-12-02	8
3	2025-12-03	7
4	2025-12-04	43
5	2025-12-05	21
6	2025-12-06	10
.	.	.
.	.	.
.	.	.
30	2025-12-30	14
31	2025-12-31	38

- WITH ***cte*** AS (SELECT id, date, temperature, CEIL(id/2) *R1*, ROW_NUMBER() OVER(PARTITION BY (CEIL(id/2)) ORDER BY temperature desc) *R2* FROM temperature) SELECT * FROM ***cte*** WHERE *R2=1*;

window function - LEAD

- Get the temperature which is higher on next day

Table:- temperature

id	date	temperature
1	2025-12-01	10
2	2025-12-02	8
3	2025-12-03	7
4	2025-12-04	43
5	2025-12-05	21
6	2025-12-06	10
. . .	. . .	. . .
30	2025-12-30	14
31	2025-12-31	38

Output →

output

id	date	Current Temperature	Next Day Temperature	R1
2	2025-12-02	17	45	1
5	2025-12-05	9	29	1
6	2025-12-06	29	32	1
9	2025-12-09	23	29	1
10	2025-12-10	29	31	1
12	2025-12-12	22	44	1
. . .	. . .	. . .	. . .	. . .
28	2025-12-28	36	42	1
30	2025-12-30	18	32	1

```
WITH cte AS (SELECT id, date, temperature 'Current Temperature', LEAD(temperature) OVER(ORDER BY date) 'Next Day Temperature', temperature < LEAD(temperature) OVER(ORDER BY date) R1 FROM temperature) SELECT * FROM cte WHERE R1 = 1;
```

window function- examples

- Last N Records.
- Train Time Difference
- Train Departure Time and Arrival Time
- Which room a student stayed in the longest

```
SELECT ROW_NUMBER() OVER() R1, emp.* FROM emp;
SELECT RANK() OVER(PARTITION BY job ORDER BY sal) R1, ename, sal, job FROM emp;
SELECT DENSE_RANK() OVER(PARTITION BY job ORDER BY sal) R1, ename, sal, job FROM emp;
SELECT ordid, total, SUM(total) OVER(ORDER BY ordid) FROM ord;
SELECT * FROM (SELECT ROW_NUMBER() OVER() 'last N records', emp.* FROM emp) d WHERE `last N records` > (SELECT COUNT(*) - 2 FROM emp);  // Print last n records
SELECT id, trainID stationname, timing, TIMEDIFF(LEAD(timing) OVER(PARTITION BY trainid ORDER BY timing), timing) trainTimeDifference FROM traintimetable; // train time difference between to stations.
SELECT id, trainID, stationname 'From Station', timing  'Departure Time' ,  LEAD(stationname) OVER(PARTITION BY trainid ORDER BY id) 'To Station', LEAD(timing) OVER(PARTITION BY trainid ORDER BY id) 'Arrival Time' FROM traintimetable;
SELECT *, DATEDIFF(endDate, startDate) AS "Stay Duration", RANK() OVER (ORDER BY DATEDIFF(endDate, startDate) DESC) AS room_rank FROM student a,  room_assigned b WHERE a.studentID = b.studentID; // Which room a student stayed in the longest
```

- Running total.

window function - with SUM

```
SUM(column_name) OVER (
   PARTITION BY expr1, expr2,  . . .     -- Optional
   ORDER BY expr1 [ ASC|DESC ],  . . . ])
) AS running_total
```

- SELECT custId, type, amount, CASE type WHEN 'd' THEN amount WHEN 'c' THEN -amount END *Amount* FROM transactions;
- SELECT year, quarter, amount, SUM(amount) OVER(PARTITION BY year ORDER BY quarter) *R1* FROM quarter_revenue;
- SELECT custId, type, amount, SUM(CASE type WHEN 'd' THEN -amount WHEN 'c' THEN amount END) OVER(PARTITION BY custID ORDER BY _id) *Balance* FROM transactions;
- SELECT ordid, custid, total, SUM(total) OVER(PARTITION BY custid ORDER BY ordid) *R1* FROM ord;

- TODO.

window function- LAG

Table:- cars

car	day	cumulative_distance
Car1	day1	50
Car1	day2	120
.	.	.
.	.	.
Car2	day1	175
.	.	.
.	.	.
Car3	day1	133
Car3	day2	120
.	.	.

Output

car	day	cumulative_distance
Car1	day1	50
Car1	day2	120
.	.	.
.	.	.
Car2	day1	175
.	.	.
.	.	.
Car3	day1	133
Car3	day2	120
.	.	.

window function- LAG

- Difference between first row of total column and immediate next row of total customer wise.

Table:- ord

ordid	custid	orderDate	total
606	100	1986-07-14	3.40
609	100	1986-08-01	97.50
.	.	.	.
.	.	.	.
610	101	1987-01-07	101.40
.	.	.	.
.	.	.	.
602	102	1986-06-05	56.00
603	102	1986-06-05	224.00
.	.	.	.

Output

ordid	custid	orderDate	Current Row Value	**Next Row Value**	**Difference**
606	100	1986-07-14	3.40	3.40	0.00
609	100	1986-08-01	97.50	3.40	94.10
.	.	.	.	.	.
.	.	.	.	.	.
610	101	1987-01-07	101.40	101.40	0.00
.	.	.	.	.	.
.	.	.	.	.	.
602	102	1986-06-05	56.00	56.00	168.00
603	102	1986-06-05	224.00	56.00	-179.00
.	.	.	.	.	.

- SELECT t1.*, `Current Row Value` - `Next Row Value` *"Difference"* FROM (SELECT ordid, custid, orderDate, total *"Current Row Value"*, LAG(total, 1, total) OVER(PARTITION BY custid ORDER BY ordid) *"Next Row Value"* FROM ord) T1;

- Get First Value and Last Value of partition.

window function- FIRST_VALUE

Table:- traintimetable

ID	trainID	stationname
1	22863	Mumbai central
2	22863	Surat
3	22863	Baroda
4	22863	Ahmedabad JN
5	22868	Pune JN
6	22868	Uruli
7	22868	Daund chord line
8	22868	Ahmednagar
9	22868	Belapur

Output →

Output

trainid	From Station	To Station
22863	Mumbai central	Ahmedabad JN
22868	Pune JN	Belapur
.	.	.
.	.	.
.	.	.
.	.	.

- SELECT DISTINCT trainid, FIRST_VALUE(stationname) OVER(PARTITION BY trainid ORDER BY id) *"From Station"*, LAST_VALUE(stationname) OVER(PARTITION BY trainid ORDER BY id ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING) *"To Station"* FROM traintimetable;

- Fill down data.

window function- FIRST_VALUE

Table:- job_list

ID	jobRole	skills
1	Data Engineer	SQL
2	NULL	redis
3	NULL	mongoDB
4	Web Developer	HTML
5	NULL	CSS
6	NULL	nodeJS
7	Data Scientists	Python
8	NULL	ML
9	NULL	AI

Output →

Output

ID	**jobRole**	skills
1	Data Engineer	SQL
2	Data Engineer	redis
3	Data Engineer	mongoDB
4	Web Developer	HTML
5	Web Developer	CSS
6	Web Developer	nodeJS
7	Data Scientists	Python
8	Data Scientists	ML
9	Data Scientists	AI

Type 1

- SELECT id, FIRST_VALUE(jobRole) OVER(PARTITION BY *R1* ORDER BY id) *fillJob*, skills FROM (SELECT id, jobRole, skills, COUNT(jobRole) OVER(ORDER BY id) *R1* FROM job_list) T1;

- Fill down data with ID column.

window function- FIRST_VALUE

Table:- job_list

ID	jobRole	skills
1	Data Engineer	SQL
2	NULL	redis
3	NULL	mongoDB
4	Web Developer	HTML
5	NULL	CSS
6	NULL	nodeJS
7	Data Scientists	Python
8	NULL	ML
9	NULL	AI

Output →

Output

ID	jobRole	skills
1	Data Engineer	SQL
2	Data Engineer	redis
3	Data Engineer	mongoDB
4	Web Developer	HTML
5	Web Developer	CSS
6	Web Developer	nodeJS
7	Data Scientists	Python
8	Data Scientists	ML
9	Data Scientists	AI

Type 2

```
SELECT id, FIRST_VALUE(jobRole) OVER(PARTITION BY R1 ORDER BY id) fillJob, skills FROM (SELECT id, jobRole, skills,
SUM(CASE WHEN jobRole IS NULL THEN 0 ELSE 1 END) OVER(ORDER BY id) R1 FROM job_list)  T1;
```

- Fill down data without ID column.

window function- FIRST_VALUE

Table:- brands

category	brandName
chocolates	5-star
NULL	dairy milk
NULL	perk
NULL	eclair
Biscuits	britannia
NULL	good day
NULL	boost

Output →

Output

category	brandName
chocolates	5-star
chocolates	dairy milk
chocolates	perk
chocolates	eclair
Biscuits	britannia
Biscuits	good day
Biscuits	boost

- SELECT FIRST_VALUE(category) OVER(PARTITION BY *R2* ORDER BY NULL) category, brandName FROM (SELECT category, brandName, COUNT(category) OVER(ORDER BY *R1*) *R2* FROM (SELECT ROW_NUMBER() OVER(ORDER BY NULL) *R1*, category, brandName FROM brands) T1) T1;

window function- LEAD

- next metro station.

```
LEAD(expr [, N [, default ] ] ) OVER (
 [ PARTITION BY expr1, expr2,  . . .     -- Optional
   ORDER BY expr1 [ ASC|DESC ],  . . . ]) AS R1
```

Table:- pune_metro_stations

stop_id	stop_name	.	next_stop_id
1	Vanaz	.	2
2	Anand Nagar	.	3
3	Ideal Colony	.	4
.	.	.	.
.	.	.	.
.	.	.	.
15	Kalyani Nagar	.	16
16	Ramwadi	.	NULL

Output

stop_id	current_stop	next_stop
1	Vanaz	Anand Nagar
2	Anand Nagar	Ideal Colony
3	Ideal Colony	.
.	.	.
.	.	.
.	.	.
15	Kalyani Nagar	Ramwadi
16	Ramwadi	NULL

- SELECT route_no, stop_name AS *current_stop*, LEAD(stop_name) OVER(PARTITION BY route_no ORDER BY stop_id) AS *next_stop* FROM pune_metro_stations;

user-defined variables

user-defined variables

TODO

session variable

Remember:

- A user variable name can contain other characters if you quote it as a string or identifier (for example, @'my-var', @"my-var", or @`my-var`).
- User-defined variables are session specific. A user variable defined by one client cannot be seen or used by other clients.
- All variables for a given client session are automatically freed when that client exits.
- User variable names are not case sensitive. Names have a maximum length of 64 characters.
- If the value of a user variable is selected in a result set, it is returned to the client as a string.
- If you refer to a variable that has not been initialized, it has a value of NULL and a type of string.

e.g. `SELECT @variable_name;`

user-defined variables

You can store a value in a user-defined variable in one statement and refer to it later in another statement. This enables you to pass values from one statement to another.

SET *@variable_name* = *expr* [, *@variable_name* = *expr*] . . .

Remember:

- for SET, either = or := can be used as the assignment operator.
- You can also assign a value to a user variable in statements (SELECT, …) other than SET. In this case, the assignment operator must be := and **not** = because latter is treated as the **comparison operator =**.

- set @v1 = 1001, @v2 := 2, @v3 = 'Saleel';
- set @v1 = 1001, @v2 = 2, @v3 := @v1 + @v2;
- SELECT @v1 := MIN(sal), @v2 := MAX(SAL) FROM emp;
- SELECT @v1, @v2, @v3;

```
SELECT VARIABLE_NAME , VARIABLE_VALUE FROM
PERFORMANCE_SCHEMA.USER_VARIABLES_BY_THREAD;
```

user-defined variables

Note:

- User variables are intended to provide data values. They cannot be used directly in an SQL statement as an identifier or as part of an identifier.

e.g.

```
SET @v1 = 'ENAME';              #WHERE ENAME IS COLUMN NAME.
SELECT @v1 FROM emp;
```

	@x
	ename
	ename
	ename
	ename
	ename
	ename
	ename
	ename
	ename
	ename
	ename
	ename

rownum

```
mysql> SELECT * FROM (SELECT @cnt := @cnt + 1 "R1", emp.* FROM emp, (SELECT @cnt := 0) T1) T2 WHERE "R1" > @cnt - 7;
```

select - rownum

```
mysql> SET @rank = 0;
mysql> SELECT @row := @row + 1 as rownum , emp.* FROM emp;
mysql> SELECT @row := @row + 1 as rownum , emp.* FROM emp,  (SELECT @row := 0) as E;
mysql> SELECT @row := @row + 1 ,E.* FROM (SELECT job, sal FROM emp  GROUP BY job ORDER BY SAL DESC) E , (SELECT
        @row := 0) EE;
```

ROWNUM	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO	BONUSID	USER NAME	PWD
1	7839	KING	PRESIDENT	NULL	1981-11-17 00:00:00	5000.00	NULL	10	1	KING	r50mom
2	7698	BLAKE	MANAGER	7839	1981-05-01 00:00:00	2850.00	NULL	30	1	BLAKE	sales@2017
3	7782	CLARK	MANAGER	7839	1981-06-09 00:00:00	2450.00	NULL	10	3	CLARK	r50mom
4	7566	JONES	MANAGER	7839	1981-04-02 00:00:00	2975.00	NULL	20	4	JONES	a12recmom
5	7654	MARTIN	SALESMAN	7698	1981-09-28 00:00:00	1250.00	1400.00	30	6	MARTIN	sales@2017
6	7499	ALLEN	SALESMAN	7698	1981-02-20 00:00:00	1600.00	300.00	30	4	ALLEN	sales@2017
7	7844	TURNER	SALESMAN	7698	1981-09-08 00:00:00	1500.00	0.00	30	5	TURNER	sales@2017
8	7900	JAMES	CLERK	7698	1981-12-03 00:00:00	950.00	NULL	30	2	JAMES	sales@2017
9	7521	WARD	SALESMAN	7698	1981-02-22 00:00:00	1250.00	500.00	30	1	WARD	sales@2017
10	7902	FORD	ANALYST	7566	1981-12-03 00:00:00	3000.00	NULL	20	4	FORD	a12recmom

common sql statements mistakes

Examples:

- SELECT ename, job, sal, comm FROM emp WHERE comm = NULL; #using comparison operator to check NULL
- SELECT job, COUNT(job) FROM emp; #not giving group by clause
- SELECT job, COUNT(job) FROM emp WHERE COUNT(job) > 4; #use of aggregate function in where clause
- SELECT job, deptno, COUNT(job) FROM emp GROUP BY job; #not giving all the columns in group by clause
- SELECT ename, COUNT(job) FROM emp GROUP BY ename; #grouping by a unique key
- SELECT ename, sal, sal + 1000 *R1* FROM emp WHERE *R1* > 2400; #use of alias name in where clause
- SELECT ename, sal FROM emp WHERE sal BETWEEN (1000 and 4000); #use of () in between comparison operator

r1 = { col1, col2, col3 }

- INSERT INTO ***r1*** VALUSE(10, 10); #number of values are less than the number of columns in the table
- INSERT INTO ***r1*** VALUSE(10, 10, 10, 10); #number of values are more than the number of columns in the table

Remember:

- A subquery must be enclosed in parentheses.
- Use single-row operators with single-row subqueries, and use multiple-row operators with multiple-row subqueries.
- If a subquery (inner query) returns a **NULL** value to the outer query, the outer query will not return any rows when using certain comparison operators in a WHERE clause.
- If ORDER BY occurs within a subquery and also is applied in the outer query, the outermost ORDER BY takes precedence.
- If LIMIT occurs within a subquery and also is applied in the outer query, the outermost LIMIT takes precedence.

sub-queries

A subquery is a SELECT statement within another statement.

Note:

- You may use comparison operators such as **<>**, **<**, **>**, **<=**, and **>=** with a single row subquery.
- Multiple row subquery returns one or more rows to the outer SQL statement. You may use the **IN**, **ANY**, or **ALL** operator in outer query to handle a subquery that returns multiple rows.

subqueries

A subquery is a SELECT statement within another statement.

Remember:

A subquery may occur in:

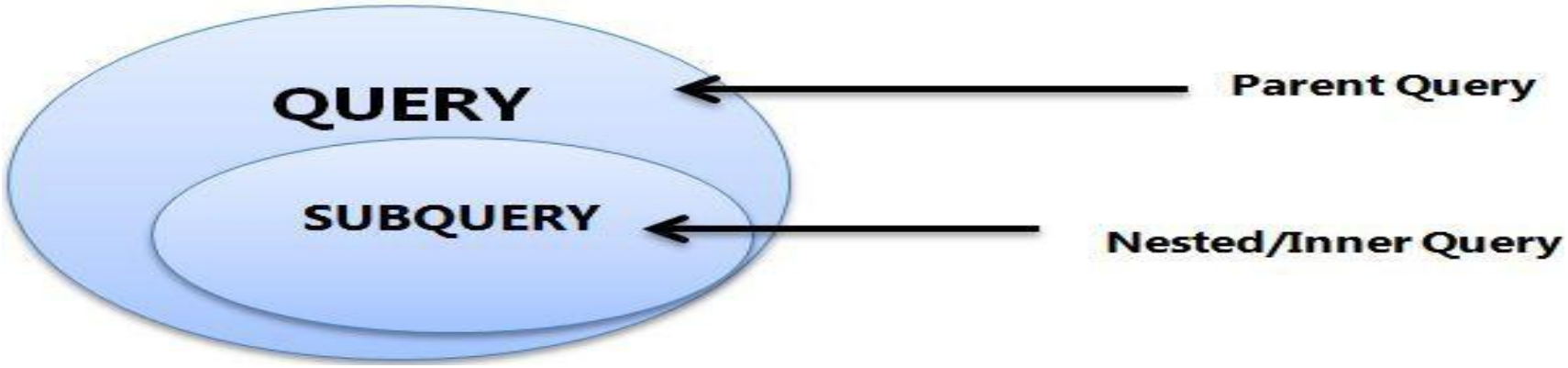
- a SELECT clause
- a FROM clause
- a WHERE clause
- a HAVING clause

Note:

A subquery's outer statement can be any one of:

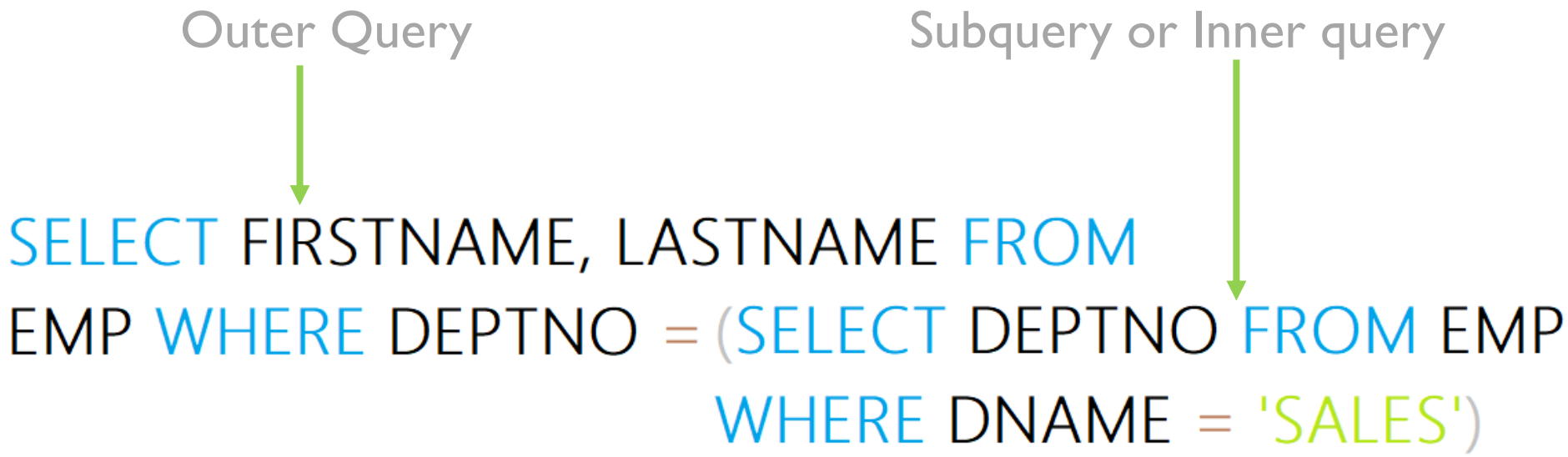
- SELECT
- INSERT
- UPDATE
- DELETE
- CREATE

- INSERT . . . SELECT . . .
- UPDATE . . . SELECT . . .
- DELETE . . . SELECT . . .
- CREATE TABLE . . . AS SELECT . . .
- CREATE VIEW . . . AS SELECT . . .
- DECLARE CURSOR . . . AS SELECT . . .
- EXPLAIN SELECT . . .



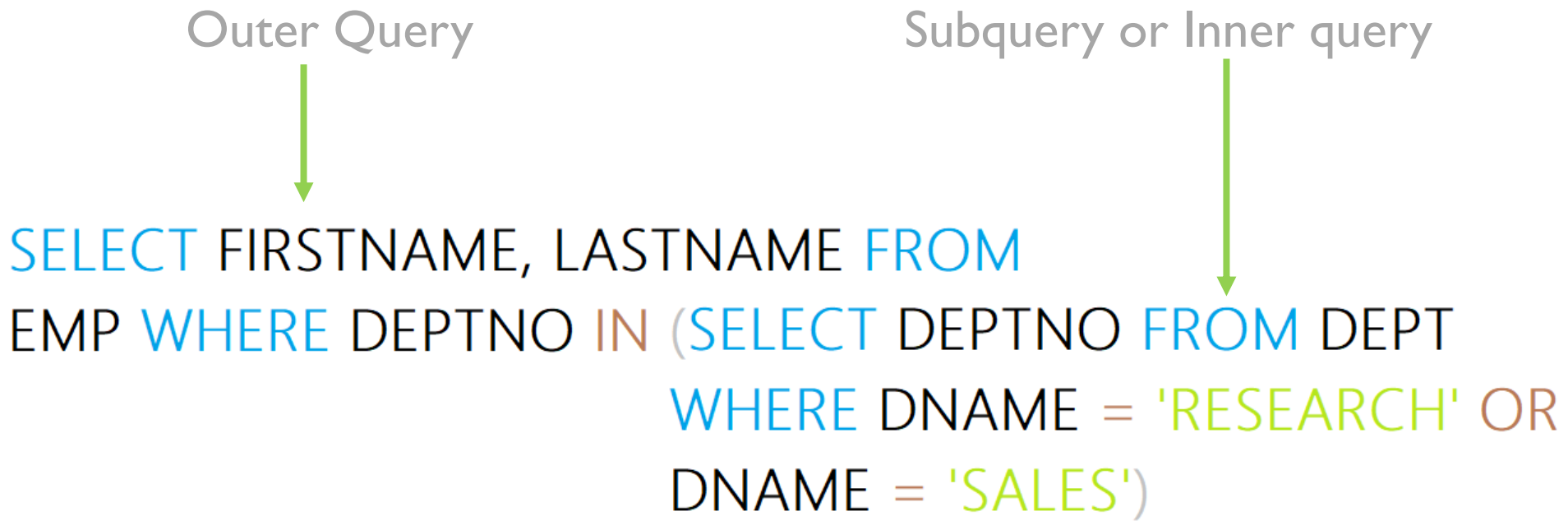
single row subqueries

A single row subquery returns ***zero or one row*** to the outer SQL statement. You can place a subquery in a ***WHERE*** clause, a ***HAVING*** clause, or a ***FROM*** clause of a SELECT statement.



multiple row and column subqueries

A multiple row subquery returns ***one or more rows*** to the outer SQL statement. You may use the ***IN***, ***ANY***, or ***ALL*** operator in outer query to handle a subquery that returns multiple rows.



subqueries with insert

TODO

INSERT INTO table_name [(column1 [, column2])] (*subquery*)

```
INSERT INTO EMPLOYEE SELECT * FROM EMP
```

```
INSERT INTO EMPLOYEE (FIRSTNAME, LASTNAME)
    SELECT FIRSTNAME, LASTNAME FROM EMP
```

subqueries with update

TODO

UPDATE table_name SET column_name = (*subquery*) [WHERE (*subquery*)]

subqueries with delete

TODO

DELETE FROM table_name [WHERE (*subquery*)]

types of subqueries

- The Subquery as Scalar Operand – SELECT clause
- Comparisons using Subqueries – WHERE / HAVING clause (*Single row subquery*)
- Subqueries in the FROM Clause – INLINE VIEWS (*Derived Tables*)
- Subqueries with ALL, ANY, IN, or SOME – WHERE / HAVING clause (*Multiple row subquery*)
- Subqueries with EXISTS or NOT EXISTS

A **scalar subquery** is a subquery that returns **exactly one value (one row, one column)** and is often used inside ***SELECT***, ***WHERE***, or ***HAVING*** clauses—just like a constant.

WITH var(param) as (SELECT) [CTE] Common Table Expressions

```
WITH a AS (p1, p2, p3, p4) AS (SELECT * FROM dept) SELECT p1, p2, p3, p4 FROM a;
```

the subquery as scalar operand

A **scalar subquery** is a subquery that returns **exactly one value (one row, one column)** and is often used inside ***SELECT***, ***WHERE***, or ***HAVING*** clauses—just like a constant.

the subquery as scalar operand

TODO

SELECT A_1, A_2, A_3, (*subquery*) as A_4, . . . FROM *r*

Remember:

- A scalar subquery is a subquery that returns **exactly one column value from one row**.
- A scalar subquery is a simple operand, and you can use it almost anywhere a single column value is legal.

Note:

- If the subquery returns 0 rows then the value of scalar subquery expression is **null**.
- if the subquery returns more than one row then MySQL returns an **error**.

Think:

- `SELECT (SELECT 1, 2); #error`
- `SELECT (SELECT ename, sal FROM emp); #error`
- `SELECT (SELECT * FROM emp); #error`
- `SELECT (SELECT NULL + 1);`
- `SELECT ename, (SELECT dname FROM dept WHERE emp.deptno = dept.deptno) R1 FROM emp ;`

the subquery as scalar operand

e.g.

- SELECT (SELECT stdprice FROM price WHERE prodid = 100890 AND enddate IS NOT NULL) AS *"Standard Price"*, (SELECT minprice FROM price WHERE prodid = 100890 AND enddate IS NOT NULL) AS *"Minimum Price"*;

	Standard Price	Minimum Price
	54.00	40.50

- SELECT (SELECT stdprice FROM price WHERE prodid = 100890 AND enddate IS NOT NULL) - (SELECT minprice FROM price WHERE prodid = 100890 AND enddate IS NOT NULL) AS *"Price Difference"*;

	Price Difference
	13.50

- SELECT MAX(CASE WHEN job = 'manager' THEN sal END) - MIN(CASE WHEN job = 'manager' THEN sal END) FROM emp;

the subquery as scalar operand

TODO

1. Employee Salary vs Company Average
2. Latest Order in E-commerce
3. Highest-Paid Employee
4. Product Price above average price.
5. Customer with Highest Total Purchase
6. ……

subquery in the from clause

subqueries in the from clause

TODO

SELECT A_1, A_2, A_3, (*subquery*) as A_4, . . . FROM (*subquery*) [as] name, . . .

Note:

- Every table in a FROM clause must have a name, therefore the [AS] name clause is mandatory.

- SET @x := 0;
  SELECT * FROM (SELECT @x := @x + 1 as *R1*, emp.* FROM emp) DT WHERE *R1* = 5;
- SELECT * FROM (SELECT @cnt := @cnt + 1 *R1*, MOD(@cnt, 2) *R2*, emp.* FROM emp, (SELECT @cnt:=0) DT1) DT2 WHERE *R2* = 0;
- SELECT MIN(*R1*) FROM (SELECT COUNT(job) *R1* FROM emp GROUP BY job) *DT*;
- SELECT MAX(*R1*) FROM (SELECT COUNT(*) *R1* FROM actor_movie GROUP BY actorid) *DT*

	MAX(R1)
	5

- SELECT AVG(SUM(column1)) FROM t1 GROUP BY column1; //ERROR
- SELECT AVG(sum_column1)
  FROM (SELECT SUM(column1) AS *sum_column1*
  FROM t1 GROUP BY column1) AS *t1*;

subqueries in the from clause

- SELECT * FROM (SELECT job, hiredate, DENSE_RANK() OVER(PARTITION BY job ORDER BY hiredate) *"First Employee"*, DENSE_RANK() OVER(PARTITION BY job ORDER BY hiredate DESC) *"Last Employee"* FROM emp GROUP BY job, hiredate ORDER BY job, hiredate) t1 WHERE `First Employee` = 1 OR `Last Employee` = 1;
- SELECT custid, SUM(total) FROM (SELECT custid, total, ROW_NUMBER() OVER(PARTITION BY custid ORDER BY total DESC) *R1* FROM ord) t1 WHERE *R1* <= 2 GROUP BY custid; // SUM of first two orders.
- SELECT * FROM (SELECT ordid, custid, orderdate, total, DENSE_RANK() OVER(PARTITION BY custid ORDER BY ordid) *R1* FROM ord) T1 WHERE *R1* = 1; //First order of every customers

- SELECT ename, t1.* FROM (SELECT * FROM (SELECT job, hiredate, DENSE_RANK() OVER(PARTITION BY job ORDER BY hiredate) *"First Employee"*, DENSE_RANK() OVER(PARTITION BY job ORDER BY hiredate DESC) *"Last Employee"* FROM emp GROUP BY job, hiredate ORDER BY job, hiredate) t1 WHERE *`First Employee`* = 1 OR *`Last Employee`* = 1) t1, emp WHERE (t1.job, t1.hiredate) = (emp.job, emp.hiredate) ORDER BY job, hiredate;

- top 3 densely populated cities.

subqueries in the from clause

TODO

Table:- population

ID	City	area	. . .	population
1	Ahmedabad	530	. . .	8253000
2	Surat	462	. . .	7490000
3	Vadodara	220	. . .	2233000
4	Rajkot	170	. . .	1934000
. . .	. . .	. . .	. . .	. . .
31	Deesa	20	. . .	111160

output

city	area	population	**Density**	**rnk**
Botad	10	169000	16900.0000	1
Surat	462	7490000	16212.1212	2
Ahmedabad	530	8253000	15571.6981	3

- SELECT * FROM (SELECT city, area, population, population/area Density, RANK() OVER(ORDER BY population/area desc) *rnk* FROM population) t1 WHERE *rnk* <= 3;

- TODO

subqueries in the from clause

TODO

Table:- category

Category	SubCategory	Price
Chips	Bingo	10
Chips	Lays	40
Chips	Kurkure	60
Chocolate	Dairy Milk	120
Chocolate	Five Star	40
Chocolate	Perk	25
Chocolate	Munch	5
Biscuits	Oreo	120
Sauce	Kissan	30

Output

Category	SubCategory	Price	Top	cnt
Biscuits	Oreo	120	1	1
Chips	Kurkure	60	1	3
Chips	Lays	40	2	3
Choclate	Dairy Milk	120	1	4
Chocolate	Five Star	40	2	4

Q1. Write a query selects the top 2 priced items per category *if* there are multiple items in the category, and includes items priced over 50 if they are the only item in the category.

- SELECT category, subcategory, price, top, *cnt* FROM (SELECT category, subcategory, price, DENSE_RANK() OVER(PARTITION BY category ORDER BY price DESC) Top, COUNT(subcategory) OVER(PARTITION BY category) *cnt* FROM category) t1 WHERE (top <= 2 AND cnt > 1) or (*cnt* = 1 AND price > 50);

control flow functions - case

Table:- year_of_service

ID	year_of_service
1	2021-05-30
2	2022-09-12
3	2023-11-19
4	2024-02-01
5	2025-08-04
6	2024-07-21
7	2022-12-18
8	2020-02-07
9	. . .
10	2020-09-23

Output

ID	year_of_service	**service_bucket**
1	2021-05-30	4-5 year
2	2022-09-12	2-3 year
3	2023-11-19	2-3 year
4	2024-02-01	0-1 year
5	2025-08-04	0-1 year
6	2024-07-21	0-1 year
7	2022-12-18	2-3 year
8	2020-02-07	4-5 year
9	. . .	. . .
10	2020-09-23	4-5 year

Q1. Write a query to create service buckets based on the number of years of service.

If year is 2025

- SELECT id, year_of_service, CASE WHEN *R2* BETWEEN 0 AND 1 THEN '0-1 year' WHEN *R2* BETWEEN 2 AND 3 THEN '2-3 year' WHEN *R2* BETWEEN 4 AND 5 THEN '4-5 year' END *service_bucket* FROM (SELECT id, year_of_service, YEAR(CURDATE()) - YEAR(year_of_service) *R2* FROM year_of_service WHERE YEAR(CURDATE()) - YEAR(year_of_service) >= 0) t1;

- TODO

subqueries in the from clause

Table:- emp

ename	job
SCOTT	ANALYST
FORD	ANALYST
SUPRIYA	ANALYST
SMITH	CLERK
AARAV	CLERK
THOMAS	CLERK
ADAMS	CLERK
JAMES	CLERK
MILLER	CLERK
NATASHA	CLERK
GITA	Compliance officers
VRUSHALI	Compliance officers
SANGITA	Compliance officers

Output

Row to columns

job	Col1	Col2	Col3	Col4	Col5	Col6	Col7
ANALYST	SCOTT	FORD	SUPRIYA	NULL	NULL	NULL	NULL
CLERK	SMITH	AARAV	THOMAS	ADAMS	JAMES	MILLER	NATASHA
COMPLIANCE OFFICERS	GITA	VRUSHALI	SANGITA	NULL	NULL	NULL	NULL
MANAGER	JONES	BLAKE	CLARK	HOFFMAN	ALEX	NULL	NULL
PRESIDENT	KING	NULL	NULL	NULL	NULL	NULL	NULL
PUBLIC RELATION	SHARMIN	BANDISH	NULL	NULL	NULL	NULL	NULL
SALESMAN	ALLEN	WARD	MARTIN	TURNER	GRASS	AARUSH	NULL
VICE PRESIDENT	FRED	NULL	NULL	NULL	NULL	NULL	NULL

```
SELECT job, MAX(CASE WHEN R1 = 1 THEN ename END) Col1, MAX(CASE WHEN R1 = 2 THEN ename END) Col2, MAX(CASE WHEN R1 = 3 THEN ename END) Col3, MAX(CASE WHEN R1 = 4 THEN ename END) Col4, MAX(CASE WHEN R1 = 5 THEN ename END) Col5, MAX(CASE WHEN R1 = 6 THEN ename END) Col6, MAX(CASE WHEN R1 = 7 THEN ename END) Col7 FROM (SELECT ename, job, ROW_NUMBER() OVER(PARTITION BY job) R1 FROM emp) t1 GROUP BY job;
```

subqueries in the from clause

Table:- ord

custid	orderdate	total
106	1986-05-01	2.40
102	1986-06-05	56.00
102	1986-06-05	224.00
106	1986-06-15	698.00
106	1986-07-14	8324.00
100	1986-07-14	3.40
104	1986-07-18	5.60
104	1986-07-25	35.20
100	1986-08-01	97.50
101	1987-01-07	101.40
.	.	.
.	.	.
.	.	.

Output

First and Last Order for every customer

custid	orderdate	total	Order Type
100	1986-07-14	3.40	First Order
100	1987-03-15	730.00	Last Order
101	1987-01-07	101.40	First Order
102	1986-06-05	56.00	First Order
102	1986-06-05	224.00	First Order
102	1987-02-15	3510.50	Last Order
103	1987-02-03	764.00	First Order
104	1986-07-18	5.60	First Order
104	1987-02-23	232.87	Last Order
.	.	.	.
.	.	.	.

- SELECT custid, orderdate, total, CASE WHEN `First Order` = 1 THEN *"First Order"* WHEN `Last Order` = 1 THEN *"Last Order"* END *"Order Type"* FROM (SELECT custid, orderdate, total, DENSE_RANK() OVER(PARTITION BY custid ORDER BY orderdate) *'First Order'*, DENSE_RANK() OVER(PARTITION BY custid ORDER BY orderdate DESC) *'Last Order'* FROM ord ORDER BY custid, orderdate) t1 WHERE `First Order` = 1 OR `Last Order` = 1;

- TODO

subqueries in the from clause

Combine city with all other city

Table:- city

ID	cityName
1	Baroda
2	Surat
3	Rajkot
4	Anand

Output →

Output

cityName	cityName
Baroda	Surat
Baroda	Bharuch
Baroda	Rajkot
Baroda	Jamnagar
Surat	Bharuch
Surat	Rajkot
Surat	Jamnagar
Bharuch	Rajkot
Bharuch	Jamnagar
Rajkot	Jamnagar

- SELECT c1.cityName, c2.cityName FROM city c1, city c2 WHERE c1.id < c2.id ORDER BY c1.id;

- TODO

subqueries in the from clause

Combined teams with all remaining teams

Table:- ipl

Team_Names
PBKS
RCB
GT
MI

Output →

Output

Team_Names	Team_Names
PBKS	RCB
RCB	GT
PBKS	GT
GT	MI
RCB	MI
PBKS	MI

- SELECT T1.Team_Names, T2.Team_Names FROM (SELECT ROW_NUMBER() OVER(ORDER BY NULL) *R1*, ipl.* FROM ipl) T1, (SELECT ROW_NUMBER() OVER(ORDER BY NULL) *R1*, ipl.* FROM ipl) T2 WHERE T1.*R1* < T2.*R1*;

subqueries in the from clause

Input Table:- Order1

CustomerId	ItemId	ItemName	Quantity	Price
1	1	Butter	3	35
1	2	Bread	1	20
1	3	Soap	4	23
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

Running Total

Calculate Total = Quantity * Price

Output

CustomerId	ItemId	ItemName	Quantity	Price	**Total Price**	**Running Total**
1	1	Butter	3	35	105	105
1	2	Bread	1	20	20	125
1	3	Soap	4	23	92	217
1	4	Biscuit	2	15	30	247
2	10	Chocolate	8	10	80	80
3	13	Coke	8	25	200	970

- SELECT t1.*, SUM(*TotalPrice*) OVER(PARTITION BY customerid ORDER BY ItemID) *'Running Total'* FROM (SELECT customerid, ItemID, itemname, quantity, price, quantity * price *TotalPrice* FROM order1) t1;

- TODO

subqueries in the from clause

Swap cityName

Table:- swapCity

ID	ename	cityName
1	Saleel	Baroda
2	Sharmin	Rajkot
3	Vrushali	Surat
4	Ruhan	Bharuch
5	Nitish	Jamnagar
6	Deep	Anand
7	Neel	Bhuj

Output →

Output

ID	ename	**cityName**
1	Saleel	Rajkot
2	Sharmin	Baroda
3	Vrushali	Bharuch
4	Ruhan	Surat
5	Nitish	Anand
6	Deep	Jamnagar
7	Neel	Bhuj

- SELECT ename, cityName, CASE WHEN id%2 = 1 THEN LEAD(cityName, 1, cityName) OVER(ORDER BY id) WHEN id%2 = 0 THEN LAG(cityName, 1, cityName) OVER(ORDER BY id) END *'Swap City'* FROM swapCity;

- SELECT ename, cityName, CASE WHEN *R1*%2 = 1 THEN LEAD(cityName, 1, cityName) OVER(ORDER BY *R1*) WHEN *R1*%2 = 0 THEN LAG(cityName, 1, cityName) OVER(ORDER BY R1) END *'Swap City'* FROM (SELECT ename, cityName, ROW_NUMBER() OVER(ORDER BY NULL) *R1* FROM swapCity) T1;

- TODO

subqueries in the from clause

Swap items

Table:- cafe

order_id	item
1	Chow Mein
2	Pizza
3	Veg Nuggets
4	Paneer Butter Masala
5	Spring Rolls
6	Veg Burger
7	Paneer Tikka
8	Paneer Cheese Masala
9	Onion Spring Rolls
10	Momos
11	Paneer Tikka Masala
12	Vada Pav
13	Masala Dosa

Output →

Output

order_id	item	**Output**
1	Chow Mein	Pizza
2	Pizza	Chow Mein
3	Veg Nuggets	Paneer Butter Masala
4	Paneer Butter Masala	Veg Nuggets
5	Spring Rolls	Veg Burger
6	Veg Burger	Spring Rolls
7	Paneer Tikka	Paneer Cheese Masala
8	Paneer Cheese Masala	Paneer Tikka
9	Onion Spring Rolls	Momos
10	Momos	Onion Spring Rolls
11	Paneer Tikka Masala	Vada Pav
12	Vada Pav	Paneer Tikka Masala
13	Masala Dosa	Masala Dosa

subqueries in the from clause

3 Consecutive numbers

Table:- cNumbers

ID	num
1	1
2	1
3	1
4	2
5	1
6	2
7	2
8	7
9	7
10	7
11	5
12	5
13	2
14	4
15	4
16	4

Output →

Output1

id	3 Consecutive num
1	1
2	1
3	1
8	7
9	7
10	7
14	4
15	4
16	4

Output2

3 Consecutive num
1
7
4

- SELECT * FROM (SELECT id, num, COUNT(num) OVER(PARTITION BY *R1*) `*3 Consecutive num*` FROM (SELECT id, num, id - ROW_NUMBER() OVER(PARTITION BY num ORDER BY id) *R1* FROM cnumbers) t1) t2 WHERE `*3 Consecutive num*` >= 3;

- SELECT t1.num FROM (SELECT num, LAG(num, 1) OVER (ORDER BY id) AS *prev1*, LAG(num, 2) OVER (ORDER BY id) AS *prev2* FROM cNumbers) t1 WHERE t1.num = t1.prev1 AND t1.num = t1.prev2;

subqueries in the from clause

3 Consecutive Seats

Table:- hall_ticket

ID	status
1	1
2	0
3	1
4	1
5	1
6	0
7	1
8	1
9	0
10	1
11	0
12	1
13	0
14	1
15	1

Output →

Output

ID	> 3 Consecutive Seats
3	3
4	3
5	3
17	4
18	4
19	4
20	4
27	3
28	3
29	3

- SELECT id, `*Consecutive Seats*` FROM (SELECT id, COUNT(*R1*) OVER(PARTITION BY *R1* ORDER BY *R1*) `*Consecutive Seats*` FROM (SELECT id, id - ROW_NUMBER() OVER(ORDER BY id) *R1* FROM hall_ticket WHERE status = 1) t1) t2 WHERE `*Consecutive Seats*` >= 3;

comparisons using subquery

comparisons using subqueries

Comparison Operators like : =, !=/<>, >, >=, <, <= ,<=>

TODO

SELECT A_1, A_2, A_3, (*subquery*) as A_4, . . . FROM *r* WHERE *p* = (*subquery*)

Remember:

- A subquery can be used before or after any of the comparison operators.
- The subquery can return **at most one value**.
- The value can be the result of an arithmetic expression or a function.

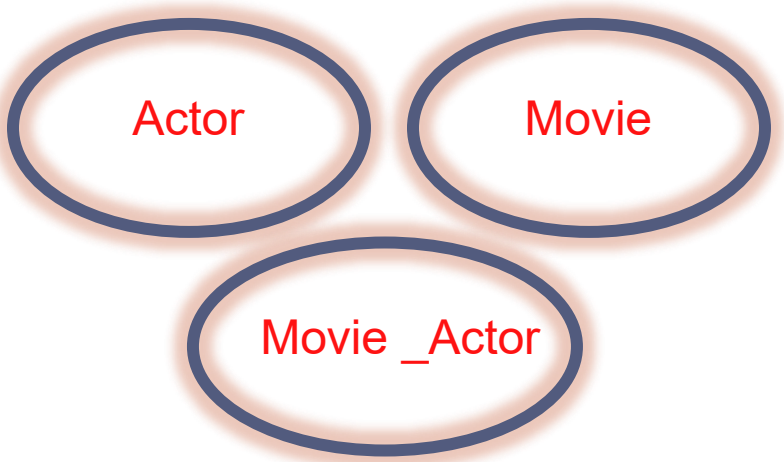
- SELECT * FROM emp WHERE deptno = (SELECT 5 + 5);
- SELECT * FROM emp WHERE sal = (SELECT MAX(sal) FROM emp);
- SELECT MAX(sal) FROM emp WHERE sal < (SELECT MAX(sal) FROM emp);
- SELECT * FROM emp WHERE sal > (SELECT sal FROM emp GROUP BY sal ORDER BY sal DESC limit 3, 1) ORDER BY sal DESC;

Following statements will raise an error.

- SELECT * FROM emp WHERE deptno = (SELECT deptno FROM dept WHERE deptno IN(10, 20));

comparisons using subqueries

movie : (movieid, name, release_date)
actor : (actorid, name)
actor_movie : (actorid, movieid)



- SELECT *a*.actorid, *a*.name FROM actor *a*, actor_movie *am* WHERE *a*.actorid = *am*.actorid GROUP BY *am*.actorid HAVING COUNT(*) = (SELECT MAX(*R1*) FROM (SELECT COUNT(*) *"R1"* FROM actor_movie GROUP BY actorid) *M*);

	actorid	name
	2	Akshay Kumar
	3	Salman Khan
	7	Madhuri Dixit

Remember:

- When used with a subquery, the word IN is an alias for =ANY
- NOT IN is not an alias for <>ANY, but for <>ALL

subquery with in, all, any, some

subqueries with in, all, any, some

- *operand comparison_operator* ANY (*subquery*)
- *operand* IN (*subquery*)
- *operand comparison_operator* SOME (*subquery*)
- *operand comparison_operator* ALL (*subquery*)

The word SOME is an alias for ANY.

- The **ANY** keyword, which must follow a comparison operator, means return TRUE if the comparison is TRUE for ANY of the values in the column that the subquery returns.
- The word **ALL**, which must follow a comparison operator, means return TRUE if the comparison is TRUE for ALL of the values in the column that the subquery returns.
- **IN** and **=ANY** are ***not synonyms*** when used with an expression list. **IN** can take an expression list, but **=ANY** cannot.

```
SELECT * FROM emp WHERE deptno IN (5 + 5, 10 + 10);
SELECT * FROM emp WHERE job =ANY (SELECT job FROM emp WHERE deptno IN(10, 20));
SELECT * FROM emp WHERE deptno =ANY (10, 20);    //error
```



subqueries with in, all, any, some

TODO

SELECT A_1, A_2, A_3, (*subquery*) as A_4, . . . FROM (*subquery*) [as] alias_name WHERE *p* IN (*subquery*)
SELECT A_1, A_2, A_3, (*subquery*) as A_4, . . . FROM (*subquery*) [as] alias_name WHERE *p* ANY (*subquery*)
SELECT A_1, A_2, A_3, (*subquery*) as A_4, . . . FROM (*subquery*) [as] alias_name WHERE *p* ALL (*subquery*)

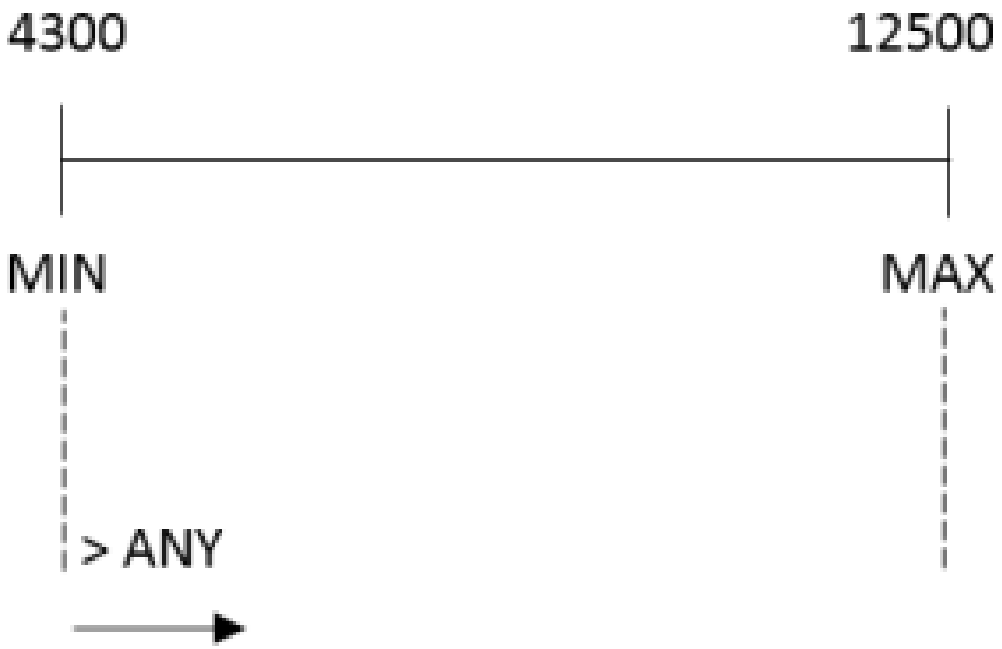
- Remember:

- SELECT * FROM emp WHERE deptno = SELECT deptno FROM dept WHERE dname = 'SALES’; // error
- SELECT * FROM emp WHERE deptno IN SELECT deptno FROM dept WHERE dname = 'SALES'; // error
- SELECT * FROM emp WHERE deptno IN SELECT * FROM dept WHERE dname = 'SALES’; // error

With ANY (*subquery*)

any / some

- "x = ANY (...)": The value must match one or more values in the list to evaluate to TRUE.
- "x != ANY (...)": The value must not match one or more values in the list to evaluate to TRUE.
- "x > ANY (...)": The value must be greater than the smallest value in the list to evaluate to TRUE.
- "x < ANY (...)": The value must be smaller than the biggest value in the list to evaluate to TRUE.
- "x >= ANY (...)": The value must be greater than or equal to the smallest value in the list to evaluate to TRUE.
- "x <= ANY (...)": The value must be smaller than or equal to the biggest value in the list to evaluate to TRUE.

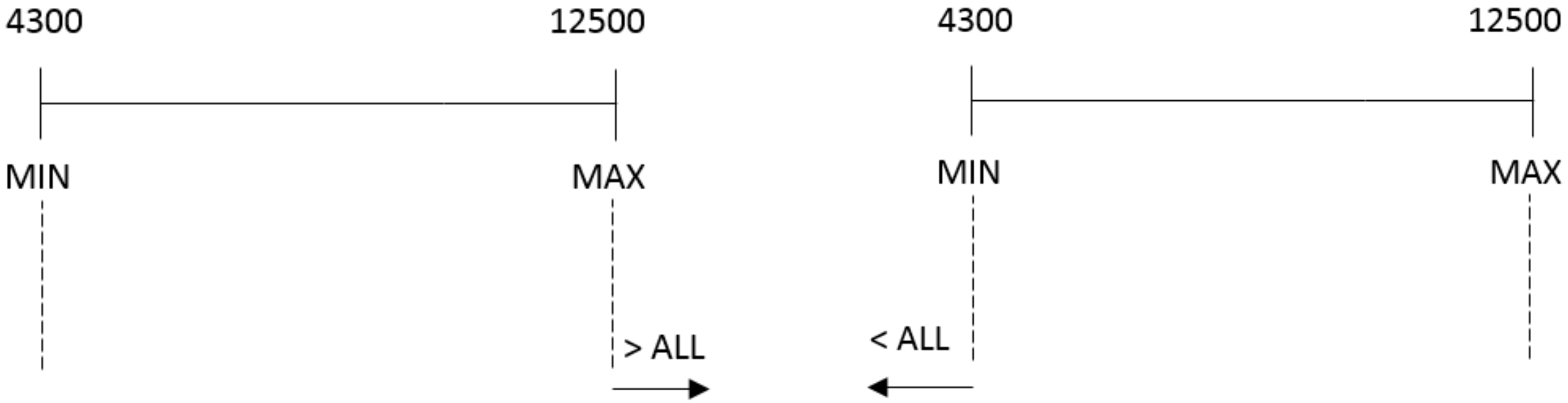




With ALL (*subquery*)

all

- "x = ALL (...)": The value must match all the values in the list to evaluate to TRUE.
- "x != ALL (...)": The value must not match any values in the list to evaluate to TRUE.
- "x > ALL (...)": The value must be greater than the biggest value in the list to evaluate to TRUE.
- "x < ALL (...)": The value must be smaller than the smallest value in the list to evaluate to TRUE.
- "x >= ALL (...)": The value must be greater than or equal to the biggest value in the list to evaluate to TRUE.
- "x <= ALL (...)": The value must be smaller than or equal to the smallest value in the list to evaluate to TRUE.



all

The expression is TRUE, if table T2 is empty.




```
SELECT * FROM EMP WHERE DEPTNO >ALL (SELECT C1 FROM T2) // This statement will return all rows from EMP table.
```

subqueries with in, all, any, some

You can use a subquery after a comparison operator, followed by the keyword IN, ALL, ANY, or SOME.

- SELECT * FROM emp WHERE deptno IN (SELECT deptno FROM emp WHERE deptno = 10 OR deptno = 20);
- SELECT * FROM emp WHERE sal >ALL (SELECT sal FROM emp WHERE ename = 'ADAMS' OR ename = 'TURNER');
- SELECT * FROM emp WHERE sal >ANY (SELECT sal FROM emp WHERE ename = 'ADAMS' OR ename = 'TURNER');
- SELECT * FROM emp WHERE sal >SOME (SELECT sal FROM emp WHERE ename = 'ADAMS' OR ename = 'TURNER');
- SELECT * FROM server WHERE id =ANY (SELECT id FROM runningserver); // match all the values in the list
- SELECT * FROM server WHERE id <ALL (SELECT id FROM runningserver); // must be smaller than the smallest value
- SELECT * FROM server WHERE id <ANY (SELECT id FROM runningserver); // smaller than the biggest value

SELECT * FROM d1 WHERE c1 not in (SELECT min(c1) FROM d1 GROUP BY deptno, dname, loc, walletid) ORDER BY deptno;

row subquery

row subqueries

Scalar or column Subqueries return a single value or a column of values. A row subquery is a subquery variant that returns a single row and can thus return more than one column value. You can use = , >, <, >=, <=, <>, !=, <=>

1. SELECT * from EMP where deptno, 1 = (SELECT deptno, 1 from DEPT where deptno=10);

2. SELECT * FROM emp WHERE ROW (deptno, 1) = (SELECT deptno, 1 FROM dept WHERE deptno = 10);

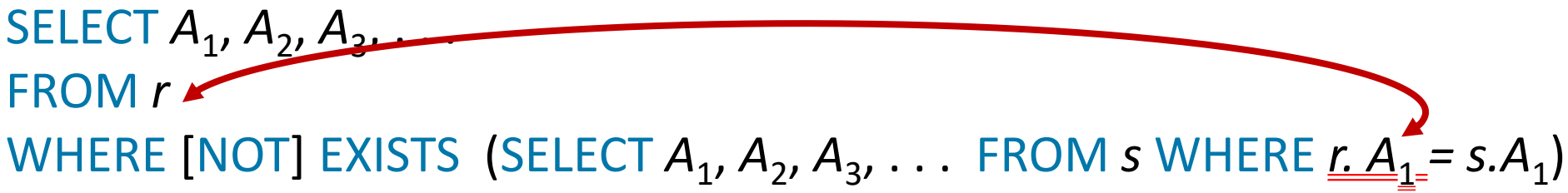
- SELECT * FROM emp WHERE EXISTS (SELECT 1);

subquery with exists or not exists

SELECT A_1, A_2, A_3, A_4, . . . FROM *r* WHERE [NOT] EXISTS (*subquery*)

subqueries with exists or not exists

The EXISTS operator tests for the existence of rows in the results set of the subquery. If a subquery row value is found, EXISTS subquery returns TRUE and in this case NOT EXISTS subquery will return FALSE.



The records will be displayed from outer SELECT statement….

- SELECT * FROM emp WHERE EXISTS (SELECT * FROM dept WHERE emp.deptno = dept.deptno);
- SELECT * FROM dept WHERE NOT EXISTS (SELECT deptno FROM emp WHERE emp.deptno = dept.deptno);
- SELECT * FROM emp *m* WHERE EXISTS (SELECT * FROM emp *e* WHERE *e*.mgr = *m*.empno);
- SELECT * FROM emp *m* WHERE NOT EXISTS (SELECT * FROM emp *e* WHERE *e*.mgr = *m*.empno);

- SELECT * FROM dept WHERE deptno IN (SELECT * FROM emp WHERE emp.deptno = dept.deptno);
- SELECT * FROM dept WHERE deptno NOT IN (SELECT * FROM emp WHERE emp.deptno = dept.deptno);

- SELECT * FROM emp *f* WHERE NOT EXISTS (SELECT * FROM emp m WHERE *f*.deptno = *m*.deptno AND gender = 'm');
- SELECT * FROM emp *m* WHERE NOT EXISTS (SELECT true FROM emp *f* WHERE *m*.deptno = *f*.deptno AND *f*.gender = 'f');

- TODO

subqueries in the from clause

Same orderID and orderDate orders

Table:- orderdetails

id	orderID	. . .	total
1	1	. . .	130
2	1	. . .	70
. . .	. . .	. . .	. . .
33	5	. . .	57
34	5	. . .	114

Output →

Output

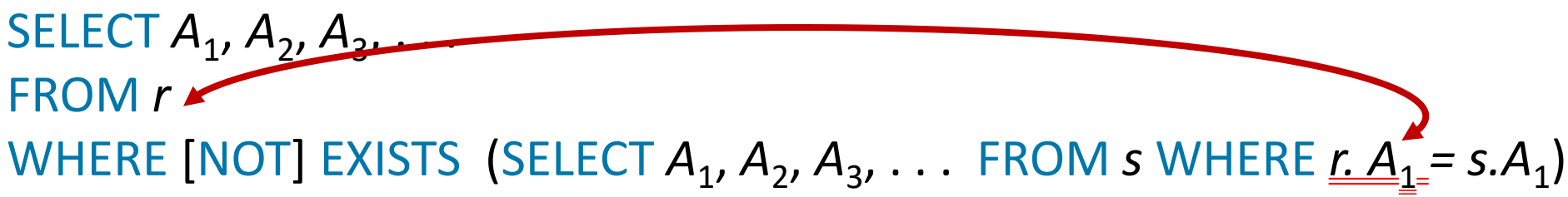
id	orderID	orderDate	. . .	total
12	1	2022-04-13	. . .	30
18	1	2022-04-13	. . .	135
. . .	. . .		. . .	. . .
33	5	2022-04-17	. . .	57
34	5	2022-04-17		114

- SELECT * FROM orderdetails *a* WHERE EXISTS (SELECT true FROM orderdetails *b* WHERE *a*.orderdate = *b*.orderdate AND *a*.orderID = *b*.orderID AND *a*.id <> *b*.ID) ORDER BY orderID, orderdate;

correlated subquery

correlated subqueries

A correlated subquery (**also known as a synchronized subquery**) is a subquery that uses values from the outer query. The subquery is evaluated once for each row processed by the outer query.



- SELECT * FROM emp *R* WHERE hiredate IN (SELECT hiredate FROM emp *S* WHERE *R*.hiredate = *S*.hiredate AND *R*.empno <> *S*.empno); // same hiredate.
- SELECT *R*.* FROM emp *R* WHERE sal IN (SELECT *S*.sal FROM emp *S* WHERE *R*.sal = *S*.sal AND *R*.empno <> *S*.empno); // same salary.
- SELECT *R*.ename, *R*.job, *R*.sal FROM emp *R* WHERE EXISTS (SELECT true FROM emp *S* WHERE *R*.job = *S*.job AND *R*.sal = *S*.sal GROUP BY job, sal HAVING COUNT(*) > 1) ORDER BY job; // same salary jobwise.

correlated subqueries

A correlated subquery (**also known as a synchronized subquery**) is a subquery that uses values from the outer query. The subquery is evaluated once for each row processed by the outer query.

Following query find all employees who earn more than the average salary in their department.

- SELECT * FROM emp *e* WHERE sal > (SELECT AVG(sal) FROM emp WHERE *e*.deptno = emp.deptno) ORDER BY deptno;
- SELECT ename, sal, job FROM emp WHERE sal > (SELECT AVG(sal) FROM emp *e* WHERE *e*.job = emp.job);
- SELECT job, MAX(sal) FROM emp WHERE sal < (SELECT MAX(sal) FROM emp *e* WHERE emp.job = *e*.job GROUP BY e.job) GROUP BY job; // 2nd highest salary jobwise.
- SELECT DISTINCTROW deptno FROM emp WHERE EXISTS (SELECT deptno FROM dept WHERE emp.deptno = dept.deptno); (***Intersect***)
- SELECT DISTINCTROW deptno FROM DEPT WHERE NOT EXISTS (SELECT deptno FROM emp WHERE emp.deptno = dept.deptno); (***except /minus***)

IS NULL
NULL

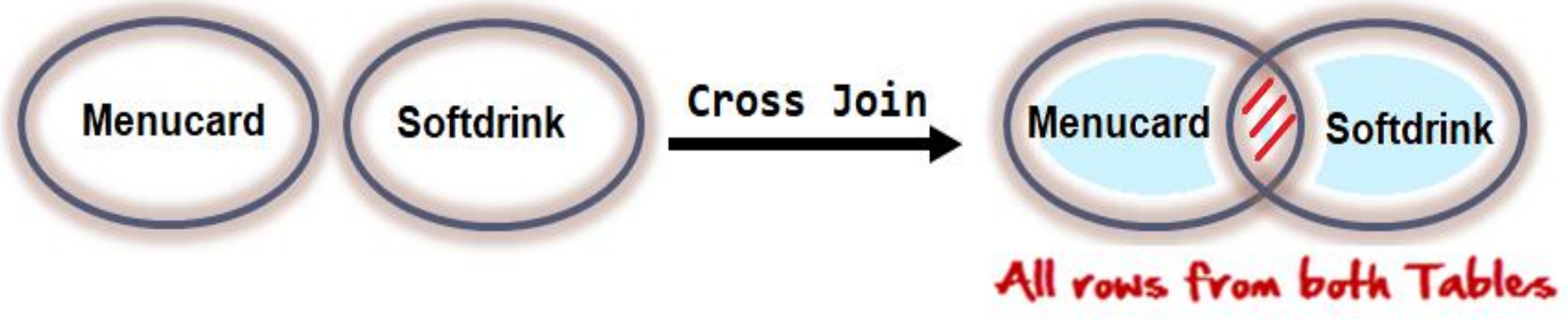
joins

JOINS are used to **retrieve data from multiple tables**.

JOIN is performed whenever **two or more tables** are joined in a SQL statement.

Type of JOINS

- Cartesian or Product Join – Cross Join
- Equijoin – Inner Join
- Natural Join
- Simple Join
- Outer Join – Right Outer Join, Left Outer Join
- Self Join



cartesian or product join (R × S)

- **Cartesian/Product means** Number of Rows present in Table 1 Multiplied by Number of Rows present in Table 2.
- **Cross Join in MySQL** does not require any common column to **join** two table.

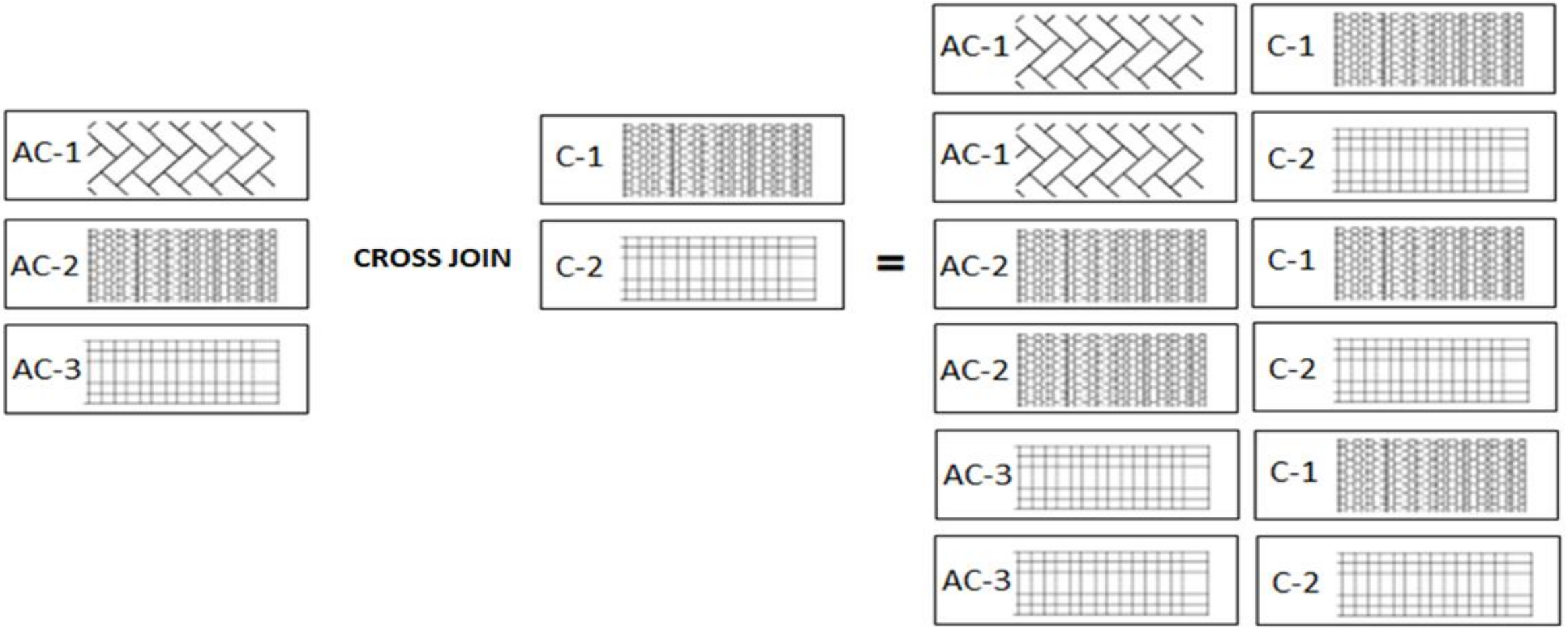
The result of $R(A_1, A_2, \ldots, A_n) \times S(B_1, B_2, \ldots, B_m)$ is a relation Q with degree n + m attributes $Q(A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_m)$, in that order.

Degree d(R×S) = d(R) + d(S).

Cardinality |R×S| = |R| * |S|

joins - cartesian or product

The CROSS JOIN gets a row from the first table (r_1) and then creates a new row for every row in the second table (r_2). It then does the same for the next row for in the first table (r_1) and so on.



joins - cartesian or product

Cartesian or Product joins are joins without a join condition. Each row of one table is combined with each row of another table. The result is referred to as a Cartesian product.

SELECT $A_1, A_2, A_3, \ldots$ FROM $r_1, r_2, \ldots$

$r_1 = \{1, 2\}$
$r_2 = \{a, b, c\}$

r_1 x r_2

R = { (1, a),
(2, a),
(1, b),
(2, b),
(1, c),
(2, c) }

$r_1 = \{1, 2, 3\}$
$r_2 = \{a, b\}$

r_1 x r_2

R = { (1, a),
(1, b),
(2, a),
(2, b),
(3, a),
(4, b) }

$r_1 = \{1, 2\}$
$r_2 = \{a, b, null\}$

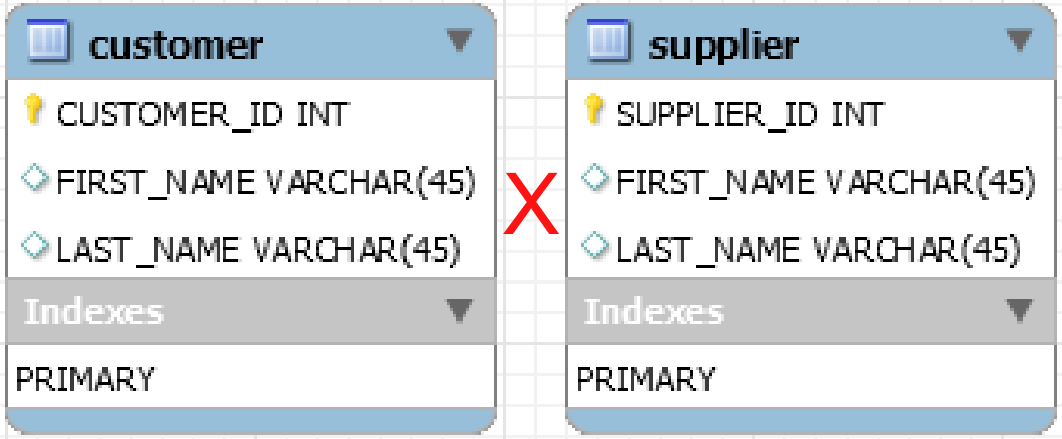
r_1 x r_2

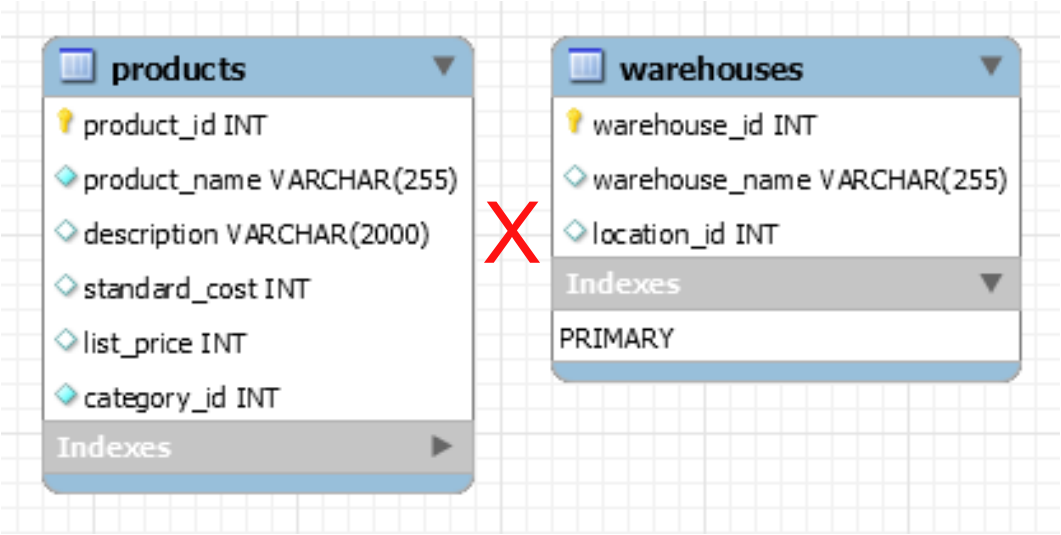
R = { (1, a),
(2, a),
(1, b),
(2, b),
(1, null),
(2, null) }

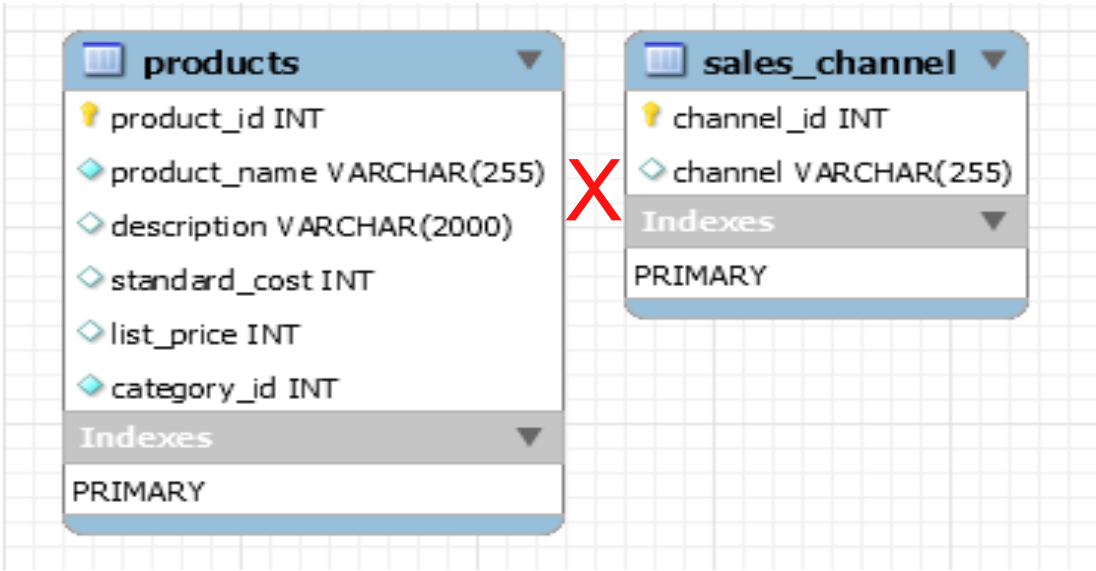
joins - cartesian or product

Cartesian or Product joins are joins without a join condition. Each row of one table is combined with each row of another table. The result is referred to as a Cartesian product.

SELECT $A_1, A_2, A_3, \ldots$ FROM $r_1, r_2, \ldots$







- Product /Warehouse
- Product/sales_channel
- Person/Cards
- Generate Calendar — Dates × Shifts
- Travel Planner — Source × Destination

joins - cartesian or product

SELECT $A_1, A_2, A_3, \ldots$ FROM $r_1, r_2, \ldots$

- SELECT * FROM menucard, softdrink;

	ID	NAME	RATE
▶	1	Extra Long Cheeseburger	100
	2	Double Stacker	125
	3	Double Cheeseburger	100
	4	Hamburger	85
	5	Classic Grilled Dog	95
	6	Chili Cheese Grilled Dog	115
	7	Flame Grilled Chicken Burger	135
	8	Original Chicken Sandwich	55
	9	McALLO TIKKI	45
	10	Veg Maharaja Mac	75
	11	Big Spicy Chicken Wrap	100
	12	McVeggie Schezwan	85

	ID	NAME	RATE
▶	1	Coca-Cola	45
	2	Mello Yello	75
	3	Diet Coke	60
	4	Frozen Fanta Cherry	65
	5	Iced Tea	35

	ID	NAME	RATE	ID	NAME	RATE
▶	1	Extra Long Cheeseburger	100	1	Coca-Cola	45
	1	Extra Long Cheeseburger	100	2	Mello Yello	75
	1	Extra Long Cheeseburger	100	3	Diet Coke	60
	1	Extra Long Cheeseburger	100	4	Frozen Fanta Cherry	65
	1	Extra Long Cheeseburger	100	5	Iced Tea	35
	2	Double Stacker	125	1	Coca-Cola	45
	2	Double Stacker	125	2	Mello Yello	75
	2	Double Stacker	125	3	Diet Coke	60
	2	Double Stacker	125	4	Frozen Fanta Cherry	65
	2	Double Stacker	125	5	Iced Tea	35
	3	Double Cheeseburger	100	1	Coca-Cola	45
	3	Double Cheeseburger	100	2	Mello Yello	75
	3	Double Cheeseburger	100	3	Diet Coke	60
	3	Double Cheeseburger	100	4	Frozen Fanta Cherry	65
	3	Double Cheeseburger	100	5	Iced Tea	35
	4	Hamburger	85	1	Coca-Cola	45
	4	Hamburger	85	2	Mello Yello	75
	4	Hamburger	85	3	Diet Coke	60
	4	Hamburger	85	4	Frozen Fanta Cherry	65
	4	Hamburger	85	5	Iced Tea	35

joins - cartesian or product

SELECT $A_1, A_2, A_3, \ldots$ FROM $r_1, r_2, \ldots$

- SELECT mnu.name, sftdrink.name, mnu.rate, sftdrink.rate, mnu.rate + sftdrink.rate as "Total" FROM menucard mnu, softdrink sftdrink;

ID	NAME	RATE
1	Extra Long Cheeseburger	100
2	Double Stacker	125
3	Double Cheeseburger	100
4	Hamburger	85
5	Classic Grilled Dog	95
6	Chili Cheese Grilled Dog	115
7	Flame Grilled Chicken Burger	135
8	Original Chicken Sandwich	55
9	McALLO TIKKI	45
10	Veg Maharaja Mac	75
11	Big Spicy Chicken Wrap	100
12	McVeggie Schezwan	85

ID	NAME	RATE
1	Coca-Cola	45
2	Mello Yello	75
3	Diet Coke	60
4	Frozen Fanta Cherry	65
5	Iced Tea	35

name	name	rate	rate	Total
Extra Long Cheeseburger	Coca-Cola	100	45	145
Extra Long Cheeseburger	Mello Yello	100	75	175
Extra Long Cheeseburger	Diet Coke	100	60	160
Extra Long Cheeseburger	Frozen Fanta Cherry	100	65	165
Extra Long Cheeseburger	Iced Tea	100	35	135
Double Stacker	Coca-Cola	125	45	170
Double Stacker	Mello Yello	125	75	200
Double Stacker	Diet Coke	125	60	185
Double Stacker	Frozen Fanta Cherry	125	65	190
Double Stacker	Iced Tea	125	35	160
Double Cheeseburger	Coca-Cola	100	45	145
Double Cheeseburger	Mello Yello	100	75	175
Double Cheeseburger	Diet Coke	100	60	160
Double Cheeseburger	Frozen Fanta Cherry	100	65	165
Double Cheeseburger	Iced Tea	100	35	135
Hamburger	Coca-Cola	85	45	130
Hamburger	Mello Yello	85	75	160
Hamburger	Diet Coke	85	60	145
Hamburger	Frozen Fanta Cherry	85	65	150
Hamburger	Iced Tea	85	35	120
Classic Grilled Dog	Coca-Cola	95	45	140
Classic Grilled Dog	Mello Yello	95	75	170
Classic Grilled Dog	Diet Coke	95	60	155

joins - cartesian or product example

- SELECT name, COUNT(*) *"Total Employees"*, rate * COUNT(*) *"Total Cost"* FROM emp, softdrink GROUP BY name;

joins – cross join

The CROSS JOIN produced a result set which is the product of rows of two associated tables when no WHERE clause is used with CROSS JOIN. In this join, the result set appeared by multiplying each row of the first table with all rows in the second table if no condition introduced with CROSS JOIN.

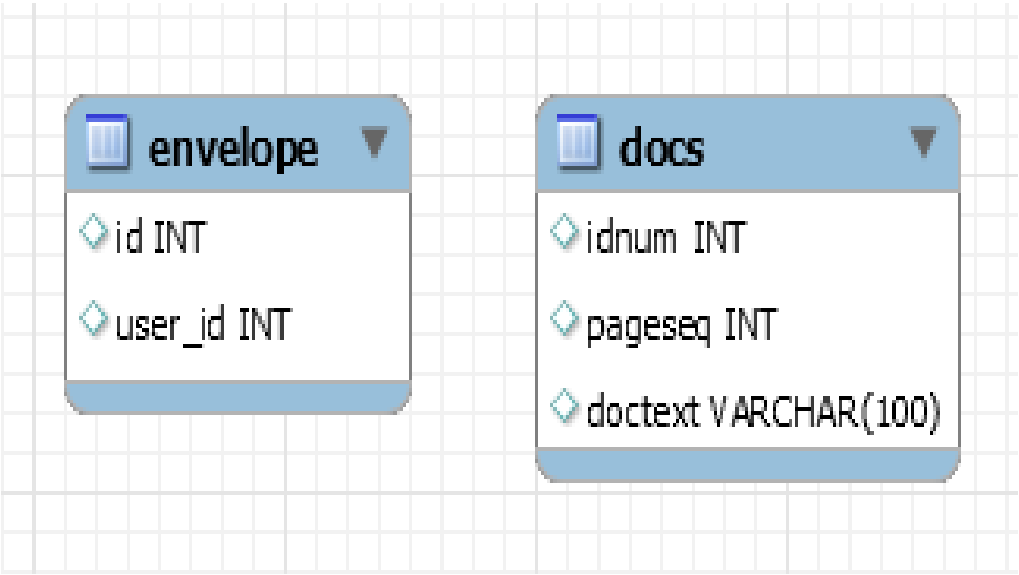
SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 CROSS JOIN $r_2, \ldots$

envelope Table

	id	user_id
▶	1	1
	2	2
	3	3

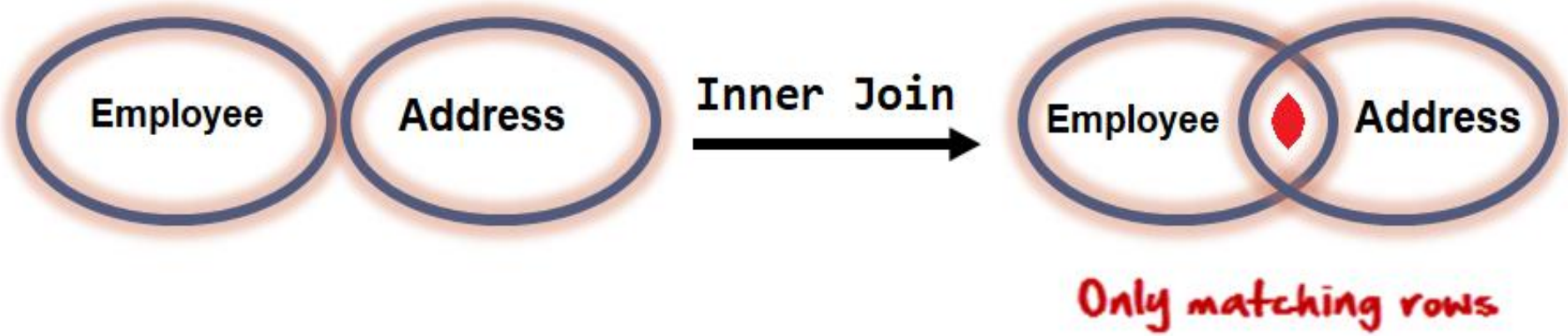
docs Table

	idnum	pageseq	doctext
▶	1	5	NULL
	2	6	NULL
	NULL	0	NULL



- SELECT * FROM envelope CROSS JOIN docs;

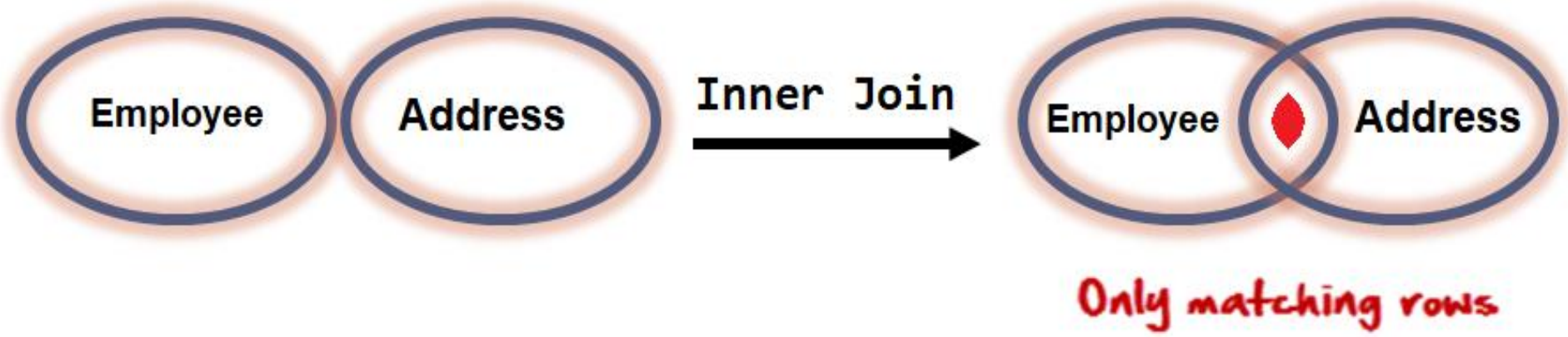
	id	user_id	idnum	pageseq	doctext
▶	1	1	1	5	NULL
	2	2	1	5	NULL
	3	3	1	5	NULL
	1	1	2	6	NULL
	2	2	2	6	NULL
	3	3	2	6	NULL
	1	1	NULL	0	NULL
	2	2	NULL	0	NULL
	3	3	NULL	0	NULL



equi join $(R \bowtie S)$

An **equi join / Inner Join** is a join with a join condition containing an equality operator.
An equijoin returns only those rows that have equivalent values for the specified columns. Rows that match remain in the result, those that don't are rejected. The match condition is commonly called the **join condition**. **equi join / Inner Join** returns rows when there is at least one match in both tables.

The result of $R(A_1, A_2, \ldots, A_n) \bowtie_{\text{<join condition>}} S(B_1, B_2, \ldots, B_m)$ is a relation Q with degree **n + m** attributes **$Q(A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_m)$, in that order**. Q has one tuple for each combination of tuples—one from R and one from S—whenever the combination satisfies the join condition.



employeeID	Name	deptNo
1	Saleel	10
2	Sharmin	20
3	Vrushali	10

deptNo	Name
10	Sales
20	Accounting
30	Manager

employeeID	Name	deptNo	deptNo	Name
1	Saleel	10	10	Sales
2	Sharmin	20	20	Accounting
3	Vrushali	10	10	Sales

- TODO

equi join example

Include null ID values in j join

Table:- null1

ID	productName
1	Bread
2	Cheese
3	Butter
NULL	Jam

Table:- null2

ID	price
1	45
2	175
3	125
NULL	120
NULL	140

Output →

Output

ID	productName	price
1	Bread	45
2	Cheese	175
3	Butter	125
NULL	Jam	120
NULL	Jam	140

Type 1

- SELECT null1.id, null1.productName, null2.price FROM null1, null2 WHERE null1.id = null2.id OR COALESCE(null1.id, -1) = COALESCE(null2.id, -1);

Type 2

- SELECT * FROM null1, null2 WHERE null1.id = null2.id OR IFNULL(null1.id, -1) = IFNULL(null2.id, -1);

equi join example

Row to column

Table:- classAndSubjects

class	subjects
A	Maths
B	English
A	Science
B	History
A	Physics
B	Maths
A	Biology
B	Social
A	English
B	Chemistry

Output →

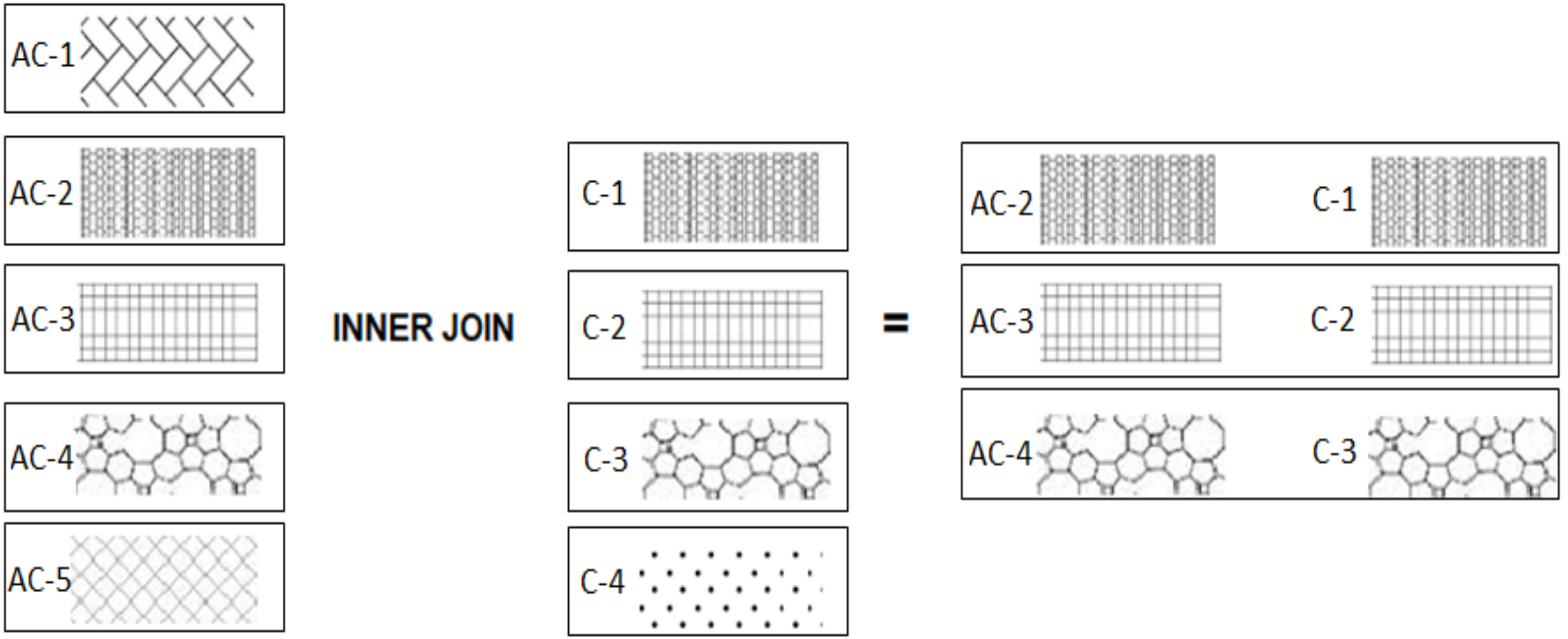
Output

class	Subject1	Subject2	Subject3	Subject4	Subject5
A	Maths	Science	Physics	Biology	English
B	English	History	Maths	Social	Chemistry

- SELECT class, MAX(CASE WHEN *R1* = 1 THEN subjects END) *Subject1*, MAX(CASE WHEN *R1* = 2 THEN subjects END) *Subject2*, MAX(CASE WHEN *R1* = 3 THEN subjects END) *Subject3*, MAX(CASE WHEN *R1* = 4 THEN subjects END) *Subject4*, MAX(CASE WHEN *R1* = 5 THEN subjects END) *Subject5* FROM (SELECT a.*, ROW_NUMBER() OVER(PARTITION BY class) *R1* FROM classandsubjects a) T1 GROUP BY class;

equi join example

The following table illustrates the inner join of two tables r_1(AC-1, AC-2, AC-3, AC-4, AC-5) and r_2(C-1, C-2, C-3, C-4). The result includes rows: (2,A), (3,B), and (4,C) as they have the same patterns.



joins – equi join

EQUI JOIN performs a JOIN against equality or matching column(s) values of the associated tables. An equal sign (=) is used as comparison operator in the where clause to refer equality.

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1, r_2 WHERE $r_1.A_1 = r_2.A_1$

$\boldsymbol{r}_1 = \{ 1, 2, 3, 4 \}$
$\boldsymbol{r}_2 = \{ (1, a), (2, b), (1, c), (3, d), (2, e), (1, f) \}$

$\boldsymbol{r}_1 = \boldsymbol{r}_2$

R = {(1,1,a),
(2,2,b),
(1,1,c),
(3,3,d),
(2,2,e),
(1,1,f)}

Remember:

A general join condition is of the form <condition> AND <condition> AND . . . AND <condition>, where each <condition> is of the form Ai θ Bj, **Ai is an attribute of R, Bj is an attribute of S**.

joins – equi join

- SELECT * FROM emp , dept WHERE emp.deptno = dept.deptno;

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO	BONUSID	USER NAME	PWD	isActive	DEPTNO	DNAME	LOC	PWD
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20	2	SMITH	a12recmpm	0	20	RESEARCH	DALLAS	a12recmpm
7415	AARAV	CLERK	7902	1981-12-31	3350.00	NULL	10	NULL	AARAV	NULL	0	10	ACCOUNTING	NEW YORK	r50mpm
7421	THOMAS	CLERK	7920	1981-07-19	1750.00	NULL	10	1	THOMAS	r50mpm	0	10	ACCOUNTING	NEW YORK	r50mpm
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30	4	ALLEN	sales@2017	1	30	SALES	CHICAGO	sales@2017
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30	1	WARD	sales@2017	1	30	SALES	CHICAGO	sales@2017
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20	4	JONES	a12recmpm	1	20	RESEARCH	DALLAS	a12recmpm
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30	6	MARTIN	sales@2017	1	30	SALES	CHICAGO	sales@2017
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30	1	BLAKE	sales@2017	1	30	SALES	CHICAGO	sales@2017
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10	3	CLARK	r50mpm	1	10	ACCOUNTING	NEW YORK	r50mpm
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20	3	SCOTT	a12recmpm	1	20	RESEARCH	DALLAS	a12recmpm
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10	1	KING	r50mpm	1	10	ACCOUNTING	NEW YORK	r50mpm
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30	5	TURNER	sales@2017	1	30	SALES	CHICAGO	sales@2017
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20	1	ADAMS	a12recmpm	1	20	RESEARCH	DALLAS	a12recmpm
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30	2	JAMES	sales@2017	1	30	SALES	CHICAGO	sales@2017
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20	4	FORD	a12recmpm	0	20	RESEARCH	DALLAS	a12recmpm
7919	HOFFMAN	MANAGER	7566	1982-03-24	4150.00	NULL	30	3	HOFFMAN	sales@2017	1	30	SALES	CHICAGO	sales@2017
7920	GRASS	SALESMAN	7919	1980-02-14	2575.00	2700.00	30	5	GRASS	sales@2017	1	30	SALES	CHICAGO	sales@2017
7934	MILLER	CLERK	7919	1982-01-23	1300.00	NULL	10	2	MILLER	r50mpm	0	10	ACCOUNTING	NEW YORK	r50mpm

Remember:

- SELECT * FROM emp, dept WHERE emp.deptno = dept.deptno AND dname = 'accounting';
- SELECT * FROM emp, dept WHERE (emp.deptno, dname) = (dept.deptno, 'accounting');

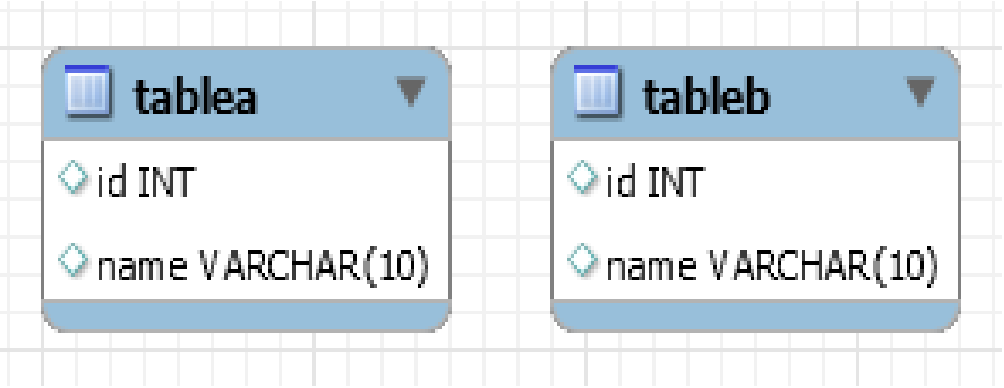
joins – equi join

tableA Table

id	name
5	aa
1	a
2	b
3	y
NULL	d
5	NULL
1	NULL
1	b
8	a

tableB Table

id	name
1	a
2	x
4	b
NULL	c
6	NULL
NULL	NULL
7	z
2	NULL
5	z
9	u



- SELECT * FROM tableA , tableB WHERE tableA.id = tableB.id;

id	name	id	name
1	a	1	a
1	NULL	1	a
1	b	1	a
2	b	2	x
2	b	2	NULL
5	aa	5	z
5	NULL	5	z

- SELECT * FROM tableA , tableB WHERE tableA.name = tableB.name;

id	name	id	name
1	a	1	a
8	a	1	a
2	b	4	b
1	b	4	b

ON Contrition

- When this join condition gets applied none of the columns of the relation will get eliminated in the result set.
- In order to apply this join condition, on any two tables they need not to have any common column.

on condition and using attribute

USING Attribute Contrition

- When all the common columns are used in the join predicate then the result would be same as Natural join.
- In the result set of the join the duplicates of the columns used in the predicate gets eliminated.
- It should not have a qualifier(table name or Alias) in the referenced columns.

Note:

- **ON** clause is optional, If not given then *INNER JOIN* works like *CROSS JOIN.*

joins – on & using clause example

The ON clause

The ON clause is used to join tables where the column names don't match in both tables.

```
SELECT * FROM EMP
INNER JOIN DEPT
ON EMP.DEPTNO = DEPT.ID
```

JOINING CONDITION

The USING clause

The USING clause is used if several columns share the same name but you don't want to join using all of these common columns. The columns listed in the USING clause can't have any qualifiers in the statement.

```
SELECT * FROM EMP
INNER JOIN DEPT
USING(DEPTNO)
```

JOINING CONDITION

inner join $(R \bowtie S)$

The inner join is one of the most commonly used joins in SQL. The inner join clause allows you to query data from two or more related tables.

INNER JOIN returns rows when there is at least one match in both tables.

joins – inner join

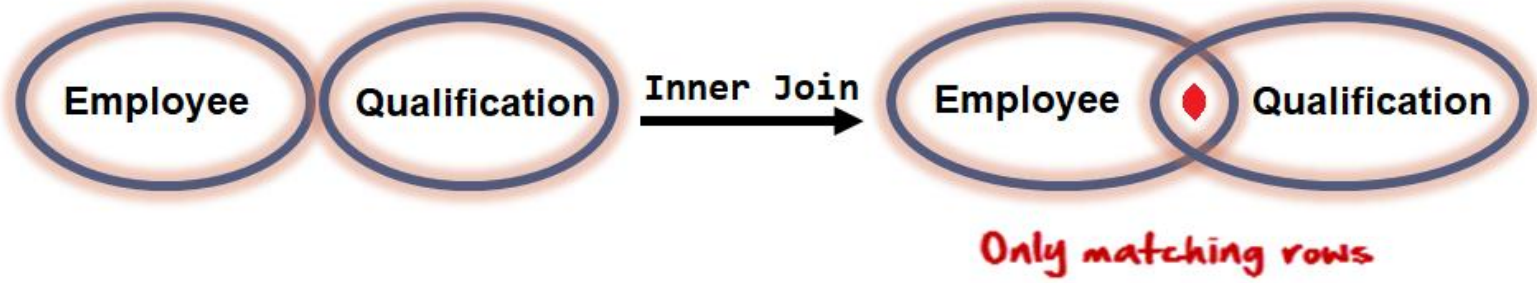
The INNER JOIN selects all rows from both participating tables as long as there is a match between the columns. An SQL INNER JOIN is same as JOIN clause, combining rows from two or more tables.

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 [INNER] JOIN r_2 ON $r_1.A_1 = r_2.A_1$

- SELECT * FROM employee *emp* INNER JOIN qualification *quali* ON *emp*.id = *quali*.employeeid;

ID	FIRSTNAME	LASTNAME	GENDER	HIREDATE	ID	EMPLOYEEID	NAME	Stream	ADDMISSIONYEAR	INSTITUTE	UNIVERSITY	YEAROFPASSING	PERCENTAGE	GRADE
1	Denis	Murphy	M	1964-06-12	1	1	10	General	1957-08-02	Alabama	Stanford University	1958	62.00	D
1	Denis	Murphy	M	1964-06-12	2	1	12	Science	1959-06-22	Alaska	Harvard University	1960	56.00	D
1	Denis	Murphy	M	1964-06-12	3	1	BE	IT	1960-06-12	Arizona	Harvard University	1964	75.00	B
2	Jenny	Ross	F	1964-10-25	4	2	10	General	1957-01-19	Alaska	University of Chicago	1958	67.00	C
2	Jenny	Ross	F	1964-10-25	5	2	12	Commerce	1959-10-23	New York	Yale University	1960	67.00	C
2	Jenny	Ross	F	1964-10-25	6	2	B.Com	Accounting	1960-06-12	Arkansas	Yale University	1964	69.00	C
3	David	Ross	M	1964-10-25	7	3	10	General	1957-11-25	Arizona	Yale University	1958	86.00	A
3	David	Ross	M	1964-10-25	8	3	12	Science	1959-02-17	California	California University	1960	57.00	D
3	David	Ross	M	1964-10-25	9	3	BE	IT	1960-06-12	Florida	University of Florida	1964	85.00	A
4	Fred	NULL	M	1965-10-31	10	4	10	General	1958-03-19	Idaho	Pennsylvania University	1959	89.00	A
4	Fred	NULL	M	1965-10-31	11	4	12	Commerce	1960-05-21	New Ham...	Yale University	1961	96.00	A+
4	Fred	NULL	M	1965-10-31	12	4								
5	Helen	Taylor	F	1965-01-12	13	5								

ON clause is optional, If not given then INNER JOIN works like CROSS JOIN



joins – inner join

INNER JOIN returns rows when there is at least one match in both tables.

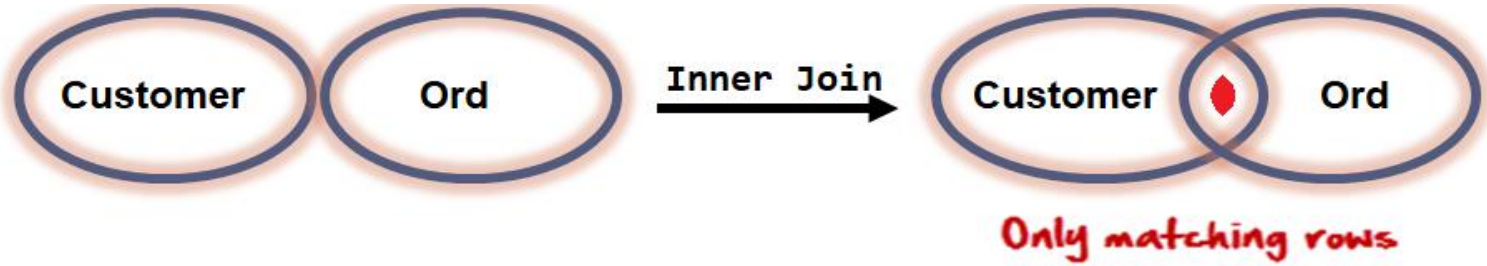
The INNER JOIN selects all rows from both participating tables as long as there is a match between the columns. An SQL INNER JOIN is same as JOIN clause, combining rows from two or more tables.

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 [INNER] JOIN r_2 ON $r_1.A_1 = r_2.A_1$

- SELECT * FROM customer INNER JOIN ord USING (custid);

CUSTID	NAME	ADDRESS	CITY	STATE	ZIP	AREA	PHONE	REPID	CREDITLIMIT	COMMENTS	ORDID	ORDERDATE	COMMPLAN	SHIPDATE	STA
106	SHAPE UP	908 SEQUOIA	PALO ALTO	CA	94301	415	364-9777	7521	6000.00	Support intensive. Orders small a...	601	1986-05-01 00:00:00	A	1986-05-30 00:00:00	In Pr
102	VOLLYRITE	9722 HAMILTON	BURLINGAME	CA	95133	415	644-3341	7654	7000.00	Company doing heavy promotion ...	602	1986-06-05 00:00:00	B	1986-06-20 00:00:00	On H
102	VOLLYRITE	9722 HAMILTON	BURLINGAME	CA	95133	415	644-3341	7654	7000.00	Company doing heavy promotion ...	603	1986-06-05 00:00:00	NULL	1986-06-05 00:00:00	Canc
106	SHAPE UP	908 SEQUOIA	PALO ALTO	CA	94301	415	364-9777	7521	6000.00	Support intensive. Orders small a...	604	1986-06-15 00:00:00	A	1986-06-30 00:00:00	Reso
106	SHAPE UP	908 SEQUOIA	PALO ALTO	CA	94301	415	364-9777	7521	6000.00	Support intensive. Orders small a...	605	1986-07-14 00:00:00	A	1986-07-30 00:00:00	Dispu
100	JOCKSPORTS	345 VIEWRIDGE	BELMONT	CA	96711	415	598-6609	7844	5000.00	Very friendly people to work with ...	606	1986-07-14 00:00:00	A	1986-07-30 00:00:00	Shipp
104	EVERY MOUNTAIN	574 SURRY RD.	CUPERTINO	CA	93301	408	996-2323	7499	10000.00	Customer with high market share ...	607	1986-07-18 00:00:00	C	1986-07-18 00:00:00	In Pr
104	EVERY MOUNTAIN	574 SURRY RD.	CUPERTINO	CA	93301	408	996-2323	7499	10000.00	Customer with high market share ...	608	1986-07-25 00:00:00	C	1986-07-25 00:00:00	Shipp
100	JOCKSPORTS	345 VIEWRIDGE	BELMONT	CA	96711	415	598-6609	7844	5000.00	Very friendly people to work with ...	609	1986-08-01 00:00:00	B	1986-08-15 00:00:00	On H
101	TKB SPORT SHOP	490 BOLI RD.	REDWOOD CITY	CA	94061	415	368-1223	7521	10000.00	Rep called 5/8 about change in or...	610	1987-01-07 00:00:00	A	1987-01-08 00:00:00	In Pr
102	VOLLYRITE	9722 HAMILTON	BURLINGAME	CA	95133	415	644-3341	7654	7000.00	Company doing heavy promotion ...	611	1987-01-11 00:00:00	B	1987-01-11 00:00:00	Shipp
104	EVERY MOUNTAIN	574 SURRY RD.	CUPERTINO	CA	93301	408	996-2323	7499	10000.00	Customer with high market share ...	612	1987-01-15 00:00:00	C	1987-01-20 00:00:00	Canc
108	NORTH WOODS ...	98 LONE PINE ...	HIBBING	MN	55649	612	566-9123	7844	8000.00	NULL	613	1987-02-01 00:00:00	NULL	1987-02-01 00:00:00	Shipp
102	VOLLYRITE	9722 HAMILTON	BURLINGAME	CA	95133	415	644-3341	7654	7000.00	Company doing heavy promotion ...	614	1987-02-01 00:00:00	NULL	1987-02-05 00:00:00	In Pr
107	WOMENS SPORTS	VALCO VILLAGE	SUNNYVALE	CA	93301	408	967-4398	7499	10000.00	First sporting goods store geared ...	615	1987-02-01 00:00:00	NULL	1987-02-06 00:00:00	In Pr
103	JUST TENNIS	HILLVIEW MALL	BURLINGAME	CA	97544	415	677-9312	7521	3000.00	Contact rep about new line of ten...	616	1987-02-03 00:00:00	NULL	1987-02-10 00:00:00	Reso
105	K + T SPORTS	3476 EL PASEO	SANTA CLARA	CA	91003	408	376-9966	7844	5000.00	Tends to order large amounts of ...	617	1987-02-05 00:00:00	NULL	1987-03-03 00:00:00	Shipp
102	VOLLYRITE	9722 HAMILTON	BURLINGAME	CA	95133	415	644-3341	7654	7000.00	Company doing heavy promotion ...	618	1987-02-15 00:00:00	A	1987-03-06 00:00:00	On H
104	EVERY MOUNTAIN	574 SURRY RD.	CUPERTINO	CA	93301	408	996-2323	7499	10000.00	Customer with high market share ...	619	1987-02-22 00:00:00	NULL	1987-02-04 00:00:00	In Pr
100	JOCKSPORTS	345 VIEWRIDGE	BELMONT	CA	96711	415	598-6609	7844	5000.00	Very friendly people to work with ...	620	1987-03-12 00:00:00	NULL	1987-03-12 00:00:00	Canc
100	JOCKSPORTS	345 VIEWRIDGE	BELMONT	CA	96711	415	598-6609	7844	5000.00	Very friendly people to work with ...	621	1987-03-15 00:00:00	A	1987-01-01 00:00:00	Shipp

ON clause is optional, If not given then INNER JOIN works like CROSS JOIN



- Assignments

joins – inner join

TODO

Table:- trains

trainID	. . . .	destination
1	. . . .	Mumbai
2	. . . .	Bhopal
3	. . . .	Kolkata
4	. . . .	Bangalore
5	. . . .	Mumbai
6	. . . .	Pune
7	. . . .	Pune

Table:- reservations

reservationID	trainID	. . . .	status
201	1	. . . .	Confirmed
202	8	. . . .	Cancelled
203	9	. . . .	Confirmed
204	5	. . . .	Confirmed
205	5	. . . .	Confirmed
206	8	. . . .	Confirmed
201	1	. . . .	Cancelled

1. Count of Confirmed vs Cancelled Reservations per Train

- SELECT trains.trainid, trainName, SUM(CASE WHEN status = 'Confirmed' THEN 1 END) *'Count confirmed status'*, SUM(CASE WHEN status = 'Cancelled' THEN 1 END) *'Count Cancelled status'* FROM trains INNER JOIN reservations ON trains.trainID = reservations.trainID GROUP BY trains.trainID, trainName;

- Assignments

joins – inner join

TODO

1. List of Trains with No Reservations

- `SELECT trains.* FROM trains INNER JOIN reservations ON trains.trainID = reservations.trainID WHERE status = 'confirmed';`
- `SELECT * FROM trains WHERE NOT EXISTS (SELECT true FROM reservations WHERE trains.trainID = reservations.trainID AND status = 'Confirmed');`

2. List of Reservations with Status 'Confirmed' Only

- `Select * from reservations INNER JOIN trains ON trains.trainID = reservations.trainID and status = 'Confirmed’;`
- `SELECT * FROM trains WHERE EXISTS(SELECT true FROM reservations WHERE trains.trainID = reservations.trainID AND status = 'Confirmed');`

In general, the join condition for NATURAL JOIN is constructed by equating each pair of join attributes that have the same name in the two relations and combining these conditions with AND.

natural join

The NATURAL JOIN is such a join that performs the same task as an INNER JOIN. NATURAL JOIN does not use any comparison operator. We can perform a NATURAL JOIN only if there is at least one common attribute that exists between two relations. In addition, the attributes must have the same name. When this join condition gets applied always the duplicates of the common columns get eliminated from the result.

Remember:

The standard definition of NATURAL JOIN requires that the two join attributes (or each pair of join attributes) have the same name in both relations. **If this is not the case, a renaming operation is applied first.**

e.g.

- `SELECT * FROM r NATURAL JOIN (SELECT a1 AS c1, a2 FROM s) t1;`

joins – natural join

Joins two tables based on common column names. Hence one must confirm the common columns before using a NATURAL JOIN

The NATURAL JOIN is such a join that performs the same task as an INNER JOIN.

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 NATURAL [INNER] JOIN r_2 NATURAL [INNER] JOIN $r_3 \ldots$

- SELECT * FROM emp NATURAL JOIN dept;

- The associated tables have one or more pairs of identically column-names.
- The columns must be of the same name.
- The columns datatype may differ.
- Don't use ON / USING clause in a NATURAL JOIN.
- When this join condition gets applied always the duplicates of the common columns get eliminated from the result.

A **NATURAL JOIN** can be used with a **LEFT OUTER** join, or a **RIGHT OUTER** join.

If the column-names are not same, then NATURAL JOIN will work as CROSS JOIN.

SELECT * FROM EMP
NATURAL JOIN DEPT

INNER Vs NATURAL

inner join vs natural join

INNER JOIN	NATURAL JOIN
Inner Join joins two table on the basis of the column which is explicitly specified in the ON clause.	Natural Join joins two tables based on same attribute name.
In Inner Join, The resulting table will contain all the attribute of both the tables including duplicate columns also	In Natural Join, The resulting table will contain all the attributes of both the tables but keep only one copy of each common column
In Inner Join, only those records will return which exists in both the tables	Same as Inner Join
SYNTAX: • SELECT * FROM $\mathbf{r}_1$ INNER JOIN $\mathbf{r}_2$ ON $\mathbf{r}_1.\mathbf{A}_1 = \mathbf{r}_2.\mathbf{A}_1$; • SELECT * FROM $\mathbf{r}_1$ INNER JOIN $\mathbf{r}_2$ USING($\mathbf{A}_1$, [$\mathbf{A}_2$]);	SYNTAX: • SELECT * FROM $\mathbf{r}_1$ NATURAL JOIN $\mathbf{r}_2$;

simple join

TODO

joins – simple join

The SIMPLE JOIN is such a join that performs the same task as an INNER JOIN.

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 SIMPLE JOIN r_2 USING $(A_1, \ldots)$

- SELECT * FROM emp SIMPLE JOIN dept USING(deptno)

```
SELECT * FROM EMP
SIMPLE JOIN DEPT
USING(DEPTNO)
```

JOINING CONDITION

The ON clause is required for a left or right outer join.

The LEFT OUTER JOIN operation keeps every tuple in the first, or left, relation R in R ⟕ S, if no matching tuple is found in S, then the attributes of S in the join result are filled or padded with NULL values.

The RIGHT OUTER JOIN keeps every tuple in the second, or right, relation S in the result of R ⟖ S, if no matching tuple is found in R, then the attributes of R in the join result are filled or padded with NULL values.

outer joins

In an outer join, along with rows that satisfy the matching criteria, we also include some or all rows that do not match the criteria.

```
CREATE TABLE r1 (id INT, c1 VARCHAR(10));
CREATE TABLE r2 (id INT, c1 VARCHAR(10));

INSERT INTO r1 VALUES (4,'AC-1'), (1,'AC-2'),(2,'AC-3'),(3,'AC-4'),(5,'AC-5');
INSERT INTO r2 VALUES (1,'C-1'), (2,'C-2'),(3,'C-3'),(7,'C-4');
```

Suppose, we want to join two tables: r1 and r2. SQL left outer join returns all rows in the left table (r1) and all the matching rows found in the right table (r2). It means the result of the SQL left join always contains the rows in the left table. . **If no matching rows found in the right table, NULL are displayed.**

left outer joins $(R \bowtie S)$

$r_1 = \{1, 2, 3, 4\}$

$r_2 = \{(1, a), (2, b), (1, c), (3, d), (2, e), (1, f), (5, z)\}$

```
SELECT * FROM r1 LEFT JOIN r2 ON r1.c1 = r2.c1;
```

r_1 left join r_2

R = {(1, 1, a),
(2, 2, b),
(1, 1, c),
(3, 3, d),
(2, 2, e),
(1, 1, f),
(4, NULL, NULL}

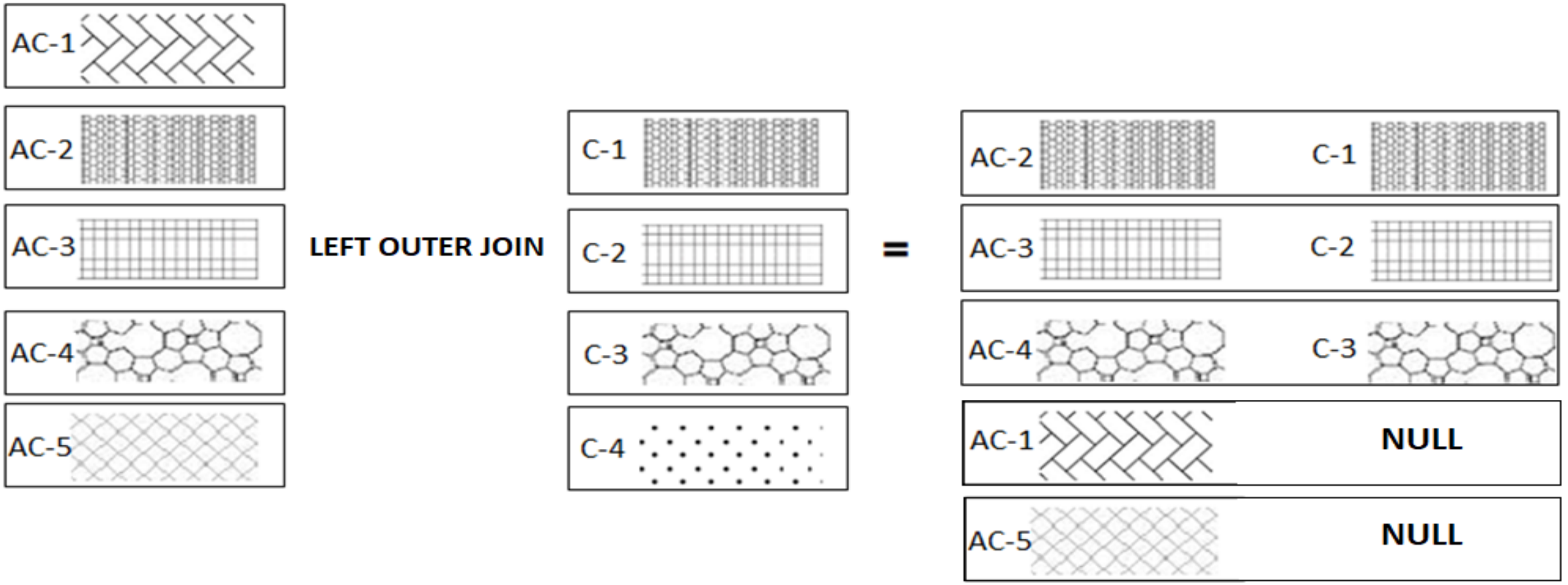
A LEFT OUTER JOIN returns:

- All rows from the left table
- Matching rows from the right table
- If there's no match in the right table → returns NULL for those columns.

	c1	c1	c2
▶	1	1	a
	2	2	b
	1	1	c
	3	3	d
	2	2	e
	1	1	f
	4	NULL	NULL

joins – left outer join

The following example the LEFT JOIN of two tables r_1(AC-1, AC-2, AC-3, AC-4, AC-5) and r_2(C-1, C-2, C-3, C-4). The LEFT JOIN will match rows from the r_1 table with the rows from r_2 table using patterns:

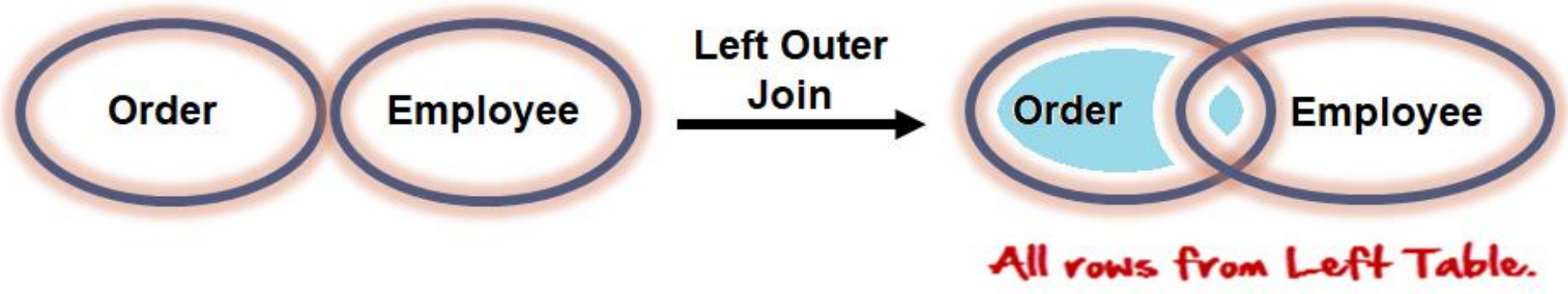


joins – left outer join

The LEFT JOIN keyword returns all rows from the left table (r_1), with the matching rows in the right table (r_2).
The result is NULL in the right side when there is no match.

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 LEFT [OUTER] JOIN r_2 ON $r_1.A_1 = r_2.A_1$

SELECT customer.custid, name, ordid, orderdate, total FROM ord LEFT OUTER JOIN customer ON ord.custid = customer.custid;



Remember:

ON :- Specifies how to join two tables (defines the relationship between them).

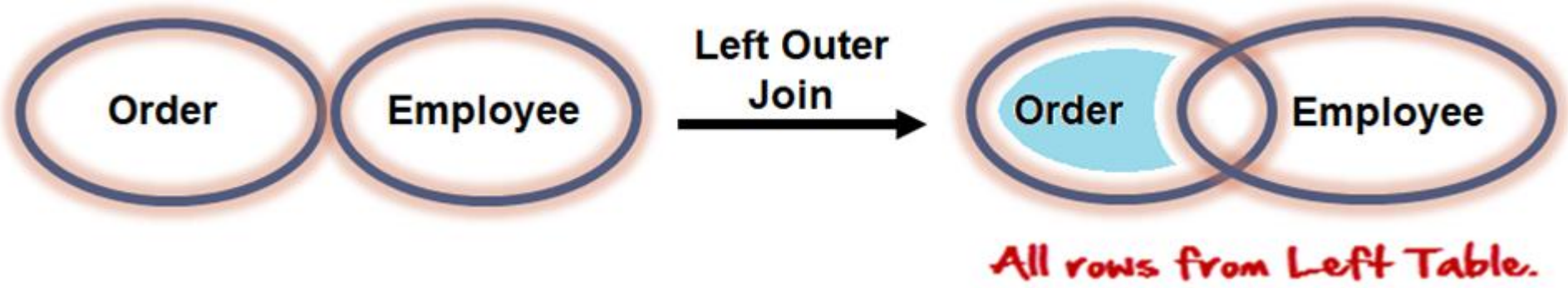
WHERE :- Filters rows after the join has been performed (applies to the result set).

joins – left outer join

The LEFT JOIN keyword returns all rows from the left table (r_1), with the matching rows in the right table (r_2).
The result is NULL in the right side table when there is no match.

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 LEFT [OUTER] JOIN r_2 ON $r_1.A_1 = r_2.A_1$ WHERE $r_2.A_1$ IS NULL

SELECT customer.custid, name, ordid, orderdate, total FROM ord LEFT OUTER JOIN customer ON ord.custid = customer.custid WHERE ord.custid IS NULL;



Remember:

ON :- Specifies how to join two tables (defines the relationship between them).

WHERE :- Filters rows after the join has been performed (applies to the result set).

- Assignments

joins – left outer join

TODO

Table:- trains

trainID	. . . .	destination
1	. . . .	Mumbai
2	. . . .	Bhopal
3	. . . .	Kolkata
4	. . . .	Bangalore
5	. . . .	Mumbai
6	. . . .	Pune
7	. . . .	Pune

Table:- reservations

reservationID	trainID	. . . .	status
201	1	. . . .	Confirmed
202	8	. . . .	Cancelled
203	9	. . . .	Confirmed
204	5	. . . .	Confirmed
205	5	. . . .	Confirmed
206	8	. . . .	Confirmed
201	1	. . . .	Cancelled

1. Find reservations with invalid trainIDs
2. List all trains along with the passenger names (if any).
3. Find trains that currently have no reservations.
4. Show each train with the count of reservations including 0.

- SELECT * FROM trains *t* LEFT OUTER JOIN reservations *r* ON *t*.trainID = *r*.trainID;
- SELECT * FROM trains *t* LEFT OUTER JOIN reservations *r* ON *t*.trainID = *r*.trainID WHERE reservationID IS NULL;
- SELECT * FROM trains *t* LEFT OUTER JOIN reservations *r* ON *t*.trainID = *r*.trainID WHERE reservationID IS NOT NULL;
- SELECT *t*.trainName, COUNT(*r*.reservationID) *total_reservations* FROM trains *t* LEFT OUTER JOIN reservations *r* ON *t*.trainID = *r*.trainID GROUP BY *t*.trainName;

joins – left outer join

The ON clause with WHERE clause for left or right outer join.

- SELECT *c*.custid, name, city, state, *o*.custid, orderdate, status, total FROM customer *c* LEFT OUTER JOIN ord *o* ON *c*.custid = o.custid;

- SELECT *c*.custid, name, city, state, *o*.custid, orderdate, status, total FROM customer c LEFT OUTER JOIN ord *o* ON *c*.custid = *o*.custid WHERE *o*.custid IS NOT NULL;

Remember:

ON :- Specifies how to join two tables (defines the relationship between them).

WHERE :- Filters rows after the join has been performed (applies to the result set).

joins – left outer join

The LEFT JOIN keyword returns all rows from the left table ($\boldsymbol{r}_1$), with the matching rows in the right table ($\boldsymbol{r}_2$).
The result is NULL in the right side table when there is no match.

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 LEFT [OUTER] JOIN r_2 USING ($A_1, \ldots$)

- SELECT * FROM emp LEFT OUTER JOIN dept USING(deptno);

SELECT $A_1, A_2, A_3, \ldots$ FROM $\boldsymbol{r}_1$ NATURAL LEFT [OUTER] JOIN $\boldsymbol{r}_2$

- SELECT * FROM emp NATURAL LEFT OUTER JOIN dept;

Suppose, we want to join two tables: r1 and r2. Right outer join returns all rows in the right table (r1) and all the matching rows found in the left table (r2). It means the result of the SQL right join always contains the rows in the right table. . If no matching rows found in the left table, NULL are displayed.

right outer joins $(R \bowtie S)$

$r_1 = \{1, 2, 3, 4\}$
$r_2 = \{(1, a), (2, b), (1, c), (3, d), (2, e), (1, f), (5, z)\}$

r_1 right join r_2

R = {(1, 1, a),
(1, 1, c),
(1, 1, f),
(2, 2, b),
(2, 2, e),
(3, 3, d),
(NULL, 5, z)}

A RIGHT OUTER JOIN returns:

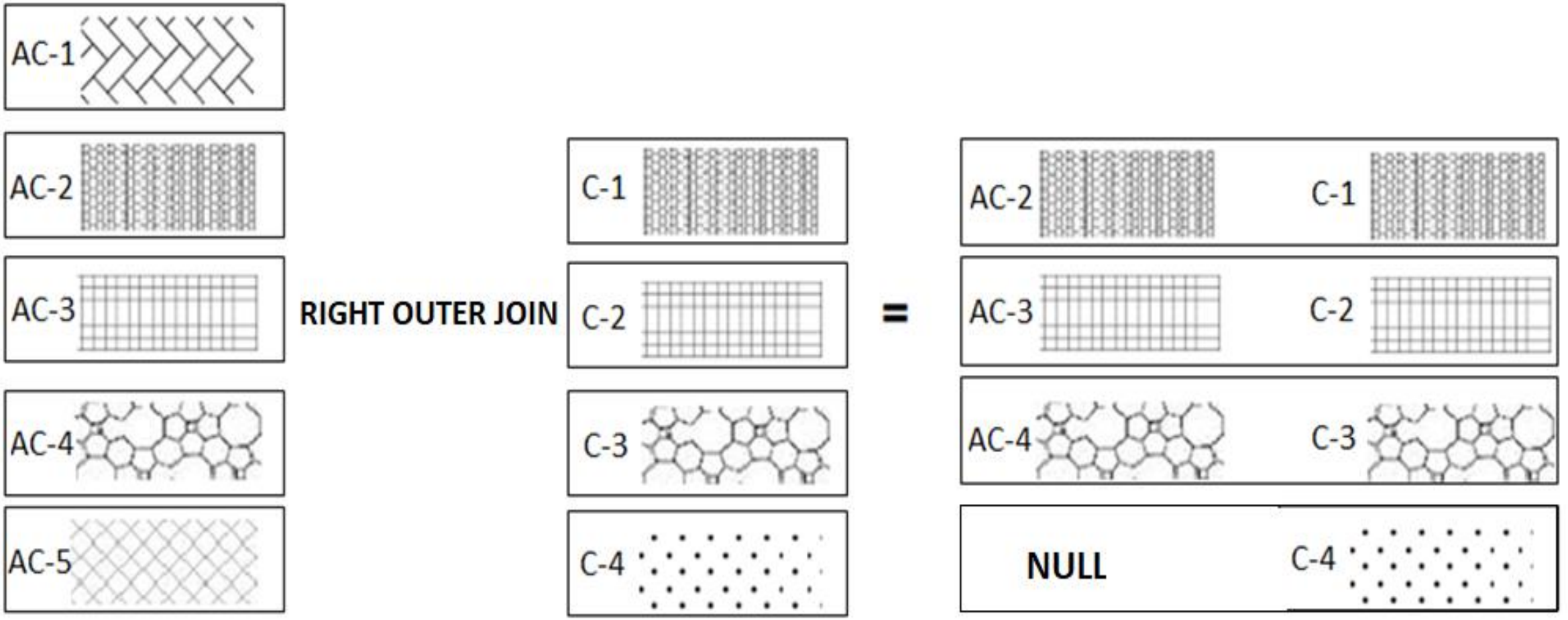
- All rows from the right table
- Matching rows from the left table
- If there's no match in the left table → returns NULL for those columns.

```
SELECT * FROM r1 RIGHT JOIN r2 ON r1.c1 = r2.c1;
```

	c1	c1	c2
▶	1	1	a
	1	1	c
	1	1	f
	2	2	b
	2	2	e
	3	3	d
	NULL	5	z

joins – right outer join

The following example the RIGHT OUTER JOIN of two tables r_1(AC-1, AC-2, AC-3, AC-4, AC-5) and r_2(C-1, C-2, C-3, C-4). The RIGHT JOIN will match rows from the r_1 table with the rows from r_2 table using patterns:

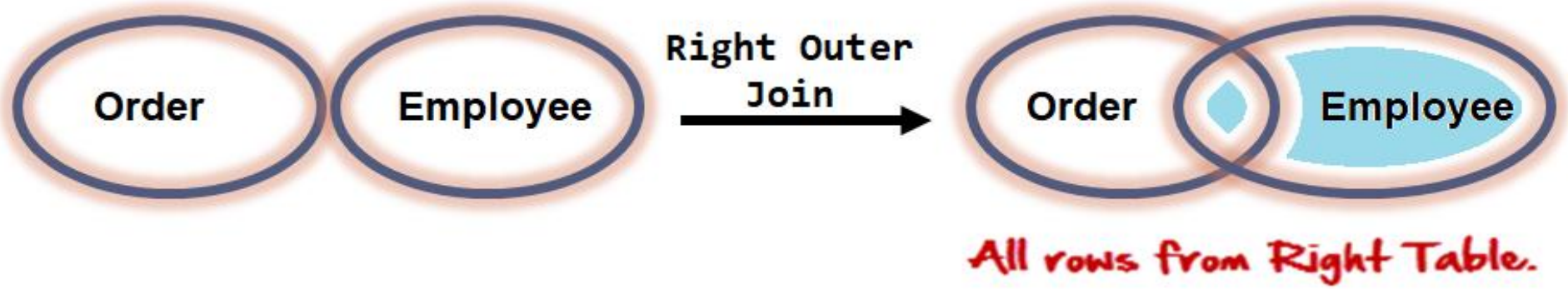


joins – right outer join

The RIGHT JOIN keyword returns all rows from the right table (r_2), with the matching rows in the left table (r_1). **The result is NULL in the left side table when there is no match.**

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 RIGHT [OUTER] JOIN r_2 ON $r_1.A_1 = r_2.A_1$

SELECT customer.custid, name, ordid, orderdate, total FROM ord RIGHT OUTER JOIN customer ON ord.custid = customer.custid;



Remember:

ON :- Specifies how to join two tables (defines the relationship between them).

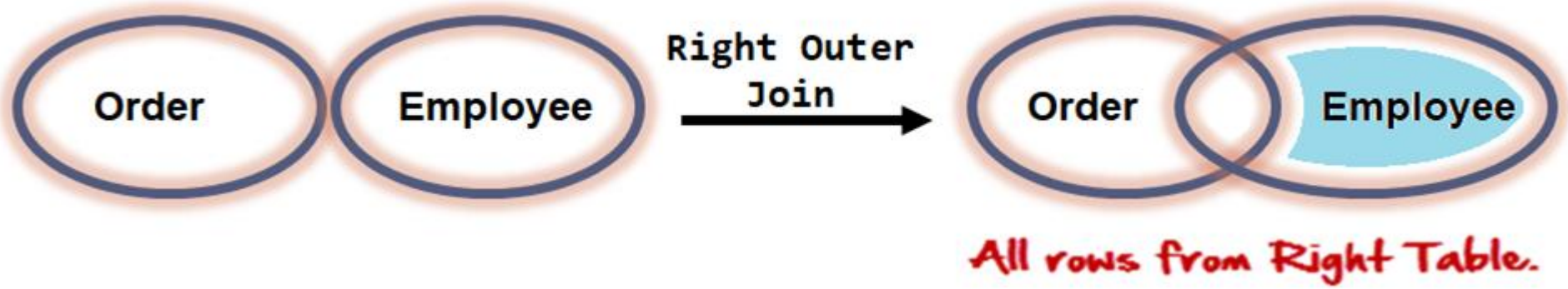
WHERE :- Filters rows after the join has been performed (applies to the result set).

joins – right outer join

The RIGHT JOIN keyword returns all rows from the right table (r_2), with the matching rows in the left table (r_1). **The result is NULL in the left side table when there is no match.**

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 RIGHT [OUTER] JOIN r_2 ON $r_1.A_1 = r_2.A_1$ WHERE $r_1.A_1$ IS NULL

SELECT customer.custid, name, ordid, orderdate, total FROM ord RIGHT OUTER JOIN customer ON ord.custid = customer.custid WHERE ordid IS NULL;



Remember:

ON :- Specifies how to join two tables (defines the relationship between them).

WHERE :- Filters rows after the join has been performed (applies to the result set).

- Assignments

joins – right outer join

TODO

Table:- trains

trainID	. . . .	destination
1	. . . .	Mumbai
2	. . . .	Bhopal
3	. . . .	Kolkata
4	. . . .	Bangalore
5	. . . .	Mumbai
6	. . . .	Pune
7	. . . .	Pune

Table:- reservations

reservationID	trainID	. . . .	status
201	1	. . . .	Confirmed
202	8	. . . .	Cancelled
203	9	. . . .	Confirmed
204	5	. . . .	Confirmed
205	5	. . . .	Confirmed
206	8	. . . .	Confirmed
201	1	. . . .	Cancelled

- SELECT * FROM trains *t* RIGHR OUTER JOIN reservations *r* ON *t*.trainID = *r*.trainID;
- SELECT * FROM trains *t* RIGHR OUTER JOIN reservations *r* ON *t*.trainID = *r*.trainID WHERE reservationID IS NULL;
- SELECT * FROM trains *t* RIGHR OUTER JOIN reservations *r* ON *t*.trainID = *r*.trainID WHERE reservationID IS NOT NULL;

joins – right outer join

The ON clause with WHERE clause for left or right outer join.

- SELECT *c*.custid, name, city, state, *o*.custid, orderdate, status, total FROM customer *c* RIGHT OUTER JOIN ord *o* ON *c*.custid = *o*.custid;

- SELECT *c*.custid, name, city, state, *o*.custid, orderdate, status, total FROM customer *c* RIGHT OUTER JOIN ord *o* ON *c*.custid = *o*.custid WHERE *o*.custid IS NOT NULL;

Remember:

ON :- Specifies how to join two tables (defines the relationship between them).

WHERE :- Filters rows after the join has been performed (applies to the result set).

joins – right outer join

The RIGHT JOIN keyword returns all rows from the right table ($\boldsymbol{r}_2$), with the matching rows in the left table ($\boldsymbol{r}_1$).
The result is NULL in the left side table when there is no match.

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 RIGHT [OUTER] JOIN r_2 USING ($A_1, \ldots$)

```
SELECT * FROM emp RIGHT OUTER JOIN dept USING(deptno);
```

SELECT $\boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{A}_3, \ldots$ FROM $\boldsymbol{r}_1$ NATURAL RIGHT [OUTER] JOIN $\boldsymbol{r}_2$

```
SELECT * FROM emp NATURAL RIGHT OUTER JOIN dept;
```

joins – left/right outer join examples

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 LEFT [OUTER] JOIN r_2 USING $(A_1, \ldots)$

SELECT $A_1, A_2, A_3, \ldots$ FROM r_1 RIGHT [OUTER] JOIN r_2 USING $(A_1, \ldots)$

- SELECT server.serverName 'All Servers', IF((server.serverName = runningserver.serverName) = 1 , 'Running Server', 'Server not in use...') FROM server LEFT OUTER JOIN runningserver ON server.id = runningserver.id;

- SELECT COUNT(IF((server.serverName = runningserver.serverName) = 1 , 0, NULL)) 'Running Server', COUNT(IF((server.serverName = runningserver.serverName) = 1 , NULL, 1)) 'Server not is use.....' FROM server LEFT OUTER JOIN runningserver ON server.id = runningserver.id;

LEFT JOIN Vs RIGHT JOIN

left join vs right join

LEFT OUTER JOIN	RIGHT OUTER JOIN	FULL OUTER JOIN
All the tuples of the left table remain in the result.	All the tuples of the right table remain in the result.	All the tuples from left as well as right table remain in the result.
The tuples of left table that does not have a matching tuple in right table are extended with NULL value for attributes of the right table.	The tuples of right table that does not have a matching tuple in left table are extended with NULL value for attributes of the left table.	The tuples of left as well as the right table that does not have the matching tuples in the right and left table respectively are extended with NULL value for attributes of the right and left tables.

Full outer join is like a left or right join except that it contains all rows from both tables. In full outer join, tuples in R that have no matching tuples in S and tuples in S that have no matching tuples in R in their common attribute name.

full outer joins $(R \bowtie S)$

MySQL does not support full outer joins like some database systems. You can get around this by using a UNION operator to join a LEFT OUTER JOIN statement and a RIGHT OUTER JOIN statement.

TODO

self joins

TODO

joins – self join

A SELF JOIN is a join in which a table is joined with itself (which is also called Unary relationships), especially when the table has a FOREIGN KEY which references its own PRIMARY KEY.

SELECT $r_x.A_1$, $r_x.A_2$, $r_y.A_1$, $r_y.A_2$, . . . FROM r_1 r_x, r_1 r_y WHERE $r_x.A_1 = r_y.A_1$

- SELECT DISTINCT *e1*.* FROM emp *e1* , emp *e2* WHERE *e1*.sal = *e2*.sal AND *e1*.empno != *e2*.empno ORDER BY *e1*.sal;
- SELECT *a*.origin, *a*.destination FROM rockets *a*, rockets *b* WHERE (*a*.origin, *a*.destination) = (*b*.destination , *b*.origin) ORDER BY 1;

Hierarchical Data

WITH RECURSIVE ***m*** AS (SELECT m.empno, m.ename, m.mgr FROM emp m WHERE m.mgr IS NULL UNION SELECT e.empno, e.ename, e.mgr FROM emp e, ***m*** WHERE ***m.empno*** = e.mgr) SELECT * FROM ***m***;

joins – self join

SELECT $r_x.A_1$, $r_x.A_2$, $r_y.A_1$, $r_y.A_2$, . . . FROM r_1 r_x, r_1 r_y WHERE $r_x.A_1 = r_y.A_1$

- Overlapping date.

joins – self join

Overlapping dates

Table:- overlapping_dates

ID	start_Date	end_Date
1	2025-01-01	2025-01-12
2	2025-01-18	2025-01-22
3	2025-01-07	2025-01-15
4	2025-01-23	2025-01-26
5	2025-01-26	2025-01-28
6	2025-01-30	2025-02-04
7	2025-02-07	2025-02-10
8	2025-02-09	2025-02-12
9	2025-02-11	2025-02-16

Output →

Output

ID	start_Date	end_Date	ID	start_Date	end_Date
1	2025-01-01	2025-01-12	3	2025-01-07	2025-01-15
7	2025-02-07	2025-02-10	8	2025-02-09	2025-02-12
8	2025-02-09	2025-02-12	9	2025-02-11	2025-02-16

This works for:

Employee shifts, Batch scheduling, Room bookings, Course batches, Leave applications, Seat reservations

To check if this date range: ('2024-07-01', '2024-07-10') overlaps:

```
start_date <= '2024-07-10' AND end_date >= '2024-07-01';
```

Two date ranges overlap if:

```
a. start_date <= b. end_date AND b. start_date <= a. end_date AND a.id < b.id;
```

- SELECT * FROM overlapping_dates a, overlapping_dates b WHERE a.start_date <= b.end_date and b.start_date <= a.end_date and a.id < b.id;

You can define multiple CTEs by separating them with commas

WITH [*RECURSIVE*]
cte_name [(*col_name* [, *col_name*] . . .)] AS (*subquery*)
[, ***cte_name*** [(*col_name* [, *col_name*] . . . AS (*subquery*)] . . .

common table expression (CTE)

In SQL, temporary result set that you can reference within a single SELECT, INSERT, UPDATE, or DELETE statement. It functions like a temporary view.

The **Anchor Member** is the **first (non-recursive)** part of a **recursive CTE**. It provides the **starting rows** (the *base case*) from which the recursion begins.

- Step 1 — Run Anchor Member
- Step 2 — Run Recursive Member (Iteration 1)
- Step 3 — Next Recursive Iteration

common table expression (CTE)

Basic CTE Syntax

```
WITH cte_name (column1, column2, ...) AS (
    -- CTE query definition (a SELECT statement)
    SELECT ...
    FROM ...
    WHERE ...
)
-- Main query that uses the CTE
SELECT * FROM cte_name;
```

- WITH ***cte*** (customer_id, total_amount) AS (SELECT cnum, SUM(amt) FROM orders GROUP BY cnum HAVING SUM(amt) > 50) SELECT c.cnum, c.cname, h.total_amount FROM customers c JOIN ***cte h*** ON c.cnum = ***h***.customer_id;

common table expression (CTE)

Recursive CTE Syntax

```
WITH RECURSIVE cte_name AS (
    -- Anchor member (non-recursive part, the base case)
    SELECT ...
    UNION ALL
    -- Recursive member (refers to the CTE name itself)
    SELECT ...
    FROM ... JOIN cte_name ON ...
    WHERE termination_condition
)
-- Main query
SELECT * FROM cte_name;
```

- WITH *RECURSIVE* ***cte*** AS (SELECT empno, ename, mgr, 1 AS level FROM emp WHERE mgr IS NULL UNION ALL SELECT e.empno, e.ename, e.mgr, ***c***.level + 1 FROM emp e, ***cte c*** WHERE e.mgr = ***c***.empno AND ***c***.level < 10) SELECT * FROM ***cte***;

common table expression (CTE) with INSERT

CTE with INSERT Syntax

```
INSERT INTO target_table (column1, column2, ...)
WITH cte_name (cte_column1, cte_column2, ...) AS (
    -- The CTE query that defines the temporary result set
    SELECT ...
    FROM ...
    WHERE ...
)
-- The final SELECT statement that pulls data from the CTE
SELECT cte_column1, cte_column2, ...
FROM cte_name;
```

- INSERT INTO x(c1) *RECURSIVE* WITH ***cte*** AS (SELECT 1 AS n UNION ALL SELECT n + 1 FROM ***cte*** WHERE n < 10) SELECT curdate() + INTERVAL n DAY FROM ***cte***;

common table expression (CTE) with UPDATE

CTE with INSERT Syntax

```
WITH cte_name (column_list) AS (
    -- The subquery that defines the CTE
    SELECT ...
    FROM ...
    WHERE ...
)
-- The UPDATE statement that uses the CTE
UPDATE table_to_be_updated AS t1
INNER JOIN cte_name AS t2
    ON t1.column_id = t2.column_id
SET t1.column_to_update = t2.new_value_from_cte
WHERE
    [condition];
```

'TOTAL

- WITH ***cte*** AS (SELECT emp.empno FROM emp, dept WHERE emp.deptno = dept.deptno AND dept.dname = 'HR') UPDATE emp SET sal = -1 WHERE empno IN (SELECT empno FROM ***cte***);

common table expression (CTE) with DELETE

CTE with INSERT Syntax

```
WITH cte_name (column_list) AS (
    -- The SELECT statement that defines the temporary result
set
    SELECT columns
    FROM table_name
    WHERE condition
)
-- The DELETE statement references the CTE in the WHERE or
JOIN clause
DELETE FROM table_name
WHERE primary_key_column IN (SELECT primary_key_column FROM
cte_name);
```

common table expression (CTE)

```
WITH [RECURSIVE]
    cte_name [(col_name [, col_name] . . .)] AS (subquery)
    [, cte_name [(col_name [, col_name] . . . AS (subquery)] . . .
```

- `WITH RECURSIVE a AS (SELECT 1 AS n UNION SELECT n + 1 FROM a WHERE n < 10)`
  `SELECT n FROM a;`
- `WITH RECURSIVE a AS (SELECT 1 AS n UNION SELECT n + 1 FROM a WHERE n < 10)`
  `SELECT NOW() + INTERVAL n DAY FROM a;`
- `WITH RECURSIVE a AS (SELECT 1 AS n UNION SELECT n + 1 FROM a WHERE n < 50)`
  `SELECT ROUND(RAND() * 50) + 50 FROM a;`
- `WITH a AS (SELECT MIN(sal) R1 FROM emp),`
  `b AS (SELECT MAX(sal) R2 FROM emp)`
  `SELECT * FROM emp WHERE sal IN ((SELECT R1 FROM a), (SELECT R2 FROM b));`
- `WITH a AS (SELECT job, COUNT(*) R1 FROM emp GROUP BY job)`
  `SELECT job FROM emp GROUP BY job HAVIN COUNT(job) = (SELECT MIN(R1) FROM a);`
- `WITH a AS (SELECT CASE WHEN sal <= 1500 THEN 'low' WHEN sal > 1501 AND sal < 3000 THEN 'medium' WHEN sal >= 3000 THEN 'high' END R1 FROM emp) SELECT *, COUNT(*) FROM a GROUP BY R1;`

common table expression (CTE)

```
WITH [RECURSIVE]
    cte_name [(col_name [, col_name] . . .)] AS (subquery)
    [, cte_name [(col_name [, col_name] . . . AS (subquery)] . . .
```

- WITH *RECURSIVE* ***m*** AS (SELECT empno, ename, mgr, 1 as ***level*** FROM emp WHERE mgr IS NULL UNION SELECT e.empno, e.ename, e.mgr, ***m.level*** + 1 FROM emp e, ***m*** WHERE ***m.empno*** = e.mgr) SELECT * FROM ***m*** ORDER BY level, ***m.mgr***;

common table expression (CTE) with insert

- INSERT INTO xx (c1) WITH *RECURSIVE* ***a*** AS (SELECT 1 *R1* UNION SELECT *R1* + 1 FROM ***a*** WHERE *R1* < 10) SELECT NOW() + INTERVAL R1 DAY D FROM ***a***;

common table expression (CTE) with update

- WITH *RECURSIVE* ***cte*** AS (SELECT 1 *n* UNION SELECT *n* + 1 FROM ***cte*** WHERE *n* <= 64) UPDATE pune_metro_stations SET distance_to_next_km = ROUND(2 * RAND() + 1 , 2);

- Generate numbers between 1 to 10

CTE

TODO

n
1
2
3
4
6
7
8
9
10

Output →

Output
1,2,3,4,5,6,7,8,9,10

- WITH *RECURSIVE* ***cte*** AS (SELECT 1 *n* UNION SELECT *n* + 1 FROM ***cte*** WHERE *n* < 10) SELECT GROUP_CONCAT(*n*) ***Output*** FROM ***cte***

- Find missing numbers.

CTE

Missing numbers

Table:- missingNumbers

id
1
2
3
4
6
.
.
.
7
8

Output →

Output (Missing number)

missingNumbers
5
10
11
12
13
.
.
.
23
24

- WITH *RECURSIVE* ***cte*** AS (SELECT 1 *n* UNION SELECT *n* + 1 FROM ***cte*** WHERE n < (SELECT MAX(id) FROM missingNumbers)) SELECT *n* missingNumbers FROM ***cte*** WHERE NOT EXISTS (SELECT True FROM missingNumbers WHERE ***cte***.*n* = missingnumbers.id);

- Find missing numbers.

CTE

Missing numbers

Table:- ord

ordid
601
602
603
.
.
.
.
631

Output →

Output (Missing number)

ordid
623

- WITH *RECURSIVE* ***cte*** AS (SELECT MIN(ordid) *n* FROM ord UNION SELECT *n* + 1 FROM ***cte*** WHERE *n* < (SELECT MAX(ordid) FROM ord)) SELECT *n* FROM ***cte*** WHERE NOT EXISTS (SELECT True FROM ord WHERE ord.ordid = *n*);

- Generate repeating numbers.

CTE

Repeating numbers

Table:- repeatingNumbers

num
1
2
3
4
. . . .
n

Output →

Output (Repeating number)

num
1
2
2
3
3
3
. . . .
n

- WITH *RECURSIVE* ***cte*** AS (SELECT 1 *n* UNION SELECT *n* + 1 FROM ***cte*** WHERE *n* < (SELECT MAX(num) FROM repeatingNumbers)) SELECT num FROM repeatingNumbers, ***cte*** WHERE num >= *n*;

- Find leap year.

CTE

Find Leap year

Table:- find_leap_year

c1
2000
2001
2002
. . . .
2025
Till system year

Output →

Output (Leap Year)

year	Output
2000	Leap Year
2001	Not a leap year...
2002	Not a leap year...
2003	Not a leap year...
. . . .	. . . .
2024	Leap Year
2025	Not a leap year...

- WITH *RECURSIVE* ***cte*** AS (SELECT MIN(c1) *n* FROM find_leap_year UNION SELECT *n* + 1 FROM ***cte*** WHERE *n* < (SELECT MAX(c1) FROM find_leap_year)) SELECT *n* year, CASE WHEN (*n*%4 = 0 AND *n*%100 != 0) OR (*n*%400 = 0) THEN 'Leap Year' ELSE 'Not leap year... ' END Output FROM ***cte***;

- Fill dates using INSERT statement.

CTE

Fill Dates using INSERT

Table:- x

c1

Output →

Output

c1
2025-06-15
2025-06-16
2025-06-17
2025-06-18
2025-06-19
2025-06-20
2025-06-21
2025-06-22
2025-06-23
2025-06-24

- CREATE TABLE x(c1 DATE);
- INSERT INTO x WITH *RECURSIVE* ***cte*** AS (SELECT 1 *n* UNION SELECT *n* + 1 FROM ***cte*** WHERE *n* < 10) SELECT CURDATE() + INTERVAL *n* DAY FROM ***cte***;

common table expression (CTE) assignments

WITH [*RECURSIVE*]
    cte_name [(*col_name* [, *col_name*] . . .)] AS (*subquery*)
    [, ***cte_name*** [(*col_name* [, *col_name*] . . . AS (*subquery*)] . . .

Customers and orders table.

Which customer spent the most.

```
WITH cte AS (SELECT distinct cname, ROUND(sum(amt) over(partition by o.cnum), 0) R1 FROM customers c, orders o
                WHERE c.cnum = o.cnum),
     cte1 AS (SELECT *, row_number() over(order by R1 desc) rnk FROM cte) SELECT * FROM cte1 WHERE rnk = 1;
```

Repeat Customers

common table expression (CTE)

WITH [*RECURSIVE*]
cte_name [(*col_name* [, *col_name*] . . .)] AS (*subquery*)
[, ***cte_name*** [(*col_name* [, *col_name*] . . . AS (*subquery*)] . . .

- WITH ***a*** AS (SELECT origin, destination FROM rockets),
  b AS (SELECT origin, destination FROM rockets)
  SELECT ***a***.* FROM ***a***, ***b*** WHERE (***a***.origin, ***a***.destination) = (***b***.destination,***b***.origin) ORDER BY 1;
- WITH ***a*** AS (SELECT channel FROM sales_channel),
  b AS (SELECT warehouse_Name FROM warehouse),
  c AS (SELECT channel, warehouse_Name FROM sales_channel, warehouse WHERE sales_channel.channel_ID = warehouse.channel_id)
  SELECT * FROM ***a***, ***b*** WHERE NOT EXISTS (SELECT TRUE FROM ***c*** WHERE (***a***.channel, ***b***.warehouse_Name) = (***c***.channel, ***c***.warehouse_Name)) ORDER BY 1;
- WITH ***a*** AS (SELECT DISTINCT productName FROM pivot_table),
  b AS (SELECT DISTINCT storeLocation FROM pivot_table),
  c AS (SELECT * FROM a, b)
  SELECT * FROM c WHERE NOT EXISTS (SELECT TRUE FROM pivot_Table WHERE (c.productName, c.storeLocation) = (productName, storeLocation));

common table expression (CTE)

WITH [*RECURSIVE*]
cte_name [(*col_name* [, *col_name*] . . .)] AS (*subquery*)
[, ***cte_name*** [(*col_name* [, *col_name*] . . . AS (*subquery*)] . . .

- WITH ***a*** AS (SELECT ename, deptno, gender FROM emp *e1* WHERE NOT EXISTS(SELECT True FROM emp *e2* WHERE *e1*.deptno = *e2*.deptno AND *e2*.gender = 'F')),

  b AS (SELECT ename, deptno, gender FROM emp *e3* WHERE NOT EXISTS(SELECT True FROM emp *e4* WHERE *e3*.deptno = *e4*.deptno AND *e4*.gender = 'M')),

  c AS (SELECT * FROM ***a*** UNION SELECT * FROM ***b***)
  SELECT * FROM emp *e* WHERE NOT EXISTS(SELECT True FROM ***c*** WHERE (*e*.deptno, *e*.gender) = (***c***.deptno, ***c***.gender));

common table expression (CTE)

```
WITH [RECURSIVE]
    cte_name [(col_name [, col_name] . . .)] AS (subquery)
    [, cte_name [(col_name [, col_name] . . . AS (subquery)] . . .
```

- ```
  WITH cte AS (SELECT ename, LENGTH(ename) - LENGTH(REPLACE(ename , ' ', '')) R1, REGEXP_INSTR(ename, ' ')
  firstSpace, REGEXP_INSTR(ename, ' ', REGEXP_INSTR(ename, ' ‘) + 1 ) secondSpace FROM fullname) SELECT cte.*,
  CASE WHEN R1 = 0 THEN ename ELSE SUBSTR(ename, 1, firstSpace - 1) END firstName, CASE WHEN R1 <= 1
  THEN NULL ELSE SUBSTR(ename, firstSpace + 1, secondSpace - firstSpace -1) END middleName, CASE WHEN R1 = 2
  THEN SUBSTR(ename, secondSpace + 1 ) WHEN R1 = 1 THEN SUBSTR(ename, firstSpace+1) END lastName FROM
  cte ORDER BY 2;
  ```

- ```
  WITH cte AS (SELECT sn.id sid, sn.ename sname, tn.id tid, tn.ename tname FROM sourceNames sn LEFT OUTER JOIN
  targetNames tn ON sn.id = tn.id UNION ALL SELECT sn.id sid, sn.ename sname, tn.id tid, tn.ename tname FROM
  sourceNames sn RIGHT OUTER JOIN targetNames tn ON sn.id=tn.id), cet2 AS ( SELECT cte.*, CASE WHEN
  (sid,sname)=(tid, tname) THEN "Common" WHEN sid IS NULL THEN "New in targer" WHEN tid IS NULL THEN
  "New is source" ELSE "Mismatch" END R1 FROM cte) SELECT * FROM cet2 WHERE R1 <> 'common';
  ```

Hierarchical Query (CTE)

A **Hierarchical Query** is a type of SQL query used to retrieve data that has a **parent–child relationship**, such as:

- ***Employee → Manager***
- ***Category → Subcategory***

```
SELECT C1 FROM T1 UNION SELECT C1 FROM T2
ORDER BY C1

(SELECT C1 FROM T1 ORDER BY C1) UNION
(SELECT C1 FROM T2 ORDER BY C1)

SELECT C1 FROM T1 ORDER BY C1 UNION
SELECT C1 FROM T2 ORDER BY C1   //ERROR
```

(UNION ALL, EXCEPT ALL, INTERSECT ALL)

set operation in sql

Set operators are used to join the results of two (or more) SELECT statements.

There are set union (**UNION ->** ∪), set difference (**EXCEPT ->** -), and set intersection (**INTERSECT ->** ∩) operations. The relations resulting from these set operations are sets of tuples; that is, duplicate tuples are eliminated from the result.

Remember:

- The result set column names are taken from the column names of the first SELECT statement.
- SELECT statement should have the same data type. (Not in MySQL)
- UNION: To apply ORDER BY or LIMIT to an individual SELECT, place the clause inside the parentheses that enclose the SELECT.

  e.g. (SELECT . . .) UNION (SELECT . . .)

union / intersect / except

```
SELECT name FROM students          /* Fetch the union of queries */
UNION
SELECT name FROM contacts;

SELECT name FROM students          /* Fetch the union of queries with duplicates*/
UNION ALL
SELECT name FROM contacts;

SELECT name FROM students          /* Fetch names from students */
EXCEPT                             /* that aren't present in contacts */
SELECT name FROM contacts;

SELECT name FROM students          /* Fetch names from students */
INTERSECT                          /* that are present in contacts as well */
SELECT name FROM contacts;
```

- CREATE TABLE student (
  studentID INT PRIMARY KEY,
  name VARCHAR(45)
  );

- CREATE TABLE Facebook (
  facebookID INT PRIMARY KEY,
  studentID INT,
  name VARCHAR(45),
  location VARBINARY(45)
  );

- CREATE TABLE LinkedIn (
  linkedInID INT PRIMARY KEY,
  studentID INT,
  name VARCHAR(45),
  location VARBINARY(45)
  );

1. **All students who are either having Facebook or LinkedIn.**

   - SELECT studentID FROM Facebook UNION SELECT studentID FROM LinkedIn;

   Returns unique student IDs in ***either*** Facebook OR LinkedIn.

2. **Students with both Facebook or LinkedIn.**

   - SELECT studentID FROM Facebook INTERSECT SELECT studentID FROM LinkedIn;

   Returns student IDs in ***both*** tables[Facebook AND LinkedIn]

3. **Students with a Facebook but no LinkedIn and vice versa.**

   - SELECT studentID FROM Facebook EXCEPT SELECT studentID FROM LinkedIn;

   Returns student IDs who **have a** Facebook but **did not** LinkedIn.

union

syntax

```
SELECT . . . UNION [ALL | DISTINCT]
SELECT . . . [UNION [ALL | DISTINCT]
SELECT . . .]
```

- SELECT DISTINCT * FROM duplicate;
- SELECT * FROM duplicate UNION SELECT * FROM duplicate;
- SELECT deptno, dname, loc, walletid FROM (SELECT ROW_NUMBER() OVER(PARTITION BY deptno) R1, duplicate.* FROM duplicate) T1 WHERE R1=1;

Note:

- It is used to combine two or more result sets (SELECT statements) into a single set
- it removes duplicate rows between the various SELECT statements.
- each SELECT statement within the UNION operator must have the same number of fields in the result sets.
- default behaviour for UNION is that duplicate rows are removed from the result.

- (SELECT deptno FROM emp LIMIT 1) UNION (SELECT deptno FROM dept LIMIT 1);
- SELECT 'EMP' as *'Table Name'*, COUNT(*) FROM emp UNION SELECT *'DEPT'*, COUNT(*) FROM dept UNION SELECT 'BONUS', COUNT(*) FROM bonus;
- SELECT COUNT(*) FROM customer UNION SELECT COUNT(*) FROM ord;
- SELECT * FROM emp WHERE deptno NOT IN (SELECT deptno FROM emp *m* WHERE *m*.deptno NOT IN (SELECT deptno FROM emp *f* WHERE gender = 'F') UNION SELECT deptno FROM emp *m* WHERE *m*.deptno NOT IN (SELECT deptno FROM emp *f* WHERE gender = 'M'));

union

books

bookID	bookName	Type	Cost
1	DS	Hardcover	950
1	DS	Hardcover	950
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

newbooks

bookID	bookName	Type	Cost
1	Redis	Paperback	850
1	Redis	Paperback	850
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

Output

bookID	bookName	Type	Cost
1	DS	Hardcover	950
2	JavaScript	Paperback	700
1	Redis	Paperback	850

❖ ***r*** UNION ***s*** includes distinct/unique rows appearing in both result sets ***r*** and ***s***.

Duplicate rows not repeated in result set

- SELECT * FROM books UNION SELECT * FROM newbooks;
- SELECT * FROM emp WHERE sal IN (SELECT MAX(sal) FROM emp UNION SELECT MIN(sal) FROM emp);

Note:

The following statement will give an error

- SELECT bookName, type FROM books ORDER BY bookname
  UNION
  SELECT bookName, type FROM newbooks;

union all

books

bookID	bookName	Type	Cost
1	DS	Hardcover	950
1	DS	Hardcover	950
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

newbooks

bookID	bookName	Type	Cost
1	Redis	Paperback	850
1	Redis	Paperback	850
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

Output

bookID	bookName	Type	Cost
1	DS	Hardcover	950
1	DS	Hardcover	950
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700
1	Redis	Paperback	850
1	Redis	Paperback	850
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

❖ ***r*** UNION ***s*** includes all rows appearing in both result sets ***r*** and ***s***, including duplicates.

Duplicate rows are repeated in result set

- SELECT * FROM books UNION ALL SELECT * FROM newbooks;
- SELECT * FROM emp UNION ALL SELECT * FROM emp;
- SELECT bookname, COUNT(*) FROM (SELECT bookname FROM books UNION ALL SELECT bookname FROM newbooks) *b* GROUP BY bookname;

union & union all

todo

Count number of employees per job title, including total row

Table:- emp

Empno	Ename	Job	. . .	isActive
6001	GITA	Compliance officers	. . .	. . .
6129	VRUSHALI	Compliance officers	. . .	. . .
6473	SHARMIN	Public Relation	. . .	. . .
6781	BANDISH	Public Relation	. . .	. . .
. . .	. . .	. . .	. . .	. . .
9400	SANGITA	Compliance officers	. . .	. . .
9473	SUPRIYA	ANALYST	. . .	. . .

Output

job	COUNT(*)
Compliance officers	3
Public Relation	2
CLERK	7
SALESMAN	6
MANAGER	5
ANALYST	3
PRESIDENT	1
vice PRESIDENT	1
TOTAL	28

Output →

- SELECT job, COUNT(*) FROM emp GROUP BY job UNION ALL SELECT 'TOTAL', COUNT(*) FROM emp;

subqueries in the from clause with union & union all

Count all records from delivery carriers(FEDEX, DHL, and UPS)

todo

Table:- DHL / UPS / FEDEX (same structure with different dataset)

tracking_id	carrier	status	delivery_date
DHL100021	DHL	Delivered	2025-07-04
DHL100022	DHL	In Transit	2025-07-05
DHL100023	DHL	Dispatched	2025-07-06
DHL100024	DHL	Out for Delivery	2025-07-07
.	.	.	.
.	.	.	.
.	.	.	.
DHL100025	DHL	Delivered	2025-07-08
DHL100026	DHL	In Transit	2025-07-03

Output →

Output

FedEx	DHL	UPS
7	8	10

- WITH **cte** AS (SELECT * FROM fedex UNION SELECT * FROM dhl UNION SELECT * FROM ups) SELECT COUNT(CASE WHEN carrier = 'FedEx' THEN 1 END) FedEx, COUNT(CASE WHEN carrier = 'DHL' THEN 1 END) DHL, COUNT(CASE WHEN carrier = 'UPS' THEN 1 END) UPS FROM **cte**;

- SELECT COUNT(CASE WHEN carrier = 'FedEx' THEN 1 END) *FedEx*, COUNT(CASE WHEN carrier = 'DHL' THEN 1 END) *DHL*, COUNT(CASE WHEN carrier = 'UPS' THEN 1 END) *UPS* FROM (SELECT * FROM FEDEX UNION SELECT * FROM DHL UNION SELECT * FROM ups) t1;

subqueries in the from clause with union & union all

Aassignment's → Try-it

todo

Table:- DHL / UPS / FEDEX (same structure with different dataset)

tracking_id	carrier	status	delivery_date
DHL100021	DHL	Delivered	2025-07-04
DHL100022	DHL	In Transit	2025-07-05
DHL100023	DHL	Dispatched	2025-07-06
DHL100024	DHL	Out for Delivery	2025-07-07
.	.	.	.
.	.	.	.
.	.	.	.
DHL100025	DHL	Delivered	2025-07-08
DHL100026	DHL	In Transit	2025-07-03

Use Cases

1. Show all dispatched shipments
2. Count delivered vs. pending
3. Track shipments scheduled for today
4. Combine with other carriers using UNION ALL
5. Check All Delayed Shipments
6. Shipments Scheduled for Delivery Today
7. Count of Shipments by Status
8. Total Shipments Per Day
9. Latest Shipment Record
10. List All Pending Shipments

DHL { *tracking_id, carrier, status, delivery_date* }

UPS { *tracking_id, carrier, status, delivery_date* }

FedEx { *tracking_id, carrier, status, delivery_date* }

DHL { *tracking_id, carrier, status, delivery_date* }

UPS { *tracking_id, carrier, status, delivery_date* }

FedEx { *tracking_id, carrier, status, delivery_date* }

union

1. SELECT * FROM (SELECT * FROM dhl UNION SELECT * FROM fedex UNION SELECT * FROM ups) t1 WHERE status = 'Dispatched';
2. SELECT status, COUNT(CASE WHEN carrier = 'DHL' THEN 1 END) *DHL*, COUNT(CASE WHEN carrier = 'UPS' THEN 1 END) *UPS*, COUNT(CASE WHEN carrier = 'FedEx' THEN 1 END) *FedEx* FROM (SELECT * FROM dhl UNION SELECT * FROM FedEx UNION SELECT * FROM ups) t1 WHERE status IN ('Delivered', 'Pending') GROUP BY status;

status	DHL	UPS	FedEx
Delivered	1	2	1
Pending	2	1	2

3. SELECT * FROM (SELECT * FROM ups UNION SELECT * FROM FedEx UNION SELECT * FROM dhl) t1 WHERE delivery_date = '2025-07-07';

tracking_id	carrier	status	delivery_date
UPS100033	UPS	Dispatched	2025-07-07
FEDEX100004	FedEx	Out for Delivery	2025-07-07
DHL100024	DHL	Out for Delivery	2025-07-07

subqueries in the from clause with union & union all

Total Payments from Two Sources

todo

Table:- cashPayments

student_id	sutdent_name	amount
27	saleel	4000
43	sharmin	2000
25	ruhan	2600
41	vrushali	2200
20	nitish	3200
7	neel	3400
14	bandish	3200
17	naresh	3400
27	saleel	2500

Table:- cardPayments

student_id	sutdent_name	amount
43	sharmin	2300
9	bhuru	3200
63	bhavin	3000
39	boy	3400
20	nitish	1200
27	saleel	4000
27	saleel	3500

cashPayments.amount + cardPayments.amount

Output with UNION

Output →

totalPayment
29900

Output with UNION ALL

Output →

totalPayment
47100

- SELECT SUM(amount) totalPayment FROM (SELECT amount FROM cashpayments UNION SELECT amount FROM cardpayments) t1;
- SELECT SUM(amount) totalPayment FROM (SELECT amount FROM cashpayments UNION ALL SELECT amount FROM cardpayments) t1;

subqueries in the from clause with union

todo

Table:- sourceCity

ID	cityName
1	Baroda
2	Surat
3	Rajkot
4	Anand

Table:- targetCity

ID	cityName
1	Baroda
2	Surat
4	Bharuch
5	Bhuj

Output →

Output

sID	sCityName	tID	tCityName	**Output**
1	Baroda	1	Baroda	Matching in both table
2	Surat	2	Surat	Matching in both table
3	Rajkot	NULL	NULL	New in source table
4	Anand	4	Bharuch	Mismatch Cities
NULL	NULL	5	Bhuj	New in target table

- SELECT T1.*, CASE WHEN (sid, scityname) = (tid, tcityname) THEN 'Matching in both table' WHEN tid IS NULL THEN 'New in source table' WHEN sid IS NULL THEN 'New in target table' WHEN (sid, scityname) <> (tid, tcityname) THEN 'Mismatch Cities' END Output FROM (SELECT s.id *sid*, s.cityName *sCityName*, t.id *tid*, t.cityName *tCityName* FROM sourceCity *S* LEFT OUTER JOIN targetCity *T* ON s.id = t.id UNION SELECT s.id *sid*, s.cityname *sCityname*, t.id *tid*, t.cityname *tCityname* FROM sourceCity s RIGHT OUTER JOIN targetCity t on s.id = t.id) T1;

Table:- onlineOrders

orderID	customerID	product	category	quantity	orderDate	amount
1001	C001	Laptop	Electronics	1	2025-07-01	83000
1002	C002	Smartphone	Electronics	2	2025-07-02	47000
1003	C003	Headphones	Electronics	1	2025-07-02	2500
1004	C004	T-shirt	Clothing	3	2025-07-03	1200
1005	C005	Sneakers	Footwear	1	2025-07-03	3200
1006	C006	Book	Books	2	2025-07-04	800
1007	C001	Power Bank	Electronics	1	2025-07-04	1700

Table:- fromStoreOrders

orderID	customerID	product	category	quantity	orderDate	amount
2001	C001	T-shirt	Clothing	2	2025-07-01	1200
2002	C003	Smartphone	Electronics	1	2025-07-01	35000
2003	C007	Formal Shoes	Footwear	1	2025-07-02	2000
2004	C004	Jeans	Clothing	1	2025-07-03	3500
2005	C008	Book	Books	3	2025-07-04	700
2006	C002	Laptop	Electronics	1	2025-07-04	93500
2007	C009	Power Bank	Electronics	1	2025-07-05	1800
2008	C021	Mouse	Electronics	7	2025-07-09	350

onlineOrders { *orderID, customerID, product, category, quantity, orderDate, amount* }

union

fromStoreOrders { *orderID, customerID, product, category, quantity, orderDate, amount* }

Use Cases

1. Show all customers who made online or from-store purchases
2. Customers who purchased both online and from-store
3. Customers who ordered online but not from-store
4. Show all customers with product details who made online or from-store purchase with the status ‘Purchased online’ or ‘Purchased from store’
5. Show all customers who made online of from-store purchase of ‘Laptop’
6. Count total sales made online or from-store
7. Show total sales done either from online or from-store with appropriate message.
8. Show all customers who made online purchase only
9. Show all customers who made form-store purchase only
10. Show total online orders
11. Show total from-store orders
12. Show total online amount
13. Show total from-store amount
14. Show total online orders
15. Show total from-store orders

onlineOrders { *orderID, customerID, product, category, quantity, orderDate, amount* }

fromStoreOrders { *orderID, customerID, product, category, quantity, orderDate, amount* }

union

1. SELECT customerID, product, category, 'Online Purchase' FROM onlineOrders UNION SELECT customerID, product, category, 'Store Purchase' FROM fromStoreOrders;
2. SELECT customerID, 'Both Online and Store Purchase' FROM onlineOrders INTERSECT SELECT customerID, 'Both Online and Store Purchase' FROM fromStoreOrders;
3. SELECT customerID, 'Purchased Online' AS "Purchase Type" FROM (SELECT customerID FROM onlineOrders EXCEPT SELECT customerID FROM fromStoreOrders) t1;
4. SELECT OO.customerID, OO.product, OO.amount, 'Purchased online' *status* FROM onlineOrders OO UNION SELECT SO.customerID, SO.product, SO.amount, 'Purchased from store' *status* FROM fromStoreOrders SO;
5. SELECT * FROM (SELECT OO.customerID, OO.product, 'Purchased online' *status* FROM onlineOrders OO UNION SELECT SO.customerID, SO.product, 'Purchased from store' *status* FROM fromStoreOrders SO) t1 WHERE product = 'Laptop';
6. SELECT COUNT(*), 'Total sales done online' *status* FROM onlineOrders OO UNION SELECT COUNT(*), 'Total sales done from store' *status* FROM fromStoreOrders SO;

- TODO

union all

Table:- cricketwinnerteam

Team1	Team2	Winner
India	SL	India
SL	Aus	Aus
SA	Eng	Eng
Eng	NZ	NZ
Aus	India	India

Output →

Output

Team Name	**Matches played**	**Matches won**	**Matches lost**
India	2	2	0
SL	2	0	2
SA	1	0	1
Eng	2	1	1
Aus	2	1	1
NZ	1	1	0

- SELECT team1 *"Team Name"*, COUNT(*) *"Matches played"*, SUM(Winner) *"Matches won"*, COUNT(*) - SUM(Winner) *"Matches lost"* FROM (SELECT team1, CASE WHEN team1 = winner THEN 1 ELSE 0 END *Winner* FROM cricketwinnerteam UNION ALL SELECT team2, CASE WHEN team2 = winner THEN 1 ELSE 0 END FROM cricketwinnerteam) T1 GROUP BY team1;

intersect / intersect all

It is used to combine two result sets and returns the data which are common in both the result set.

books

bookID	bookName	Type	Cost
1	DS	Hardcover	950
1	DS	Hardcover	950
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

newbooks

bookID	bookName	Type	Cost
1	Redis	Paperback	850
1	Redis	Paperback	850
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

Output

bookID	bookName	Type	Cost
2	JavaScript	Paperback	700

Output

bookID	bookName	Type	Cost
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

❖ ***r*** INTERSECT ***s*** includes only rows appearing in both result sets ***r*** and ***s***.

- SELECT * FROM books INTERSECT SELECT * FROM newbooks;
- SELECT product_id FROM VendorProducts WHERE vendor_id = 'A' INTERSECT SELECT product_id FROM VendorProducts WHERE vendor_id = 'B'; //same products sold by different vendors [A and B].
- SELECT custid FROM ord WHERE YEAR(orderdate) = 1986 INTERSECT SELECT custid FROM ord WHERE YEAR(orderdate) = 1987;
- SELECT bookName, type FROM books WHERE EXISTS (SELECT bookname, type FROM newbooks WHERE books.bookName = newbooks.bookName);
- SELECT bookName, type FROM books WHERE bookName IN (SELECT bookName FROM newbooks);

difference between intersect and inner join

todo

Difference between INTERSECT and INNER JOIN

Table:- managers

emp_id	name
1	saleel
2	sharmin
3	vrushali

Table:- trainers

emp_id	name
1	ruhan
2	sharmin
3	saleel

Output →

Output

emp_id	name
2	sharmin

- SELECT * FROM managers INTERSECT SELECT * FROM trainers;

Table:- managers

emp_id	name
1	saleel
2	sharmin
3	vrushali

Table:- trainers

emp_id	name
1	ruhan
2	sharmin
3	saleel

Output →

Output

emp_id	name	emp_id	name
1	saleel	1	ruhan
2	sharmin	2	sharmin
3	vrushali	3	saleel

- SELECT * FROM managers, trainers WHERE managers.emp_id = trainers.emp_id;

intersect

todo

Payments done by students through Cash and Card both the ways

Table:- cashPayments

student_id	sutdent_name	amount
27	saleel	4000
43	sharmin	2000
25	ruhan	2600
41	vrushali	2200
20	nitish	3200
7	neel	3400
14	bandish	3200
17	naresh	3400
27	saleel	2500

Table:- cardPayments

student_id	sutdent_name	amount
43	sharmin	2300
9	bhuru	3200
63	bhavin	3000
39	boy	3400
20	nitish	1200
27	saleel	4000
27	saleel	3500

Output →

Output with INTERSECT

student_id	Remark
27	Payment done . . .
43	Payment done . . .
20	Payment done . . .

Output with INTERSECT ALL

student_id	Remark
27	Payment done . . .
27	Payment done . . .
43	Payment done . . .
20	Payment done . . .

- SELECT student_id, 'Payment done by Case as well as Card' *Remark* FROM cashpayments INTERSECT SELECT student_id, 'Payment done by Case as well as Card' FROM cardpayments;
- SELECT student_id, 'Payment done by Case as well as Card' *Remark* FROM cashpayments INTERSECT ALL SELECT student_id, 'Payment done by Case as well as Card' FROM cardpayments;

subqueries in the from clause with intersect & intersect all

Total (SUM) of Payments done by Cash and Card both ways

todo

Table:- cashPayments

student_id	sutdent_name	amount
27	saleel	4000
43	sharmin	2000
25	ruhan	2600
41	vrushali	2200
20	nitish	3200
7	neel	3400
14	bandish	3200
17	naresh	3400
27	saleel	2500

Table:- cardPayments

student_id	sutdent_name	amount
43	sharmin	2300
9	bhuru	3200
63	bhavin	3000
39	boy	3400
20	nitish	1200
27	saleel	4000
27	saleel	3500

Output with INTERSECT

Output →

totalPayment
10600

Output with INTERSECT ALL

Output →

totalPayment
10600

cashPayments.amount + cardPayments.amount

- SELECT SUM(amount) totalPayment FROM (SELECT amount FROM cashpayments INTERSECT SELECT amount FROM cardpayments) t1;
- SELECT SUM(amount) totalPayment FROM (SELECT amount FROM cashpayments INTERSECT ALL SELECT amount FROM cardpayments) t1;

except / except all

It is used to combine two result sets and returns the data from the first result set which is not present in the second result set.

books

bookID	bookName	Type	Cost
1	DS	Hardcover	950
1	DS	Hardcover	950
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

newbooks

bookID	bookName	Type	Cost
1	Redis	Paperback	850
1	Redis	Paperback	850
2	JavaScript	Paperback	700
2	JavaScript	Paperback	700

Output

bookID	bookName	Type	Cost
1	DS	Hardcover	950

Output

bookID	bookName	Type	Cost
1	DS	Hardcover	950
1	DS	Hardcover	950

❖ ***r*** EXCEPT ***s*** returns only those rows from result set ***r*** which do not appear in ***s***.

- SELECT * FROM books EXCEPT SELECT * FROM newbooks;
- SELECT * FROM books EXCEPT ALL SELECT * FROM newbooks;

- SELECT bookName, type FROM books WHERE NOT EXISTS (SELECT bookName, type FROM newbooks WHERE books.bookName = newbooks.bookName);
- SELECT bookName, type FROM newbooks WHERE NOT EXISTS (SELECT bookName, type FROM books WHERE books.bookName = newbooks.bookName);

Note:

except

- There is no MINUS operator in MySQL, you can easily simulate this type of query using either the EXCEPT , NOT IN clause or the NOT EXISTS clause.

1.
```
SELECT * FROM books          /* Fetch  everything from books */
EXCEPT                       /* that are not present in newbooks */
SELECT * FROM newbooks;
```

2.
```
SELECT * FROM newbooks  /* Fetch everything  from newbooks */
EXCEPT                  /* that are not present in books */
SELECT * FROM books;
```

except

todo

Customers who have never placed an order.

Table:- customers

CNUM	CNAME	. . .
1001	Saleel	. . .
1037	Nitish	. . .
2001	Santosh	. . .
2002	Joe	. . .
2003	Raj	. . .
.	.	. . .
.	.	
.	.	

Table:- orders

ONUM	CNUM	. . .
2322	NULL	. . .
2364	NULL	. . .
2475	NULL	. . .
3001	2008	. . .
3002	2007	. . .
.	.	. . .
.	.	
.	.	

Output →

Output

CNUM
1001
1037
2009
2010

- SELECT cnum FROM customers EXCEPT SELECT cnum FROM orders;

rows to columns(pivot) and columns to rows (unpivot)

- **pivot** operation involves transforming rows into columns.
- **unpivoting** operation involves transforming columns into rows.

rows to columns (pivot table)

pivot operation involves transforming rows into columns.

syntax

```
SELECT
   row_identifier_column,
   SUM(CASE WHEN column_to_pivot = 'value1' THEN value_column ELSE 0 END) AS value1,
   SUM(CASE WHEN column_to_pivot = 'value2' THEN value_column ELSE 0 END) AS value2,
   SUM(CASE WHEN column_to_pivot = 'value3' THEN value_column ELSE 0 END) AS value3
FROM table_name
GROUP BY
   row_identifier_column;
```

- TODO

rows to columns (pivot table)

pivot operation involves transforming rows into columns.

Table:- fruits

product	amount	country
banana	1000	usa
carrots	1500	usa
beans	1600	usa
orange	2000	usa
banana	400	india
carrots	1200	india
beans	1500	india
orange	4000	india
banana	2000	canada
carrots	2000	canada
beans	2000	mexico

Output →

Output

country	banana	carrots	beans	orange
usa	1000	1500	1600	2000
india	400	1200	1500	4000
canada	2000	2000	NULL	NULL
mexico	NULL	NULL	2000	NULL

- SELECT country, MAX(CASE WHEN product = 'banana' THEN amount END) *'banana'*, MAX(CASE WHEN product = 'carrots' THEN amount END) *'carrots'*, MAX(CASE WHEN product = 'beans' THEN amount END) *'beans'*, MAX(CASE WHEN product = 'orange' THEN amount END) *'orange'* FROM fruits GROUP BY country;

rows to columns (pivot table)

pivot operation involves transforming rows into columns.

```
SELECT custid,
  CASE WHEN MONTH(orderdate) = 1 THEN total END January,
  CASE WHEN MONTH(orderdate) = 2 THEN total END February,
  CASE WHEN MONTH(orderdate) = 3 THEN total END March,
  CASE WHEN MONTH(orderdate) = 4 THEN total END April,
  CASE WHEN MONTH(orderdate) = 5 THEN total END May,
  CASE WHEN MONTH(orderdate) = 6 THEN total END June,
  CASE WHEN MONTH(orderdate) = 7 THEN total END July,
  CASE WHEN MONTH(orderdate) = 8 THEN total END August,
  CASE WHEN MONTH(orderdate) = 9 THEN total END September,
  CASE WHEN MONTH(orderdate) = 10 THEN total END October,
  CASE WHEN MONTH(orderdate) = 11 THEN total END November,
  CASE WHEN MONTH(orderdate) = 12 THEN total END December FROM ord;
```

```
SELECT custid,
 COUNT(CASE WHEN QUARTER(orderdate) = 1 THEN total END) "First Quarter",
 COUNT(CASE WHEN QUARTER(orderdate) = 2 THEN total END) "Second Quarter",
 COUNT(CASE WHEN QUARTER(orderdate) = 3 THEN total END) "Third Quarter",
 COUNT(CASE WHEN QUARTER(orderdate) = 4 THEN total END) "Forth Quarter" FROM ord GROUP BY custid;
```

rows to columns (pivot table)

pivot operation involves transforming rows into columns.

```
SELECT custid,
  SUM(CASE WHEN MONTH(orderdate) = 1  THEN total END) January,
  SUM(CASE WHEN MONTH(orderdate) = 2 THEN total END) February,
  SUM(CASE WHEN MONTH(orderdate) = 3 THEN total END) March,
  SUM(CASE WHEN MONTH(orderdate) = 4 THEN total END) April,
  SUM(CASE WHEN MONTH(orderdate) = 5 THEN total END) May,
  SUM(CASE WHEN MONTH(orderdate) = 6 THEN total END) June,
  SUM(CASE WHEN MONTH(orderdate) = 7 THEN total END) July,
  SUM(CASE WHEN MONTH(orderdate) = 8 THEN total END) August,
  SUM(CASE WHEN MONTH(orderdate) = 9 THEN total END) September,
  SUM(CASE WHEN MONTH(orderdate) = 10 THEN total END) October,
  SUM(CASE WHEN MONTH(orderdate) = 11 THEN total END) November,
  SUM(CASE WHEN MONTH(orderdate) = 12 THEN total END) December FROM ord GROUP BY custid;

SELECT custid,
REPLACE(COUNT(CASE WHEN QUARTER(orderdate) = 1 THEN total END), 0, "no purchase") "First Quarter",
REPLACE(COUNT(CASE WHEN QUARTER(orderdate) = 2 THEN total END), 0, "no purchase") "Second Quarter",
REPLACE(COUNT(CASE WHEN QUARTER(orderdate) = 3 THEN total END), 0, "no purchase") "Third Quarter",
REPLACE(COUNT(CASE WHEN QUARTER(orderdate) = 4 THEN total END), 0, "no purchase") "Forth Quarter" FROM ord
GROUP BY custid;
```

rows to columns (pivot table)

pivot operation involves transforming rows into columns.

columns to rows (unpivot table)

unpivoting operation involves transforming columns into rows.

syntax

```
SELECT row_identifier, 'column1_name' AS column_name, column1 AS value FROM table_name
UNION ALL
SELECT row_identifier, 'column2_name' AS column_name, column2 AS value FROM table_name
UNION ALL
SELECT row_identifier, 'column3_name' AS column_name, column3 AS value FROM table_name;
```

columns to rows (unpivot table)

unpivoting operation involves transforming columns into rows.

create table … like statement

create table ... like

Use CREATE TABLE ... LIKE to create an empty table based on the definition of another table, including any column attributes and indexes defined in the original table.

```
CREATE TABLE [IF NOT EXISTS] new_tbl LIKE orig_tbl;
```

- LIKE works only for base tables, not for VIEWS.
- You cannot execute CREATE TABLE or CREATE TABLE ... LIKE while a LOCK TABLES statement is in effect.

- `CREATE TABLE e LIKE emp;`

MEMORY tables are visible to another client, but TEMPORARY tables are not visible to another client.

create temporary table ... like statement

create temporary table ... like

Use CREATE TABLE ... LIKE to create an empty table based on the definition of another table.

```
CREATE TEMPORARY TABLE [IF NOT EXISTS] new_tbl LIKE orig_tbl;
```

- LIKE works only for base tables, not for VIEWS.
- You cannot execute CREATE TABLE or CREATE TABLE ... LIKE while a LOCK TABLES statement is in effect.

You can use the TEMPORARY keyword when creating a table. A TEMPORARY table is visible only to the current session, and is dropped automatically when the session is closed.

```
CREATE TEMPORARY TABLE e like emp;
```

Use TEMPORARY table with the same name as the original can be useful when you want to try some statements that modify the contents of the table, without changing the original table.

create table … select statement

create table ... select

You can create one table from another by adding a SELECT statement at the end of the CREATE TABLE statement.

```
CREATE TABLE new_tbl(column_definition) [as] SELECT * FROM orig_tbl;
```

You cannot use FOR UPDATE as part of the SELECT in a statement such as CREATE TABLE new_table SELECT ... FROM old_table .... FOR UPDATE. If you attempt to do so, the statement fails.

- `CREATE TABLE e SELECT * FROM emp;`
- `CREATE TABLE e as SELECT * FROM emp;`
- `CREATE TABLE e (a INT, b INT DEFAULT 200) SELECT * FROM emp;` `//Add new column('s) with existing columns.`
- `CREATE TABLE e as SELECT 1+1, ename FROM emp;`
- `CREATE TABLE e as SELECT 1+1 "R1", ename FROM emp;`
- `CREATE TABLE e as SELECT * FROM emp WHERE 1=2;`

Note:

- By default, this statement does not copy all column attributes such as AUTO_INCREMET

- SET autocommit = { 0 | 1 } 0 - disable; 1 - enable

- show variables like 'auto%';
- SELECT @@AUTOCOMMIT;

Turning the autocommit off globally. Open **my.ini** file and add the following statement.

```
[mysqld]
autocommit = 0
```

After adding the line re-start the MySQL server.

commit and rollback

Remember:

- When you **UPDATE** or **DELETE** a row, that row is automatically locked for other users.
- ROW LOCKING IS AUTOMATIC IN MYSQL AND ORACLE.
- When you **UPDATE** or **DELETE** a row, that row becomes READ ONLY for other users.
- Other users can **SELECT** from that table; they will view the old data before your changes.
- Other users can **INSERT** rows into that table.
- Other users can **UPDATE** or **DELETE** "other" rows of that table.
- No other user can **UPDATE** or **DELETE** your locked row, till you have issued a Rollback or Commit.
- To lock the rows manually you require **SELECT** statement with a **FOR UPDATE** clause.
- **LOCKS ARE AUTOMATICALLY RELEASED WHEN YOU ROLLBACK OR COMMIT.**

commit and rollback

To undo MySQL statements you use the ROLLBACK statement. To write the changes into the database within a transaction, you use the COMMIT statement.

```
START TRANSACTION
    [transaction_characteristic]

transaction_characteristic:
  | READ WRITE
  | READ ONLY
```

Note:

- It is important to note that MySQL automatically commits the changes to the database by default.

```
START TRANSACTION READ ONLY;
INSERT INTO dept VALUES (60, 'HRD', 'Pune' );
COMMIT
```

commit and rollback

To undo MySQL statements you use the ROLLBACK statement. To write the changes into the database within a transaction, you use the COMMIT statement.

```
BEGIN [work]
COMMIT [work]
SAVEPOINT identifier
ROLLBACK [work] TO [SAVEPOINT] identifier
SET autocommit = {0 | 1}
```

Note:

- It is important to note that MySQL automatically commits the changes to the database by default.

- `BEGIN WORK`
- `INSERT INTO dept VALUES (60, 'HRD', 'Pune');`
- `ROLLBACK / COMMIT WORK`

- `SAVEPOINT pt0;`
- `INSERT INTO dept VALUES (60, 'HRD', 'Pune');`
- `SAVEPOINT pt1;`
- `INSERT INTO dept VALUES (61, 'HRD', 'baroda' );`
- `ROLLBACK TO SAVEPOINT pt1;`

select .... for update

select - for update

Sets a shared mode lock on any rows that are read. Other sessions can read the rows, but cannot modify them until your transaction commits. If any of these rows were changed by another transaction that has not yet committed, your query waits until that transaction ends and then uses the latest values.

- SELECT * FROM emp FOR UPDATE;

truncate table

Remember:

- DROP and TRUNCATE are DDL commands, whereas DELETE is a DML command.
- DELETE operations can be rolled back, while DROP and TRUNCATE operations cannot be rolled back.
- The TRUNCATE TABLE statement removes all the data/rows of a table and resets the auto-increment value to zero.

truncate table

Logically, TRUNCATE TABLE is similar to a DELETE statement that deletes all rows, or a sequence of DROP TABLE and CREATE TABLE statements.

TRUNCATE [TABLE] tbl_name

Remember:

- Truncate operations drop and re-create the table, which is much faster than deleting rows one by one.
- Truncate operations cause an implicit commit, and so cannot be rolled back.
- Truncate does not cause ON DELETE triggers to fire.
- Truncate cannot be performed for parent-child foreign key relationships.
- Truncate retain Identity and reset to the seed (start value) value.
- Cannot truncate a table referenced in a foreign key constraint.
- Any AUTO_INCREMENT value is reset to its start value [0].

DELETE Vs TRUNCATE

delete vs truncate

DELETE	TRUNCATE
You can specify the tuple that you want to delete.	It deletes all the tuples from a relation.
DELETE is a Data Manipulation Language command.	TRUNCATE is a Data Definition Language command.
DELETE command can have WHERE clause.	TRUNCATE command do not have WHERE clause.
DELETE command activates the trigger applied on the table and causes them to fire.	TRUNCATE command does not activate the triggers to fire.
DELETE command eliminate the tuples one-by-one.	TRUNCATE delete the entire data page containing the tuples.
DELETE command lock the row/tuple before deleteing it.	TRUNCATE command lock data page before deleting table data.
DELETE command acts slower as compared to TRUNCATE.	TRUNCATE is faster as compared to DELETE.
DELETE records transaction log for each deleted tuple.	TRUNCATE record transaction log for each deleted data page.
DELETE command can be followed either by COMMIT or ROLLBACK.	TRUNCATE command can't be ROLLBACK.

DROP Vs TRUNCATE

drop vs truncate

DROP	TRUNCATE
The DROP command is used to remove table definition and its contents.	Whereas the TRUNCATE command is used to delete all the rows from the table.
In the DROP command, VIEW of table does not exist.	In the TRUNCATE command, VIEW of table exist.
In the DROP command, integrity constraints will be removed.	In the TRUNCATE command, integrity constraints will not be removed.
In the DROP command, INDEX associated to the table will be removed.	In the TRUNCATE command, INDEX associated to the table will not be removed.
In the DROP command, TRIGGER associated to the table will execute.	In the TRUNCATE command, TRIGGER associated to the table will not execute.
In the DROP command, TRIGGER associated to the table will be removed.	In the TRUNCATE command, TRIGGER associated to the table will be not removed.

rename table

Change the table name.

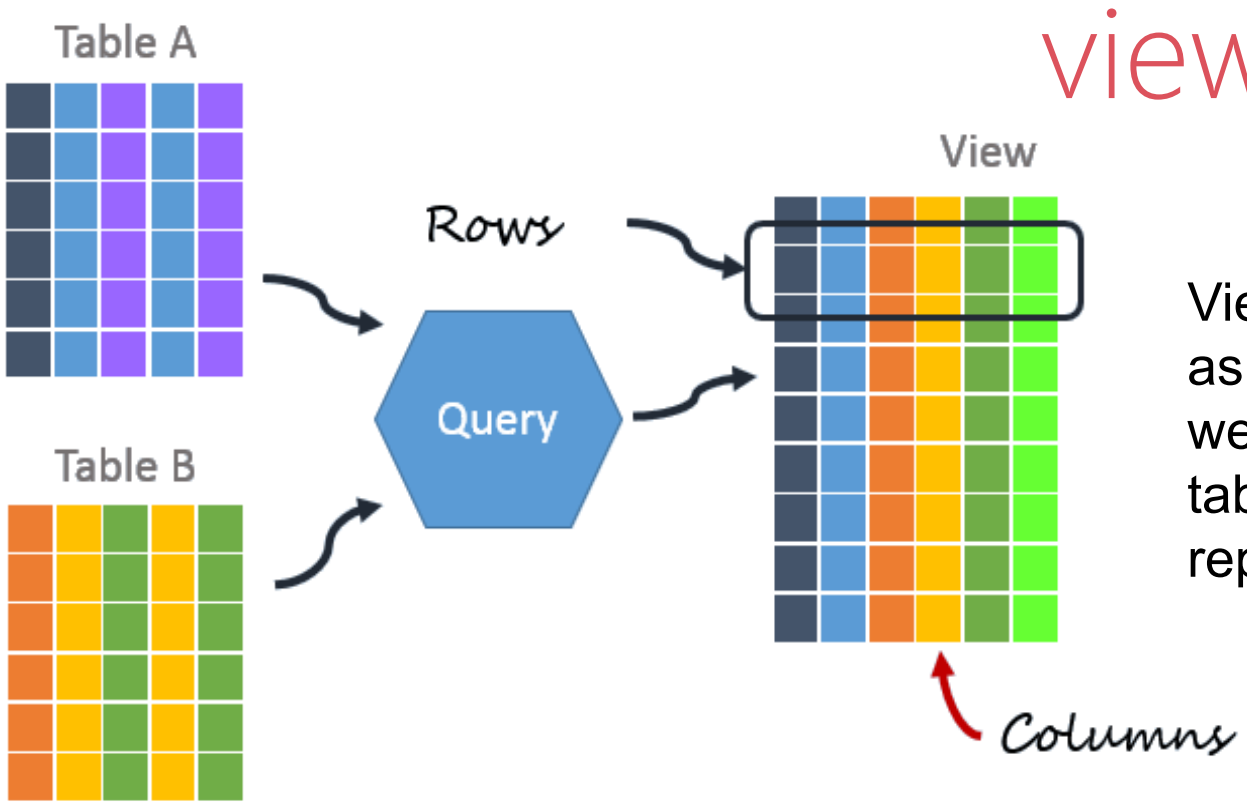
RENAME TABLE old_tbl_name TO new_tbl_name

- RENAME TABLE emp TO employee;

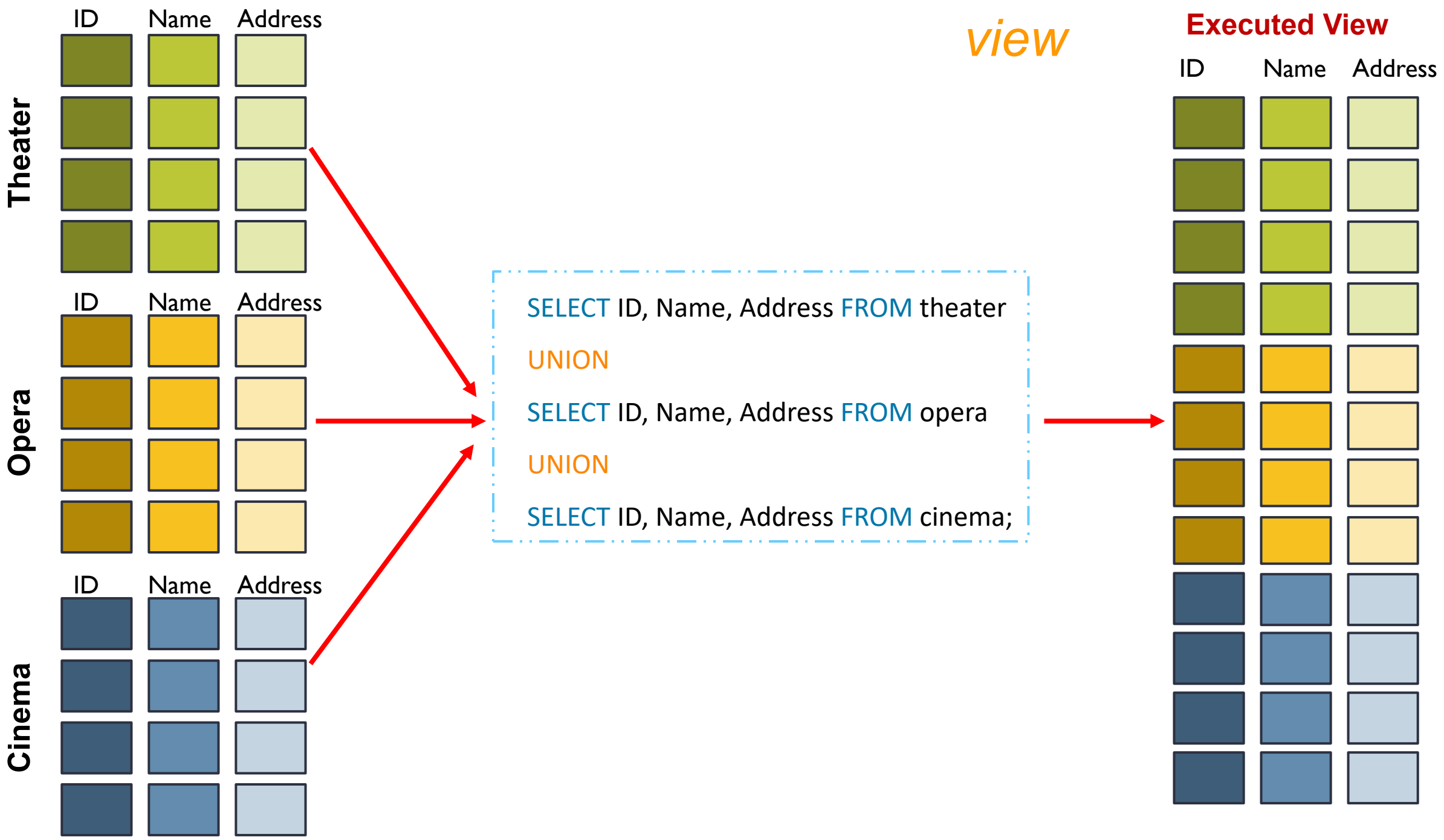
A ***VIEW*** in SQL as a logical subset of data from one or more tables. Views are used to restrict data access. A ***VIEW*** contains no data of its own but its like window through which data from tables can be viewed or changed. The table on which a View is based are called BASE Tables.

There are 2 types of Views in SQL:

- **Simple View** : Simple views can only contain a single base table.
- **Complex View** : Complex views can be constructed on more than one base table. In particular, complex views can contain: join conditions, a group by clause, a order by clause.



Views are created in order to store your queries as virtual table. If there are a lot of columns and we don't want to use all columns and make the table simpler, we can create a view and reuse it repeatedly.


ID
Name
Address
Theater
ID
Name
Address
Opera
ID
Name
Address
Cinema
view
SELECT ID, Name, Address FROM theater
UNION
SELECT ID, Name, Address FROM opera
UNION
SELECT ID, Name, Address FROM cinema;
Executed View
ID
Name
Address

Remember:

view

- it can be described as a **virtual/derived** table which derived its data from one or more than one base table.
- View names may appear in a query in any place where a relation name may appear.
- it is stored in the database.
- it can be created using tables of same database or different database.
- it is used to implement the security mechanism in the SQL.
- It can have max 64 char (view name).

Rules:

- If a VIEW is defined as SELECT * on a table, new columns added to the base table later do not become part of the VIEW, and columns dropped from the base table will result in an error when selecting from the VIEW.
- A VIEW must have unique column names with no duplicates, just like a base table. By default, the names of the columns retrieved by the SELECT statement are used for the VIEW column names.
- The VIEW definition cannot refer to a TEMPORARY table, and you cannot create a TEMPORARY VIEW.
- You cannot associate a TRIGGER with a VIEW.

Note:

- If we drop the BASE TABLE, the VIEW will not be dropped.
- A VIEW definition (structure) is not permanently stored as part of the database.

- DESC dept;
- CREATE VIEW v1 AS SELECT * FROM dept;
- DROP TABLE dept;
- DESC v1;

Try this

Uses of a View: A good database should contain views due to the following reasons:

- **Restricting data access –** Views provide an additional level of table security by restricting access to a predetermined set of rows and columns of a table.
- **Hiding data complexity –** A view can hide the complexity that exists in multiple tables join.
- **Simplify commands for the user –** Views allow the user to select information from multiple tables without requiring the users to actually know how to perform a join.
- **Store complex queries –** Views can be used to store complex queries.
- **Rename Columns –** Views can also be used to rename the columns without affecting the base tables provided the number of columns in view must match the number of columns specified in select statement. Thus, renaming helps to hide the names of the columns of the base tables.
- **Multiple view facility –** Different views can be created on the same table for different users.

Views are not updatable in the following cases:

- A table in the FROM clause is reference by a subquery in the WHERE statement.
- There is a subquery in the SELECT clause.
- The SQL statement defining the view joins tables.
- One of the tables in the FROM clause is a non-updatable view.
- The SELECT statement of the view contains an aggregate function such as SUM(), COUNT(), MAX(), MIN(), and so on.
- The keywords DISTINCT, GROUP BY, HAVING clause, LIMIT clause, UNION, or UNION ALL appear in the defining SQL statement.

difference between simple and complex view

SIMPLE VIEW	COMPLEX VIEW
Contains only one base table or is created from only one table, it may include WHERE clause and ORDER BY clause.	Contains more than one base tables or is created from more than one tables.
We cannot use group functions like MAX(), COUNT(), etc.	We can use group functions.
Does not contain groups of data.	It can conatin groups of data.
DML operations could be performed through a simple view.	DML operations could not always be performed through a complex view.
INSERT, DELETE and UPDATE are directly possible on a simple view.	We cannot apply INSERT, DELETE and UPDATE on complex view directly.
Simple view does not contain group by, , having clause, limit clause, distinct, pseudo column like rownum, columns defined by expressions.	It can contain group by, having clause, limit clause, distinct, pseudocolumn like rownum, columns defined by expressions.
Does not include NOT NULL columns from base tables.	NOT NULL columns that are not selected by simple view can be included in complex view.

create view

The select_statement is a SELECT statement that provides the definition of the view. The select_statement can select from base tables or other views.

CREATE [OR REPLACE] VIEW view_name [(*column_list*)]
as *select_statement* [WITH CHECK OPTION]

UPDATE AND DELETE on VIEW (with check option given on view) will work only when the DATA MATCHES IN WHERE CLAUSE.

- CREATE VIEW v1 as SELECT * FROM dept;
- CREATE VIEW v1(A_1, A_2) as SELECT deptno, dname FROM dept;
- CREATE or REPLACE VIEW v1 as SELECT * FROM dept WITH CHECK OPTION;

desc INFORMATION_SCHEMA.VIEWS;

licence table

create view

ID	customerID	licenceClass	licenceType	licenceNumber	ValidFrom	ValidTo	agentID
1	1	LMV	Light motor vehicles including motorcars, jeeps, taxis, delivery vans	RQFY8GZNDF9E	********	********	1
2	2	LMV	Light motor vehicles including motorcars, jeeps, taxis, delivery vans	ZFVHYH4OAHGS	********	********	2
3	1	HMV	Heavy Motor Vehicles	20CMZQ6ERXJ4	********	********	3
4	2	HMV	Heavy Motor Vehicles	DNIE1K9R5BG6	********	********	1
5	3	LMV	Light motor vehicles including motorcars, jeeps, taxis, delivery vans	Y3CNWQ6MKPCG	********	********	2
6	3	HMV	Heavy Motor Vehicles	Y6DC2IRD2EWZ	********	********	1
7	4	HGMV	Heavy Goods Motor Vehicle	2EFFSXY04OAW	********	********	3
8	5	HMV	Heavy Motor Vehicles	GC1BMWKCTVQ4	********	********	2
9	6	LMV	Light motor vehicles including motorcars, jeeps, taxis, delivery vans	2148PC50IR3P	********	********	1
10	7	HPMV	Heavy passenger motor vehicle/Heavy transport vehicle	N4NODNCFJENP	********	********	2
11	8	HGMV	Heavy Goods Motor Vehicle	EBPW9M34FKOI	********	********	2
12	9	HPMV	Heavy passenger motor vehicle/Heavy transport vehicle	940MGEL4HO00	********	********	3
13	10	LMV	Light motor vehicles including motorcars, jeeps, taxis, delivery vans	JWRDAETHDU5Z	********	********	1
14	11	HGMV	Heavy Goods Motor Vehicle	Z95XCS7W9NAU	********	********	2
15	12	HGMV	Heavy Goods Motor Vehicle	1YJ12NQX8HAJ	********	********	1
16	13	LMV	Light motor vehicles including motorcars, jeeps, taxis, delivery vans	1XMTETH8E6JS	********	********	1
17	14	HPMV	Heavy passenger motor vehicle/Heavy transport vehicle	0WEMMTPVTBPM	********	********	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

all are Simple View

- CREATE or REPLACE VIEW agent1_view AS SELECT * FROM licence WHERE agentID = 1 ORDER BY validTo;
- CREATE or REPLACE VIEW agent2_view AS SELECT * FROM licence WHERE agentID = 2;
- CREATE or REPLACE VIEW agent3_view AS SELECT * FROM licence WHERE agentID = 3;

create user and grant/revoke privileges

create user

TODO

```
CREATE USER [IF NOT EXISTS] 'user_name' @'localhost' IDENTIFIED BY 'password'
SET PASSWORD FOR  'user_name'@'localhost' = 'auth_string'
DROP USER [IF EXISTS] 'user_name' @'localhost'
```

- CREATE USER 'saleel'@'localhost' IDENTIFIED BY 'saleel';
- SET PASSWORD FOR 'saleel'@'localhost' = '*sharmin*';
- DROP USER 'saleel'@'localhost';

grant/revoke privileges

TODO

GRANT *priv_type* ON *object_type* TO '*user_name*'@'*localhost*'

REVOKE *priv_type* ON *object_type* FROM '*user_name*'@'*localhost*'

Privilege	Privilege
ALL [PRIVILEGES]	SELECT
CREATE	INSERT
ALTER	UPDATE
DROP	DELETE
EXECUTE	

- GRANT ALL PRIVILEGES ON db1.* TO 'saleel'@'localhost';
- GRANT ALL PRIVILEGES ON *.* TO 'saleel'@'localhost';

- GRANT INSERT, UPDATE ON emp TO 'saleel';
- REVOKE INSERT, UPDATE ON emp FROM 'saleel';

Try Out

create view

```
CREATE TABLE customer (
    ID INT PRIMARY KEY ,
    name VARCHAR(50),
    description TEXT
);

CREATE TABLE orders (
     orderID INT ,
     customerID INT,
     orderDate DATE,
     amount INT,
     constraint fk_customerID FOREIGN KEY(customerID) REFERENCES customer(ID));


INSERT INTO customer VALUES (1, 'saleel', 'description Text 1'), (2, 'vrushali', 'description Text 2'), (3, 'sharmin',  'description Text 3'), (4, 'ruhan', 'description Text 4');

INSERT INTO orders VALUES(1, 1, NOW(), 3000), (2, 1, NOW(), 1000), (3, 2, NOW(), 100), (4, 2, NOW(), 300), (5, 2, NOW(), 200);
```

Try Out

create view

- CREATE or REPLACE VIEW v1 AS SELECT * FROM customer, orders WHERE ID = customerID;
- CREATE or REPLACE VIEW v2 AS SELECT customer.* FROM customer, orders WHERE ID = customerID;
- CREATE or REPLACE VIEW v3 AS SELECT orders.* FROM customer, orders WHERE ID = customerID;

- INSERT INTO v1(ID, name, description, orderID, customerID, orderdate, amount) VALUES (10,'S10',"D1", 10, 1, NOW(), 111); // Error: Can not modify more than one base table through a join view 'db5.v1'

// We may get Error: Can not insert into join view 'db5.v2' without fields list.

- INSERT INTO v1(ID, name, description) VALUES(1, 'neel', 'description Text 5');
- INSERT INTO v1(orderID, customerID, orderDate, amount) VALUES(1, 1, NOW(), 7000);
- INSERT INTO v2(ID, name, description) VALUES(5, 'neel', 'description Text 5');
- INSERT INTO v3(orderID, customerID, orderdate, amount) VALUES(6, 1, NOW(), 200);

SHOW CREATE VIEW Syntax

```
SHOW CREATE VIEW view_name
```

```
show create VIEW v1;
```

```
SHOW [FULL] TABLES [ { FROM | IN } db_name ]
    [ LIKE 'pattern' | WHERE expr ]
```

```
show full tables where table_type like 'VIEW';
```

alter / drop view

This statement changes the definition of a view, which must exist.

```
ALTER VIEW view_name [(column_list)]
    AS select_statement
    [WITH CHECK OPTION]
```

e.g.

- ALTER VIEW studentview AS SELECT namefirst, namelast, emailid FROM student;

DROP VIEW removes one or more views.

```
DROP VIEW [IF EXISTS]
   view_name [, view_name,  . . . ]
```

e.g.

- DROP VIEW studentview;
- DROP VIEW studentid10view, studentviewwithcheck;
- DROP VIEW studentTotalMarksView, studentAddressView;

MySQL indexes (PRIMARY KEY, UNIQUE, INDEX, and FULLTEXT) are stored in B-trees.

index

- it is a schema object.
- it is used by the server to speed up the retrieval of rows.
- it reduces disk I/O(input/output) process.
- it helps to speed up select queries where clauses, but it slows down data input, with the update and the insert statements.
- it can be created or dropped with no effect on the data.
- multiple columns index may consist of up to 16 columns.

index

Indexes are used to find rows with specific column values quickly. Without an index, MySQL must begin with the first row and then read through the entire table to find the relevant rows. If the table has an index for the columns in question, MySQL can quickly determine the position (**from the index file**) to seek to in the middle of the data file without having to look at all the data. This is much faster than reading every row sequentially.

- To find the rows matching a WHERE clause quickly.
- If the table has a multiple-column index, any leftmost prefix of the index can be used by the optimizer to look up rows. For example, if you have a three-column index on (col1, col2, col3), you have indexed search capabilities on (col1), (col1, col2), and (col1, col2, col3).

Note:

- Index name can have max 64 char.
- It is not possible to create an INDEX on a VIEW.
- If we drop the BASE TABLE, the INDEX will be dropped automatically.

create index

Indexes are used to find rows with specific column values quickly.

```
CREATE [UNIQUE] INDEX index_name
   ON tbl_name (index_col_name, . . .)  { visible   | invisible }
```

e.g.

- CREATE INDEX indexOnName ON emp(ename);
- CREATE INDEX indexOnUniversity ON student_qualifications(university);
- CREATE UNIQUE INDEX uniqueIndexOnName ON emp(ename);

- CREATE INDEX ind1 ON emp(job) VISIBLE;
- CREATE INDEX ind1 ON emp(job) INVISIBLE;

- ALTER TABLE emp ALTER INDEX ind1 VISIBLE;
- ALTER TABLE emp ALTER INDEX ind1 VISIBLE;

clustered and non-clustered index

Indexing in MySQL is a process that helps us to return the requested data from the table very fast. If the table does not have an index, it scans the whole table for the requested data.

MySQL allows two different types of Indexing:

- Clustered Index
- Non-Clustered Index

Clustered Index:-

- Clustered index is used to optimize the speed of most common lookups and DML operations like INSERT, UPDATE, and DELETE command.
- Clustered indexes sort and store the data rows in the table based on their key (primary key) values that can be sorted in only one direction.
- If the table column contains a **primary key** or **not null** and **unique key**, MySQL creates a clustered index.
- Data retrieval is faster than non-cluster index.
- Slower inserts/updates if the indexed column values change frequently, as it may require rearranging rows to maintain the order.

clustered and non-clustered index

Indexing in MySQL is a process that helps us to return the requested data from the table very fast. If the table does not have an index, it scans the whole table for the requested data.

MySQL allows two different types of Indexing:

- Clustered Index
- Non-Clustered Index

Non-Clustered Index:-

- The indexes other than PRIMARY indexes (clustered indexes) called a non-clustered index.
- The non-clustered indexes are also known as secondary indexes.
- A secondary index may be created on one or more virtual columns or on a combination of virtual columns and regular columns.
- The non-clustered index and base table data are both stored in different places, so it does not affect the physical row order in the base table.
- It is not sorted (ordering) the table data.
- Extra space is required to store logical structure
- Data update is faster than clustered index

SHOW INDEX Syntax

To get the index of a table, you specify the table name after the FROM keyword. The statement will return the index information associated with the table in the current database.

```
SHOW { INDEX | INDEXES | KEYS }
    { FROM | IN } tbl_name
    [{ FROM | IN } db_name]
    [WHERE expr]
```

e.g.

- SHOW INDEX FROM emp;
- SHOW INDEX FROM student_qualifications;

SHOW INDEX returns table index information.

drop index

DROP INDEX drops the index named index_name from the table tbl_name.

```
DROP INDEX index_name ON tbl_name
```

e.g.

- DROP INDEX indexOnName ON emp;
- DROP INDEX indexOnUniversity ON student_qualifications;
- DROP INDEX uniqueIndexOnName ON emp;

- UPDATE j SET doc = JSON_SET(doc, '$.ename', 'sharmin');

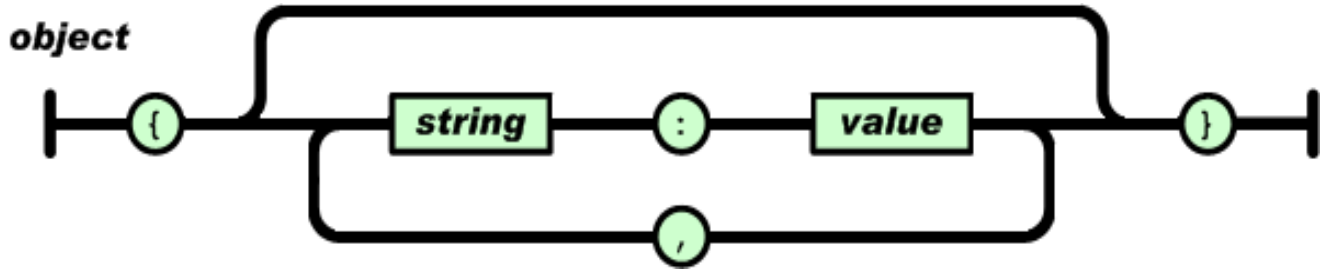
json

datatype – json

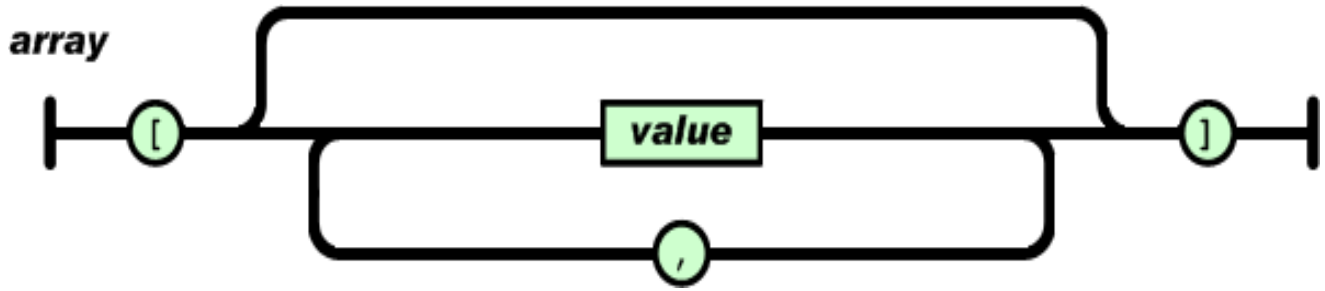
JSON (JavaScript Object Notation) documents.

- Data is in name/value pairs
- Data is separated by comm
- Curly braces hold objects
- Square brackets hold arrays

An object is an unordered set of name/value pairs.



An array is an ordered collection of values.



dml- insert json object… values

INSERT inserts new rows into an existing table. The INSERT ... VALUES

```
JSON_OBJECT("key1", 1, "key2", "abc" , "key3", JSON_ARRAY( value1, "value2" , . . . ) , . . . );
'{ "key1" : 1, "key2" : "abc" , "key3" : [ value1, "value2" , . . . ] , . . . }'
```

```
CREATE TABLE customer (
   ID INT PRIMARY KEY AUTO_INCREMENT,
   customer_details VARCHAR(1000),
   order_details JSON );
```

e.g.

- INSERT INTO customer VALUES(default,'{"CID": 1001,"name":"saleel", "city":"pune"}' , JSON_OBJECT("orderID", 1, "item", "computer", "qty", 1, "rate", 75000));
- INSERT INTO customer VALUES(default,'{"CID": 1002,"name":"saleel", "city": "pune", "phone":[11111, 22222] }' , JSON_OBJECT("orderID", 1, "item", "computer", "qty", 1, "rate", 75000));
- INSERT INTO customer VALUES(default,'{"CID": 1003,"name":"sharmin", "city":"pune"}' , JSON_OBJECT("orderID", 1, "item", JSON_ARRAY("maggi", "burger", "coffee"), "qty", JSON_ARRAY(4, 2, 2), "rate", JSON_ARRAY(45, 375, 125)));
- INSERT INTO customer VALUES(default, JSON_OBJECT("CID", 1004, "name", "ruhan", "city","Pune", "phone", JSON_ARRAY(1111, 2222)), JSON_ARRAY(JSON_OBJECT("orderID", 1,"item", "mouse", "qty", 2, "rate", 1200), JSON_OBJECT("orderID", 2, "item", "pd", "qty", 1, "rate", 1600)));

json_extract

Returns data from a JSON document, selected from the parts of the document matched by the path arguments.

json_extract (select)

MySQL supports a native JSON data type that enables efficient access to data in JSON (JavaScript Object Notation) documents.

```
JSON_EXTRACT(json_doc, path[, path] . . .
```

- `SELECT JSON_EXTRACT(document, '$.*') FROM empj;`
- `SELECT JSON_EXTRACT(document, '$.empID') FROM empj;`
- `SELECT JSON_EXTRACT(document, '$.phone[1]') FROM empj;`
- `SELECT JSON_EXTRACT(document, "$.address.street") FROM empj;`
- `SELECT JSON_EXTRACT(document, "$.empID", "$.ename") FROM empj;`

-> / ->> … (select)

MySQL supports a native JSON data type that enables efficient access to data in JSON (JavaScript Object Notation) documents.

- SELECT id, document->'$.ename', document->'$.sal' FROM empJ;
- SELECT id, document->'$.ename', document->'$.sal', document->'$.comm' FROM empJ;
- SELECT id, document->'$.ename', document->'$.sal', document->'$.comm', document->'$.sal' + document->'$.comm' FROM empJ;
- SELECT id, document->'$.address.coord[0]' FROM empJ;
- SELECT id, document->'$.phone[0]' FROM empJ;
- SELECT author_details->>'$. authorName', book_details->'$[1].*' FROM author_book;
- SELECT id, document->"$.phone", JSON_LENGTH(document, "$.phone") FROM empJ;

json_pretty … (select)

Print a JSON document in human-readable format

- SELECT oid, JSON_PRETTY(orderheader) FROM orderj;
- SELECT oid, JSON_PRETTY(order_items) FROM orderitemsj;
- SELECT orderj.oid, JSON_PRETTY(orderheader), JSON_PRETTY(order_items) FROM orderj, orderitemsj WHERE orderj.oid = orderitemsj.oid;

Note:

- TODO

json functions

TODO

json function

Syntax	Result
->	Return value from JSON column after evaluating path; equivalent to JSON_EXTRACT().
->> / JSON_UNQUOTE(json_val)	Return value from JSON column after evaluating path and unquoting the result.
JSON_EXTRACT(json_doc, path[, path] . . .)	Returns data from a JSON document, selected from the parts of the document matched by the path arguments.
JSON_ARRAY([val[, val] . . .])	Evaluates a list of values and returns a JSON array containing those values.
JSON_OBJECT([key, val[, key, val] . . .])	Create JSON object.
JSON_INSERT(json_doc, path, val[, path, val] . . .)	INSERT a key, if not present [with UPDATE]
JSON_PRETTY(json_val)	Provides pretty-printing of JSON values

json function

Syntax	Result
JSON_REPLACE(json_doc, path, val[, path, val] . . .)	Replace values in JSON document [with UPDATE]
JSON_SET(json_doc, path, val[, path, val] . . .)	Insert data into JSON document [with UPDATE]
JSON_REMOVE(json_doc, path[, path] . . .)	Remove data from JSON document [with UPDATE]
JSON_KEYS(json_doc[, path])	Array of keys from JSON document
JSON_ARRAY_APPEND(json_doc, path, val[, path, val] . . .)	Append data to JSON document [with UPDATE]
JSON_LENGTH(json_doc[, path])	Number of elements in JSON array/document

dml- update/remove json object… values

Inserts or Updates data in a JSON document and returns the result.

JSON_SET(json_doc, path, value1[, path, value2] , . . .)

JSON_REMOVE(json_doc, path[, path] , . . .)

```
CREATE TABLE customer (
   ID INT PRIMARY KEY AUTO_INCREMENT,
   customer_details VARCHAR(1000),
   order_details JSON
);
```

e.g.

- `UPDATE customer SET order_details = JSON_SET(order_details, '$.state', 'MH');`
- `UPDATE customer SET order_details = JSON_SET(order_details, '$.city', 'baroda') WHERE id = 1;`
- `UPDATE customer SET order_details = JSON_REMOVE(order_details, "$. state") WHERE id = 1;`

e.g.

pk/fk in json

- CREATE TABLE orderJ(oid INT PRIMARY KEY,
  orderheader JSON);
- INSERT INTO orderJ VALUES(1, JSON_OBJECT('customerID', 1001, 'customerName', 'saleel', 'mobile', 9850884228, 'city', 'baroda'));
- INSERT INTO orderJ VALUES(2, JSON_OBJECT('customerID', 1002, 'customerName', 'sharmin', 'mobile', 9850884229, 'city', 'pune'));
- INSERT INTO orderJ VALUES(3, JSON_OBJECT('customerID', 1003, 'customerName', 'vrushali', 'mobile', 9850884230, 'city', 'pune'));
- INSERT INTO orderj VALUES(4, JSON_OBJECT('customerID', 1004, 'customerName' ,'ruhan', 'mobile', JSON_ARRAY(11111111, 22222222), 'city', 'pune'));

- CREATE TABLE orderItemsJ(oid INT,
  order_items JSON,
  FOREIGN KEY(oid) REFERENCES orderJ(oid));
- INSERT INTO orderItemsJ VALUES(1, JSON_OBJECT('itemID', 1001, 'itemName', 'butter', 'rate', 175, 'qty', 5));
- INSERT INTO orderItemsJ VALUES(1, JSON_OBJECT('itemID',1002, 'itemName', 'cheese', 'rate', 225, 'qty', 2));
- INSERT INTO orderItemsJ VALUES(2, JSON_ARRAY(JSON_OBJECT('itemID',1001, 'itemName', 'butter', 'rate', 175, 'qty', 2), JSON_OBJECT('itemID',1003, 'itemName', 'bread', 'rate', 45, 'qty', 2), JSON_OBJECT('itemID',1002, 'itemName', 'cheese', 'rate', 225, 'qty', 4)));

lock / unlock table

lock / unlock

MySQL enables client sessions to acquire table locks to prevent from modifying tables

Sessions holding a READ lock, DROP TABLE and TRUNCATE TABLE operations are not permitted.

```
LOCK TABLES tbl_name lock_type [, tbl_name]
lock_type] ...

lock_type:
   READ |WRITE

UNLOCK TABLES
```

- LOCK TABLE DEPT READ;
- LOCK TABLE DEPT WRITE;
- UNLOCK TABLE

select … into

select … into

An INTO clause should not be used in a nested SELECT because such a SELECT must return its result to the outer context.

The SELECT ... INTO form of SELECT enables a query result to be stored in variables or written to a file.

SELECT . . . INTO *var_list*

SELECT . . . INTO OUTFILE

SELECT . . . INTO DUMPFILE

- Selects column values and stores them into variables.
- Writes the selected rows to a file. Column and line terminators can be specified to produce a specific output format.
- Writes a single row to a file without any formatting.

select … into var_list

select … into var_list

The SELECT ... INTO form of SELECT enables a query result to be stored in variables or written to a file.

SELECT . . . INTO var_list

- The selected values are assigned to the variables.
- The number of variables must match the number of columns.
- The query should return a single row.
- If the query returns no rows, a warning with error code 1329 occurs (No data), and the variable values remain unchanged.
- If the query returns multiple rows, error 1172 occurs (Result consisted of more than one row).
- The statement may retrieve multiple rows, you can use LIMIT 1 to limit the result set to a single row.

select … into var_list

The SELECT ... INTO form of SELECT enables a query result to be stored in variables or written to a file.

SELECT . . . INTO var_list

- ```
  SET @x = 0;
  SET @y = null;
  SELECT empno, ename INTO @x, @y FROM emp WHERE empno = 7788;
  ```

- ```
  SET @x = 0;
  SET @y = null;
  SELECT MAX(sal), MIN(sal) INTO @x, @y FROM emp WHERE empno = 7788;
  ```

```
+-----------------------+-----------------------------------------------------------+
| variable_name         | value                                                     |
+-----------------------+-----------------------------------------------------------+
| secure_file_priv      | C:/ProgramData/MySQL/MySQL Server 8.0/Uploads  |
+-----------------------+-----------------------------------------------------------+
```

In this example, I can only read files from the / Uploads / directory.

select ... into outfile

If not working then do changes in ***my.ini*** file.

```
secure_file_priv = ""
SHOW VARIABLES LIKE "secure_file_priv";
```

select … into outfile

Writes the selected rows to a file. Column and line terminators can be specified to produce a specific output format.

SELECT . . . INTO OUTFILE

- SELECT identifier.* FROM emp INTO OUTFILE "/tmp/emp.csv" ;
- SELECT identifier.* FROM emp INTO OUTFILE "/tmp/emp.txt" ;
- SELECT identifier.* FROM emp INTO OUTFILE "/tmp/emp.csv" FIELDS TERMINATED BY ',';
- SELECT identifier.* FROM emp INTO OUTFILE "/tmp/emp.csv" FIELDS TERMINATED BY ',';
- SELECT identifier.* FROM emp INTO OUTFILE "/tmp/emp.csv" FIELDS TERMINATED BY ',' LINES TERMINATED by '\n';
- SELECT identifier.* FROM emp INTO OUTFILE "/tmp/emp.csv" FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"' LINES TERMINATED BY '\n';
- SELECT "DEPT", "DNAME", "LOC", "PWD", "STARTEDON" UNION SELECT * FROM dept;

- TABLE emp INTO OUTFILE "/tmp/emp.csv";
- TABLE emp INTO OUTFILE "/tmp/emp.txt";

select … into dumpfile

select … into dumpfile

If you use INTO DUMPFILE instead of INTO OUTFILE, MySQL writes only one row into the file, without any column or line termination and without performing any escape processing. This is useful if you want to store a BLOB value in a file.

SELECT . . . INTO DUMPFILE

- `SELECT identifier.* FROM emp WHERE empno = 7788 INTO DUMPFILE "/tmp/emp.txt";`

- mysqldump -uroot -p db1 >> d:\bk.sql
- mysqldump -u[username] -p[password] --all-databases > d:\db_backup.sql
- mysqldump -P 3306 -h 192.168.100.26 -usaleel -psaleel **db_name** > db_backup.sql
- mysqldump -P 3306 -h 192.168.100.26 -usaleel -psaleel **db_name table_name** > db_backup.sql
- mysqldump -P 3306 -h 192.168.100.26 -usaleel -psaleel **db_name table_name --WHERE ="deptno=10"** > db_backup.sql
- mysqldump -P 3306 -h 192.168.100.26 -usaleel -psaleel **db_name table_name --WHERE ="job='manager'"** > db_backup.sql
- mysqldump --column-statistics=0 -P 3306 -h 192.168.100.26 -usaleel -psaleel db_name > db_backup.sql
- mysql -P 3306 –h192.168.100.74 -uroot -proot db2 < d:\ backup_fileName.sql
- mysql -P 3306 –h192.168.100.74 -uroot -proot < d:\ backup_fileName.sql db2

---

snapshots

In MySQL, a "snapshot" refers to a point-in-time copy of a database or a dataset within it. This copy captures the state of the data at a specific moment, preserving it for various purposes, primarily related to backup and recovery.

```
CountryName,CapitalName,CapitalLatitude,CapitalLongitude,CountryCode,ContinentName,remark
Somaliland,Hargeisa,9.55,44.05,NULL,Africa,
South Georgia and South Sandwich Islands,King Edward Point,-54.283333,-36.5,GS,Antarctica,
French Southern and Antarctic Lands,Port-aux-Français,-49.35,70.216667,TF,Antarctica,
Palestine,Jerusalem,31.76666667,35.233333,PS,Asia,
Aland Islands,Mariehamn,60.116667,19.9,AX,Europe,
Nauru,Yaren,-0.5477,166.920867,NR,Australia,
Saint Martin,Marigot,18.0731,-63.0822,MF,North America,
Tokelau,Atafu,-9.166667,-171.833333,TK,Australia,
Western Sahara,El-Aaiún,27.153611,-13.203333,EH,Africa,
```

import .csv / .tsv file

```
CREATE TABLE countries (
CountryName VARCHAR(45),
CapitalName VARCHAR(45),
CapitalLatitude VARCHAR(45),
CapitalLongitude VARCHAR(45),
CountryCode VARCHAR(45),
ContinentName VARCHAR(45),
remark VARCHAR(45)
);
```

```
LOAD DATA INFILE 'path/file-name.csv' INTO TABLE countries
FIELDS TERMINATED BY ','  or FIELDS TERMINATED BY '\t'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```

- ```
  SELECT  "DEPT", "DNAME", "LOC", "PWD", "STARTEDON" UNION
  SELECT * FROM dept;
  ```
  ```

Import .CSV file with dateFormat

```
moduleID,moduleName,duration,startedON
1,OracleCR,6 months,23-10-2022
2,PIG,2 months,12-10-2022
3,Neo4j,4 months,20-10-2022
4,Cassandra,7 months,03-10-2022
5,HIVE,3 months,09-10-2022
6,MongoDB,2 months,25-10-2022
7,Redis,1 months,06-10-2022
```

```
CREATE TABLE module (
moduleID INT,
moduleName VARCHAR(45),
duration VARCHAR(45),
startedON DATE
);
```

```
LOAD DATA INFILE 'path/file-name.csv' INTO TABLE module
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(moduleID,modulename,moduleDuration,@startedOn)
SET startedOn = STR_TO_DATE(@startedOn,'%d-%m-%Y');
```

INSERT an Entire CSV Without Cleaning Data

Anti-pattern

- Here, LOAD DATA INFILE 'path/file-name.csv' INTO TABLE countries
  FIELDS TERMINATED BY ','

Why it's bad:

- blank rows, invalid emails, wrong phone numbers, . . .

Better approach:

- Here, Clean/validate before inserting.

Product.JSON file

{"productId": "P1004","name": "LED Smart TV 50-inch","description": "4K Ultra HD, HDR, WiFi enabled.","category": "Electronics","brand":"VisionPro","price": { "currency": "USD", "amount": 499.99 },"stock": { "available": false, "quantity": 0 },"specifications": { "resolution": "4K", "screenSize": "50 inch", "hdmiPorts": 3 }}
{"productId": "P1005","name": "Wooden Coffee Table","description": "Classic design with oak finish.","category": "Furniture","brand": "HomeCraft","price": { "currency": "USD", "amount": 149.99 },"stock": { "available": true, "quantity": 35 },"specifications": { "material": "Oak Wood", "dimensions": "120x60x45 cm" }}
{"productId": "P1006","name": "Stainless Steel Water Bottle","description": "Insulated bottle keeps drinks hot/cold for 12 hours.","category": "Home & Kitchen","brand": "HydraCool","price": { "currency": "USD", "amount": 19.99 },"stock": { "available": true, "quantity": 500 },"specifications": { "capacity": "750ml", "color": "Silver" }}

import .JSON file

```
CREATE TABLE product (
    id INT PRIMARY KEY AUTO_INCREMENT,
    document JSON
);
```

```
LOAD DATA INFILE 'path/file-name.json' INTO TABLE product
(@json)
SET document = @json;
```

cluster

A cluster comprises multiple interconnected computers or servers that appear as if they are one server to end users and applications.

data dictionary

A data dictionary contains metadata i.e data about the database. The data dictionary is very important as it contains information such as what is in the database, who is allowed to access it, where is the database physically stored etc. The users of the database normally don't interact with the data dictionary, it is only handled by the database administrators.

The data dictionary in general contains information about the following −

- Names of all the database tables and their schemas.
- Details about all the tables in the database, such as their owners, their security constraints, when they were created etc.
- Physical information about the tables such as where they are stored and how.
- Table constraints such as primary key attributes, foreign key information etc.
- Information about the database views that are visible.

data dictionary - information_schema

Data Dictionary
INFORMATION_SCHEMA.COLUMNS
INFORMATION_SCHEMA.TABLES
INFORMATION_SCHEMA.TABLE_CONSTRAINTS
INFORMATION_SCHEMA.STATISTICS
INFORMATION_SCHEMA.KEY_COLUMN_USAGE
INFORMATION_SCHEMA.ROUTINES
INFORMATION_SCHEMA.PARAMETERS
INFORMATION_SCHEMA.TRIGGERS